Structures and Functions of the Periodontium - Part II

Structure • epidemiology • microbiology • biofilm • pathogenesis

University of Pikeville Tanner College of Dental Medicine
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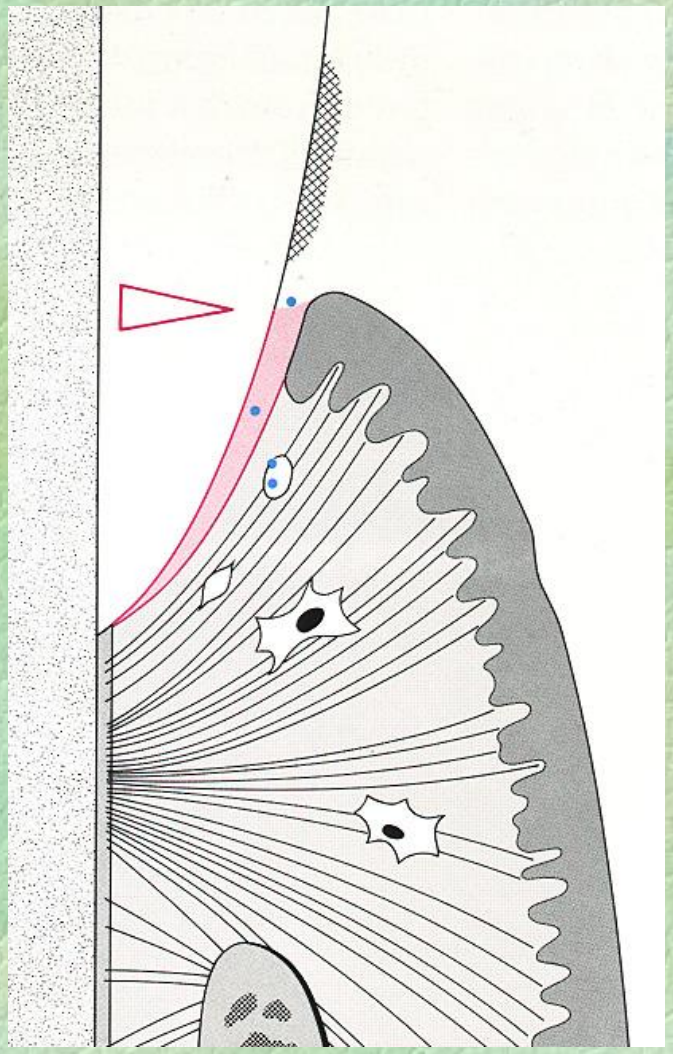
DEN 624 • Periodontology I • Tanner College of Dental Medicine

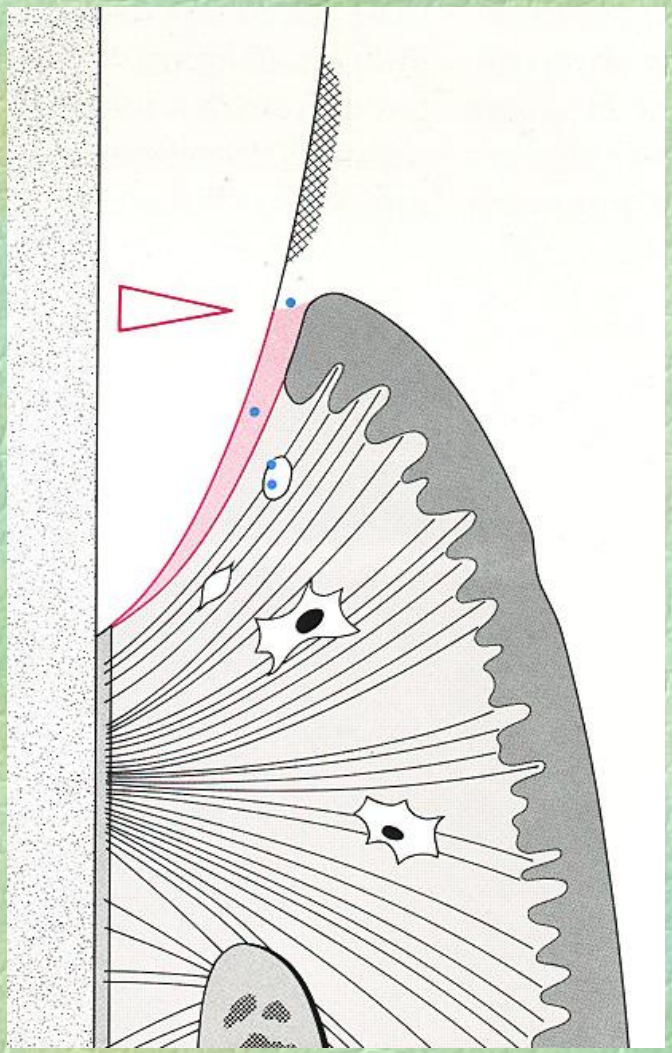
Healthy Gingiva



- Healthy Gingiva
- Absence of plaque or very little
- Normal JE (pink)
- Minimal sulcus depth (red arrow).

Healthy Gingiva

- Few PMNs (blue dots) transmigrate from the JE in the direction of the sulcus bottom.
- Dense collagenous fiber system; intact fibroblast



Healthy Gingiva

- Microflora in health is simple and sparse
- Most bacteria are gram positive and facultative anaerobic rods tend to be filamentous forms such as actinomyces
- Approximately 2/3 are cocci morphotypes

Disease Classification:

- The Old 1999' classification system and nomenclature used in the November 1999 *Annuls of Periodontolgy*
- The National Board will NOT use this classification system!
- You will **NOT** see case types I, II, III, or IV, you will see the categories described in the following slides

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1999 and 2017 Periodontal Classification

AAP classification compared with the 2017 World Workshop framework

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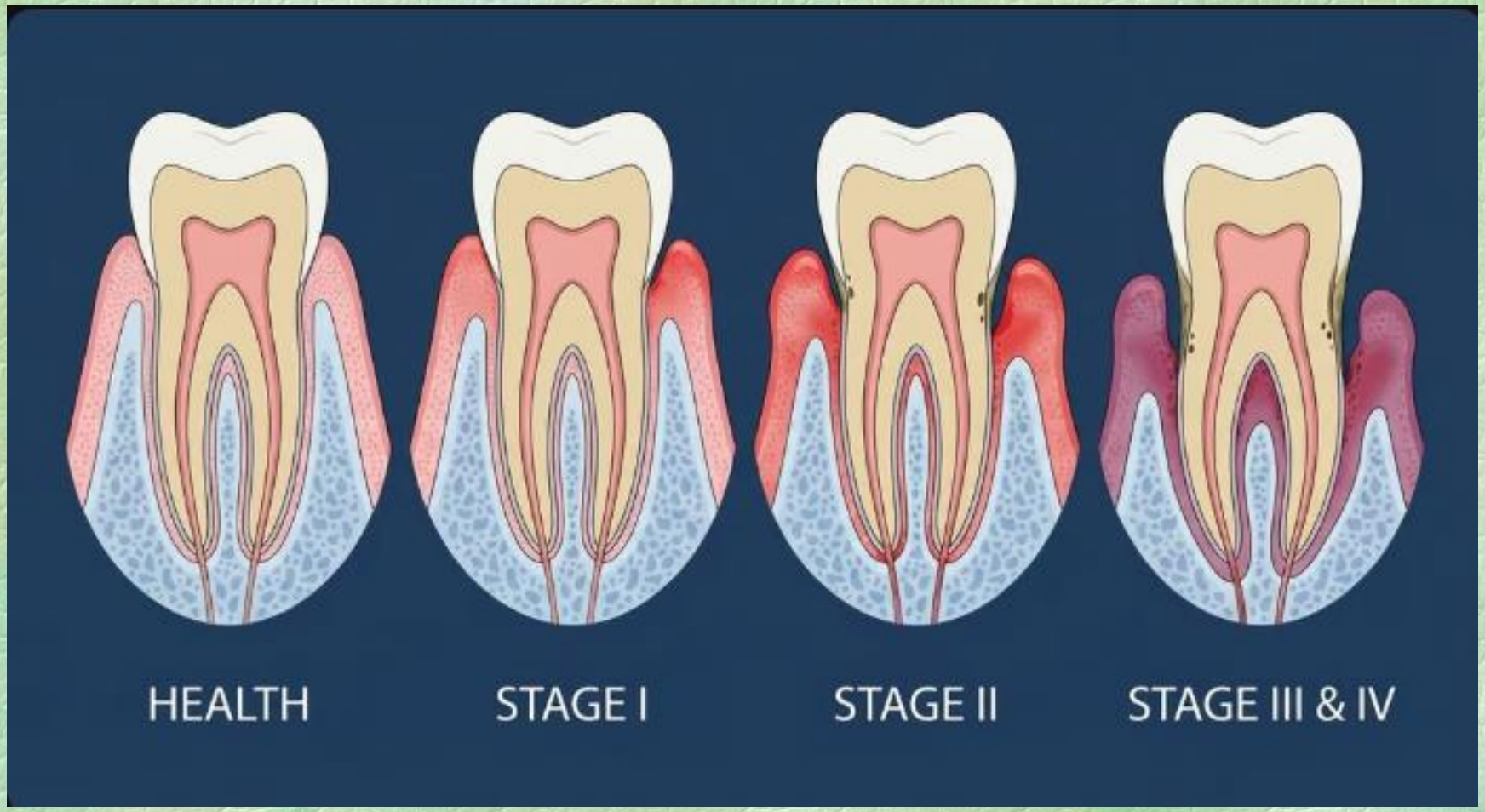
https://www.periospot.com/blog/the-2018-aapefp-periodontal-classification-a-clinicians-complete-guide-to-staging-and-grading



https://www.periospot.com/blog/the-2018-aapefp-periodontal-classification-a-clinicians-complete-guide-to-staging-and-grading

Staging and Grading Periodontitis

Periodontitis: STAGING

	Periodontitis	Stage I	Stage II	Stage III	Stage IV
Severity	**Interdental CAL** (at site of greatest loss)	1-2mm	3-4mm	≥ 5mm	≥ 5mm
	RBL (radiographic bone loss)	Coronal third (<15%)	Coronal third (15%-33%)	Extending to middle third of root and beyond	Extending to middle third of root and beyond
	Tooth loss (due to periodontitis)	No tooth loss	No tooth loss	≤ 4 teeth	≤ 5 teeth
Complexity	**Local**	- Maximum probing depth ≤ 4mm - Mostly horizontal bone loss	- Maximum probing depth ≤ 5mm - Mostly horizontal bone loss	In addition to Stage II complexity: - Probing depths ≥ 6mm - Vertical bone loss ≥ 3mm - Furcation involvement Class II or III - Moderate ridge defects	In addition to Stage III complexity: - Need complex rehabilitation due to: - Masticatory dysfunction - Secondary occlusal trauma (tooth mobility degree ≥ 2) - Severe ridge defects - Bite collapse, drifting, flaring - < 20 remaining teeth (10 opposing pairs)
Extent and Distribution	Add to stage as descriptor	For each stage, describe extent as:	Localized (< 30% of teeth involved); Generalized; or Molar/incisor pattern		

Source: American Academy of Periodontology – 2018

Periodontitis: GRADING

	Progression		Grade A: Slow rate	Grade B: Moderate rate	Grade C: Rapid rate
Primary criteria Whenever available, direct evidence should be used.	Direct evidence of progression	Radiographic bone loss or CAL	No loss over 5 years	< 2mm over 5 years	≥ 2mm over 5 years
	Indirect evidence of progression	% bone loss/age	< 0.25	0.25 to 1.0	> 1.0
		Case phenotype	Heavy biofilm deposits with low levels of destruction	Destruction commensurate with biofilm deposits	Destruction exceeds expectations given biofilm deposits; specific clinical patterns suggestive of periods of rapid progression and/or early onset disease
Grade modifiers	Risk factors	Smoking	Non-smoker	< 10 cigarettes/day	≥ 10 cigarettes/day
		Diabetes	Normoglycemic/no diagnosis of diabetes	HbA1c < 7.0% in patients with diabetes	HbA1c ≥ 7.0% in patients with diabetes

Source: American Academy of Periodontology – 2018

The four stages of periodontitis based on severity and complexity.

This is the patient who shows up with a few 4 mm pockets, some bleeding on probing, but nothing scary on the radiographs. You can manage this without referral. Scaling, root planing, good home care, and they're usually golden

- **Stage I: Initial Periodontitis**
  - This is where you catch it early—

if you're paying attention during perio charting.

- **The defining features:**
  - Clinical attachment loss (CAL): 1-2 mm
  - Radiographic bone loss: Coronal third of the root (<15%)
  - Probing depths: Maximum 4 mm
  - No tooth loss due to periodontitis

The four stages of periodontitis based on severity and complexity.

The bone loss is more visible now. Still manageable with non-surgical therapy in most cases, but you're watching these patients more closely. This is the stage where good perio charting habits pay off—catch the progression early, intervene appropriately.

- **Stage II Moderate Periodontitis**
  - Now we're getting into territory where you need to be more careful.
- **The defining features:**
  - CAL: 3-4 mm
  - Radiographic bone loss: Coronal third (15-33%)
  - Probing depths: Maximum 5 mm
  - No tooth loss due to periodontitis

The four stages of periodontitis based on severity and complexity.

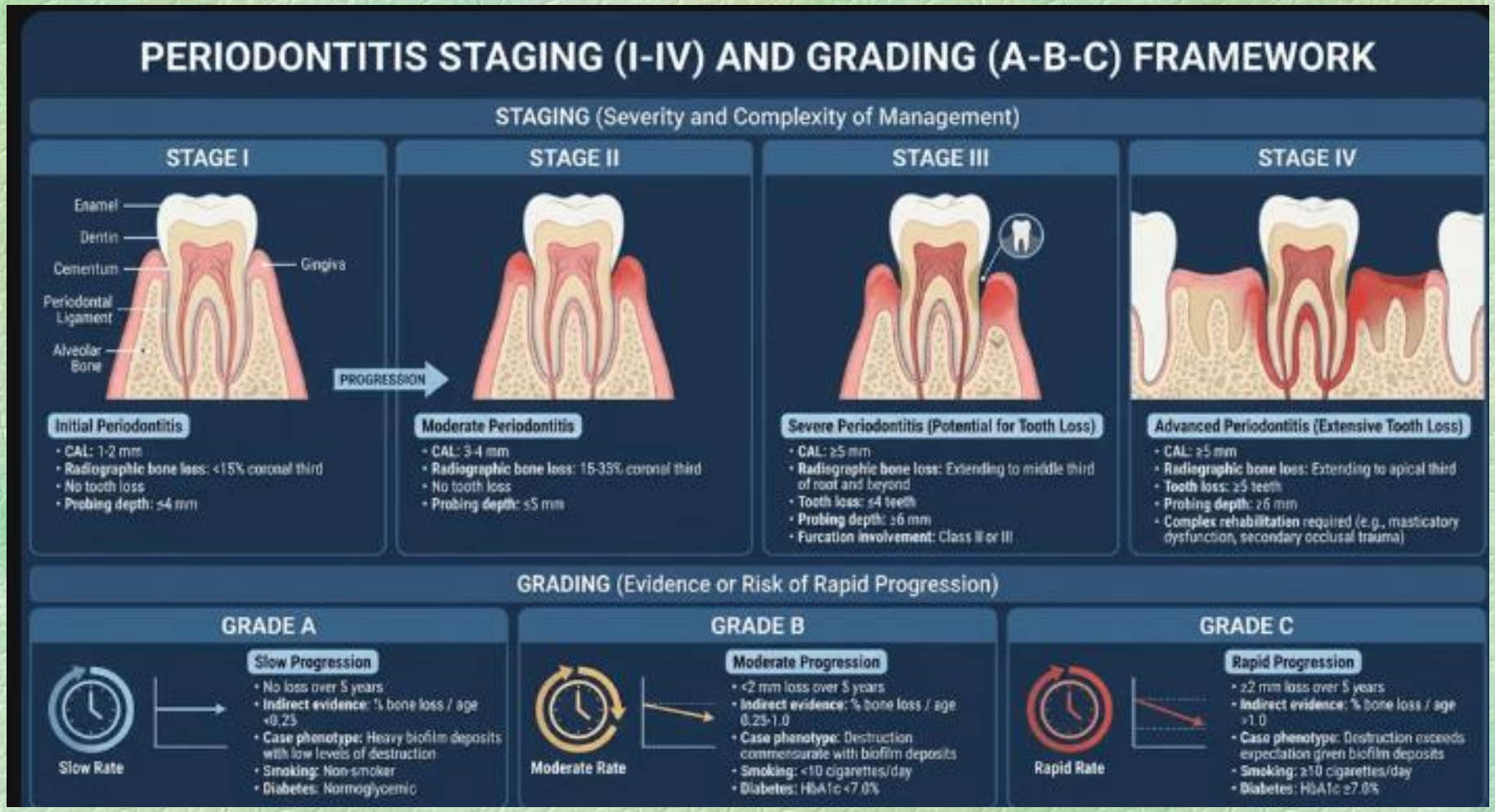
- **Stage III: Complexity factors that can bump :**
  - Vertical bone loss ≥3 mm
  - Furcation involvement Class II or III
  - Moderate ridge defects
- This patient needs more than scaling. You're looking at surgical intervention, regenerative procedures, potential extractions. If you're a general dentist, this is where you seriously consider referral.

The four stages of periodontitis based on severity and complexity.

- **Stage IV: Advanced Periodontitis with Extensive Tooth Loss**
  - The bad news stage.
- **The defining features:**
  - Everything from Stage III, plus:
  - Tooth loss: ≥5 teeth due to periodontitis
  - Masticatory dysfunction
  - Secondary occlusal trauma
  - Severe ridge defects
  - Bite collapse, drifting, flaring

Stage IV isn't just about periodontitis anymore—it's about rehabilitation. You're dealing with functional problems that require complex treatment planning involving prosthodontics, possibly orthodontics, and significant time and resources.

Staging defines severity while grading defines progression rate.



Understanding Grading: How Fast Is It Progressing?

- Here's where the 2018 classification really shines. Staging tells you how bad things are NOW. Grading tells you how FAST the disease is likely to progress.
- This is crucial information. Two patients can both have Stage III periodontitis, but if one is Grade A (slow progression) and the other is Grade C (rapid progression), your treatment approach and prognosis are completely different.

Understanding Grading: How Fast Is It Progressing?

- **Grade A: Slow Progression**
- **The profile:**
  - Bone loss / age ratio: <0.25
  - Heavy biofilm deposits with low destruction levels
  - No bone loss over 5 years (if longitudinal data available)
  - Non-smoker, no diabetes
- This is the patient who's had periodontitis for 30 years but still has most of their teeth. The disease is there, but it's moving slowly. With proper management, they can keep their teeth for a long time.

Understanding Grading: How Fast Is It Progressing?

- **Grade B: Moderate Progression**
- **The profile:**
  - Bone loss / age ratio: 0.25 to 1.0
  - Destruction commensurate with biofilm deposits
  - Less than 2 mm bone loss over 5 years
  - Smoker (<10 cigarettes/day) or HbA1c <7% in diabetics
- This is your "typical" periodontitis patient. The disease is progressing at a moderate rate, and they need consistent maintenance and monitoring.

Understanding Grading: How Fast Is It Progressing?

- **Grade C: Rapid Progression**
- **The profile:**
  - Bone loss / age ratio: >1.0
  - Destruction exceeds what you'd expect from biofilm
  - More than 2 mm bone loss over 5 years
  - Heavy smoker (≥10 cigarettes/day) or HbA1c ≥7% in diabetics
- This is the 35-year-old with bone loss that looks like a 60-year-old's. The disease is aggressive. These patients need intensive treatment and very close monitoring.

Understanding Grading: How Fast Is It Progressing?

- **The Bone Loss / Age Ratio: A Practical Tool**
  - One of the best additions in the 2018 classification is the bone loss / age ratio. It's simple math that gives you incredible insight.
- **How to calculate it:**
  - Measure the worst site's bone loss as a percentage of root length
  - Divide by the patient's age

Understanding Grading: How Fast Is It Progressing?

- **Example:**
  A 40-year-old patient with 50% bone loss at the worst site.
  Ratio = 50/40 = 1.25
- That's Grade C territory—rapid progression.

Understanding Grading: How Fast Is It Progressing?

- A 70-year-old patient with 40% bone loss at the worst site.
  Ratio = 40/70 = 0.57
- That's Grade B—moderate progression.
- Same amount of bone loss, ***but completely different disease behaviors***. This is why the grading system is so powerful.

Accurate periodontal probing is the foundation of proper classification.

- **Step 1: Complete Periodontal Examination**
- You can't classify what you don't measure. This means:
- Full-mouth probing (6 sites per tooth, minimum)
- Bleeding on probing assessment
- Recession measurements
- Furcation assessment on all molars
- Full-mouth radiographic series

- Don't cut corners here. A bite-wing series won't show you interproximal bone loss on posterior teeth. You need periapicals or a CBCT for complex cases.

Step 2: Calculate Clinical Attachment Loss

- Remember, CAL = Probing Depth + Recession
- If the gingival margin is coronal to the CEJ, you subtract that distance from the probing depth. If it's apical (recession), you add it.
- This is the number that drives staging. Probing depths alone can be misleading—a 4 mm pocket with 2 mm of recession is actually 6 mm of CAL.

Step 3: Assess Radiographic Bone Loss

- Look at your radiographs and estimate the percentage of root length with bone loss. The coronal third, middle third, and apical third each represent roughly 33%.
- **Pro tip**: Use the CEJ to the root apex as your reference. Measure from where the bone should be (2 mm below CEJ) to where it actually is.

Step 4: Identify Complexity Factors

- **Check for:**
  - Furcation involvement (especially Class II and III)
  - Vertical bone defects
  - Tooth mobility and secondary occlusal trauma
  - Ridge defects
  - Number of teeth lost to periodontitis
- These factors can bump a patient from one stage to the next, regardless of the primary criteria.

Step 5: Determine Grade

Look at:

- Age of the patient
- Amount of bone loss
- Available longitudinal data (if any)
- Risk factors (smoking, diabetes)

When in doubt between grades, choose the higher one. It's better to be more aggressive with treatment than to underestimate disease progression.

Step 6: Document Everything

Your perio chart should include:

- Stage (I, II, III, or IV)
- Grade (A, B, or C)
- Extent (localized <30% of teeth or generalized ≥30%)
- Distribution pattern (molar-incisor pattern if applicable)

- A complete diagnosis might read: "Stage III, Grade C, Generalized Periodontitis" or "Stage II, Grade B, Localized Periodontitis."

The Extent and Distribution Descriptors

The 2018 classification also clarified how we describe the extent and distribution of periodontitis:

Extent:

- **Localized:** <30% of teeth affected
- **Generalized:** ≥30% of teeth affected
- **Distribution:**
- **Molar-incisor pattern:** A specific pattern primarily affecting first molars and incisors (the old "localized aggressive periodontitis" pattern)

Risk Factor Modification in the New System

- The 2018 classification handles risk factors. Smoking and diabetes aren't just clinical notes anymore—they directly influence the grade.
- This forces a conversations with patients about systemic health. "Mrs. Johnson, your periodontitis is progressing rapidly—partly because of your uncontrolled diabetes. We need to work with your physician to get your HbA1c below 7%.“ (Interprofessional Collaboration)
- That's powerful. The classification system becomes a communication tool.

Common Mistakes to Avoid

Classification recurring errors over the years:

- **1. Staging based on probing depth alone**
  Probing depth is important, but CAL and radiographic bone loss drive staging. A 3 mm pocket with 4 mm of recession is Stage II or III, not Stage I.
- **2. Ignoring complexity factors**
  A patient with 4 mm CAL but Class III furcation involvement is Stage III, not Stage II. The complexity factors matter.

Common Recurring Errors to Avoid

- **3. Not calculating the bone loss/age ratio**
  This is such a simple calculation that gives you so much information. Don't skip it.
- **4. Defaulting to Grade B**
  When you're unsure, you're supposed to assume Grade B. But don't let this become a lazy default. Calculate the ratio and assess risk factors.
- **5. Forgetting about extent**
  The extent (localized vs. generalized) is part of the complete diagnosis. Include it.

Why This Classification Actually Matters for Your Patients

- Here's the thing—the 2018 classification isn't just academic navel-gazing. It changes how we treat patients.
- **For Stage I patients:** We can confidently manage them with non-surgical therapy and regular maintenance, knowing that with proper care, they're unlikely to progress.
- **For Stage III Grade C patients:** We know aggressive intervention is needed NOW, not after three more years of "watchful waiting."

Why This Classification Actually Matters for Your Patients

- **For diabetic patients:** The direct incorporation of HbA1c into grading means we're forced to coordinate care with their physicians.
- **For smokers:** The grading system gives us concrete evidence to show patients how smoking accelerates their disease.
- The classification is a communication tool between clinicians, between you and your patients, and between you and insurance companies trying to deny coverage

The 1999 system organized disease into eight categories

Category	1999 classification
I	Gingival diseases
II	Chronic periodontitis
III	Aggressive periodontitis
IV	Periodontitis as a manifestation of systemic disease
V	Necrotizing periodontal diseases
VI	Abscesses of the periodontium
VII	Periodontitis associated with endodontic lesions
VIII	Developmental or acquired deformities and conditions

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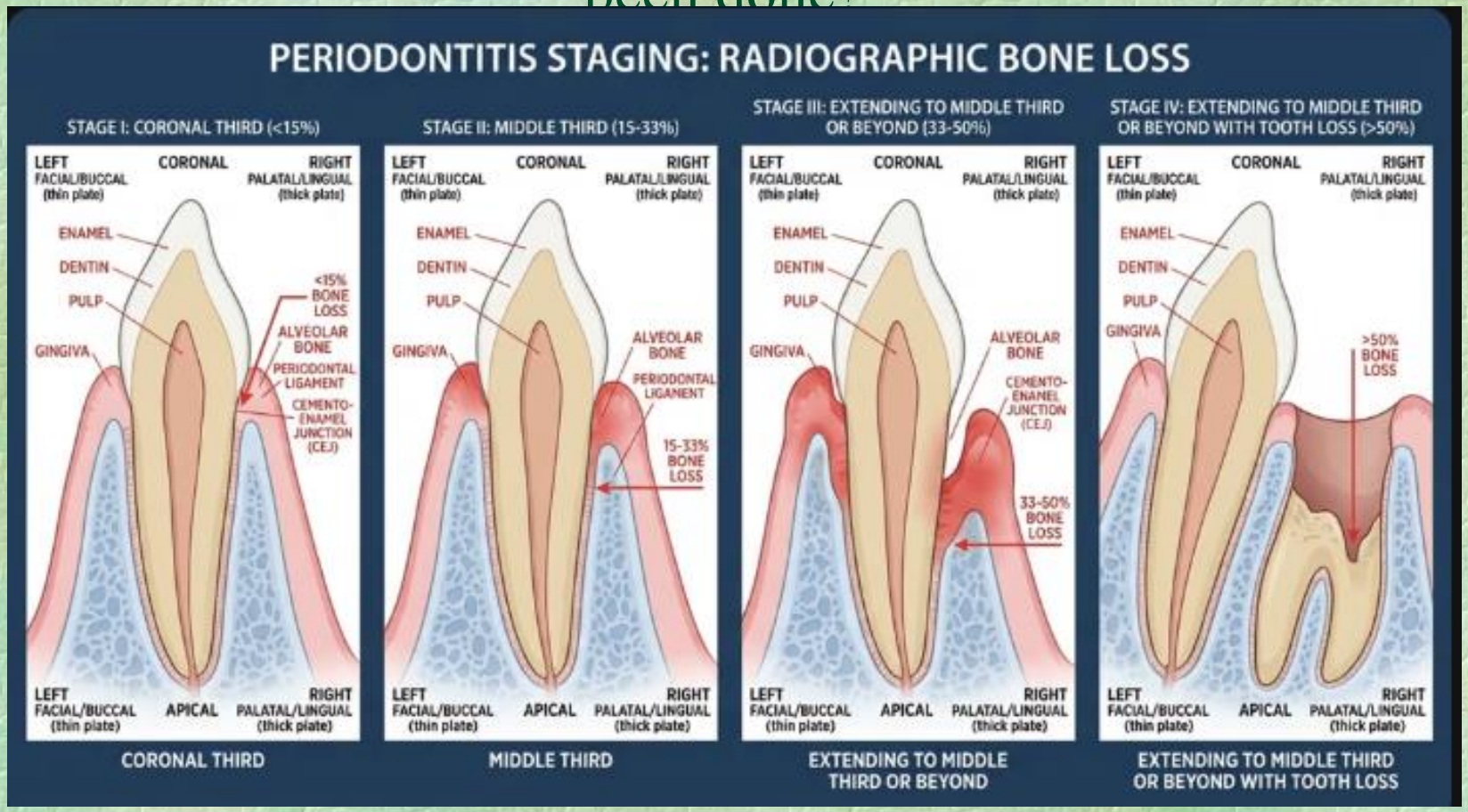
Why the Old Classification Needed to Go

- **First,** it was built on flimsy scientific foundations.
- The distinction between "chronic" and "aggressive" periodontitis was based primarily on the rate of disease progression—something we often couldn't determine on a first visit.
  - You'd see a 35-year-old with significant bone loss and think "aggressive periodontitis," when the disease had been slowly progressing since they were 20 but had never been treated.
- **Second,** the overlap was maddening.
- Was "aggressive periodontitis" localized or generalized?
- Was it different from "early-onset periodontitis" from the 1989 classification?
- Were we even speaking the same language?

Staging tells you the **severity** and **complexity** of the disease. It answers the question: "How much damage has been done?"



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The 2017 World Workshop created four broad domains

Domain	2017/2018 classification
1	Periodontal health, gingival diseases, and gingival conditions
2	Periodontitis
3	Other conditions affecting the periodontium
4	Peri-implant diseases and conditions

The workshop occurred in 2017. The consensus reports were published in 2018.

Several foundational distinctions remain

Retained concept	How the current system uses it
Plaque-related gingival inflammation	Dental biofilm-induced gingivitis remains distinct from non-biofilm-induced disease.
Gingivitis versus periodontitis	Gingivitis lacks periodontitis-related attachment loss. Periodontitis includes destruction of supporting tissues.
Clinical attachment and bone loss	CAL and radiographic bone loss remain central to diagnosis and severity assessment.
Distribution of disease	Both systems distinguish localized from generalized involvement.
Distinct disease families	Necrotizing disease, systemic manifestations, abscesses, endodontic-periodontal lesions, and mucogingival conditions remain recogniz

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Chronic and aggressive periodontitis became one category

1999	2017/2018	Why the change matters
Chronic periodontitis	Periodontitis	The diagnosis no longer assumes a distinct disease type from age or presentation alone.
Aggressive periodontitis	Periodontitis	Rapid progression is expressed through grade when evidence supports it.
Slight, moderate, severe	Stages I to IV	Stage incorporates accumulated damage and complexity of management.
Chronic or aggressive course	Grades A to C	Grade estimates progression and incorporates smoking and diabetes control.
Localized or generalized by sites	Extent by teeth or molar/incisor pattern	The current diagnosis describes how widely affected teeth are distributed.

Former aggressive periodontitis does not automatically equal Grade C.

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Stage, grade, and extent answer different questions

PERIODONTITIS DIAGNOSIS

Diagnostic element	Question answered	Primary evidence
Stage I to IV	How much damage has occurred, and how complex is management?	CAL, radiographic bone loss, periodontitis-related tooth loss, probing depth, furcations, mobility, and functional complexity
Grade A to C	How quickly is disease progressing, and what raises future risk?	Longitudinal change, bone loss-to-age ratio, disease phenotype, smoking, and diabetes control
Extent/ distribution	How widely are affected teeth distributed?	Localized: <30% of teeth; generalized: ≥30%; or molar/incisor pattern

Complete format: Periodontitis, Stage ___, Grade ___, localized/generalized or molar/incisor pattern

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Most 1999 categories were retained but reorganized

DIAGNOSTIC MAPPING

1999 category	2017/2018 placement
Chronic periodontitis	Periodontitis with stage, grade, and extent/distribution
Aggressive periodontitis	Periodontitis with stage, grade, and extent/distribution
Periodontitis as a manifestation of systemic disease	Retained and classified according to the primary systemic disease
Necrotizing ulcerative gingivitis/periodontitis	Necrotizing gingivitis or necrotizing periodontitis

The current periodontitis diagnosis describes severity, progression risk, and distribution.

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Abscesses, combined lesions, and deformities were reorganized

DIAGNOSTIC MAPPING

1999 category	2017/2018 placement
Periodontal abscess	Other conditions affecting the periodontium
Combined periodontal-endodontic lesion	Endodontic-periodontal lesion
Developmental or acquired deformities	Mucogingival, occlusal, tooth-related, prosthesis-related, and other conditions
No comprehensive peri-implant category	Peri-implant health, mucositis, peri-implantitis, and site deficiencies

Reorganization gives each condition a clearer diagnostic location.

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Health and gingivitis now have clearer case definitions

ADDITIONAL CHANGES

Area	1999 emphasis	2017/2018 change
Periodontal health	No comprehensive separate case definition	Health defined on intact and reduced periodontia, including successfully treated periodontitis
Gingivitis	Descriptive categories	Bleeding on probing provides the primary case threshold
Systemic modifiers	Separate or associated categories	Rare causal disorders remain systemic manifestations; diabetes and smoking modify grade

A successfully treated periodontitis patient remains a periodontitis patient and requires lifelong supportive care.

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Terminology and peri-implant diagnoses were expanded

ADDITIONAL CHANGES

Area	1999 emphasis	2017/2018 change
Mucogingival terminology	Periodontal biotype	Periodontal phenotype and a revised recession framework
Occlusal terminology	Excessive occlusal force	Traumatic occlusal force describes the force; occlusal trauma describes the lesion
Implants	No comprehensive parallel system	Peri-implant health, mucositis, peri-implantitis, and site deficiencies added

The 2017 World Workshop introduced the first comprehensive peri-implant classification.

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The diagnostic statement now carries more clinical information

CLINICAL TRANSLATION

Clinical presentation	Likely 1999 wording	Current diagnostic wording
Generalized CAL 3–4 mm, no complexity factors, average progression	Generalized moderate chronic periodontitis	Generalized periodontitis, Stage II, Grade B
Molar/incisor pattern, CAL ≥5 mm, no Stage IV factors, rapid progression	Localized aggressive periodontitis	Periodontitis, Stage III, Grade C, molar/incisor pattern
Severe destruction caused by Papillon-Lefèvre syndrome	Periodontitis as a manifestation of systemic disease	Periodontitis as a manifestation of Papillon-Lefèvre syndrome

Every current diagnosis must be supported by the patient's findings. These examples illustrate wording, not automatic conversion rules.

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Classification knowledge check

FORMATIVE CHECK

1. Which two 1999 periodontitis categories were combined in the current system?

2. Which element describes accumulated destruction and treatment complexity: stage, grade, or extent?

3. A former aggressive-periodontitis diagnosis automatically becomes Grade C. True or false?

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1999 versus 2017-2018 Periodontal Status Classification

Component	1999 AAP classification	2017/2018 AAP–EFP classification	Clinical significance
Overall organization	Eight major categories of periodontal diseases and conditions	Four broad domains covering periodontal and peri-implant health, diseases, and conditions	More patient-centered and clinically adaptable
Periodontal health	Not comprehensively defined as a separate diagnostic category	Defines health on an intact periodontium and on a reduced periodontium	A successfully treated periodontitis patient can be clinically stable but remains a periodontitis patient requiring lifelong surveillance

1999 versus 2017-2018 Periodontal Status Classification

Component	1999 AAP classification	2017/2018 AAP–EFP classification	Clinical significance
Gingivitis	**Dental plaque-induced or non-plaque-induced gingival disease**	**Dental biofilm-induced gingivitis or non-dental-biofilm-induced gingival disease**	**Preserves the basic distinction but provides clearer case definitions**
Gingivitis measurement	Descriptive clinical categories; no universally emphasized case threshold	Bleeding on probing is the primary parameter for defining gingival health and gingivitis	Improves diagnostic consistency

1999 versus 2017-2018 Periodontal Status Classification

Component	1999 AAP classification	2017/2018 AAP–EFP classification	Clinical significance
Gingivitis measurement	**Descriptive clinical categories; no universally emphasized case threshold**	**Bleeding on probing is the primary parameter for defining gingival health and gingivitis**	**Improves diagnostic consistency**
Periodontitis categories	Chronic periodontitis and aggressive periodontitis were separate diseases	Chronic and aggressive periodontitis are combined as one disease: periodontitis	Avoids unreliable classification based primarily on age or presumed progression pattern

1999 versus 2017/2018 classification

Component	1999 AAP classification	2017/2018 AAP–EFP classification	Clinical significance
Severity	**Slight, moderate, or severe, primarily based on clinical attachment loss**	**Stages I–IV, based on severity and treatment complexity**	**Staging incorporates CAL, bone loss, tooth loss, probing depth, vertical defects, furcations, mobility, and functional concerns**
Progression	Implied primarily by "chronic" versus "aggressive" disease labels	Grades A–C, estimating progression and future risk	Incorporates longitudinal evidence, bone loss-to-age ratio, smoking, and diabetes control

1999 versus 2017/2018 classification

Component	1999 AAP classification	2017/2018 AAP–EFP classification	Clinical significance
Extent	**Localized: ≤30% of sites; generalized: >30% of sites**	**Localized: <30% of teeth; generalized: ≥30% of teeth; or molar/incisor pattern**	**Uses teeth rather than periodontal sites when describing periodontitis extent**
Systemic disease	"Periodontitis as a manifestation of systemic disease" was a separate category	Retained for uncommon systemic disorders that directly produce periodontal destruction	Classified according to the primary systemic disease

1999 versus 2017/2018 classification

Component	1999 AAP classification	2017/2018 AAP–EFP classification	Clinical significance
Diabetes and smoking	**Recognized as systemic or environmental risk factors**	**Incorporated particularly as grade modifiers**	**Their influence becomes part of the formal diagnostic description**
Necrotizing disease	Necrotizing ulcerative gingivitis and necrotizing ulcerative periodontitis	Necrotizing gingivitis, necrotizing periodontitis, and necrotizing stomatitis	Removes "ulcerative" from the preferred terminology and reorganizes the conditions

1999 versus 2017/2018 classification

Component	1999 AAP classification	2017/2018 AAP–EFP classification	Clinical significance
Periodontal abscesses and endodontic-periodontal lesions	**Separate major categories**	**Grouped under other conditions affecting the periodontium**	**Emphasizes their distinct diagnostic and treatment implications**
Mucogingival conditions	Developmental or acquired deformities and conditions	Revised recession and mucogingival classification	"Periodontal biotype" was replaced by periodontal phenotype

1999 versus 2017/2018 classification

Component	1999 AAP classification	2017/2018 AAP–EFP classification	Clinical significance
Occlusal forces	**Occlusal trauma included under developmental/acquired conditions**	**Distinguishes traumatic occlusal force from the resulting lesion, occlusal trauma**	**Clarifies that traumatic occlusal forces do not independently initiate plaque-induced periodontitis**
Peri-implant conditions	No complete peri-implant classification	Adds peri-implant health, peri-implant mucositis, peri-implantitis, and implant-site deficiencies	First comprehensive peri-implant classification

Direct mapping from the old system to the new system

1999 diagnosis	2017/2018 placement
Chronic periodontitis	Periodontitis + stage + grade + extent/distribution
Aggressive periodontitis	Periodontitis + stage + grade + extent/distribution
Periodontitis as a manifestation of systemic disease	Retained and classified according to the primary systemic condition
Necrotizing ulcerative gingivitis	Necrotizing gingivitis
Necrotizing ulcerative periodontitis	Necrotizing periodontitis

Direct mapping from the old system to the new system

Periodontal abscess	Other conditions affecting the periodontium
Combined periodontal-endodontic lesion	Endodontic-periodontal lesion
Developmental/acquired deformities	Mucogingival, occlusal, tooth-related, prosthesis-related, and other periodontal conditions
No comparable comprehensive category	Peri-implant health, diseases, and conditions

What remained consistent

- Both systems:
- Recognize plaque/biofilm-induced and non-plaque-induced gingival conditions.
- Distinguish reversible gingivitis from destructive periodontitis.
- Use clinical attachment loss, bone loss, distribution, and clinical findings in diagnosis.
- Recognize necrotizing diseases, systemic influences, periodontal abscesses, endodontic-periodontal lesions, and mucogingival conditions.

Most important statement

- 1999 classified periodontitis primarily by disease type.
- The 2017/2018 system classifies the patient by severity and treatment complexity—stage—and by progression risk—grade.

Former Diagnosis Conversion

- A former diagnosis of aggressive periodontitis should not automatically be converted to **Grade C**.
- **Grade C** must be supported by direct or indirect evidence of rapid progression and relevant risk modifiers. The current staging and grading criteria are detailed in the AAP/EFP staging and grading consensus report.

Necrotizing Periodontal Disease

- A. **Necrotizing ulcerative periodontitis (NUP)**
- B. **Necrotizing ulcerative gingivitis (NUG)**
  - In patients with no known systemic disease or immune dysfunction, NUP appears to share many of the clinical and etiologic characteristics of NUG except that patients with NUP demonstrate loss of clinical attachment and alveolar bone at affected sites

Necrotizing Periodontal Disease

- In NUP patients there may be a sequela of a single or multiple episodes of NUG or may be the result of the occurrence of necrotizing disease at a previously periodontically involved site

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Necrotizing Ulcerative Periodontal diseases

- The microbes MOST often associated with NUP NUG are
  - Spirochetes (especially *Treponema denticola*) **** (Swimmers)
  - Fusobacterium
  - *Prevotella Intermedia* (think pregnancy & perio disease)
  - *Porphymonas gingivalis* (think perio diseases)

Necrotizing Ulcerative Periodontal Diseases

- Antibiotic therapy is often warranted
- Tetracycline is often the drug of choice because it concentrates in the gingival crevicular fluid aka Doxcycline
- Common Clinical findings of NUP/NUG
  - Punched out papilla (hallmark sign)
  - Psuedomembrane
  - Fetid odor
  - Pain
  - Severe inflammation

Abscesses of the periodontium

- A. gingival abscess
- B. periodontal abscess
- C. pericoronal abscess
- D. Periapical abscess

Abscesses of the periodontium ('new' category)

- It is important to differentiate between types of dental abscesses (for the most part, it is their **point of origin** that defines them):
  - **Gingival Abscess:** results from the injury to or infection of the surface gingival tissue
  - **Periapical abscess:** results from pulp infection (usually secondary to deep tooth decay)

Abscesses of the periodontium

- **Periodontal Abscess**(‘lateral periodontal abscess’) aka –lateral abscess; results when infection spreads deep into periodontal pockets and drainage is blocked (food impaction, calculus, etc.)
- **Pericoronal abscess:** develops in in inflamed dental follicular tissue overlying the crown of a partially erupted tooth (most often mandibular third molars)

- **Periapical abscess** requires a **periapical film**; a periodontal abscess will NOT generally show up on a radiograph
- **Reversible pulpitis** is a mild pulpal inflammation usually resulting from placement of a deep restoration. It is associated with episodic temperature-related pain which improves with deposition of reparative dentin. It is also referred to as '**pulp hyperemia**'. Know that **REVERSIBLE PULPITIS** is related **HYPEREMIA**

- Frequently the maxillary sinus will ‘pneumatize’ or extend into old extraction sites. Know that **PNEUMATIZATION** is a condition in which there is a connection from the SINUS to an EXTRACTION SITE

Development of acquired deformities and conditions

- Localized tooth related factors that increase the risk of developing plaque-induced gingivitis and periodontitis
- Occlusal trauma
- Mucogingival deformities and conditions around the teeth
  - Gingival recession
  - Lack of attached gingival
  - Frenal position – hi frenum contributes to recession
  - Gingival excess

Gingivitis description

- Acute vs. Chronic
- Extent Localized, generalized)
- Distribution (marginal, papillary
- Color
- Contour (bulbous, festooned
- Textured (stippling)-not always a sign of health.

Gingivitis

- **Acute-** vascular
- Rapid development
- Obvious inflammation
- Pain
- Redness

- **Chronic**-celluar reaction
- Slow development
- May appear ‘normal’ b/c usually stippled
- May not cause pain

Gingival Color

- “Coral Pink” usually used to describe health
- Pigmentation (melanin – Brown to Black)
- Erythema (redness associated with inflammation)
- Cyanosis (bluish hue; highly vascular, generally seen around crowns) deoxygenated blood)
- **Pallor** (lighter than normal tissue associated with leukemia, fibrotic tissue, anemia, etc.)

Gingival Contour

- Normal
- Clefted
- Bulbous –enlarged rounded
- Festooned- life saver
- Receded
- Blunted
- Hyperplastic
- Board alert: know and be able to visualize gingival conditions and recognize whether acute or chronic

Resources

- *Periodontology,* Newman and Carranza, 2023 Elsevier
- *Periodontology for the Dental Hygienist*, Perry, Beemster & Taggart, 2001, W. B Saunders Company
- *The Periodontic Syllabus*, Fedi & Vernino, 1995, Williams & Wilkins, 3rd Edition.
- *Fundamentals of Periodontics*, Wilson & Kornman, 1996, Quintessence Publishing Co., Inc.
- Clinical Periodontology, Carranza, 2006, Saunders printed by Elsevier, 10th Edition.
- *Periodontology for the Dental Hygienist*, Perry, Beemsterboer, 2007, Saunders, an imprint of Elsevier, Inc.

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Radiographic Interpretation of Periodontal Disease

DEN 624 Periodontology

Ben Bartlett, DDS, MS

Lesson Objectives

1

Radiographic Features

Identify radiographic appearance of restorations, appliances, implants, and foreign objects.

2

Bone Morphology and Trabecular Changes

Describe caries appearance, detection limitations, classification, and mimicking conditions.

3

Periodontal Disease Classification

Interpret bone loss patterns, severity, ADA classifications, and contributing factors.

4

Differential Interpretation

Identify types of periodontal disease, and processes that may mimic.

Periodontium & Periodontal Disease

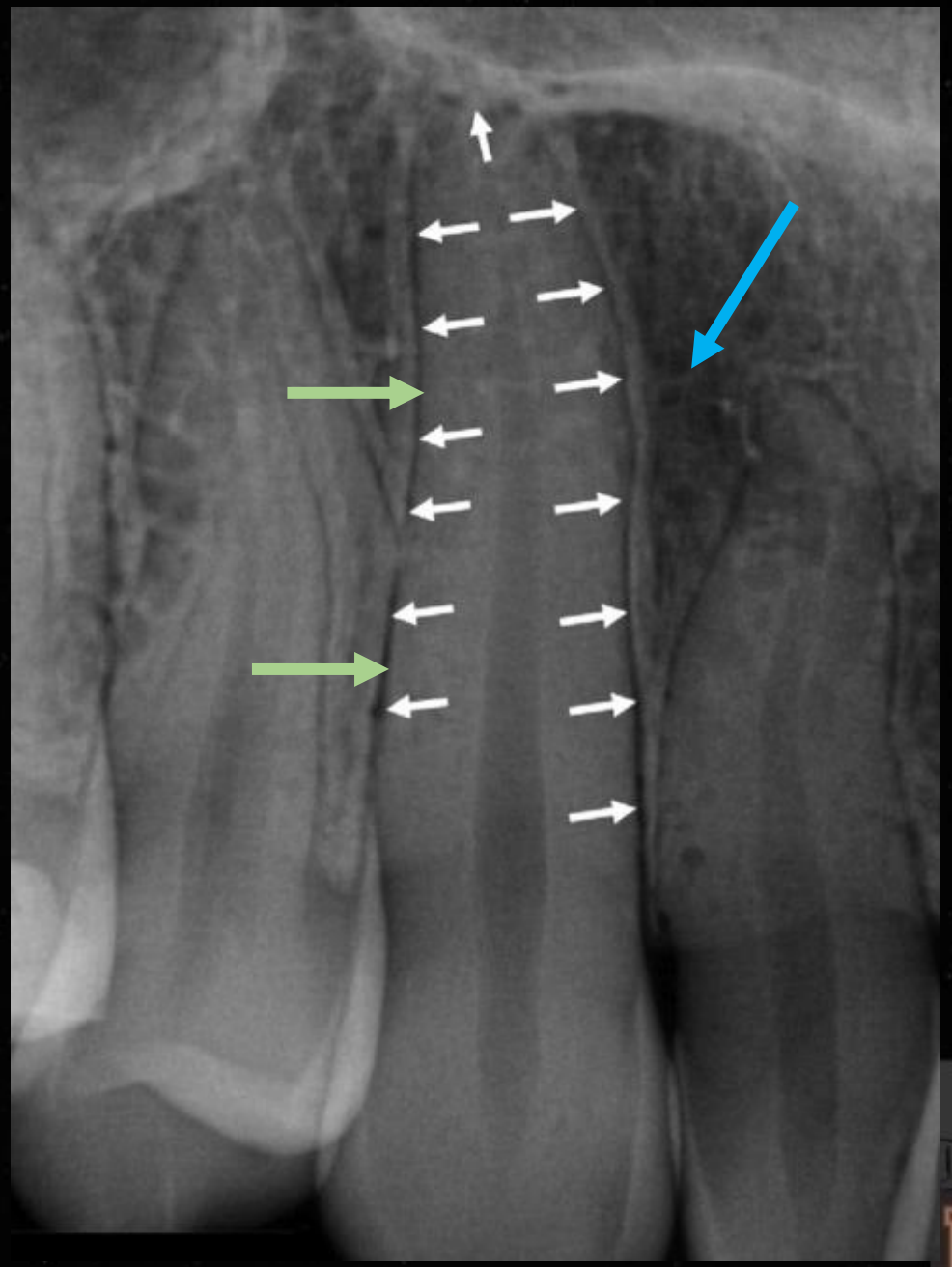
What is the Periodontium?

Periodontium: supports the teeth and includes:

- Gingiva
- Periodontal ligament (PDL)
- Cementum
- Alveolar bone

FIG. 4.2
Normal gingiva in a young adult. Note the demarcation (mucogingival line) *(arrows)* between the attached gingiva and the alveolar mucosa.

Courtesy Tun-Jan Wang.



Healthy Periodontium: Radiographic Anatomy

Lamina Dura
– thin radiopaque line around root
– continuous with intact cortical bone of socket

Alveolar crest
– smooth, intact bone between teeth

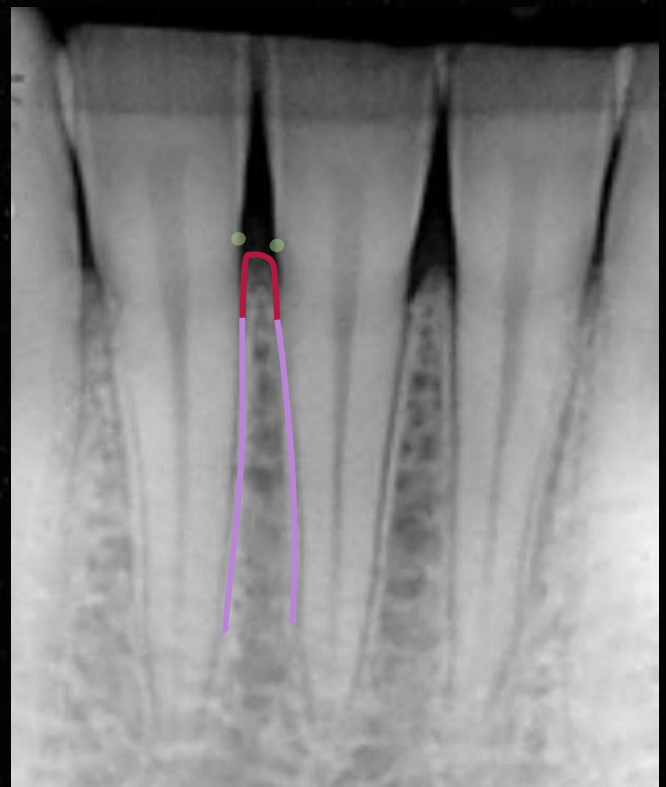
Periodontal ligament (PDL) space
– thin, uniform radiolucent line around root
– surrounds root evenly

Healthy Alveolar crest

- Crest height: ~0.5–2.0 mm apical to CEJ
- Posterior: crest runs flat/parallel to adjacent CEJs
- Anterior: crest is typically pointed and corticated
- Alveolar crest is continuous with the lamina dura
- PDL space: thin, uniform, radiolucent around the root

Important

- Well-corticated crest = health
- Lack of visible cortication does not necessarily mean disease
- Thin buccal/lingual bone may appear more radiolucent near the crest → normal variation
- Slight cervical PDL widening can be normal, especially in adolescents



What is Periodontal Disease?

- Inflammatory disease of supporting tissues
- Gingivitis → periodontitis
- Untreated disease → attachment/bone loss → possible tooth loss

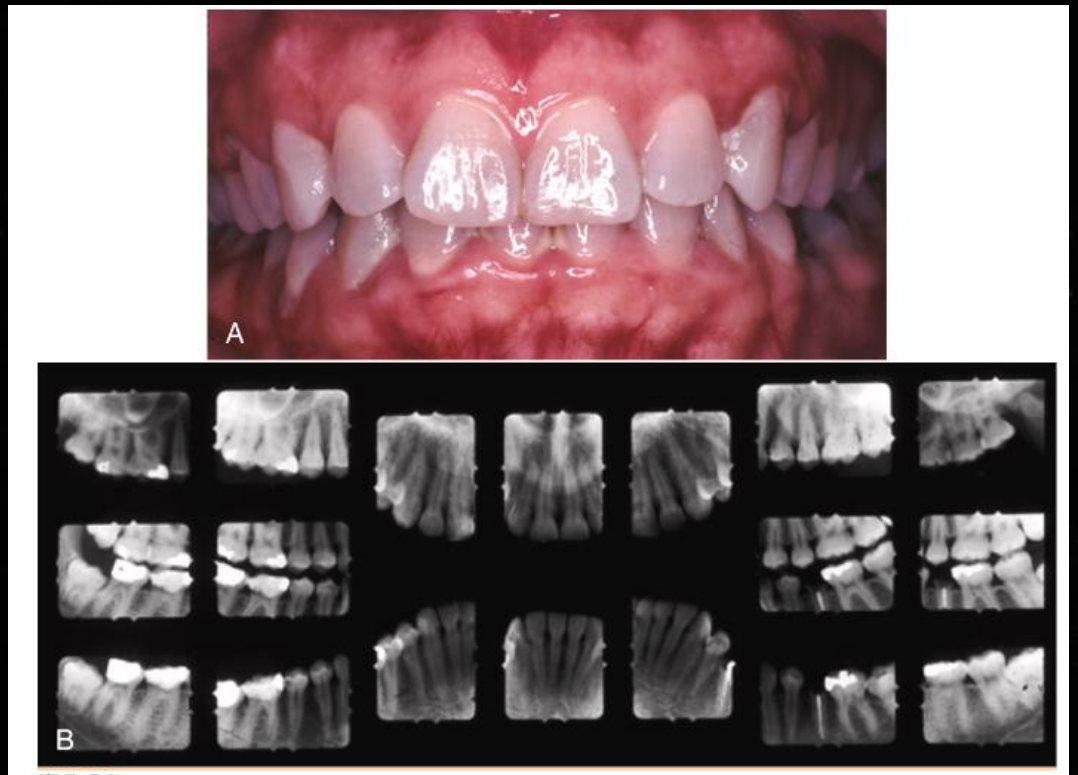


FIG. 5.1
(A) Dental biofilm-induced gingivitis depicts marginal and papillary inflammation, with 1- to 4-mm probing depths and generalized zero clinical attachment loss, except recession in tooth #28. (B) Radiographic images of the patient.

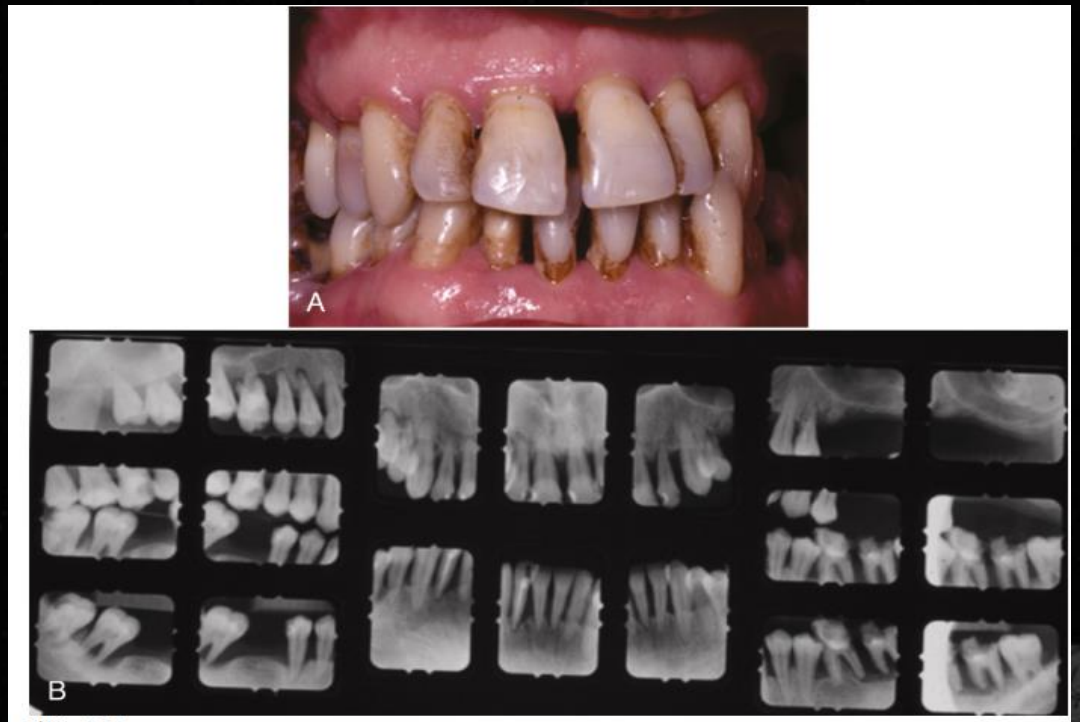


FIG. 5.17
(A) Clinical image of what we used to diagnose as aggressive periodontitis with 3- to 13-mm periodontal diseases and 7- to 15-mm clinical attachment loss in a 32-year-old male. Following the 2018 American Academy of Periodontology/European Federation of Periodontology classification system, this patient is diagnosed with Generalized Stage IV Grade C periodontitis. (B) Radiographic images of the patient.

How Periodontal Disease Develops

1. Initiation

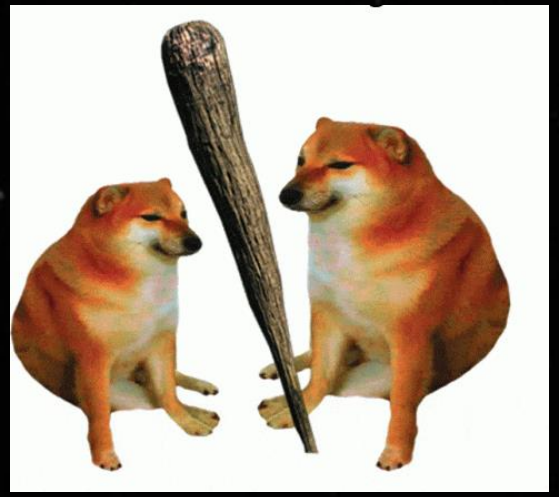
Dental plaque biofilm/bacteria directly damage tissues through toxins.

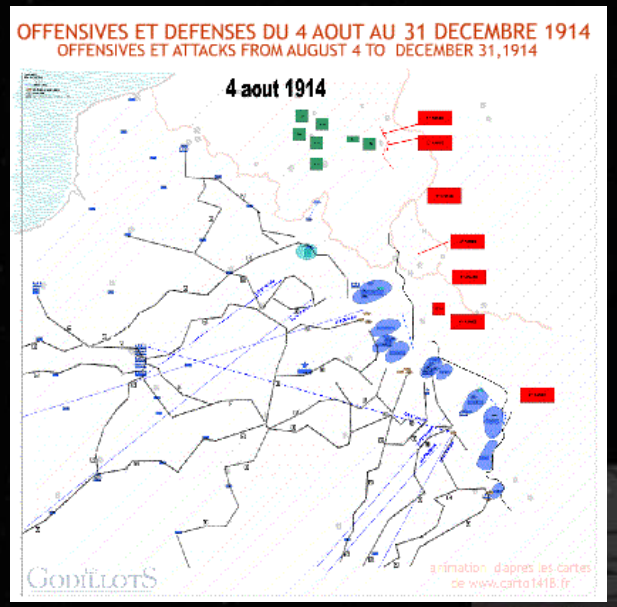
2. Host Inflammatory Response

Damage stimulates host inflammatory response using mediators →
Soft-tissue destruction →
Osteoclastic bone resorption →
Loss and apical migration of the junctional epithelium

3. Periodontal Pocket Formation

Junctional epithelium migrates apically →
Periodontal pocket develops →
Increased bacterial colonization →
Further inflammation and tissue destruction





Progression of Periodontal Disease

Periodontal disease often progresses in **bursts**

Alternates between:

- **Active phases:** inflammation and tissue destruction
- **Quiescent phases:** little or no appreciable change

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Risk & Modifying Factors

- **Diabetes / hyperglycemia**
- **Genetic/developmental disorders**
- **Hematological conditions**
- **Hormonal changes:**
  - Menstrual cycle
  - Menopause
  - Oral contraceptives
  - Pregnancy
  - Puberty
- **Immunosuppression / altered immune function**

- **Medications / pharmacological agents**
- **Malnutrition / nutritional factors**
- **Oral hygiene / plaque accumulation**
- **Oral dryness**
- **Psychological stress**
- **Smoking**
- **Socioeconomic factors / limited dental care**
- **Systemic diseases / conditions**
- **Tooth-related and local factors**
  - Dental restorations/prostheses
  - Tooth anatomy and position
  - Open interproximal contacts
  - Root fractures, grooves, proximity, etc.
- **Trauma / traumatic factors**

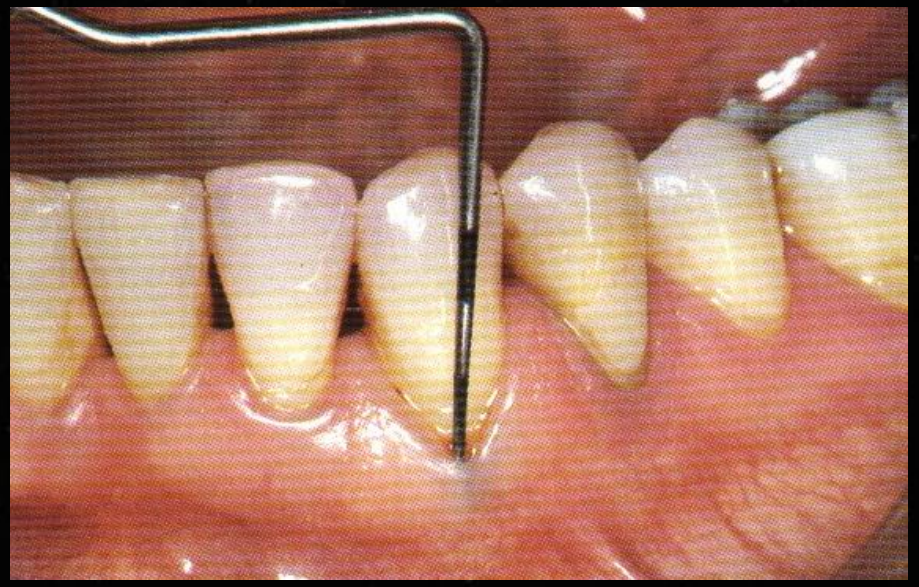
Newman and Carranza's Clinical Periodontology and Implantology, Chapter 5

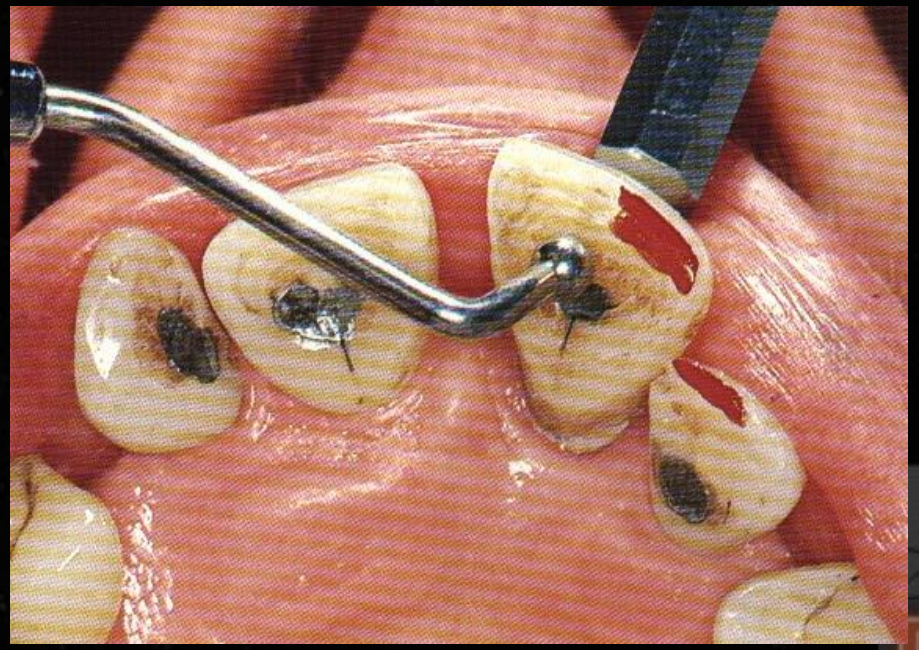
Clinical Exam + Imaging

Clinical Examination Is First

Clinical exam shows:

- Bleeding
- Clinical attachment loss
- Gingiva
- Mobility
- Periodontal pockets
- Periodontal probing depths
- Pus
- Recession
- Redness
- Swelling





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Imaging Examination Is Second

Imaging shows:

- Bone levels, crests & furcations
- PDL space
- Roots & crown-to-root ratio
- Calculus and local irritants
- Anatomic/pathologic findings

Follow-up imaging frequency should be based on clinically determined disease activity.

Check for updates

ADA CE

Supplemental material is available online.

American Dental Association and American Academy of Oral and Maxillofacial Radiology patient selection for dental radiography and cone-beam computed tomography

Box 2. Indications.

2.0 Imaging Recommendations for Periodontal Disease

2.1 The frequency of radiographic examination for the patient with periodontal disease is dependent on the clinical findings and treatment response.[3,17]

2.2 Radiographic documentation of the disease sites before periodontal treatment is important for diagnosis, baseline documentation, and treatment planning.[3,17]

2.3 Radiographic evidence of the presence and severity of periodontal bone loss does not indicate active disease.[3,17]

2.4 For evaluating periodontal structures after comprehensive clinical examination, 2D† full-mouth radiographic series, including vertical bite-wing radiographs as necessary, is recommended.[3,17]

2.5 Management of periodontal disease, in general, does not warrant use of CBCT, although CBCT may be indicated for treatment planning of complex cases.[3,17]

2.6 For patients with history of, or with active, periodontal disease, vertical bite-wing radiographs are recommended to assess bone levels in the permanent dentition.[3,17]

Box 3. Pediatric recommendations.

2.0 Intraoral Radiography–Specific Recommendations

2.1 For patients in the primary, mixed, or early adolescent dentition with severe gingival inflammation as a result of systemic conditions, maxillary and mandibular occlusal or periapical and bite-wing radiographs shall be obtained along with mobility testing as part of the initial assessment in lieu of periodontal probing of hemorrhagic gingiva.[3,27]

2.1.1 For the assessment of bone levels in adolescents with periodontal disease, in the permanent dentition, vertical bite-wing radiographs should be used.

Imaging Examination Is Second

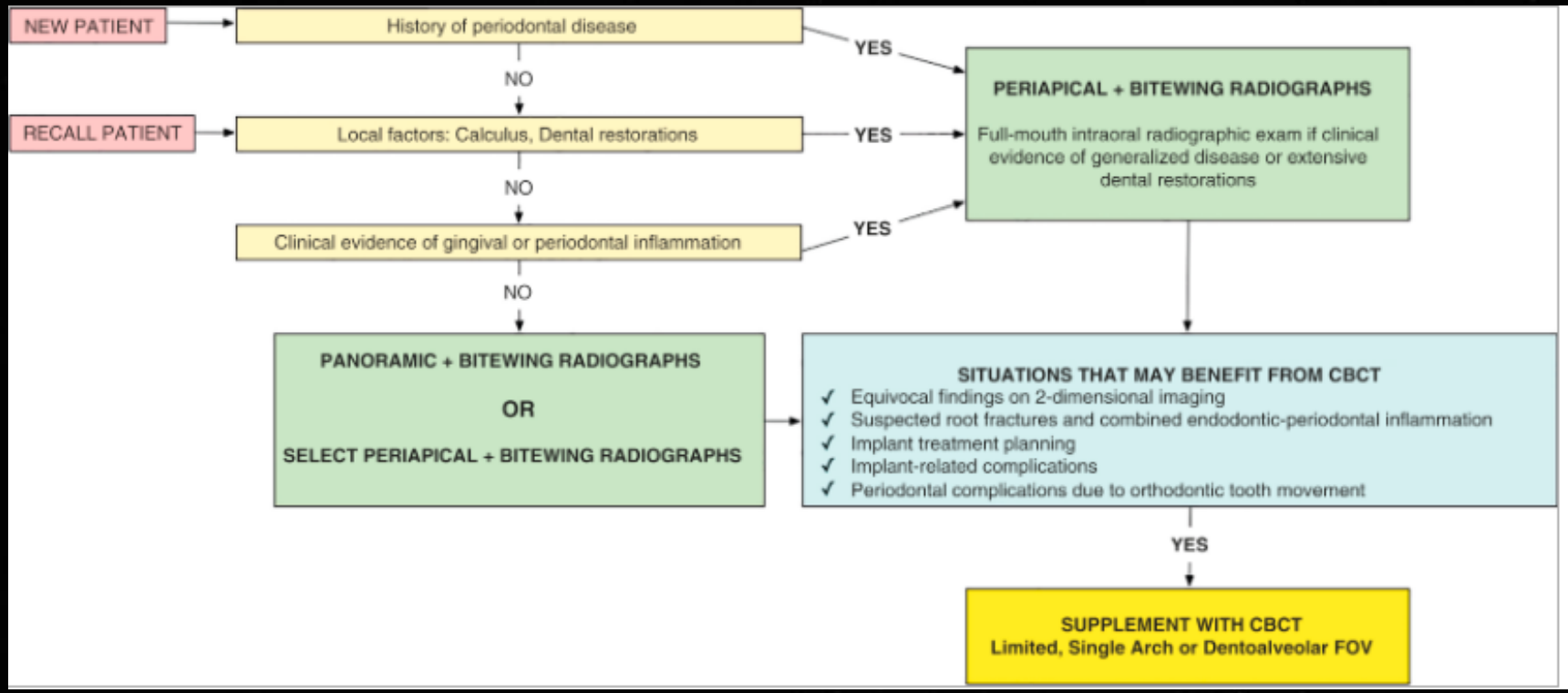


FIG 39.10

Flowchart to guide imaging evaluation of teeth and periodontal structures. The goal of this flowchart is to provide a broad guideline with respect to selection of radiographs in the management of periodontal conditions. The need and selection must be assessed on a case-by-case basis. Information collated from the recommendations, position statements, and best evidence reviews from the American Association of Endodontists, 7 the American Association of Oral and Maxillofacial Radiology, 7 , 33 the American Academy of Periodontology Best Evidence Reviews, 14 , 17 , 18 , 29 and the American Dental Association. 1

Sanjay M. Mallya; Sotirios Tetradis, Chapter 39, 520-536.e1

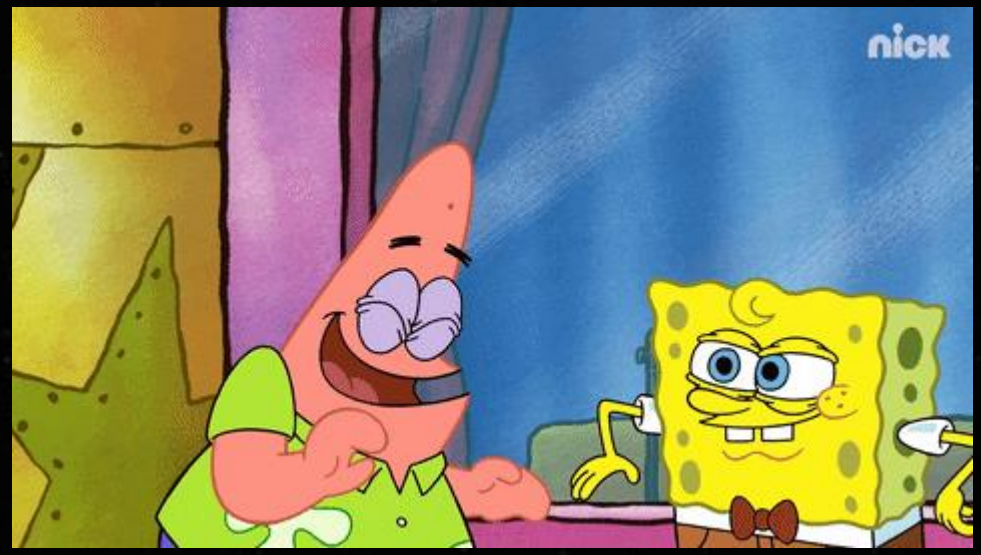
The Dynamic Duo

Key Principle:
Clinical exam + Appropriate Imaging = Complete Periodontal Assessment

Clinical exam = soft tissue
Imaging = hard tissue

Neither replaces the other.

- Provides a permanent baseline for monitoring and helps with diagnosis, prognosis, and treatment planning.
- Follow-up imaging frequency should be based on clinically determined disease activity.



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Imaging Modalities for the Assessment of Periodontal Disease

Intraoral Imaging

Highest spatial resolution → detects subtle, early periodontal changes.

- Includes bitewing and periapical radiographs.

Optimal Image Quality

- Receptor parallel to tooth long axis.
- X-ray beam perpendicular to receptor/tooth.

Correct positioning should show:

- Open interproximal contacts
- No root overlap
- Superimposed buccal/lingual molar cusps

BW

Cropped Pano

Intraoral Imaging: Bitewing vs. Periapical

Bitewing → Best for bone levels

- Excellent for CEJ → alveolar crest distance
- Best for interproximal bone levels
- Vertical bitewings are preferred when significant bone loss prevents the crest from being captured on a standard horizontal bitewing

Intraoral Imaging: Bitewing vs. Periapical

Periapical → Best for the whole tooth/root

- Is recommended in the textbook for periodontal evaluation when the paralleling technique is executed correctly due to accurate crestal bone representation
- More susceptible to distortion, especially in the maxilla
  - Especially with bisecting technique
- Shows entire tooth and root
- Useful for estimating percentage of root bone loss

Limitations of Intraoral Images

- 2D images of 3D anatomy → buccal/lingual bone can be superimposed and hide defects.
- Early bone loss may be missed → radiographs can underestimate actual destruction.
- Soft tissues are not shown → cannot determine periodontal pocket depth.
- Multiple angulations can help localize buccal vs lingual bone loss using the buccal object rule.

Key point: Intraoral radiographs are essential but must be interpreted alongside clinical examination.

Panoramic Imaging

Clinical Use

- May be useful with selected intraoral images
- Not recommended as the primary modality for periodontal assessment when intraoral imaging is available

Advantages

- Fast and comfortable
- Provides a single-image overview of teeth and jaws
- Can demonstrate periodontal bone changes

Limitations

- Lower resolution than intraoral imaging
- Superimposition & distortion
- Less useful for fine periodontal detail, especially anteriorly
- Patient positioning errors can further reduce image quality
- Does not overcome limitations of 2D imaging

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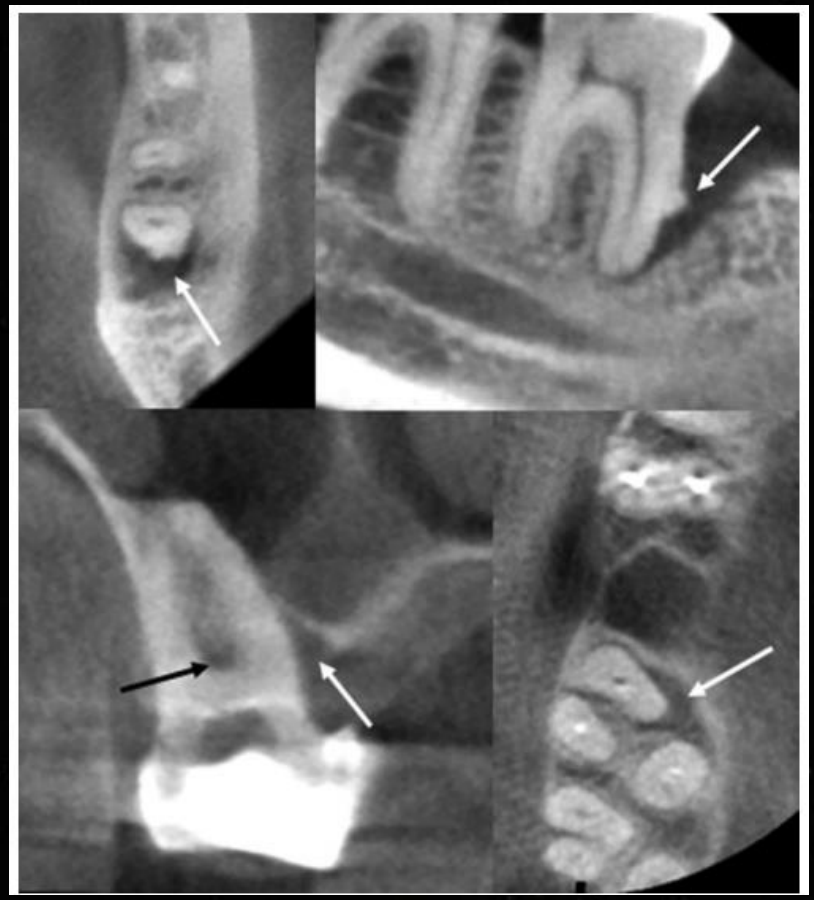
Cone Beam Computed Tomography

Clinical Use

Not recommended routinely for periodontal assessment

Consider small-volume, high-resolution CBCT for selected complex defects, particularly when surgical



Advantages

- 3D visualization of supporting bone from any angle
- Eliminates anatomic superimposition

Better demonstrates:

- Complex vertical defects & craters
- Furcation defects
- Buccal/lingual cortical plate loss
- Root morphology & fenestrations
- Coexisting periapical pathology

Limitations

- Metal artifacts can obscure bone
- Lower contrast resolution may limit visualization of low-density bone
- Higher radiation dose and cost than intraoral imaging

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Imaging Features of Periodontal Disease

Imaging Features of Periodontal Disease

- **REMEMBER** in **Gingivitis** → bone remains **radiographically normal** because inflammation is limited to gingiva.

- **REMEMBER** in **Periodontitis** → produces **radiographic changes** in bone morphology, density, and pattern
  - Bone loss → increased radiolucency (which is called **rarefaction**)
  - Bone formation → increased radiopacity **(sclerosis)**

Periodontitis may show **both**.

Patterns
- **Early/acute lesions:** predominantly bone loss
- **Chronic lesions:** greater component of bone sclerosis

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AAP Periodontal Disease Classification

STAGE I	STAGE II	STAGE III	STAGE IV
1 - 2 mm	3 - 4 mm	≥5 mm	≥5 mm
MILD > MODERATE		MODERATE > SEVERE	
Coronal third (<15%)	Coronal third (15-33%)	Extends beyond 33% of root	Extends beyond 33% of root
None	None	≤4 teeth	≥5 teeth

AAP Periodontal Disease Classification

Key Periodontitis Staging Elements

Adapted from Tonetti MS, Greenwell H, Kornman KS. Staging and grading of periodontitis: framework and proposal of a new classification and case definition. *J Periodontol*. 2018;89(Suppl 1):S159–S172, with permission.

	Periodontitis	Stage I	Stage II	Stage III	Stage IV
Severity	Clinical attachment loss	1–2 mm	3–4 mm	≥5 mm	≥5 mm
	Radiographic bone loss	Coronal third of the root (<15%)	Coronal third of the root (15%–33%)	Middle or apical third of the root	Middle or apical third of the root
	Tooth loss due to periodontitis	No tooth loss		≤4 teeth	≥5 teeth
Complexity		• PD ≤4 mm • Mostly horizontal bone loss	• PD ≤5 mm • Mostly horizontal bone loss	In addition to Stage II: • PD ≥6 mm • Vertical bone loss ≥3 mm • Class II or III furcation involvement • Moderate ridge defects	In addition to Stage III: • Need for complex rehabilitation due to masticatory dysfunction, tooth mobility, bite collapse, pathologic migration, <20 remaining teeth
Extent and distribution		• Localized (<30% of the teeth involved) • Generalized (≥30% of the teeth involved) • Molar-incisor pattern			

Detection of Periodontal Disease

Clinical Exam

+

Radiographs

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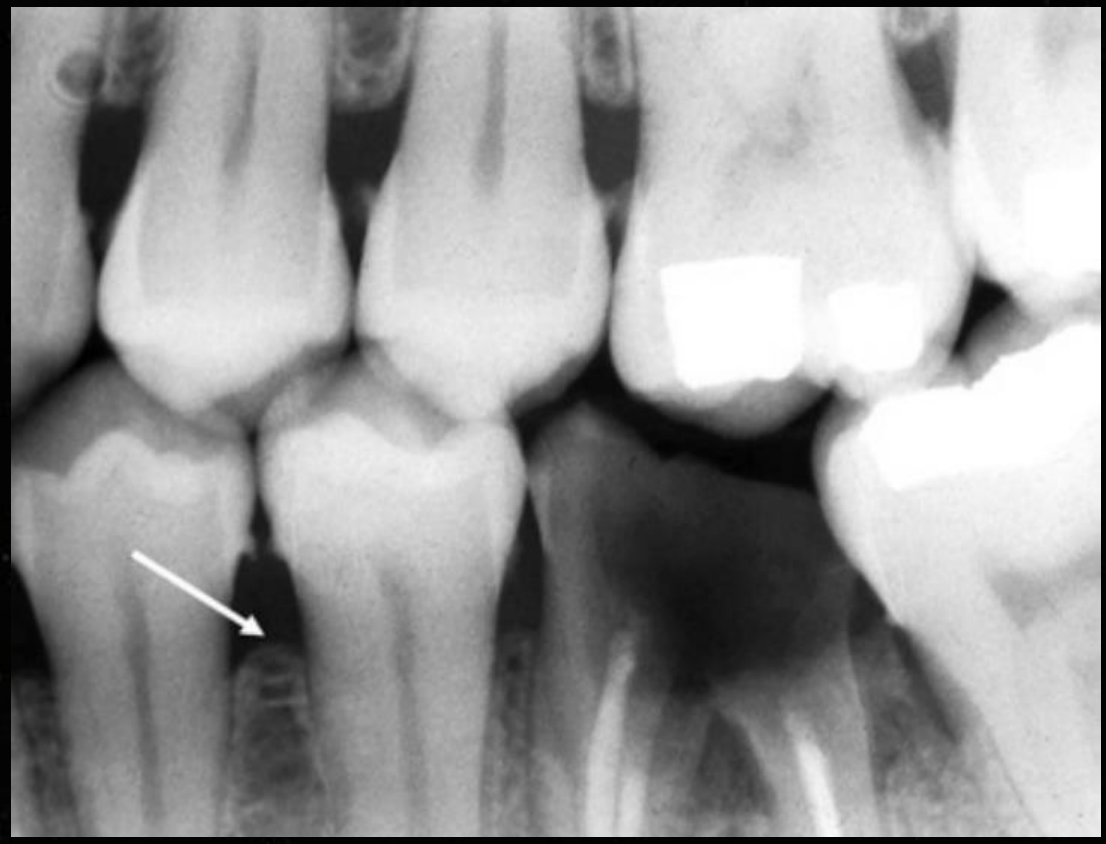
Changes in Morphology of Alveolar Processes

Early Bone Changes

- Localized erosion of the interproximal alveolar crest
- Anterior: blunting + slight loss of crestal height
- Posterior: loss of the flat shape → sharper angle
- Crest may appear rounded, irregular, and diffuse
- Early bone loss can be difficult to detect, especially buccal/lingual
- Radiographic appearance may be affected by x-ray beam angulation



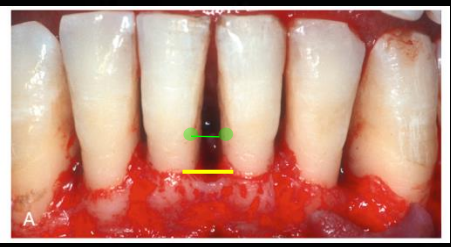
If Disease Progresses
Bone destruction may develop into:

- Horizontal bone loss
- Vertical/angular defects
- Interdental craters
- Furcation defects
- Buccal/lingual cortical plate loss

STOP THE DOMINO EFFECT

Horizontal Bone Loss

- **Loss of alveolar process height** with the crest remaining **horizontal/parallel to adjacent CEJs**
- Crest is **>2 mm apical to the CEJ** → beyond the normal range
- Resorption involves the **buccal and lingual cortical plates + interdental bone**



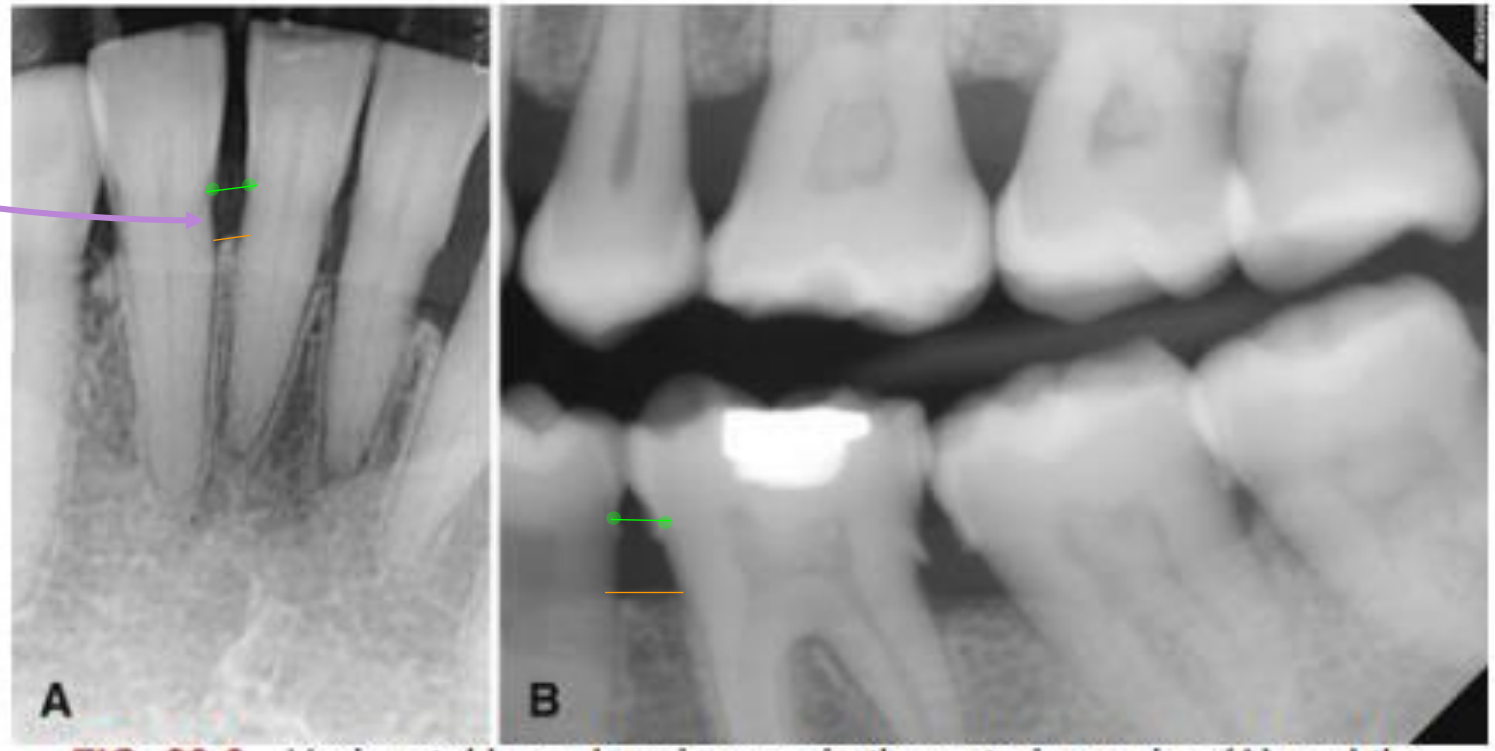


FIG. 20.6 Horizontal bone loss is seen in the anterior region (A) and the posterior region (B) as a loss of the buccal and lingual cortical plates and interdental alveolar bone.

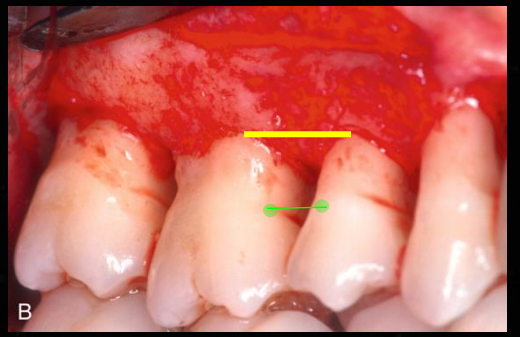


FIG. 22.37
Horizontal bone loss in the anterior (A) and posterior (B) areas. Notice the increased distance between the marginal bone and the cementoenamel junction; yet the overall bone contour is scalloped, indicating that bone resorption has affected the buccal and interproximal surfaces to a similar degree.

Buccal or Lingual Cortical Plate Loss

- **Buccal or lingual cortical plate may resorb**
- May occur **alone or with horizontal bone loss**
- Appears as increased **radiolucency near the alveolar crest**
- Typically **semicircular**, with the defect extending apically
- Can be **difficult to detect** on conventional images when interproximal bone is intact.

FIG. 20.14 (A) Loss of the lingual alveolar crest adjacent to this mandibular first premolar without associated interproximal bone loss. (B) Loss of the buccal cortical bone adjacent to the maxillary central and lateral incisors. The *black arrow* indicates the level of the buccal alveolar crest, which demonstrates more profound loss relative to the lingual alveolar crest *(white arrow)*.

Vertical Bone Changes

Angular/Infrabony Bone Defects:

- Early finding: widening of the PDL space at the alveolar crest
- Bone loss progresses down the root, deepening the periodontal pocket
- Appears as a V- or triangular-shaped defect extending apically from the alveolar crest
- Remaining alveolar bone has an oblique angulation relative to adjacent CEJs
- Typically associated with Stage III & IV periodontitis

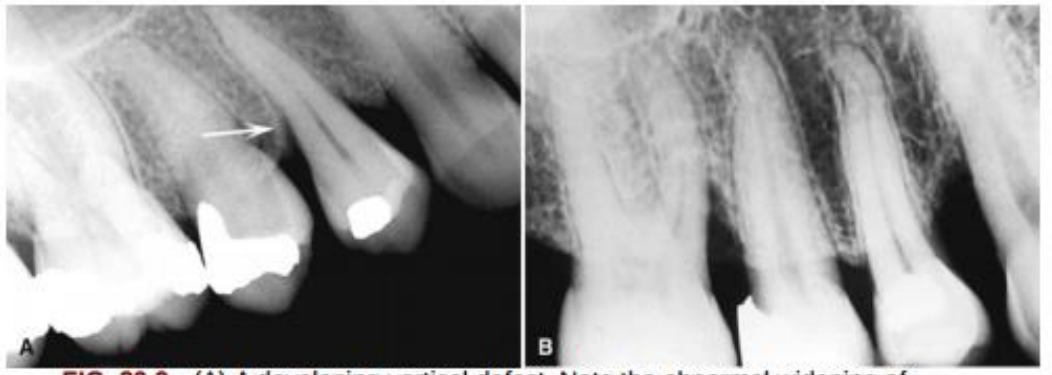


FIG. 20.9 (A) A developing vertical defect. Note the abnormal widening of the periodontal ligament space close to the crest *(arrow)*. (B) Maxillary periapical image reveals two examples of more severe vertical defects affecting the mesial surface of the first molar and the distal surface of the canine.

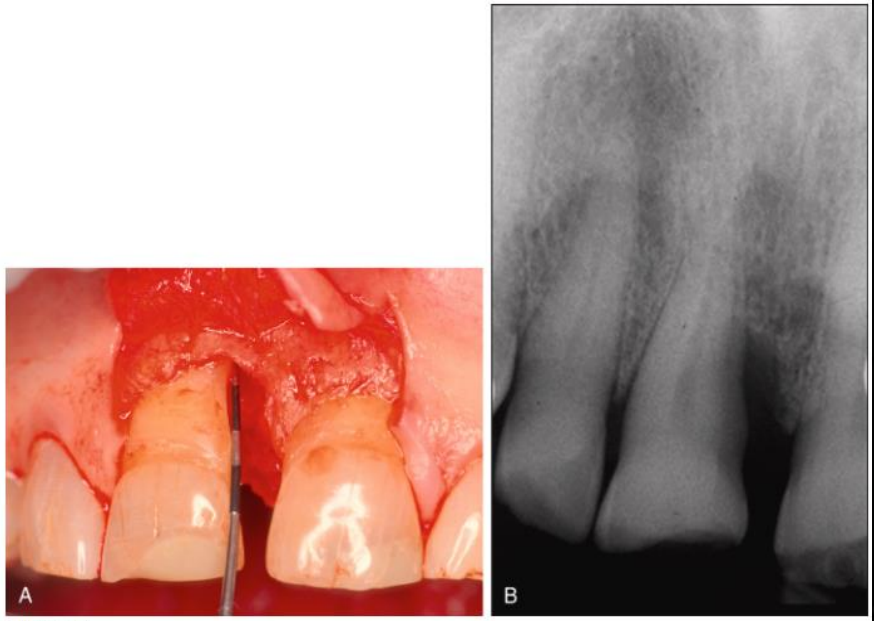


FIG. 22.38
Clinical (A) and radiographic (B) views of a vertical (angular) defect mesial of the upper right central incisor. Note the periodontal probe in a more apical position than the surrounding walls of the bony defect.

Vertical Bone Changes

- **Vertical bitewings** may be needed to show the full extent of bone loss.

- **Gutta-percha** can help visualize the depth of the defect.

- **CBCT** may better characterize complex defects but is **not routinely indicated**.

FIG. 20.10 Typical Vertical Bone Loss in Stage IV Periodontitis. Note the bone loss is confined to the region of the first molars. (Courtesy Dr. T. Charbeneau, Dallas, TX.)

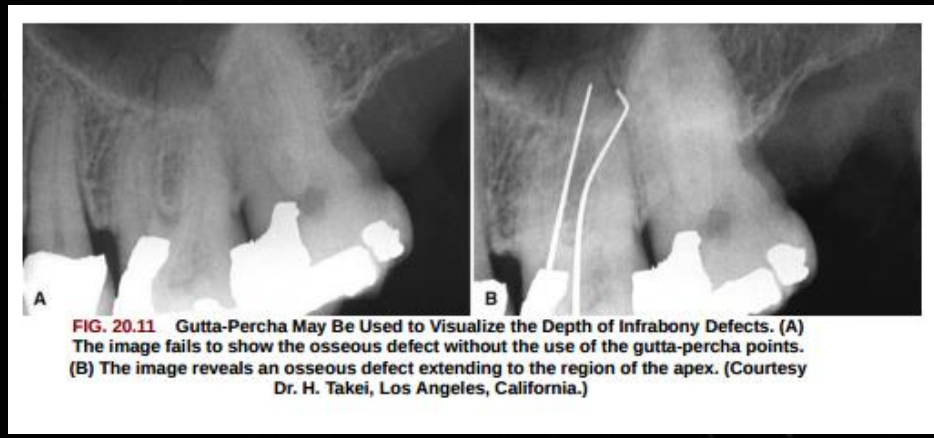
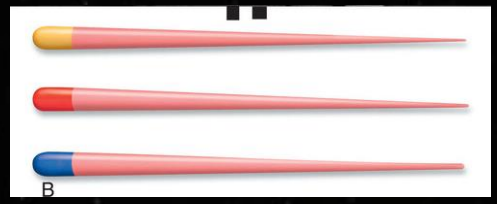


FIG. 20.11 Gutta-Percha May Be Used to Visualize the Depth of Infrabony Defects. (A) The image fails to show the osseous defect without the use of the gutta-percha points. (B) The image reveals an osseous defect extending to the region of the apex. (Courtesy Dr. H. Takei, Los Angeles, California.)



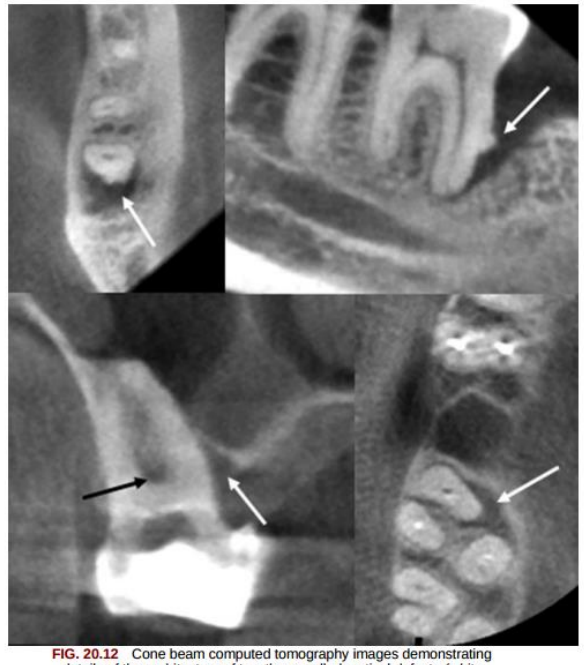
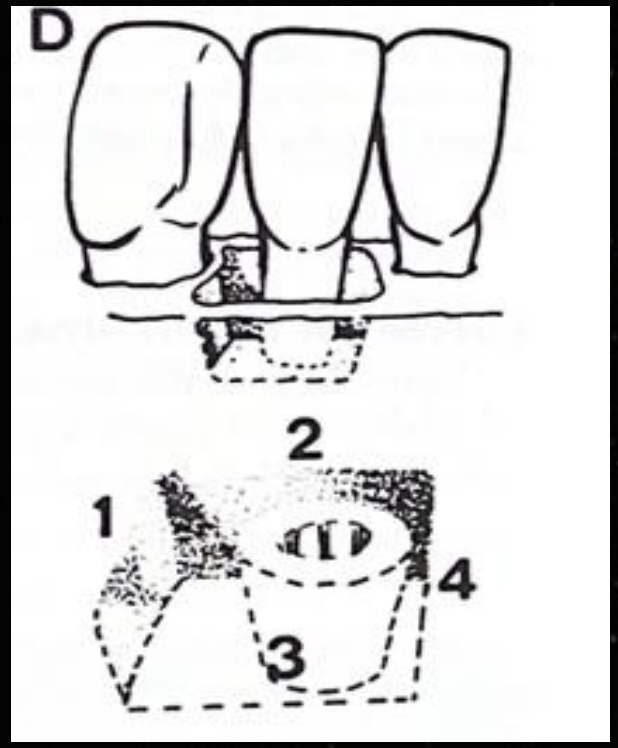


FIG. 20.12 Cone beam computed tomography images demonstrating details of the architecture of two three-walled vertical defects *(white arrows)*. Axial *(upper left)* and parasagittal *(upper right)* reconstructions show a deep vertical defect on the distal surface of the mandibular left second molar. Calculus is seen on the root surface. Coronal *(lower left)* and axial *(lower right)* images of another case show a three-walled defect on the palatal aspect of the mesiobuccal root of a maxillary molar. Early furcation involvement is also detected on this tooth *(black arrow)*.

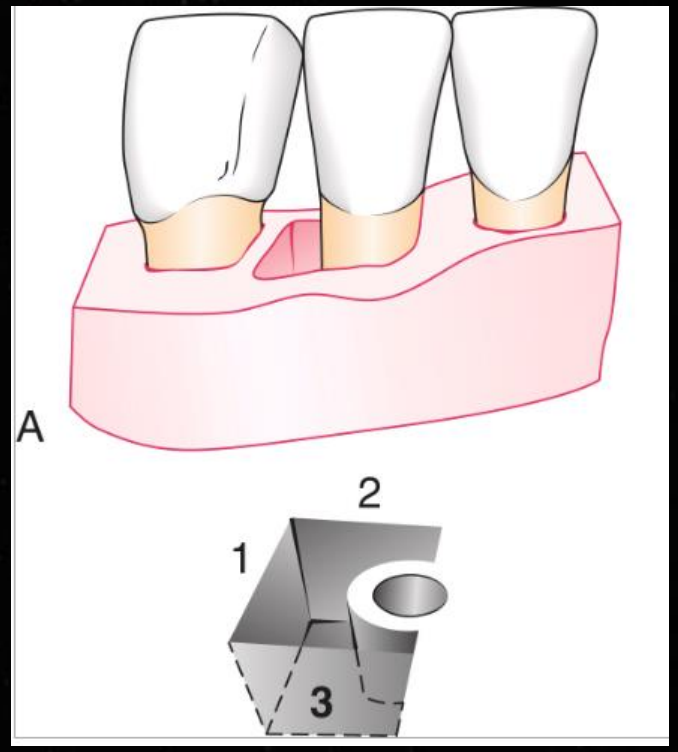
Osseous/Interdental Craters

- Goldman and Cohen classified angular defects on the basis of the number of osseous walls
- Cancellous bone is resorbed, while buccal and lingual cortical walls remain intact
- Appears as a radiolucent band or trough at the crest
- More common in the posterior jaws due to the greater buccolingual width of the alveolar crest.
- Clinical/surgical inspection is best for determining the number of remaining walls.



4 Walled

All four bony walls are intact, with the fourth wall extending past/behind the root of the tooth



3 Walled

Buccal + Lingual cortical plates intact

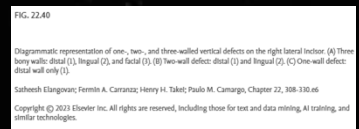


FIG. 22.40

Diagrammatic representation of one-, two-, and three-walled vertical defects on the right lateral incisor. (A) Three bony walls: distal (1), lingual (2), and facial (3). (B) Two-wall defect: distal (1) and lingual (2). (C) One-wall defect: distal wall only (1).

Satheesh Elangovan; Fermin A. Carranza; Henry H. Takei; Paulo M. Camargo, Chapter 22, 308-330.e6

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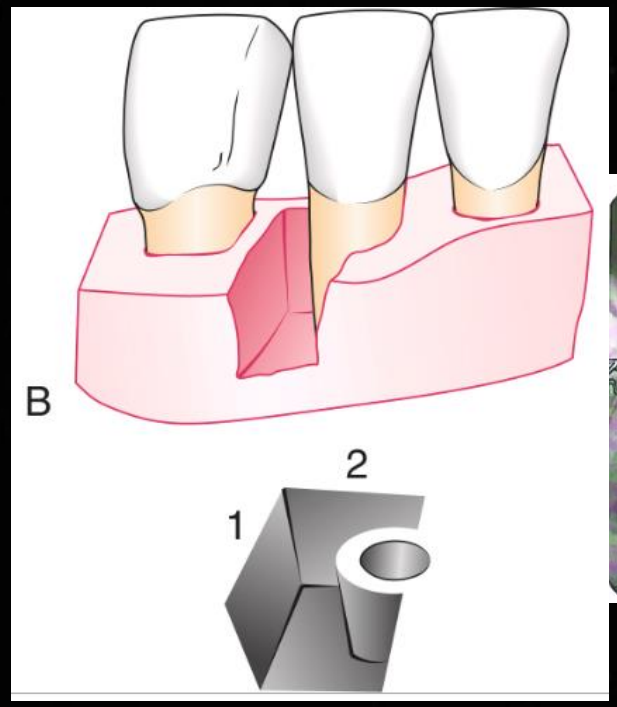
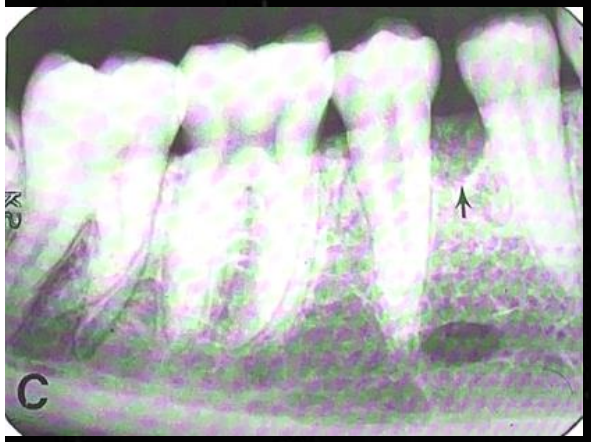
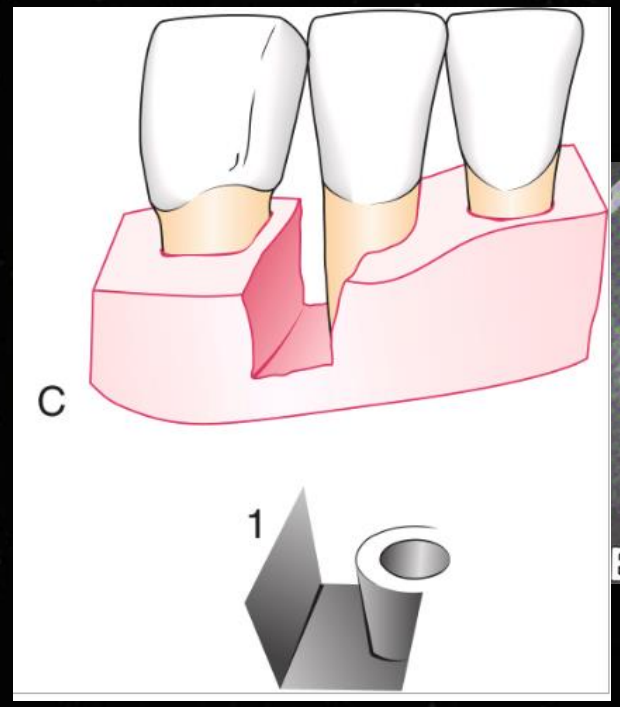
Osseous/Interdental Craters

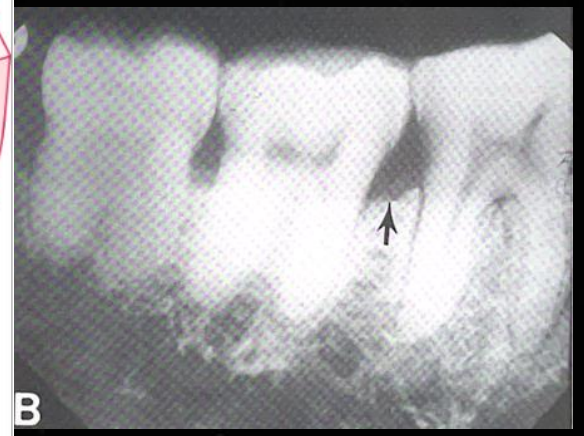




2 Walled

One cortical plate resorbed





1 Walled

Both cortical plates lost

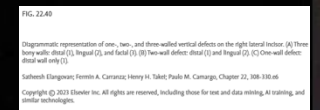


FIG. 22.40

Diagrammatic representation of one-, two-, and three-walled vertical defects on the right lateral incisor. (A) Three bony walls: distal (1), lingual (2), and facial (3). (B) Two-wall defect: distal (1) and lingual (2). (C) One-wall defect: distal wall only (1).

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Osseous/Interdental Craters

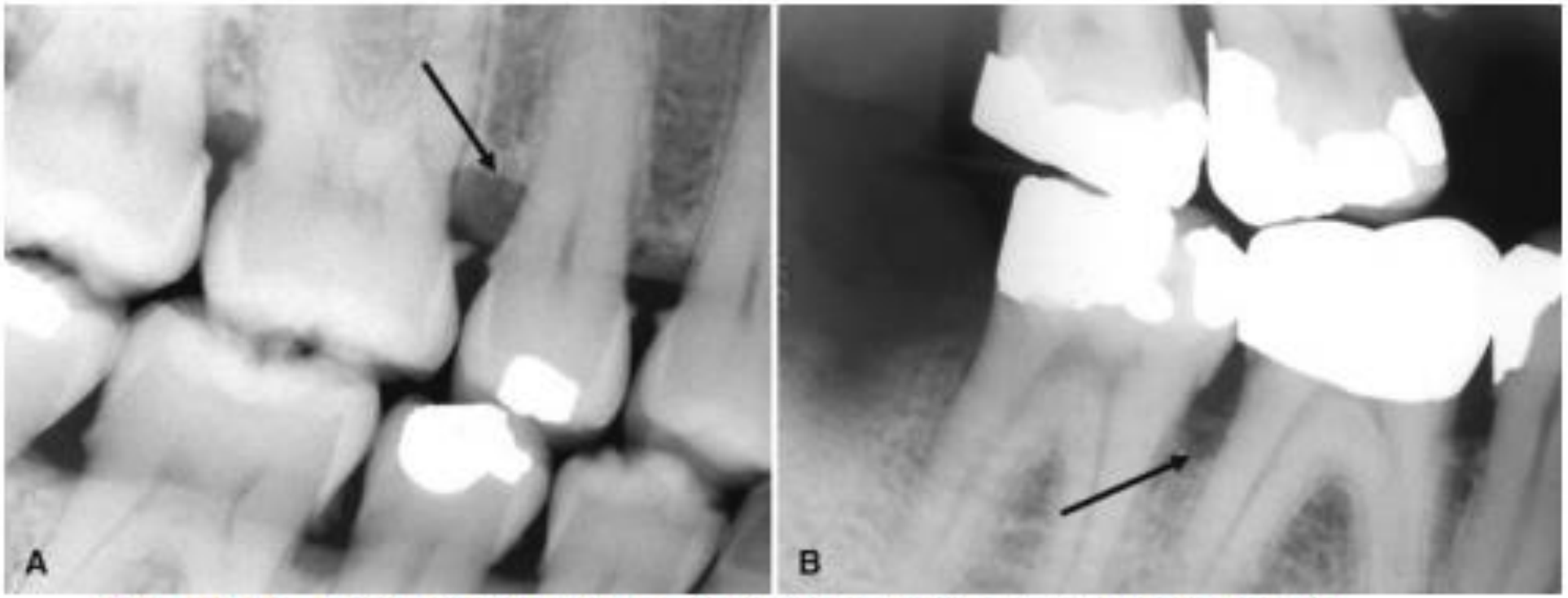


FIG. 20.13 Interproximal craters, existing as defects between the buccal and lingual cortical plates, seen as a radiolucent band (A) or trough (B) apical to the level of the crestal edges. The *arrows* indicate the base of the craters.

Important Considerations

- An angled-looking crest is not necessarily a vertical defect; compare it with the imaginary line between adjacent CEJs.

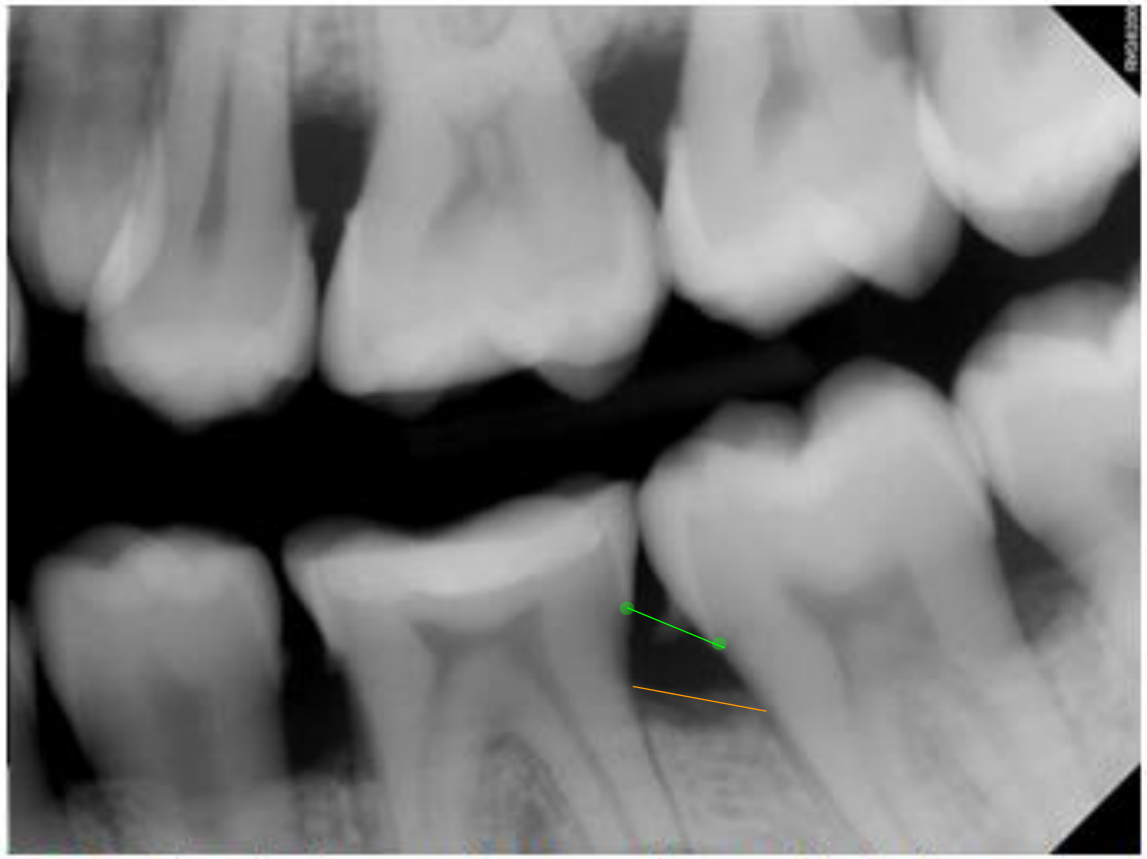


FIG. 20.7 The alveolar crest between the mandibular first and second molars appears angled but is approximately parallel to the imaginary plane created by the adjacent cementoenamel junctions, differentiating it from a vertical (angular) bone defect. There is an interproximal crater and reactive bone sclerosis present at this site.

Important Considerations cont.

- Supraeruption & severe attrition/passive eruption = makes CEJ-to-bone measurements misleading.
- The amount of bone loss on one image does not indicate current disease activity.
- Reestablishment of crestal cortication suggests stabilization after treatment.

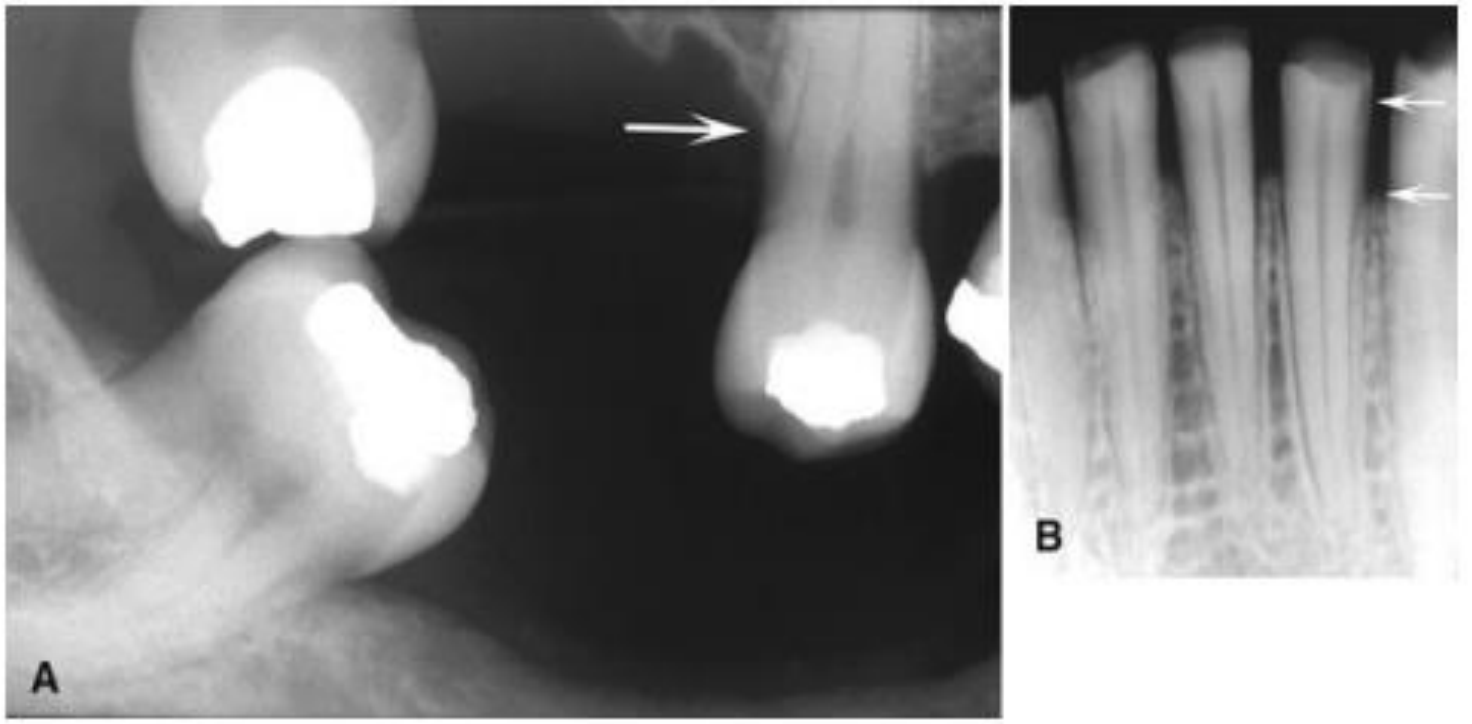


FIG. 20.8 (A) This maxillary second premolar is supraerupted; the etiology of the low bone level *(arrow)* relative to the cementoenamel junction (CEJ) is not necessarily the result of periodontal disease. (B) An example of passive eruption related to severe attrition resulting in the apparent increase in the distance from the CEJ to the bone height *(arrows)* cannot be attributed to periodontal disease. However, the resultant change in bone level relative to the CEJ still may be clinically significant.

Osseous Deformities in the Furcations of Multirooted Teeth

- Periodontal bone loss may extend into the furcation of multirooted teeth.
- Early sign: widening of the PDL space at the interradicular crest.
- More extensive loss → radiolucency in the furcation.
- May involve the buccal or lingual cortical plate alone.
- Buccal object rule with different angulations can help identify which cortical plate is lost.

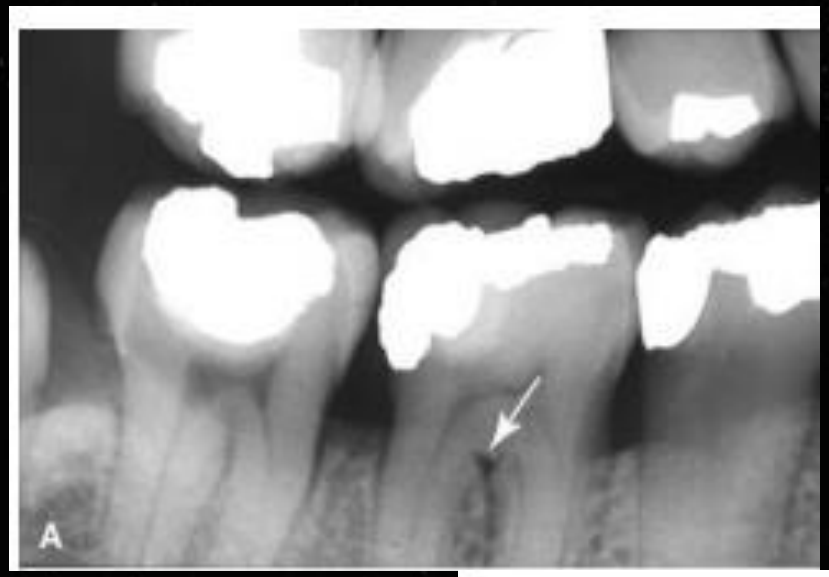


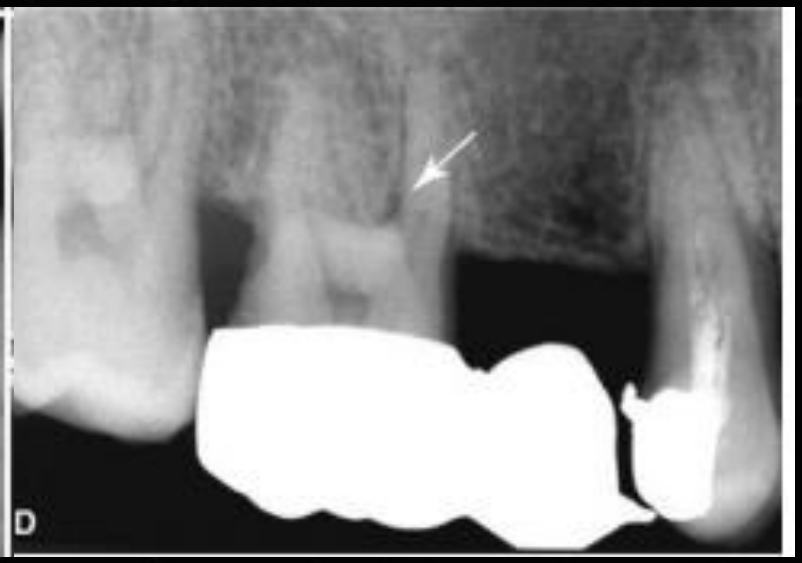
FIG. 20.15 (A) Periapical image revealing very early furcation involvement of a mandibular molar characterized by slight widening of the periodontal ligament (PDL) space in the furcation region *(arrow)*. (B) Periapical image revealing a profound radiolucent lesion within the furcation region *(arrow)* resulting from loss of bone in the furcation region and the buccal and lingual cortical plates. (C) Large furcation defect associated with the

Osseous Deformities in the Furcations of Multirooted Teeth

Important Imaging Limitations:

- External oblique ridge may obscure mandibular third-molar furcation defects.
- Convergent roots may obscure furcation defects in second and third molars.
- Maxillary molar defects are often less sharply defined due to palatal root superimposition.
- May appear as an inverted "J-shaped" radiolucency* or radiolucent triangle.

Key point: Complex furcation defects require clinical examination ± surgical exploration; CBCT may aid treatment planning when further characterization is needed.



lingual cortical plates. (C) Large furcation defect associated with the mandibular right third molar, which is less easily visualized due to superimposition by the external oblique ridge. There is also a combined periodontic-endodontic lesion of the first molar *(arrow)*. (D) Angulation of this periapical view of a maxillary first molar projected the palatal root away from the trifurcation region revealing early widening of the furcation PDL space *(arrow)*. (E) Example of an inverted "J-shape" radiolucent lesion *(arrow)* resulting from bone destruction extending into the trifurcation region of a three-rooted maxillary first bicuspid.

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Furcation Involvement in Periodontal Bone Loss

Grade I: Early furcation involvement; usually not visible radiographically

Grade II: Partial furcation bone loss; a radiolucency may be present but can be difficult to detect

Grade III: Through-and-through bone loss between the roots; the furcation radiolucency is clearly evident and appears as a “crotch” radiolucency

Grade IV: Same radiographic appearance as Grade III, but clinical gingival recession exposes the furcation, making it clinically visible

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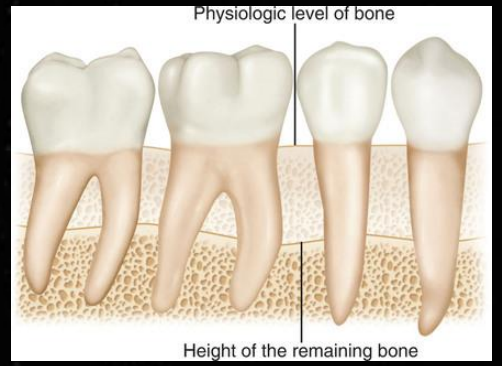
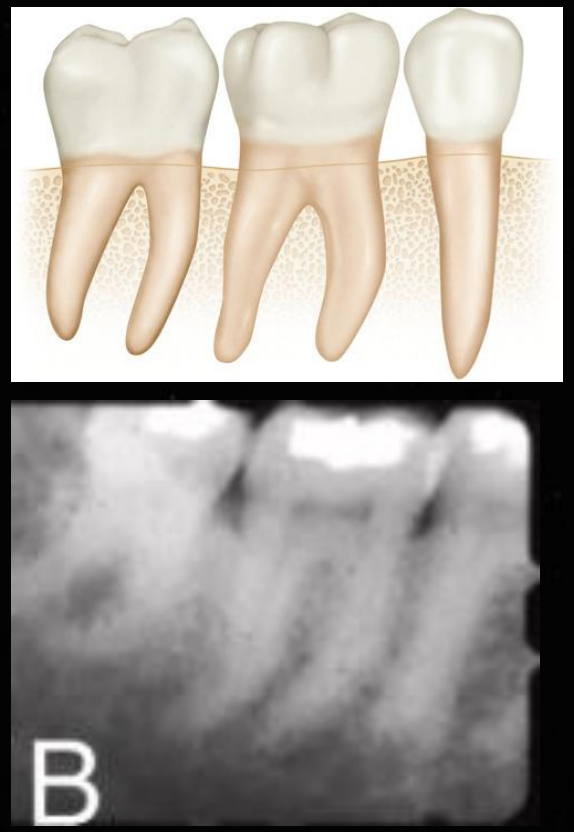
Severity of Periodontal Bone Loss

Measure the distance from the CEJ to the current alveolar bone crest and relate it to the root length to estimate the percentage of radiographic bone loss.



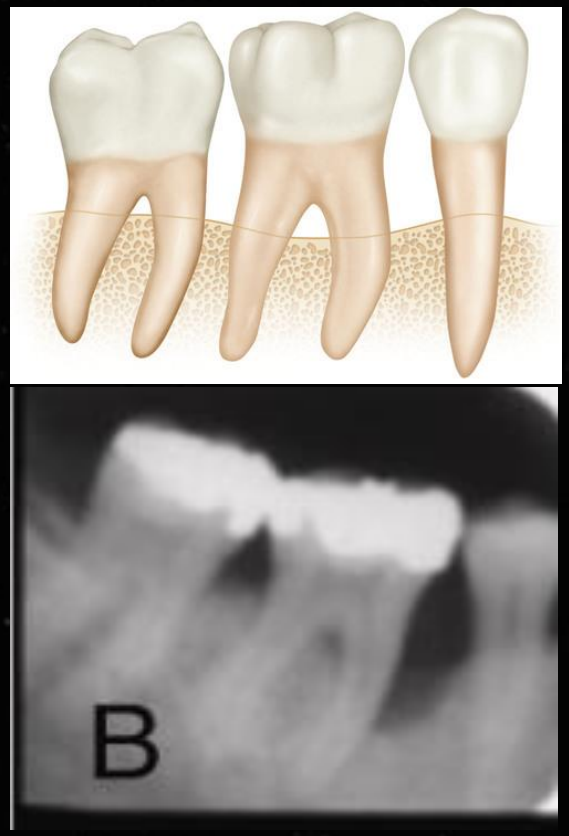


Stage I (Mild):

≤15% root length

Stage II (Moderate):

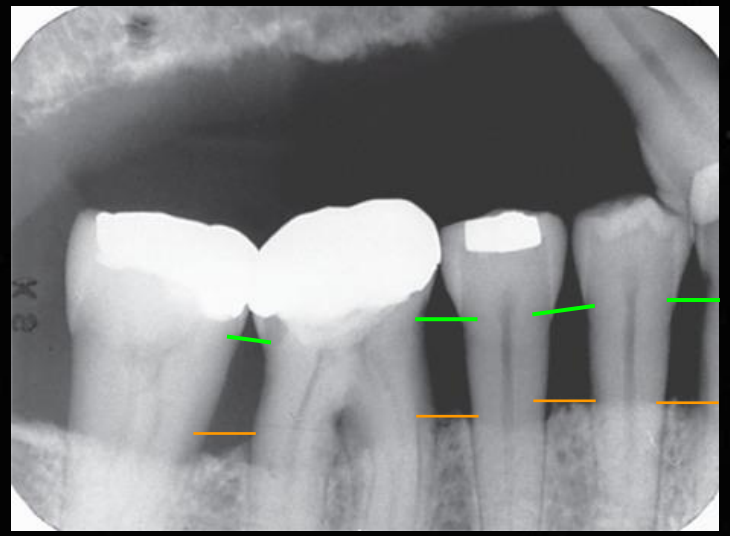
15–33% root length



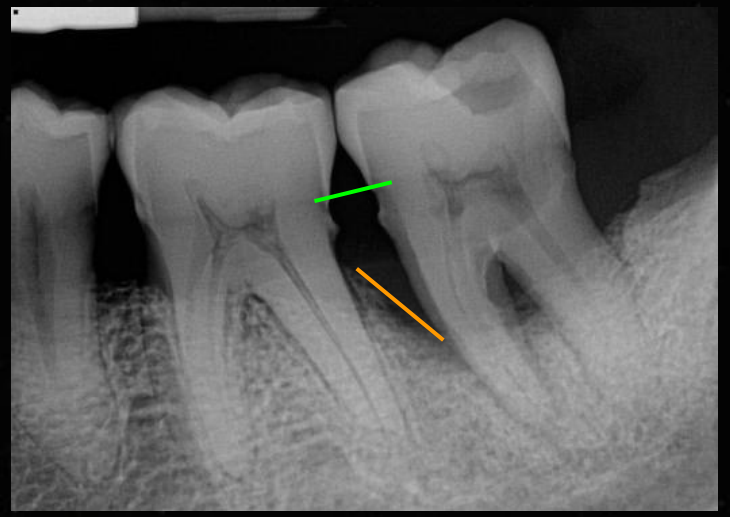
Stages III-IV (Severe):

extends to middle third of root or beyond

Recognize the Pattern of Periodontal Bone Loss

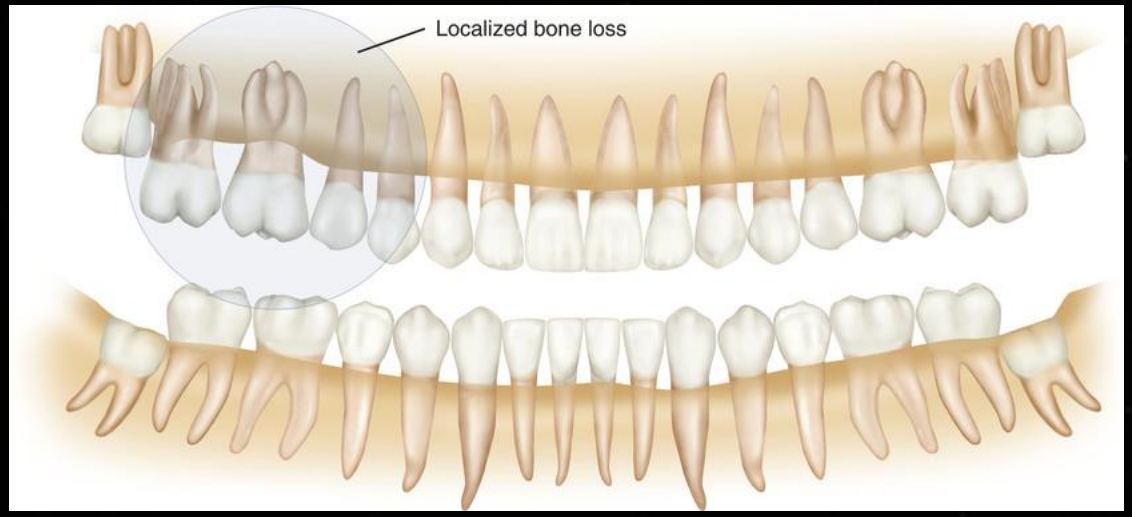


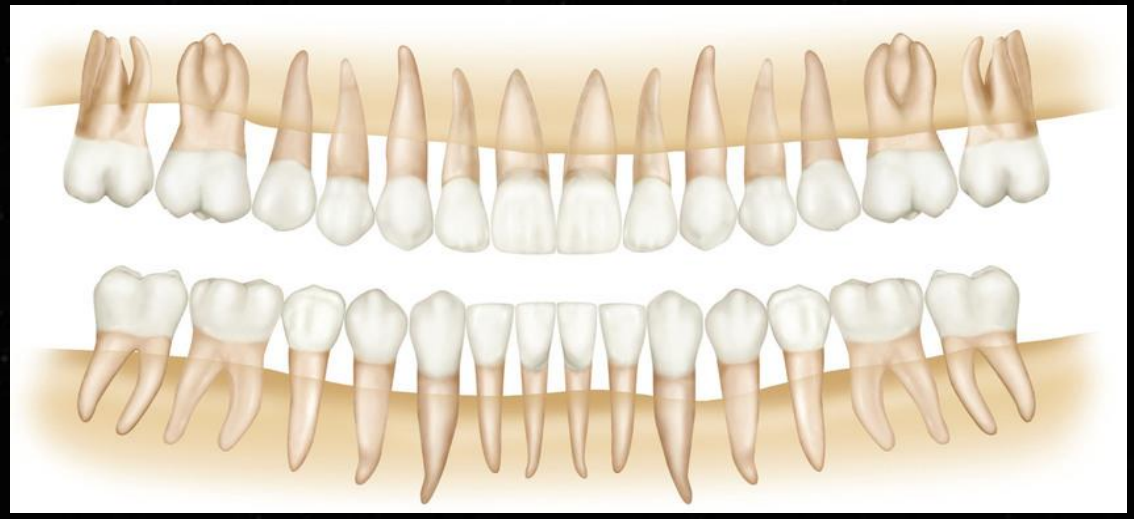
Horizontal



Vertical

Distribution of Periodontal Bone Loss





Localized: <30% of sites

Generalized: >30% of sites

End of 1st hour

Changes to the Internal Density and Trabecular Pattern of Bone

Bone Reaction to Periodontal Inflammation

- Radiolucency (rarefaction): decreased density of bone → more faint trabeculae; more common in early or acute lesions.
- Radiopacity (sclerosis): increased bone deposition → thicker, denser trabeculae; more common in chronic, low-grade inflammation
- Mixed pattern: combination of rarefaction and sclerosis is most common.

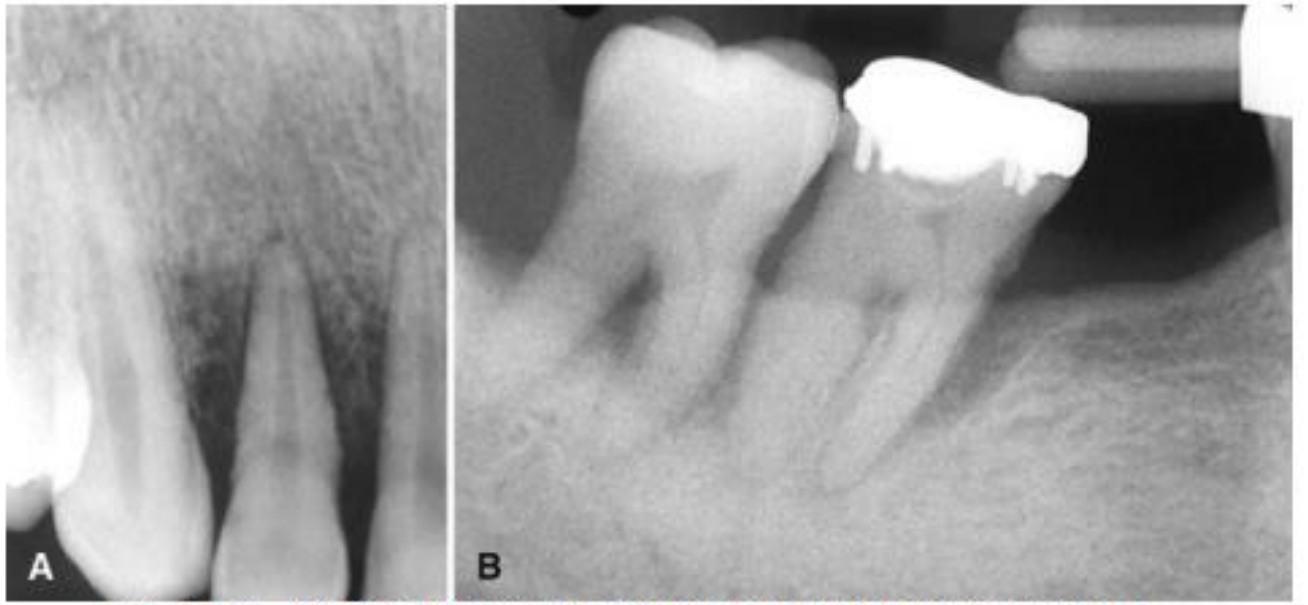


FIG. 20.16 (A) Example of a primarily radiolucent reaction around this maxillary lateral incisor. The trabeculae toward the alveolar crest on the mesial and distal aspect of the tooth are barely perceptible, and the marrow spaces are enlarged. (B) Periapical image revealing a predominantly sclerotic bone reaction resulting from the periodontal disease involving the mandibular molars. The trabeculae are thickened, and the marrow spaces are barely perceptible.

Bone Reaction to Periodontal Inflammation

- Other effects: inflammation may extend to the maxillary sinus, causing regional mucositis; rarely, a periosteal reaction may occur.

FIG. 20.17 Periapical image revealing a localized mucositis within the maxillary sinus *(arrows)* immediately adjacent to a vertical periodontal defect.

Predisposing Factors Affecting the Periodontium

BOOK CHAPTER

The Role of Dental Calculus and Other Local Predisposing Factors

Vivek Thumbigere-Math; James E. Hinrichs

Newman and Carranza's Clinical Periodontology and Implantology, Chapter 24, 341-358.e6

Primary Etiologic Factor – Bacterial Plaque & Calculus

- Calcified plaque
- Radiopaque
- Spiky, ring-like, nodular, or smooth deposits
- Plaque-retentive → inflammation + bone loss

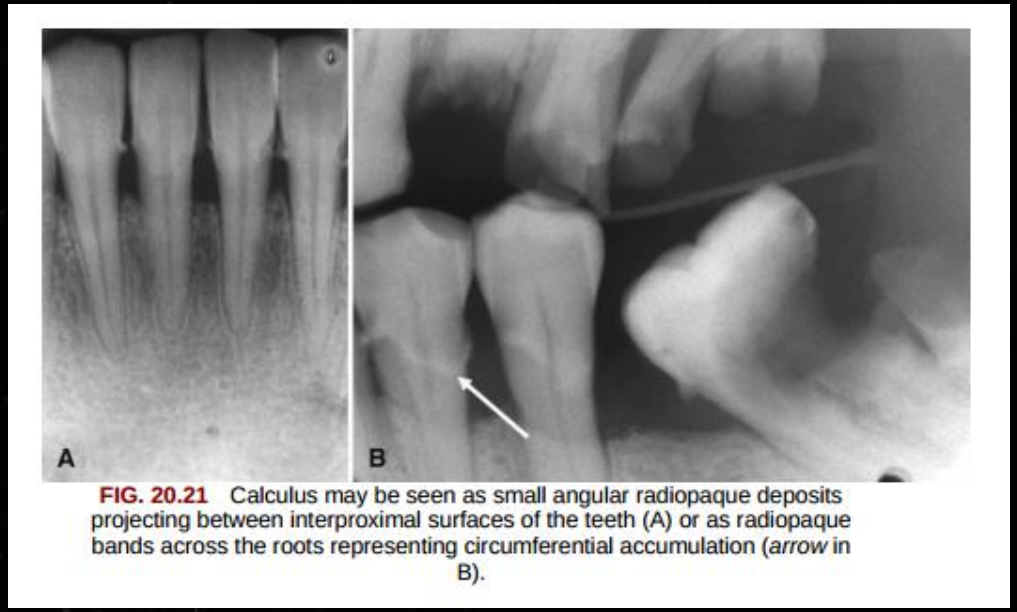


FIG. 20.21 Calculus may be seen as small angular radiopaque deposits projecting between interproximal surfaces of the teeth (A) or as radiopaque bands across the roots representing circumferential accumulation (*arrow* in B).

Local Irritating Factors – Restorative & Iatrogenic

- Deficient margins
- Open contacts
- Overcontoured crowns/pontics
- Overhanging/subgingival margins
- Partial denture clasps retaining plaque → Plaque retention + food impaction

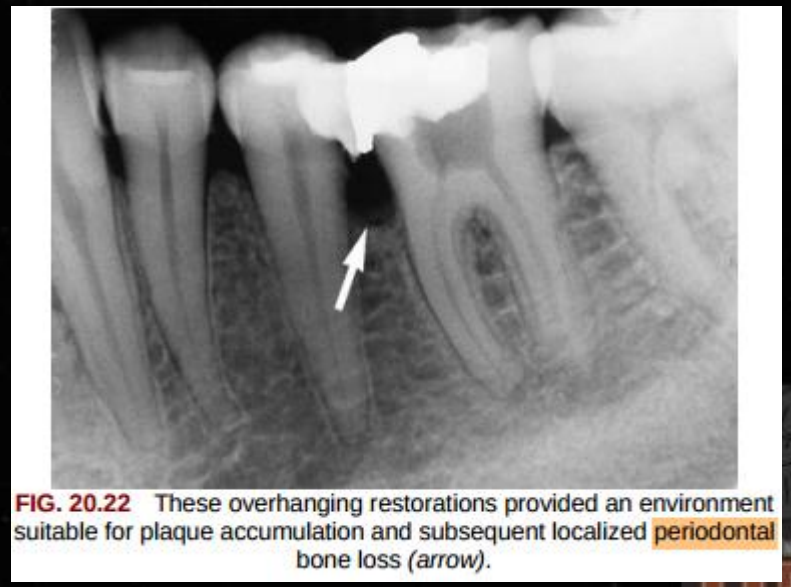


FIG. 20.22 These overhanging restorations provided an environment suitable for plaque accumulation and subsequent localized periodontal bone loss *(arrow)*.

Local Irritating Factors – Restorative & Iatrogenic cont.

- Poor adaptation
- Poor contours, rough surfaces
- Retained cement / poorly positioned implants
- Traumatic restorative procedures

FIG. 24.15
(A) A fenestration defect *(arrow)* is noted after placement of a dental implant too far labial and apical, yielding inadequate bone and blood supply to support the surrounding gingiva. (B) Radiograph of the malpositioned dental implant.

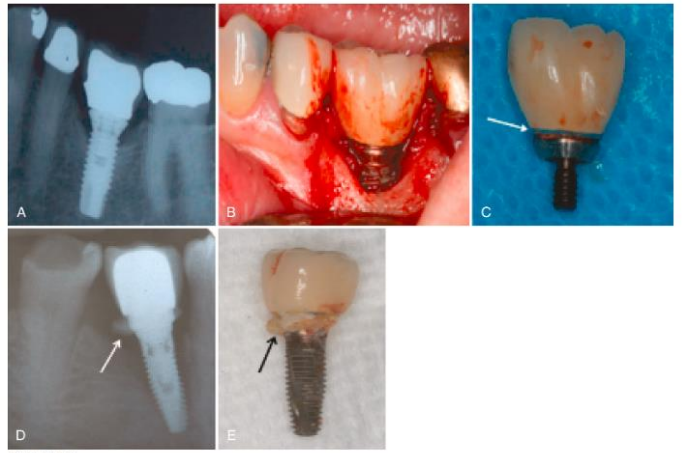


FIG. 24.20
(A) Radiograph of a moat-like bony defect at a peri-implantitis site associated with retained cement and a crown that is not fully seated. (B) Moat-like bony defect and retained excessive cement. (C) The *arrow* points to visibly retained excessive cement between the crown margin and the abutment that prevents the full seating of the crown and leads to hyperocclusion. (D) Radiograph of a peri-implantitis site shows bone loss associated with retained excessive cement *(arrow)*. (E) Retrieved implant restoration shows excessive cement remnants *(arrow)* on implant surface.

Local Irritating Factors – Anatomic & Occlusal

- Crowding/malocclusion
- Food impaction / missing teeth
- Mouth breathing
- Occlusal discrepancies
- Prominent roots & inadequate attached gingiva
- Tongue Thrusting Habit

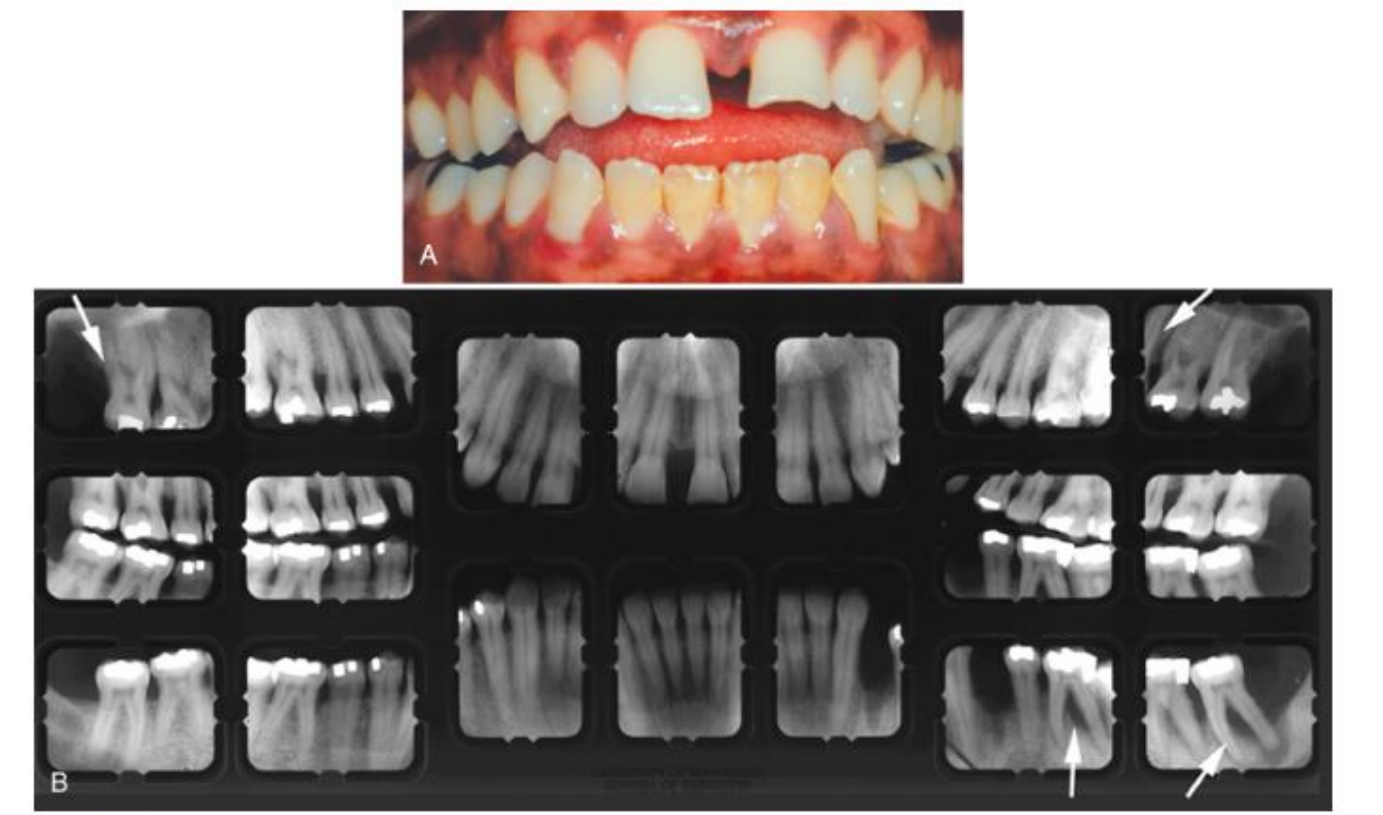
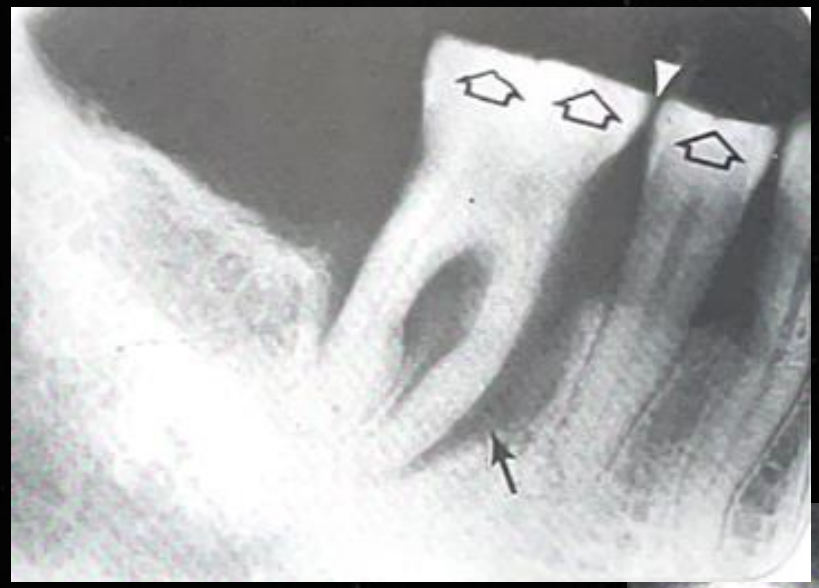


FIG. 24.24
(A) Anterior open bite with flared incisors as observed in association with a habit of tongue thrusting. (B) Radiographs show severe periodontal destruction *(arrows)* in the molar regions.

Local Irritating Factors – Food Packing Areas

- Attrition
- Caries opening contacts
- Improper restorations
- Missing adjacent teeth
- Open contacts
- Partial impactions
- Plunger cusps
- Supraeruption



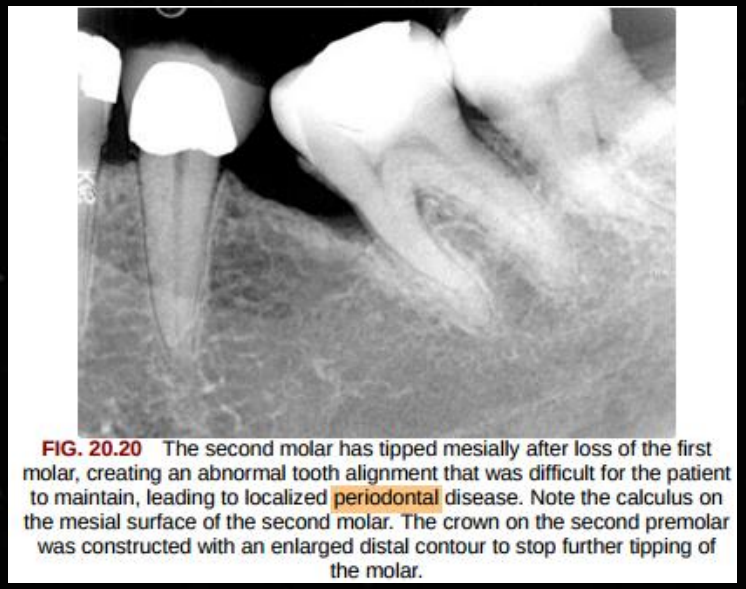
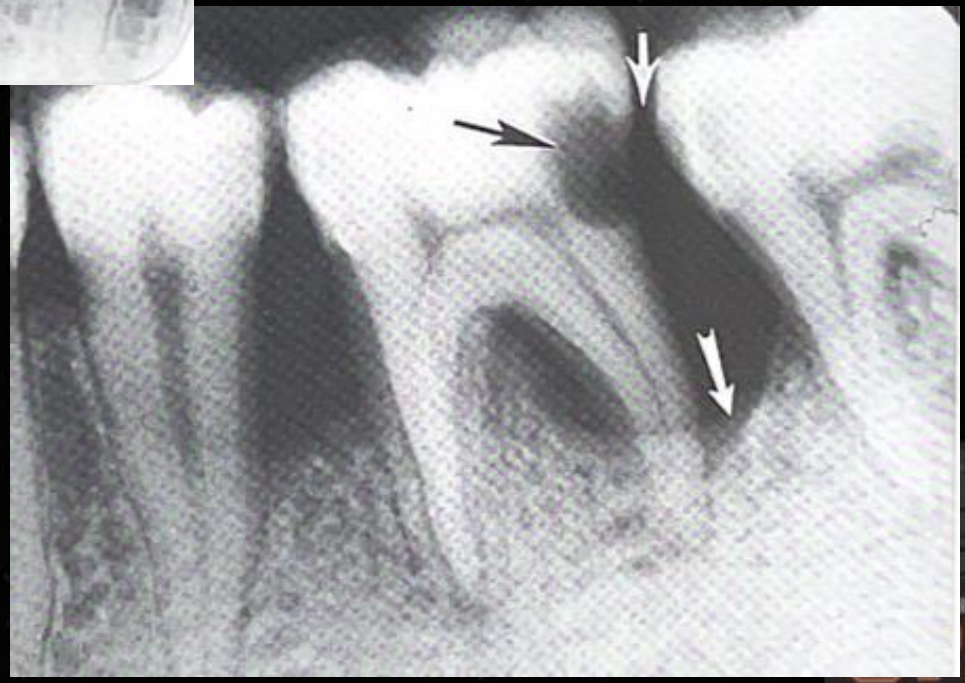


FIG. 20.20 The second molar has tipped mesially after loss of the first molar, creating an abnormal tooth alignment that was difficult for the patient to maintain, leading to localized periodontal disease. Note the calculus on the mesial surface of the second molar. The crown on the second premolar was constructed with an enlarged distal contour to stop further tipping of the molar.



Local Irritating Factors – Orthodontic

- Plaque retention around appliances
- Gingival trauma from bands
- Excessive/unfavorable forces
- Root resorption & attachment loss
- Radiographic signs of physiologic PDL space widening from orthodontic forces are commonly seen

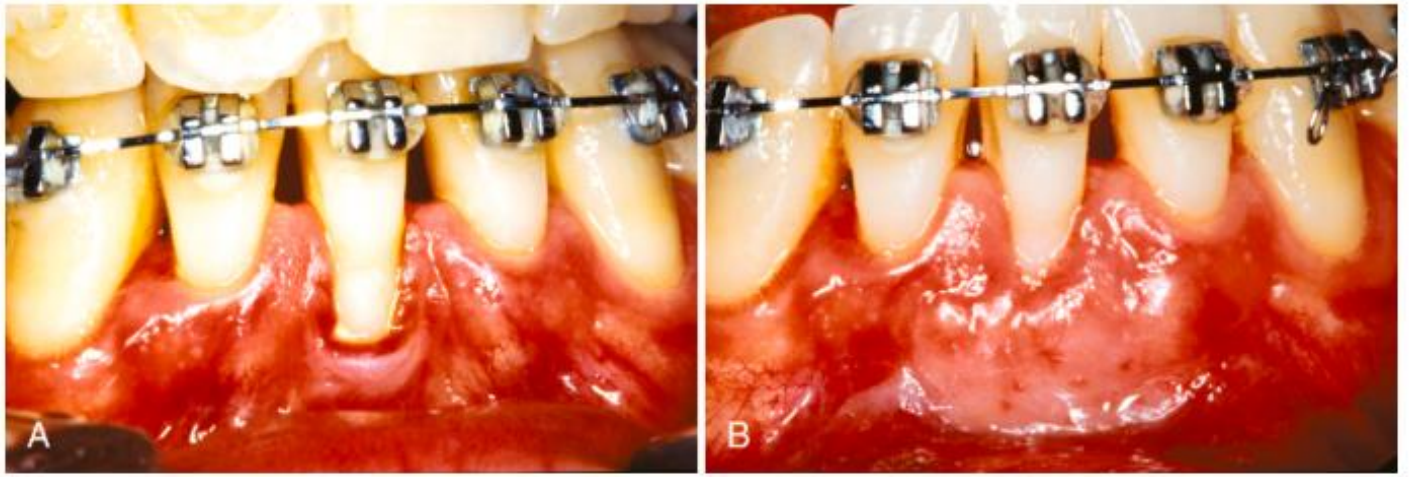


FIG. 24.23
(A) Lower incisor showing a prominent root with gingival recession and a lack of attached gingiva. (B) Appearance 3 months after the placement of a free gingival graft, which resulted in a gain in attached gingiva and a reduction in gingival recession.

Local Irritating Factors – Habits & Trauma

- Aggressive toothbrushing/flossing
- Toothpick or fingernail trauma
- Chemical irritation
- Tongue thrusting / mouth breathing
- Oral piercings

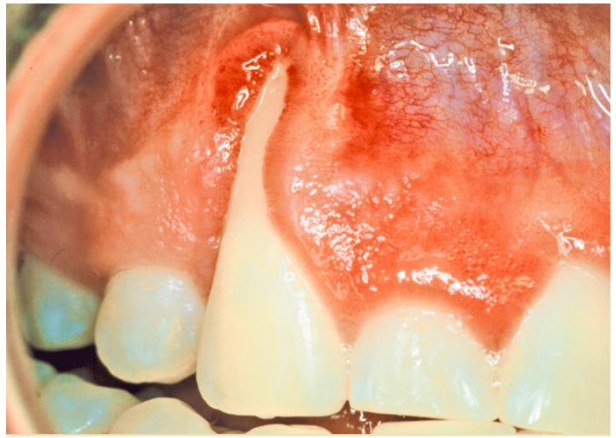


FIG. 24.32
Gingival recession on a maxillary canine caused by self-inflicted trauma from the patient's fingernail.

FIG. 24.34
The overzealous use of a toothbrush resulted in denudation of the gingival epithelial surface and exposure of the underlying connective tissue as a painful ulcer.

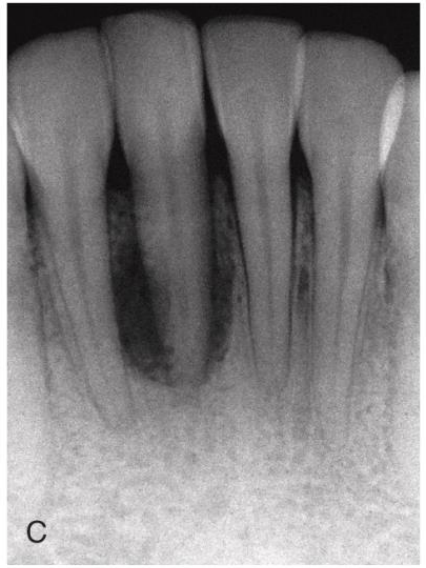


FIG. 24.33
(A) Tongue pierced with oral jewelry. (B) Probing depth of 8 mm with 10 mm of clinical attachment loss on the lingual surface of the lower central incisor adjacent to the oral jewelry in the pierced tongue. The central incisor was found to be vital. (C) Bone loss associated with a tongue pierced by oral jewelry.

Local Irritating Factors – Tobacco

- Smoking and smokeless tobacco
- Recession, attachment loss, leukoplakia
- Increased plaque/calculus retention

FIG. 24.39
Oral leukoplakia, recession, and clinical attachment loss associated with the use of smokeless tobacco.

FIG. 24.37
Oral leukoplakia in the vestibule associated with the use of smokeless tobacco.

Local Irritating Factors – Radiation Therapy

- Xerostomia & increased plaque
- Mucosal injury
- Irradiated teeth may show increased recession, attachment loss, and mobility.
- Risk of osteoradionecrosis

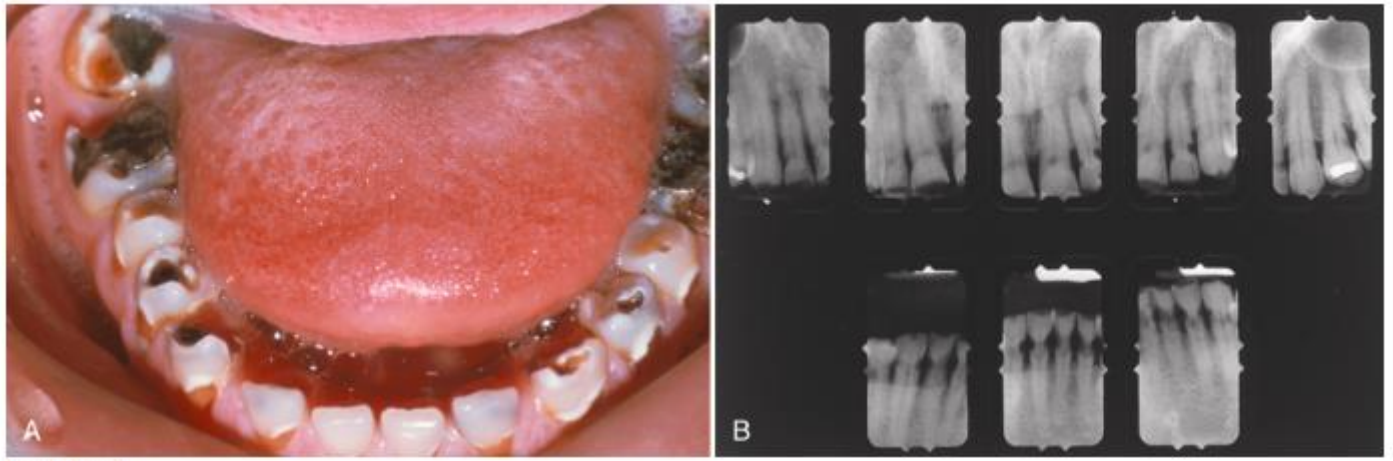


FIG. 67.4
(A) In a clinical view of a patient with radiation caries, notice how the caries primarily affected the smooth tooth surfaces, especially cervical areas and cusp tips. (B) Radiographs of anterior teeth of 52-year-old man with postradiation caries. The patient received a dose of 6000 cGy for radiation treatment of the posterior mandible and base of the tongue for squamous cell carcinoma. Radiation caries developed within 1 year after radiation treatment, affecting the cervical areas and incisal edges of the anterior teeth.

Other Conditions Affecting the Periodontium

Endodontic-Periodontic Lesions (EPL's)

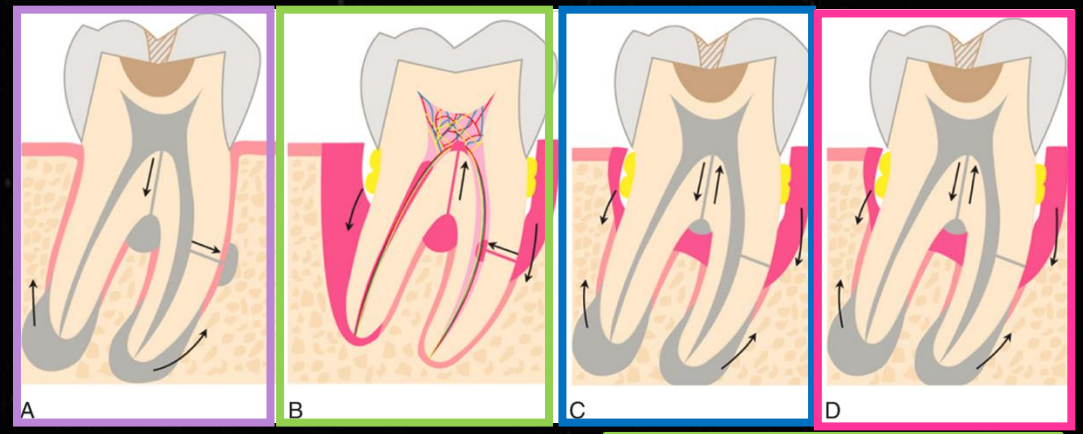
BOOK CHAPTER

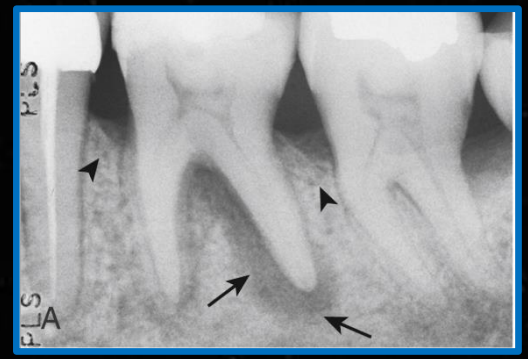
Endo-Perio Lesions : Pathogenesis, Diagnosis, and Treatment Considerations

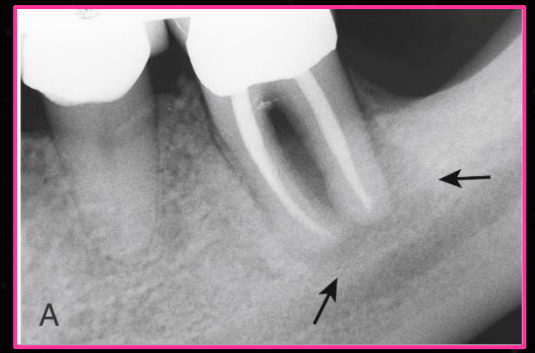
Mo K. Kang; Shebli Mehrazarin; Kenneth C. Trabert

Newman and Carranza's Clinical Periodontology and Implantology, Chapter 49, 597-609.e3

- Pulp–periodontium connection: Apical foramen, accessory canals, and dentinal tubules allow bacterial spread.
- Lesions may: Develop independently, merge, or cause one another.
- Main types:
  - A- Primary pulpal
  - B- Primary periodontal*
  - C- Independent
  - D- Combined
- Direction of spread:
  - Periodontal lesion → apex → secondary pulpal involvement
  - Periapical lesion → coronally → secondary periodontal involvement
- Pulpal origin: Pulpal necrosis more commonly causes secondary periodontal breakdown.







EPL's cont.

- Radiographic features: Deep angular defect extending to the apex and communicating with a periapical lesion.
- Typical shape: Uniform or slightly wider coronally → funnel-shaped defect.
- Extent: May involve one or multiple surfaces or surround the entire root.
- Treatment: Endodontic therapy first, followed by periodontal treatment as needed.
- Key point: Resolving pulpal infection may allow secondary periodontal defects to heal.
- Vertical tooth defects: Coronal cracks and vertical root fractures differ in location, direction, and prognosis.
- Implants: Persistent endodontic infection can cause retrograde peri-implantitis.

FIG. 20.19 Combined Endodontic-Periodontic Lesion. There is an angular bone defect extending the whole length of the mesial surface of the second molar, from the crest to the apex *(arrows)*.

*Periodontal Abscess

- Rapid, destructive lesion usually arising from a deep periodontal pocket.
- Clinical: pain, swelling ± draining fistula, pus
- Early/acute: may show no radiographic changes.
- Persistent lesion: localized radiolucency/rarefaction, often superimposed on the tooth root.
- May show loss of lamina dura with a bridge of bone separating the lesion from the alveolar crest.
- After treatment: some bone regeneration may occur.

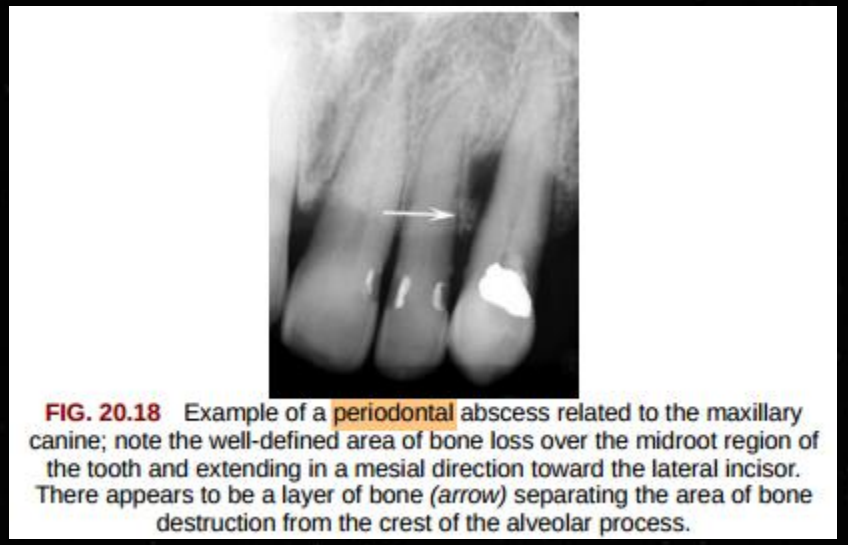
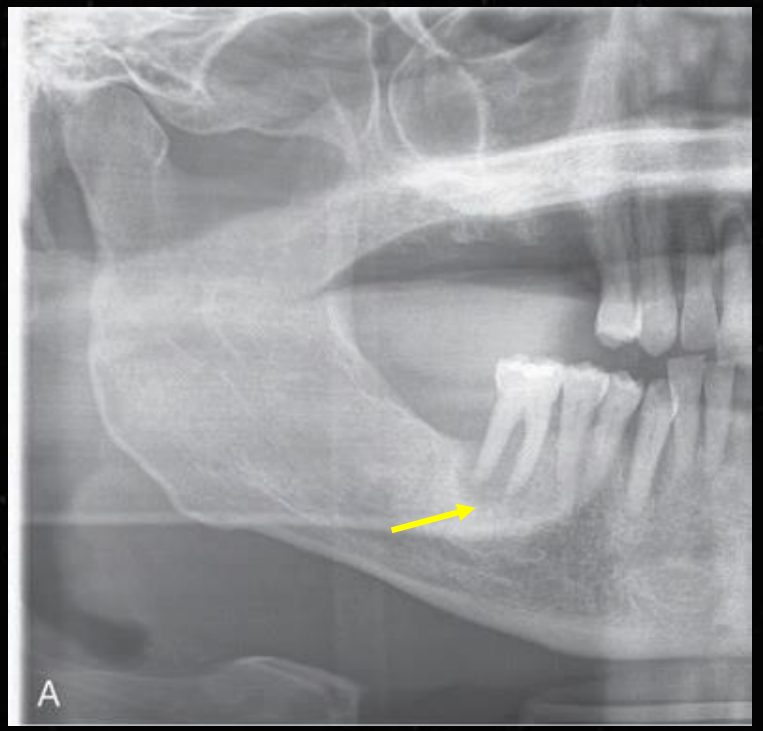


FIG. 20.18 Example of a periodontal abscess related to the maxillary canine; note the well-defined area of bone loss over the midroot region of the tooth and extending in a mesial direction toward the lateral incisor. There appears to be a layer of bone *(arrow)* separating the area of bone destruction from the crest of the alveolar process.



Occlusal Trauma & Parafunction

- Cause: Occlusal forces exceed the physiologic tolerance of supporting tissues.
- May result from excessive forces or normal forces on a periodontium weakened by bone loss.
- Parafunction: Clenching/bruxism increases occlusal forces on susceptible teeth.
- Clinical signs: Increased mobility, wear facets, percussion sensitivity, parafunctional habits.

Radiographic findings:

- Widened PDL space
- Thickened/broad, hazy lamina dura
- Bone loss and increased trabeculation
- Possible hypercementosis or root fracture

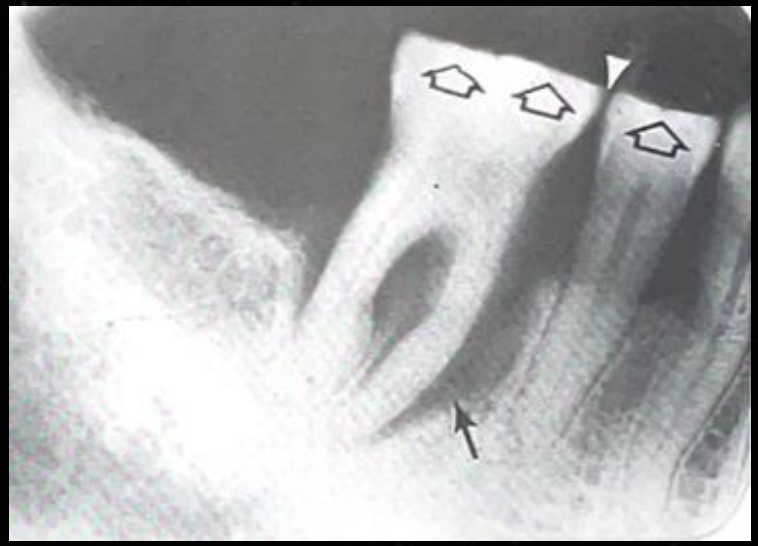
PDL widening: May occur from occlusal trauma or reduced bone support.

- Single-rooted → hourglass-shaped socket
- Multirooted → widening at apices and furcation

Role of Occlusion and TMD in Periodontal Disease Management

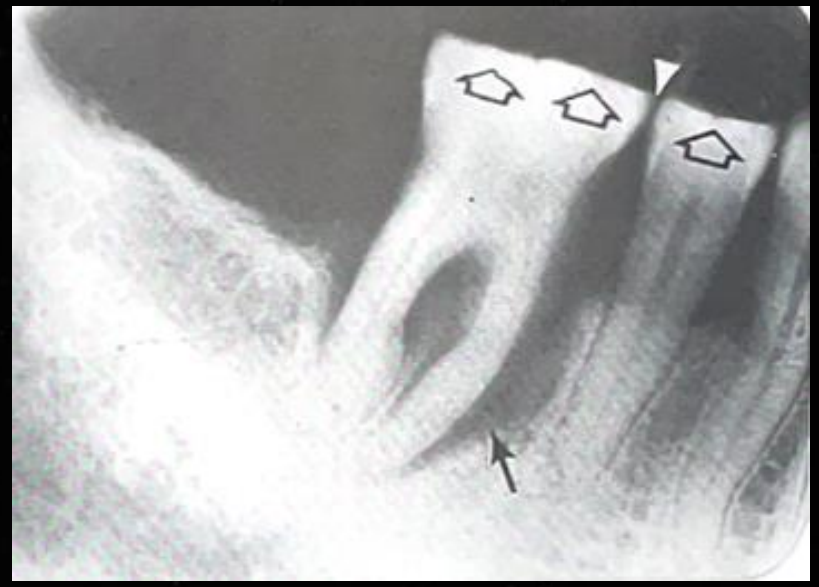
Michael J. McDevitt

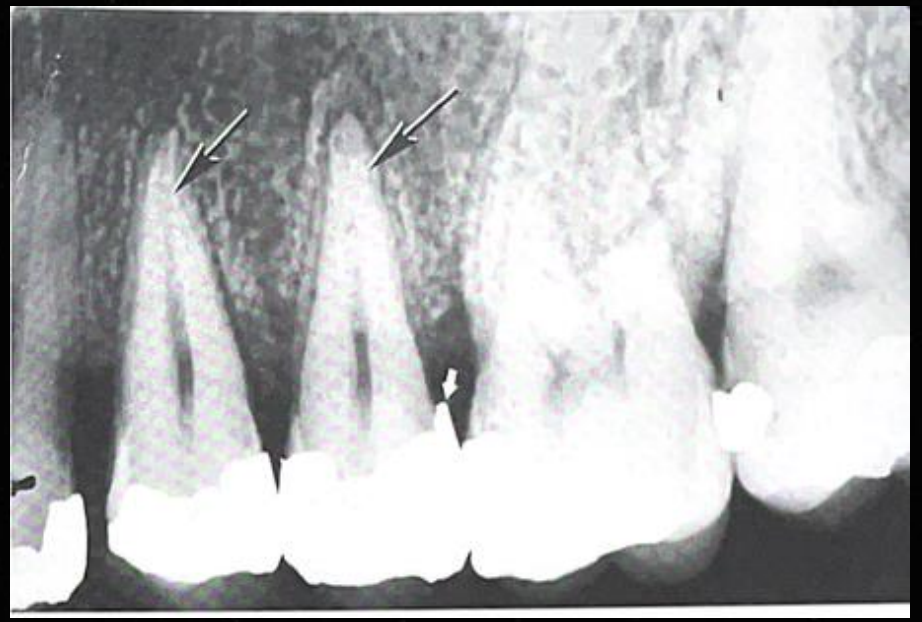
Newman and Carranza's Clinical Periodontology and Implantology, Chapter 35, 477-484.e3



Occlusal Trauma & Parafunction cont.

- Traumatic occlusion does not cause gingivitis, periodontitis, attachment loss, or pocket formation.
- May accelerate attachment/bone loss when periodontitis is already present.
- Tooth movement around an axis of rotation may contribute to PDL changes.
- Diagnosis: Clinical; radiographic findings alone are not diagnostic.





Other Modifiers of Periodontal Disease

Systemic Conditions on the Periodontium

Influence of Systemic Conditions on the Periodontium

Perry R. Klokkevold; Brian L. Mealey; Yvonne L. Hernandez-Kapila

Newman and Carranza's Clinical Periodontology and Implantology, Chapter 25, 359-379.e8

Endocrine & Metabolic Disorders

- Diabetes Mellitus (Type 1, Type 2, Gestational)
- Metabolic Syndrome (Obesity, Hypertension, Dyslipidemia)
- Hyperparathyroidism
- Hormonal Fluctuations (Puberty, Menstruation, Pregnancy, Menopause, Hormonal Contraceptives)

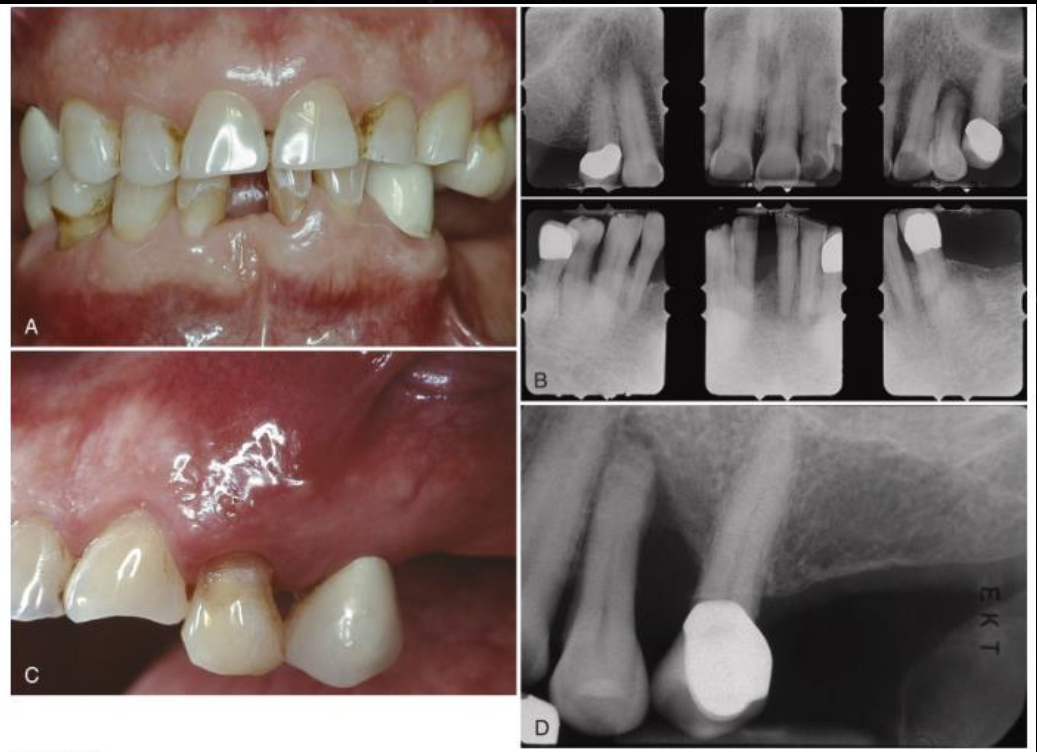


FIG. 25.2
A 60-year-old patient with a long-term history of type 2 diabetes. (A) Anterior retracted view of the patient's dental and periodontal condition. Note the missing posterior teeth, the supereruption of the premolars, and the mild generalized gingival inflammation. (B) Periapical radiographs of the remaining teeth. Note the mild generalized bone loss with localized areas of severe bone loss. The failure to replace the posterior teeth adds to the occlusal burden of the remaining dentition. (C) Clinical photograph of the maxillary premolar area presenting with abscess. Notice the diffuse erythema and inflammation surrounding the abscess area. (D) Periapical radiograph of the maxillary premolar showing extensive bone loss associated with abscess.

Systemic Conditions on the Periodontium

Hematologic (Blood) Disorders & Immune Deficiencies

- Leukemia (Acute, Chronic, Lymphocytic, Myelogenous, Monocytic)
- Neutropenia (including Agranulocytosis, Cyclic Neutropenia)
- Thrombocytopenia (including Idiopathic Thrombocytopenic Purpura)
- Anemia (Pernicious, Iron Deficiency, Sickle Cell, Aplastic)
- Antibody Deficiency Disorders (Agammaglobulinemia, Common Variable Immunodeficiency)
- ***Necrotizing Periodontal Diseases***

FIG. 25.6
Generalized periodontitis stage 4, grade C (aggressive periodontitis) in 10-year-old boy with cyclic neutropenia and agammaglobulinemia. (A) Clinical presentation of the periodontal condition. Note the severe swelling and inflammation of the marginal and papillary gingiva. There is gross migration of teeth caused by a loss of bone support. (B) Panoramic radiograph demonstrating severe bone loss around all permanent teeth that have erupted into the oral cavity.

Systemic Conditions on the Periodontium

Genetic Disorders

- Chédiak–Higashi Syndrome
- Lazy Leukocyte Syndrome
- Leukocyte Adhesion Deficiency (LAD)
- Papillon–Lefèvre Syndrome
- Down Syndrome (Trisomy 21)

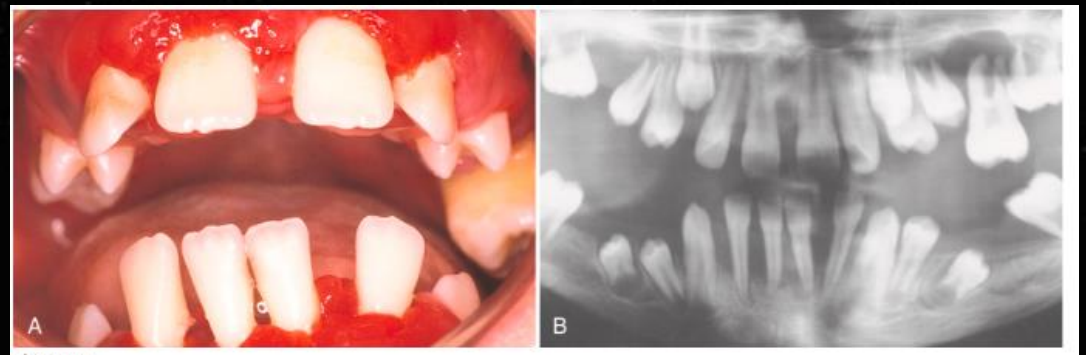


FIG. 9.2
Oral (A) and radiographic (B) appearance of a patient with leukocyte adhesion deficiency. The child was deficient in CD18 (i.e., leukocyte adhesion deficiency type I), which results in absent or severely reduced levels of the β2 integrin molecule. The patient suffered from recurrent infections of the middle ear, the tongue, and the perirectal areas as well as of the periodontium.

Systemic Conditions on the Periodontium

Psychosocial Factors

- Psychologic Stress
- Depression
- Neurotic Habits (Grinding, Clenching, Self-inflicted injury)

FIG. 25.17
Severe gingival recession localized to the labial surface of all mandibular incisors. This finding was discovered in an uncooperative, institutionalized adult with mental disorders who had been placed under general anesthesia. The patient was known to pace around the home with all four fingers inside his lower lip.

Systemic Conditions on the Periodontium

Nutritional Influences

- Fat-Soluble Vitamin Deficiencies (Vitamins A, D, E)
  - Vitamin A: Gingival hyperkeratosis, deep pockets.
  - Vitamin D: Defective alveolar bone calcification and bone loss.
  - Vitamin E: Impaired wound healing.

- Water-Soluble Vitamin Deficiencies (Vitamin B Complex, Vitamin C/Scurvy)
  - Vitamin B complex: Gingivitis, glossitis, stomatitis, and oral mucosal inflammation.
  - Vitamin C: Swollen, bluish-red, hemorrhagic/bleeding gingiva, impaired wound healing, and tooth mobility

- Protein Deficiency
  - Connective tissue degeneration, Bone loss, impaired cementum formation, delayed wound healing.

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Systemic Conditions on the Periodontium

Medications

- Bisphosphonates (e.g., Alendronate, Zoledronate)
  - Inhibit bone resorption and bone turnover
  - Associated with MRONJ

- Corticosteroids
  - Suppresses immune and inflammatory responses
  - Potential susceptibility to delayed healing and periodontal destruction

Systemic Conditions on the Periodontium

Other Systemic Conditions

- Osteoporosis
- Congenital Heart Disease (including Tetralogy of Fallot, Eisenmenger Syndrome)
- Hypophosphatasia

Dangerous Chameleons of Periodontal Disease

Most bone loss around teeth is caused by **periodontal disease**, but other diseases may mimic its appearance.

Red flag: one or a few adjacent loose teeth with **no generalized periodontal disease**.

Periodontal bone loss typically begins at the **alveolar crest** and progresses apically, with defects **larger coronally and less severe at the apical leading edge**.

Malignancy (especially squamous cell carcinoma):

- Extensive, invasive bone destruction
- **Irregular PDL widening** along its entire length
- Ragged/irregular borders
- **Lamina dura destruction**
- Ill-defined borders with no peripheral sclerosis

Langerhans cell histiocytosis:

- Localized or multiple areas of bone destruction around tooth roots
- May produce **"teeth floating in space"**
- Destruction is centered at the **midroot**, rather than the alveolar crest
- Early lesions may have an **"ice cream scoop" appearance**.

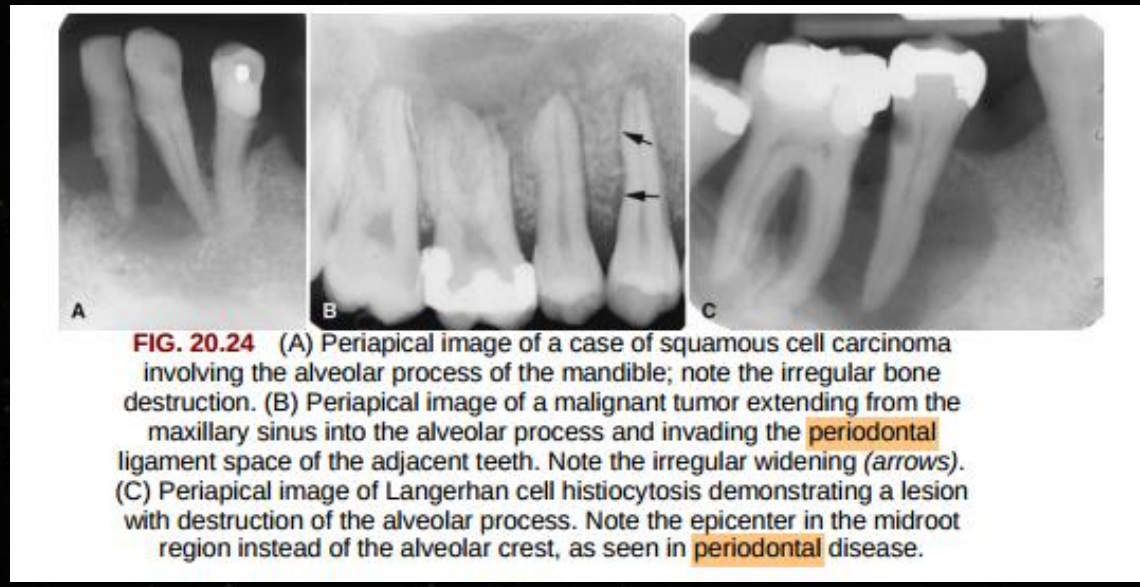
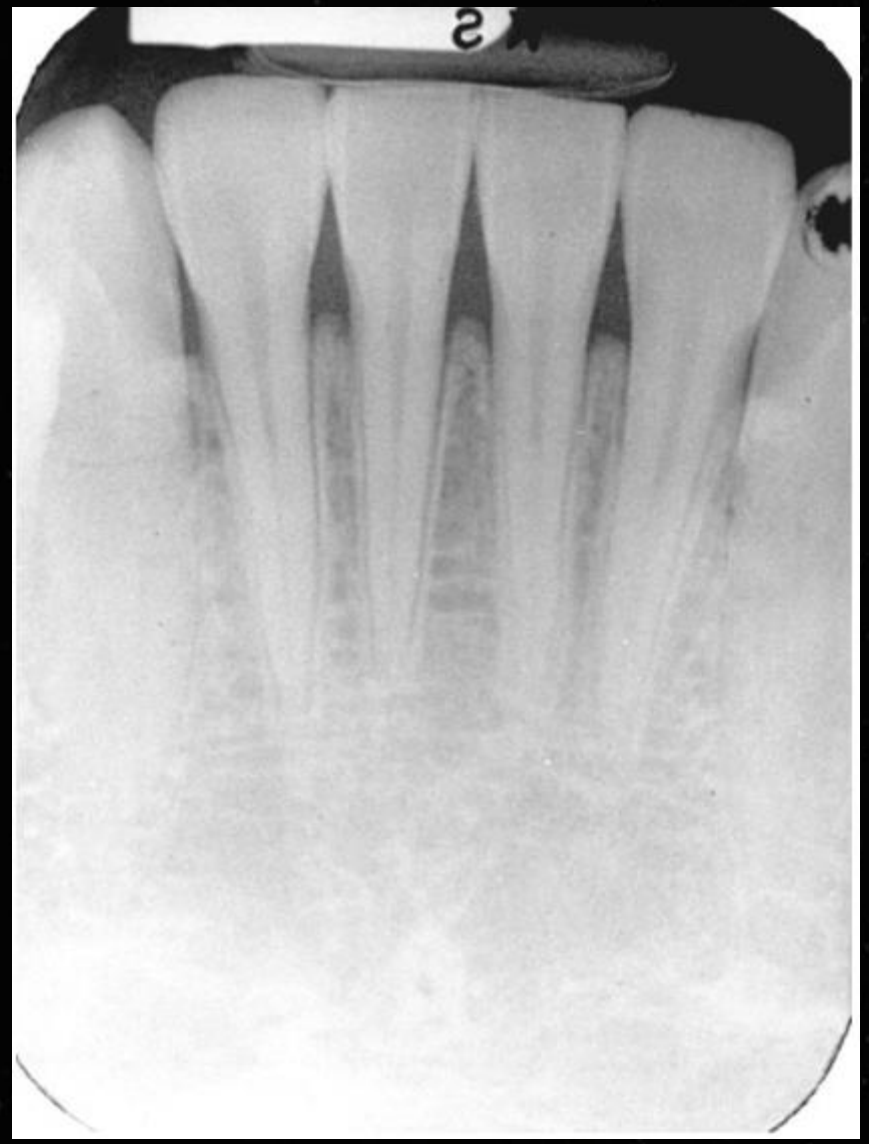


FIG. 20.24 (A) Periapical image of a case of squamous cell carcinoma involving the alveolar process of the mandible; note the irregular bone destruction. (B) Periapical image of a malignant tumor extending from the maxillary sinus into the alveolar process and invading the periodontal ligament space of the adjacent teeth. Note the irregular widening *(arrows)*. (C) Periapical image of Langerhan cell histiocytosis demonstrating a lesion with destruction of the alveolar process. Note the epicenter in the midroot region instead of the alveolar crest, as seen in periodontal disease.

Differential Interpretation

Identify the Pattern and Severity of Bone Loss - 1

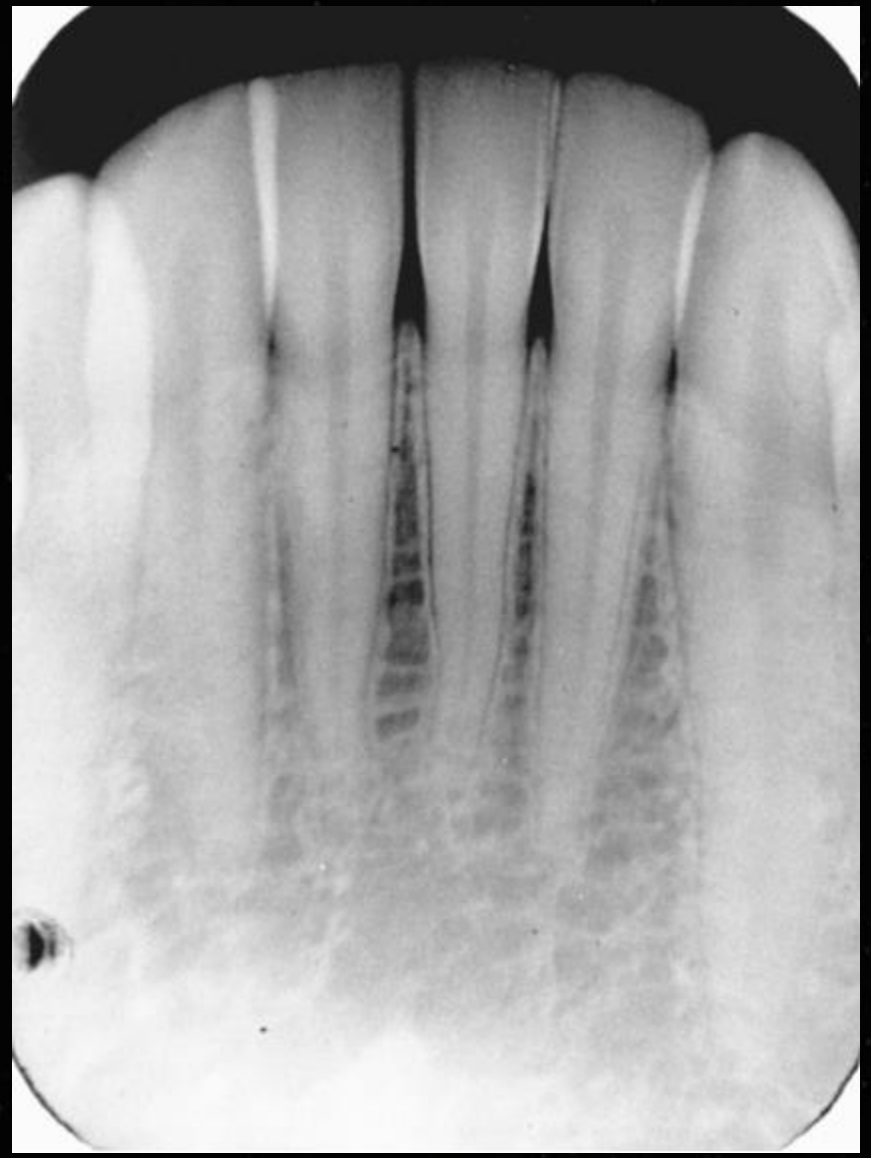
Identify the Pattern and Severity of Bone Loss - 2



Identify the Pattern and Severity of Bone Loss - 3

Identify the Pattern and Severity of Bone Loss - 4

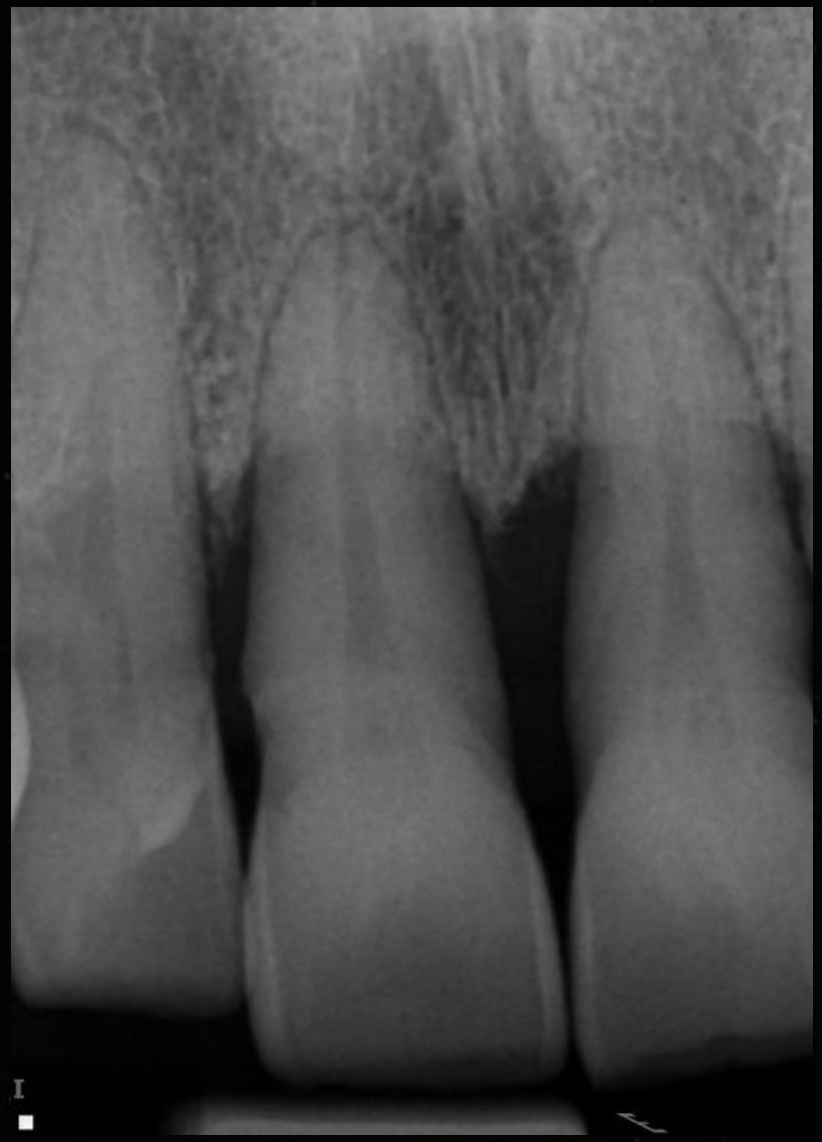
Identify the Pattern and Severity of Bone Loss - 5



Identify the Pattern and Severity of Bone Loss - 6

Identify the Pattern and Severity of Bone Loss - 7

Identify the Pattern and Severity of Bone Loss - 8

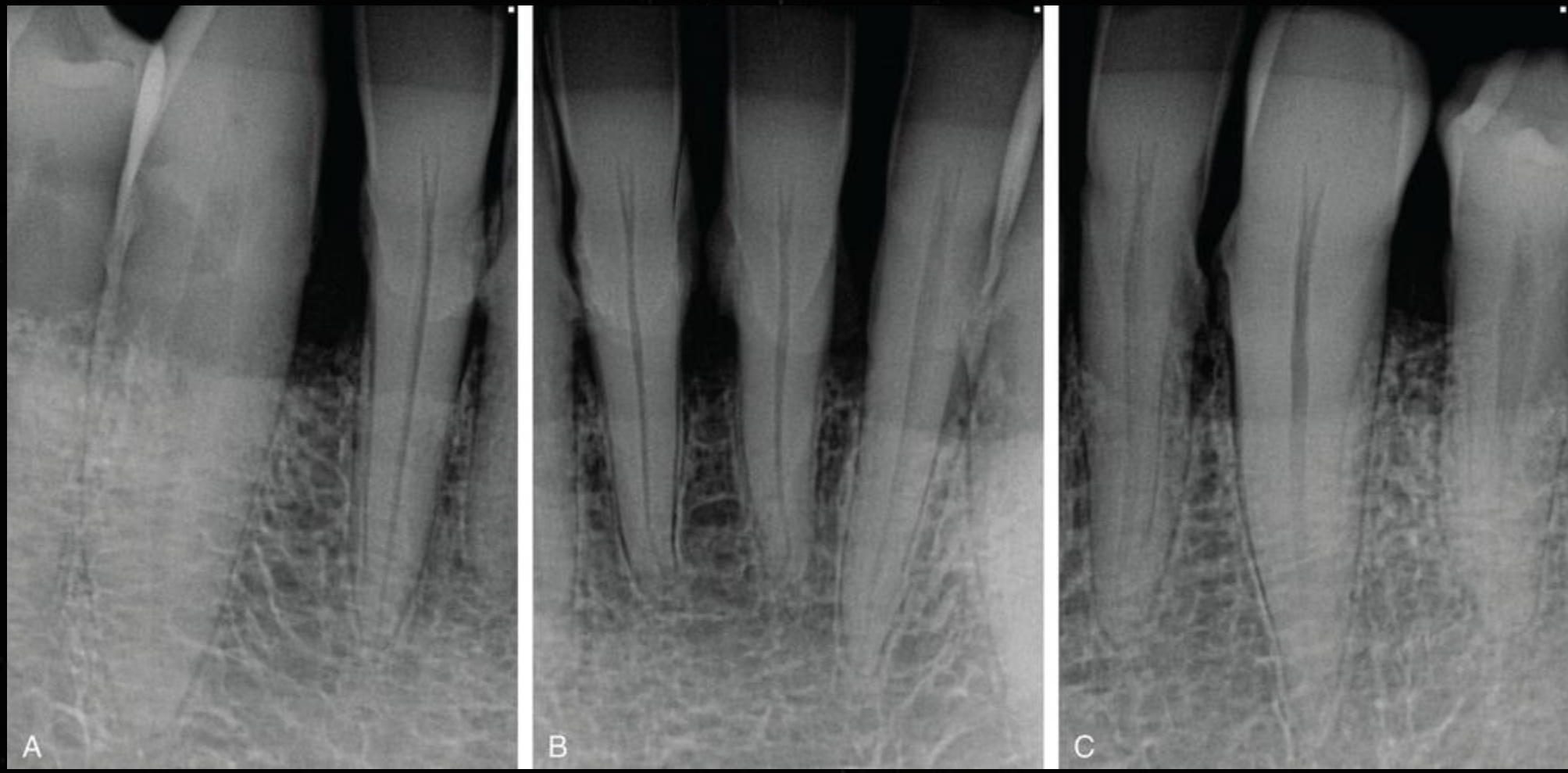


Identify the Pattern and Severity of Bone Loss - 9

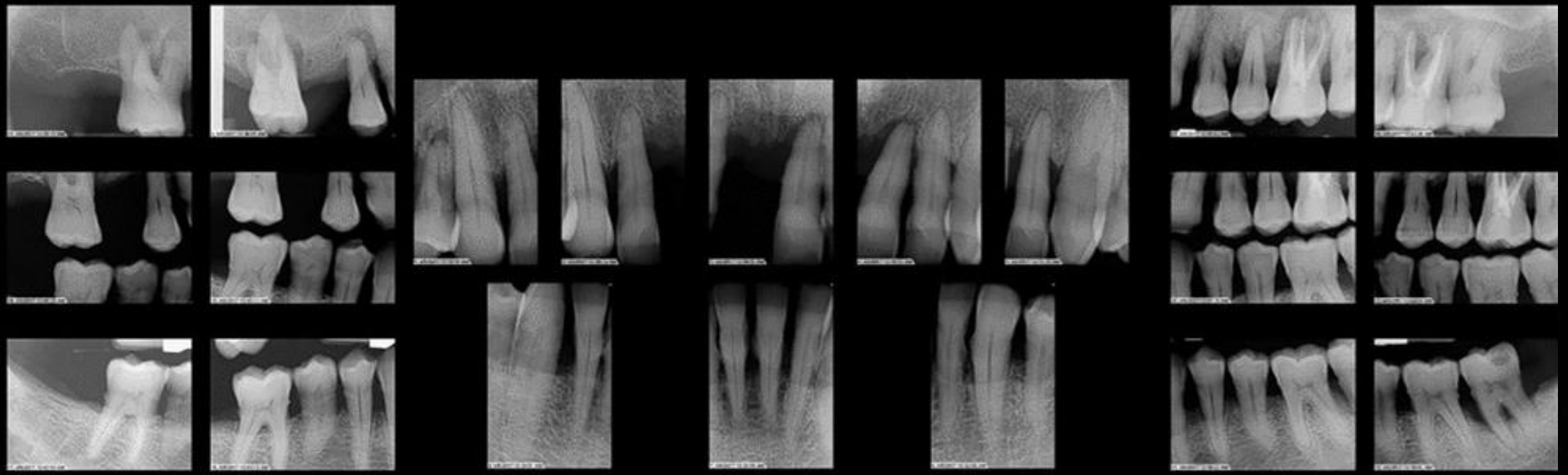
Identify the Pattern and Severity of Bone Loss - 10

Identify the Pattern and Severity of Bone Loss - 11

Identify the Pattern and Severity of Bone Loss - 12

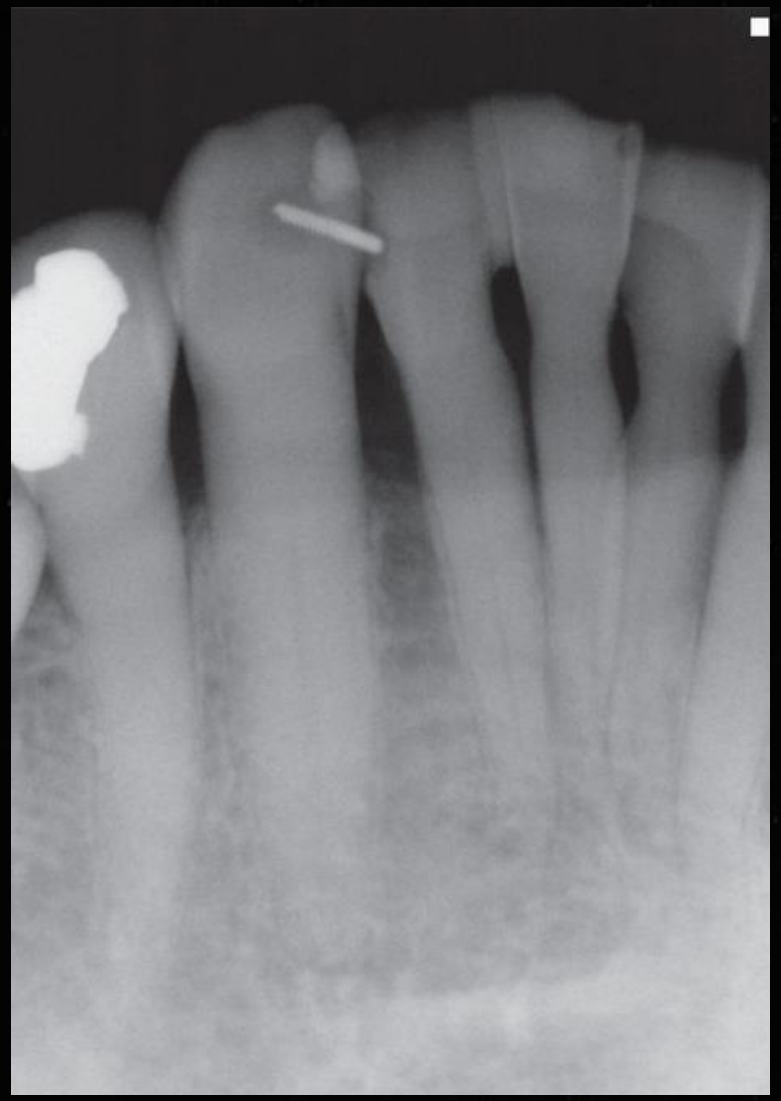


Identify the Pattern and Severity of Bone Loss - 13



Identify the Pattern and Severity of Bone Loss - 14

Identify the Pattern and Severity of Bone Loss - 15



Identify the Pattern and Severity of Bone Loss - 16

Evaluation of Periodontal Therapy

Posttreatment Changes

Table 2. Initial imaging recommendations based on type of encounter.

TYPE OF ENCOUNTER (NEW OR RECALL VISIT)	CHILD WITH PRIMARY DENTITION (BEFORE ERUPTION OF FIRST PERMANENT TOOTH)	CHILD WITH TRANSITIONAL DENTITION (AFTER ERUPTION OF FIRST PERMANENT TOOTH)	ADOLESCENT WITH PERMANENT DENTITION (BEFORE ERUPTION OF THIRD MOLARS)	ADULT, DENTATE OR PARTIALLY EDENTULOUS
New Patient Evaluation	Individualized radiographic examination with periapical or occlusal views or posterior bite-wing radiographs if proximal surfaces cannot be visualized or probed. Patients without evidence of disease and with open proximal contacts may not require a radiographic examination.	Individualized radiographic examination consisting of posterior bite-wing radiographs with panoramic examination or posterior bite-wing radiographs and selected periapical images.	Individualized radiographic examination consisting of posterior bite-wing radiographs with panoramic examination or posterior bite-wing radiographs and selected periapical images. A full-mouth intraoral radiographic examination is preferred when the patient has clinical evidence of generalized oral disease or a history of extensive dental treatment.	Individualized radiographic examination consisting of posterior bite-wing radiographs with panoramic examination or posterior bite-wing radiographs and selected periapical images. A full-mouth intraoral radiographic examination is preferred when the patient has clinical evidence of generalized oral disease or a history of extensive dental treatment.
Recall Visit, Patient With No Clinical Caries and Not at Increased Risk of Developing Caries	Posterior bite-wing examination at 12- to 24-m intervals if proximal surfaces cannot be examined visually or with a probe.	Posterior bite-wing examination at 12- to 24-m intervals if proximal surfaces cannot be examined visually or with a probe.	Posterior bite-wing examination at 18- to 36-mo intervals.	Posterior bite-wing examination at 24- to 36-mo intervals.
Recall Visit, Patient With Clinical Caries or at Increased Risk of Developing Caries	Posterior bite-wing examination at 6- to 12-mo intervals if proximal surfaces cannot be examined visually or with a probe.	Posterior bite-wing examination at 6- to 12-mo intervals if proximal surfaces cannot be examined visually or with a probe.	Posterior bite-wing examination at 6- to 12-mo intervals if proximal surfaces cannot be examined visually or with a probe.	Posterior bite-wing examination at 6- to 18-mo intervals.
Recall Visit, Patient With Periodontal Disease	Clinical judgment as to the need for and type of radiographic images for evaluation of periodontal disease. Imaging may consist of, but is not limited to, selected bite-wing or periapical images of areas where periodontal disease (other than nonspecific gingivitis) can be found clinically.	Clinical judgment as to the need for and type of radiographic images for evaluation of periodontal disease. Imaging may consist of, but is not limited to, selected bite-wing or periapical images of areas where periodontal disease (other than nonspecific gingivitis) can be found clinically.	Clinical judgment as to the need for and type of radiographic images for evaluation of periodontal disease. Imaging may consist of, but is not limited to, selected bite-wing or periapical images of areas where periodontal disease (other than nonspecific gingivitis) can be found clinically.	Clinical judgment as to the need for and type of radiographic images for evaluation of periodontal disease. Imaging may consist of, but is not limited to, selected bite-wing or periapical images of areas where periodontal disease (other than nonspecific gingivitis) can be found clinically.

Posttreatment Changes

- Sharp crestal cortex + clear lamina dura: suggests disease stabilization.
- Increased radiopacity: may indicate successful treatment/healing.
- Remineralization: can make cancellous bone visible, falsely suggesting bone growth.
- Beam angulation/exposure: can falsely suggest bone loss or gain.
- No radiographic change: does not mean treatment failed.
- Best assessment: clinical evaluation is needed because radiographs cannot show periodontal pocket resolution.
- "No radiographic signs of active alveolar bone loss, but radiographic signs of a reduced periodontium are suggested. Clinical correlation is advised."

FIG. 20.23 Example of a case in which the interproximal cortex of the alveolar crest has re-formed after successful periodontal therapy.

Clinical Attachment Level - Reinforced

- Clinical attachment level (CAL): CEJ → base of the sulcus/pocket.
- Probing depth (PD): gingival margin → base of the sulcus/pocket.
- CEJ-GM measurement/relationship (+/-)
  - (+) = gingival margin coronal to the CEJ
  - (-) = gingival margin apical to the CEJ (recession)
  - 0 = gingival margin at the CEJ
- Calculating clinical attachment level:
  - Clinical attachment level (CAL) = Probing depth – (± the gingival margin relative to the CEJ)

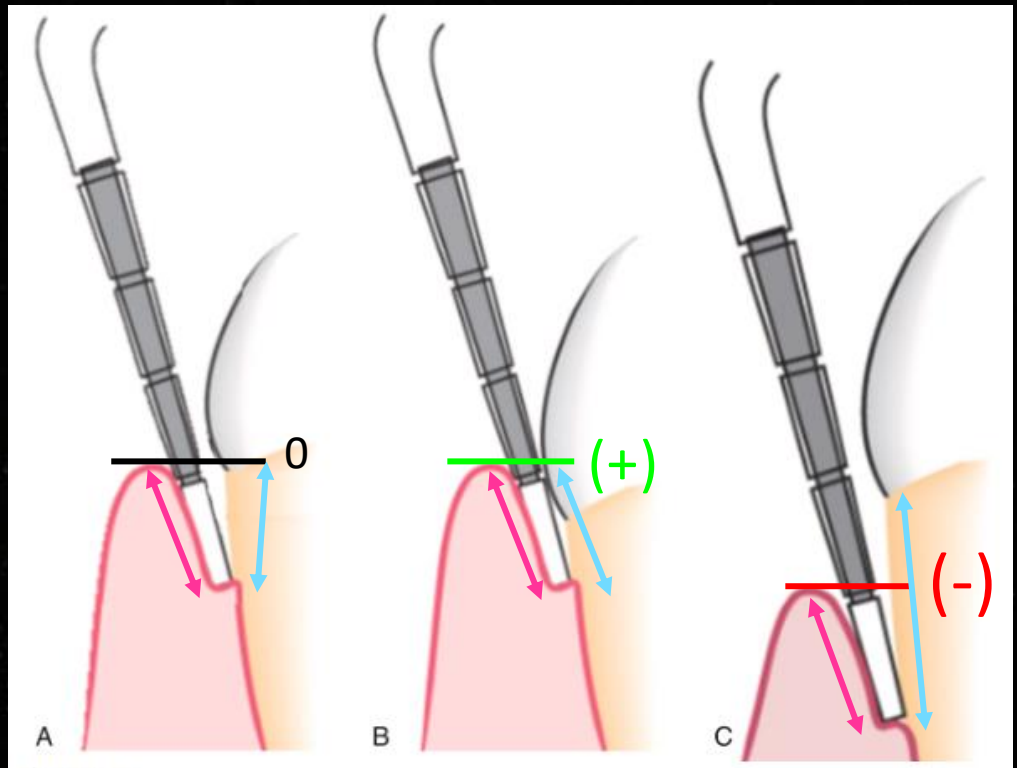


FIG. 38.18
Schematic illustration of three clinical scenarios where the probing depth is the same but with differing gingival margin locations with respect to the cementoenamel junction (CEJ), which result in different clinical attachment loss (CAL) measures. In scenario A, the gingival margin is at the level of the CEJ and, hence, the probing depth equals the CAL. In scenario B, the gingival margin is coronal to the CEJ, therefore CAL is calculated by subtracting the distance from the gingival margin to the CEJ from the probing depth. Scenario C presents a gingival recession defect. In this case, CAL is determined by adding the gingival recession measure (distance from the CEJ to the gingival margin) and the probing depth.

References

Primary Textbook

Newman, M. G., Takei, H. H., Klokkevold, P. R., & Carranza, F. A. (Eds.). (2023). *Newman and Carranza's clinical periodontology and implantology* (14th ed.). Elsevier.

Iannucci, J. and Howerton, L. (2022). Dental Radiography: Principles and Techniques, 6th Ed. Elsevier. ISBN 978-0-323-69550-3.

Supplemental Reference

Langland, O. E., & Langlais, R. P. (1997). Principles of Dental Imaging. Williams & Wilkins. ISBN 978-0683182415.

Mallya, S. M., & Lam, E. W. N. (Eds.). (2019). White and Pharoah's Oral Radiology: Principles and Interpretation (8th ed.). Elsevier.

Diagnosis and Classification Part I

Compiled and Presented by
Vickie E. Jones, DHSc, MS, RDH, FNAP

September 24, 2026 | 8:00–11:50 AM

DEN 624 • Periodontology I • Tanner College of Dental Medicine

How to prepare for exam

Step	What to do
1. Browse (15 min)	Look at each condition. Say what you see: color, contour, consistency, texture, distribution
2. Categorize	Name the AAP 2017 family: biofilm-induced gingivitis, gingivitis modified by risk factors, drug-influenced enlargement, or non-biofilm-induced condition
3. Ask what is missing	Which history, probing, or radiographic data would you need before diagnosing?
4. Self-check	Complete the self-check slide at the end
5. Bring questions	Post one "muddiest point" before class

Photographs support pattern recognition. They never establish stage, grade, or a final diagnosis on their own.

Periodontal Diseases and Conditions

Periodontal health, gingival diseases and conditions

I - Periodontal health, and gingival health	II - Gingivitis: Dental Biofilm Induced	III- Gingival disease: Non-dental biofilm induced
1. Clinical gingival health on intact periodontium 2. Clinical gingival health on reduced periodontium	1. Gingivitis is associated with dental plaque alone. 2. Gingival diseases mediated by systemic factors or risk local factors 3. Gingival diseases modified by medications	1. Gingival diseases of specific infections 2. Genetic/development disorder 3. Inflammatory immune conditions 4. Endocrine, nutritional & metabolic diseases 5. Traumatic lesions 6. Gingival pigmentation 7. Neoplasm 8. Reactive process

Periodontal Diseases and Conditions

Forms of Periodontitis

Necrotizing Periodontal Diseases	Periodontitis	Periodontitis as a manifestation of systemic diseases
1. Necrotizing gingivitis 2. Necrotizing periodontitis 3. Necrotizing stomatitis	1. Based on severity and complexity of management • Stage I • Stage II • Stage III • Stage IV 1. According to evidence and risk of rapid progression • Grade A • Grade B • Grade C 1. According to extent and distribution	Hematologic disorders: ------Acquired neutropenia –-----Leukemias Genetic disorders: -Papillon-Lefèvre syndrome -Chediak-Higashi syndrome -Down syndrome -Leukocyte adhesion deficiency -Familial & cyclic neutropenia -Histiocytosis syndrome - Glycogen Storage diseases -Infantile genetic agranulocytosis -Cohen syndrome -Ehlers-Danlos Syndrome -Hypophosphatasia

Periodontal Diseases and Conditions				
Other Conditions affecting the periodontium				
Systemic diseases & conditions affecting periodontal tissues	Periodontal abscesses & endodontic-periodontal lesions	Mucogingival deformities & conditions	Traumatic Occlusion	Tooth & Prothesis related factors
1. **Systemic disorders that have a major impact on the loss of periodontal tissues by influencing periodontal inflammation** 2. **Other systemic disorders that influence the pathogenesis of periodontal diseases.** 3. **Systemic disorders that can result in loss of periodontal tissues independent of periodontitis**	1. Periodontal abscesses 2. Endodontic periodontal lesions	1. Gingival phenotype 2. Gingival recession 3. Lack of Keratinized gingiva 4. Decrease vestibular depth 5. Aberrant frenum 6. Gingival excess 7. Abnormal color 8. Condition of exposed root surface	1. Primary occlusal trauma 2. Secondary occlusal trauma 3. Orthodontic forces	1. Localized tooth related factor 2. Localized dental protheses-related factor

Healthy Gingiva & Gingival Diseases & Conditions

Not all Gingival Inflammation is Plaque Induced

Dental Professionals MUST Differentiate and Treat Accordingly

1- Periodontal health and gingival health

- A. Clinical gingival health on an intact periodontium
- B. Clinical gingival health on a reduced periodontium.

2 – **Gingivitis, dental plaque biofilm-induced**

- A. Associated with dental biofilm only
- B. Mediated by systemic or local risk factors
- C. Drug-influenced gingival enlargement.

3 – **Gingival diseases, nondental biofilm –induced**

- A. Specific infection
- B. Genetic/developmental disorder
- C. Inflammatory and immune conditions
- D. Reactive processes
- E. Neoplasm
- F. Endocrine, nutritional, metabolic diseases
- G. Traumatic lesions
- H. Gingival Pigmentation

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Gingival Health and Periodontal Health

1- Clinical Gingival Health on Intact Periodontium

2- Clinical Gingival Health on Reduced Periodontium

Stable Treated Periodontitis Patient

Patient did not have Periodontitis. However, the Periodontium is reduced by mechanical, parafunctional or other means than bacterial infection

II- Gingivitis: Dental Biofilm Induced

1- Gingivitis Associated with Dental Plaque Alone:

Without other local contributing factors. (Poor oral hygiene)

2- Gingival Diseases Mediated by Systemic or Local Risk Factors

- Increased Gingival inflammation even with low number of local factors.
  - Vitamin C deficiency.

Swollen, inflamed interdental papillae in a patient with vitamin C deficiency.

Credit: **CDC, Public Health Image Library, image 3998 or 19449.** These CDC images permit educational use with attribution. wwwn.cdc.gov

Risk local factors (that retain plaque microorganisms) e.g. Caries

Pre-operative photograph showing Class V carious lesion, gingival redness, bleeding, and a dento-alveolar draining sinus

2- Gingival Diseases Mediated by Local Risk Factors

Risk local factors (that retain plaque microorganisms) e.g. Orthodontic brackets

Mouth breathing patient.

Lip position at rest

Left lateral region showing inflammation up to premolar region

Anterior region showing severe gingival inflammation

Right lateral region showing inflammation to the premolar region

Maxillary palatal aspect

Oral Manifestations as an Early Clinical Sign of Acute Myeloid Leukemia: A Report of Two Cases

Case 1. A and B

- An 18-year-old female reported with the complaints of oral ulcers on the tip of tongue and buccal mucosa for last 1 week.
- There was no history of previous episodes of oral ulceration or weight loss.
- Mild fever, malaise, and stomach upset accompanied the ulcers.
- Her submandibular lymph nodes were palpable and tender bilaterally and she exhibited extreme pallor.
- Intraoral examination showed the presence of several ulcers on the tip of tongue, buccal mucosa, and gingiva [Figure a & b to the left]. The ulcers were irregular and surrounded by reddish blue hemorrhagic periphery.

Case 1 C and D to the left

- There was mild enlargement of interdental gingiva in maxillary (Fig. C) and mandibular anterior teeth (Fig. D).
- There was no bleeding on probing and plaque and calculus were minimal.
  - In this case, aphthous stomatitis and HIV associated oral ulcers were considered as differential diagnosis.
  - Acute necrotizing ulcerative gingivitis and stomatitis which presents with gingival necrosis and punched out ulceration was also considered as a differential diagnosis.
  - The irregular shape and presence of reddish blue periphery around the ulcers prompted the investigation for leukemia.

Oral Manifestations as an Early Clinical Sign of Acute Myeloid Leukemia: A Report of Two Cases

- A 25-year-old male reported with the complaint of swelling in gum since 1 month.
- There was history of weakness, weight loss, and fever for 4–5 days.
- History of epilepsy, hypertension or any long-term medication was negative.
- Physical examination revealed extreme pallor and submandibular lymph nodes were enlarged and non-tender bilaterally.
- There was generalized gingival enlargement of upper and lower teeth; which was spongy in consistency, pale pink and devoid of stippling.
- There was mild evidence of reddish blue hemorrhagic areas in the anterior region [Images to the left].
- Differential diagnosis of
  - Inflammatory gingival hyperplasia was considered. However, local factors like plaque and calculus were not proportional to the severity of the disease and there was no bleeding on probing.
  - Drug induced gingival enlargement, due to phenytoin, nifedipine and cyclosporine was also ruled out by history.
- The above clinical presentation led to a provisional diagnosis of gingival enlargement due to leukemia which was further confirmed by hematological examination and peripheral smear.

Gingival diseases modified by systemic factors associated with endocrine system

- Puberty associated gingivitis

- Menstrual cycle – associated gingivitis

ure 4-3 Thirteen-year-old female with hormone exaggerated marg
papillary inflammation, with 1 to 4 mm probing depths yet min
cal attachment loss. A, Facial view. B, Lingual view.

Gingivitis due to plaque accumulation.

Figure 9-15 Conditioned gingival enlargement in puberty in a 13-year-old male.

Pyogenic granuloma, pregnancy tumor of pregnant woman.

-5 Twelve-year-old female with a primary m
that exhibits swollen/spongy gingiva.

Pyogenic granuloma. (Courtesy Dr. Sil

Fig. 17-10 A heightened gingival response to plaque during pregnancy results in pregnancy-associated gingivitis.

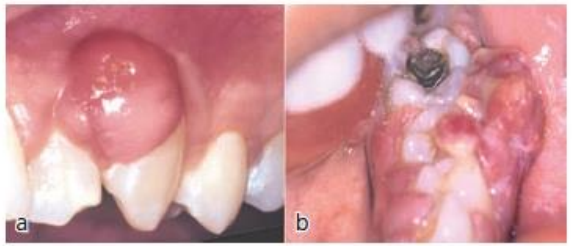

Fig. 17-11 (a) Pyogenic granuloma of pregnancy. (b) Large pyogenic granuloma of pregnancy interfering with occlusal function.

mage of pyogenic granuloma in 27-year-

Fig. 17-5 The presence of appliances, such as braces, allows for the accumulation of plaque resulting in [illegible] inflammation.

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12-year-old leukemic female

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3- Gingival diseases modified by medications:

A) Drug-influenced gingival enlargements (hyperplasia).

Anticonvulsant drugs **Phenytoin**

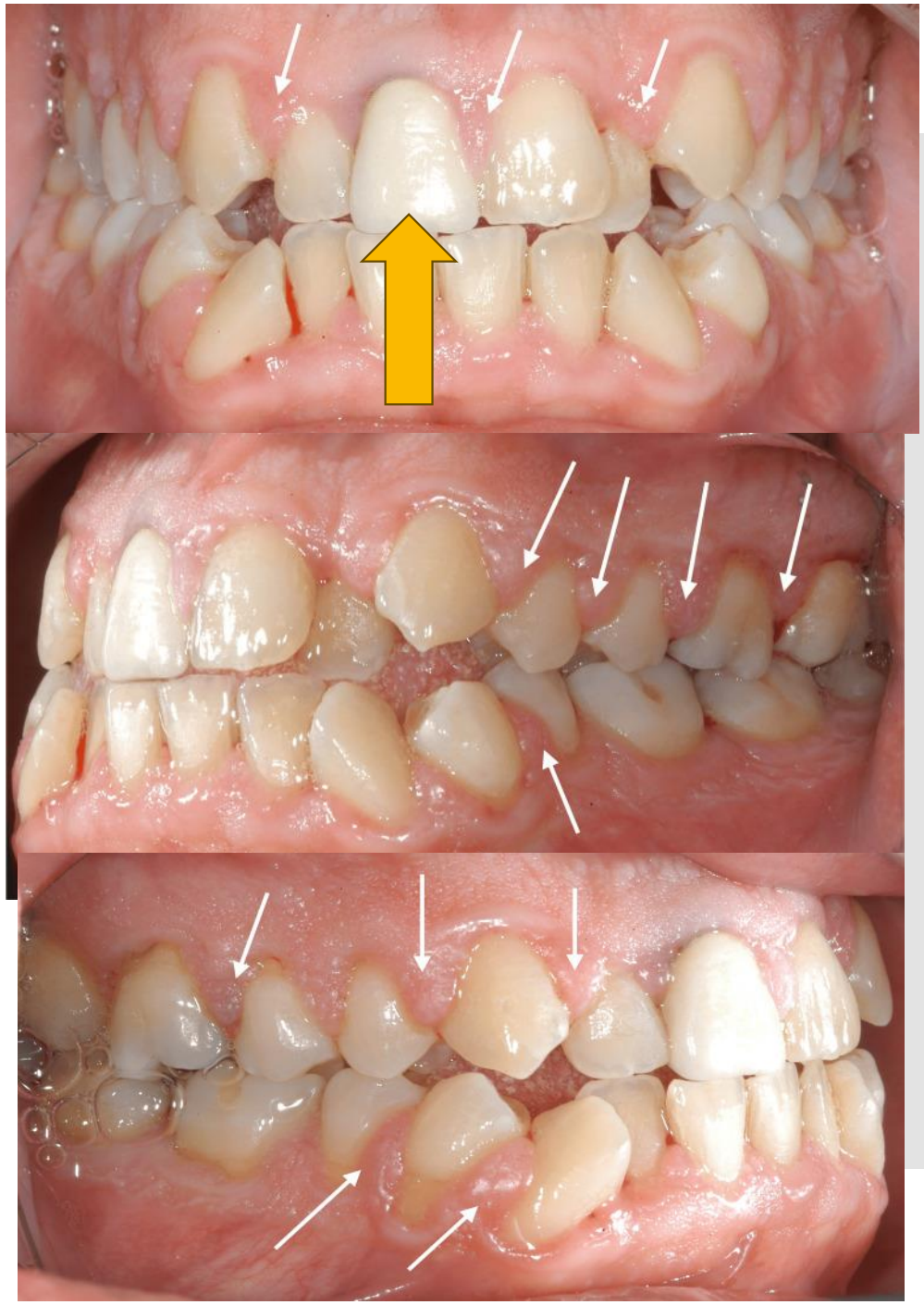
- Gingival Enlargement in patient
- taking phenytoin

Fig. 9. Gingival enlargement in a patient taking phenytoin for cerebral seizures. The gingival margins on most teeth are coronal to the cementoenamel junction (CEJ).

Gingival diseases modified by medications:

Yellow arrow presence of an unaesthetic prosthetic crown on the upper right central incisor contributing to the compromised gingival architecture and overall unaesthetic smile.

- Immunosuppressive drugs Cyclosporine
- Intraoral photography demonstrates marked hypertrophy of the interdental papilla (white arrows)



Ca-channel blockers Nifedipine –Diltiazem-Na valproate

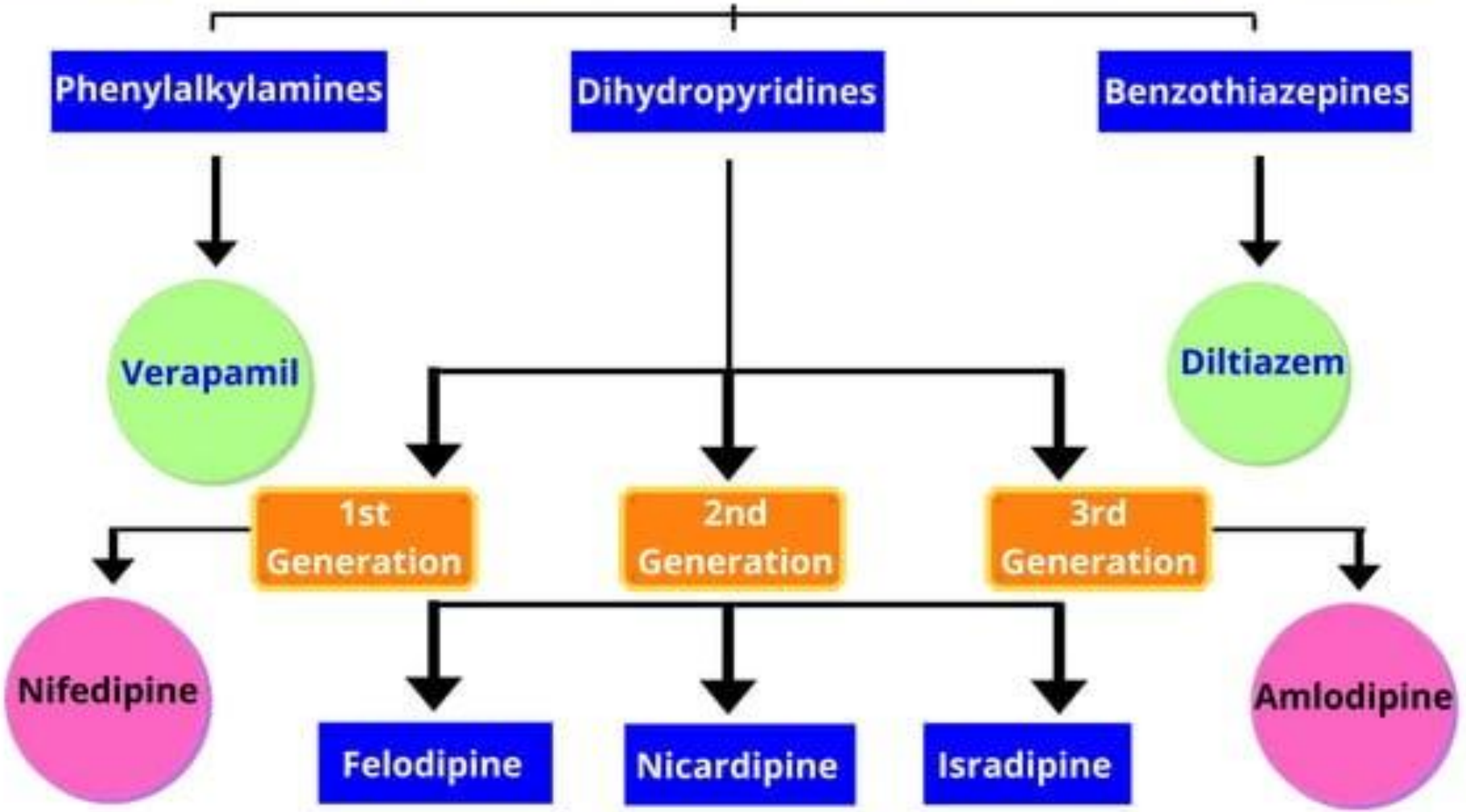
Patient with signs of inflammation of the periodontium, with increases in gingival volume and color changes

Patient on antihypertensive therapy based on calcium channel blockers

Clinical appearance of amlodipine-associated gingival overgrowth.

Gingival lobulations at the interdental papillae in the anterior segment of the labial surfaces of both lower and upper gingiva

Classification of Calcium Channel Blockers



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Gingival Diseas Modified by Medications:

B) Drug-influenced gingivitis

(oral contraceptives pills)due to elevation of hormones)

Patient taking oral contraceptive pills

Drug-influenced gingival enlargement

Nine-Year-Old Male Heart Transplant Patient under cyclosporine therapy

Hypertensive patient under Calcium channel blocker to control hypertension.

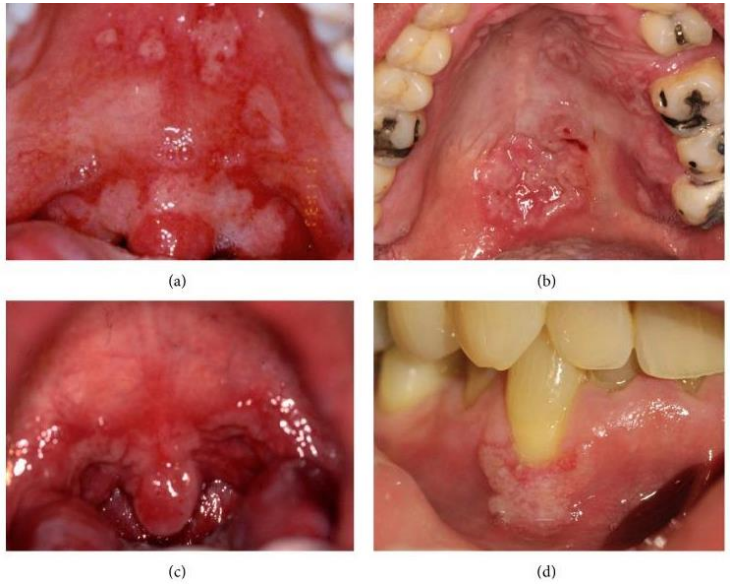
III- Gingival disease: Non dental – biofilm induced

- 1-Gingival diseases of specific infections

Bacterial origin:

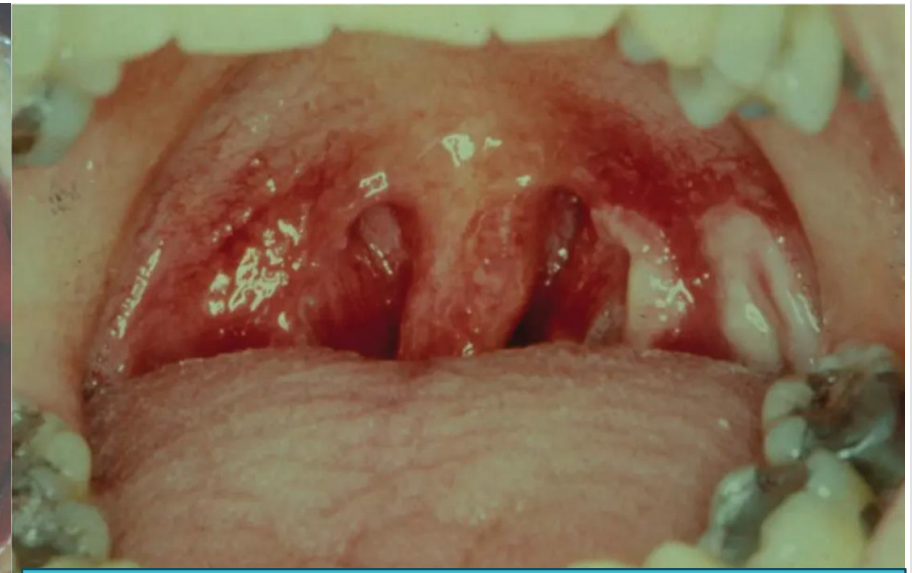
- A) Treponema pallidum Syphilis
- B) Neisseria gonorrhea Gonorrhea
- C) Streptococcal gingivitis (rare, diffuse, redness, swollen gingiva preceded by tonsillitis)

Oral Syphilis



Gonorrhea – Gonococcal Gingivitis

Gonococcal Gingivitis



Gonorrhea: Pseudomembranous in the fauces

Primary herpetic gingivostomatitis — HSV

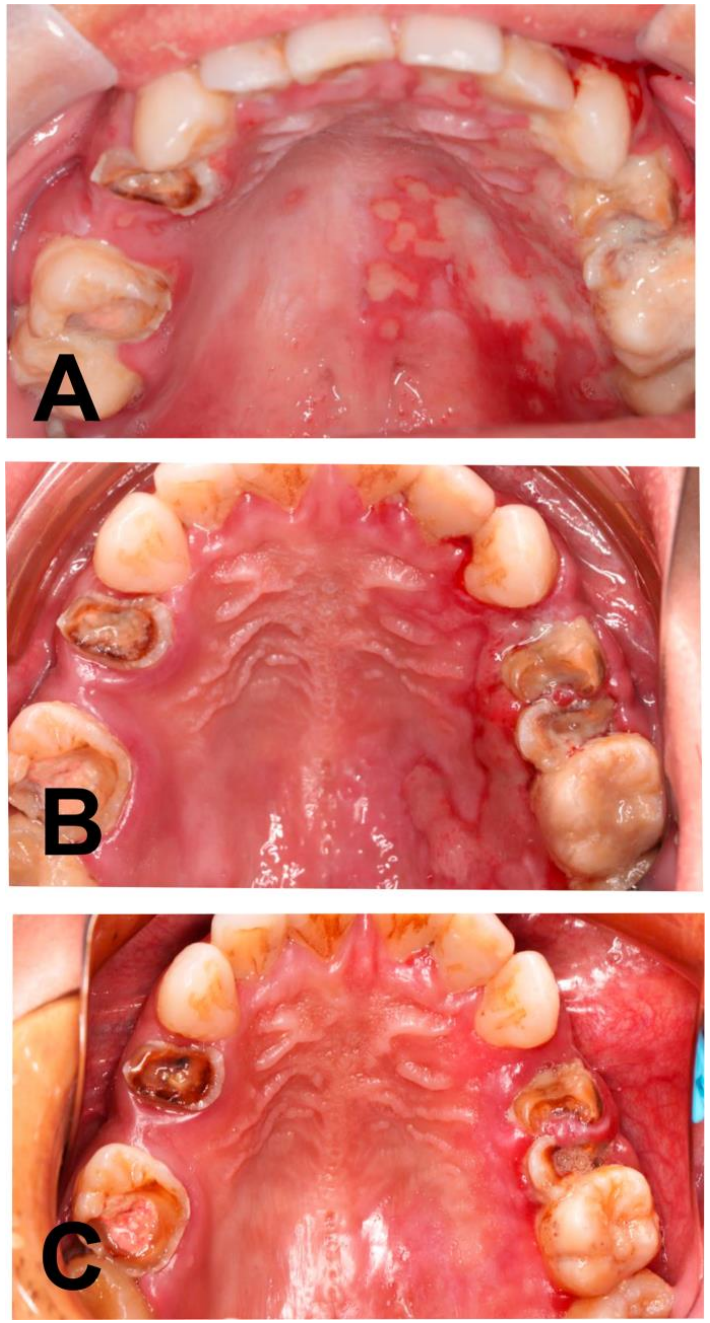
Primary HGS in a 25-year-old male patient showing (A and B) multiple vesicles, erosions, and small or large ulcerations on the whole maxillary and mandibular gingivae and parts of the hard palate.

The entire gingiva is enlarged, painful, and extremely erythematous. (C and D)

Secondary HGS in two adult male patients demonstrating a cluster of erythematous macules or ulcerations on the focal areas of hard palate. (E, F, G and H) adult patient exhibiting groups of small ulcerations on the upper labial mucosa (E), lower labial mucosa (F), ventral tongue mucosa (G), and floor of mouth mucosa (H).

Viral origin:

Varicella Zoster Infection (VZV)
Multiple ulcerations confined to **one half of the hard palate**. That **unilateral distribution is the identifying point**: reactivation follows the sensory nerve distribution, and lesions ***generally do not cross the midline.***



- Oral herpes zoster in a 33-year-old male patient.
- (A) Initial intraoral photograph exhibiting multiple white opaque ulcerations with erythematous halos on the left half (mainly the posterior two-thirds) of the hard palate of the patient.
- (B) Intraoral photograph taken one week later showing partial healing of the multiple palatal ulcerations.
- (C) Intraoral photograph taken three weeks later showing complete healing of the multiple palatal ulcerations.

III-Gingival Disease: Non-Dental-biofilm induced

<u>Viral origin:</u>

C)Human Papilloma (HPV)

- **HPV is a well-established viral cause of several wart-like/papillary lesions in the mouth**, particularly squamous papilloma.
- **Low-risk HPV-associated papillomas are different from the high-risk HPV infections linked to oropharyngeal cancer,** where HPV-16 is particularly important.

Sources: Betz, *Head and Neck Pathology* (2019), PMID **30693456**; Syrjänen, *European Journal of Oral Sciences* (2018), PMID **30178562**.

Multiple papillomas in oral cavity of a six-year-old child

D) Human Immunodeficiency (HIV – Linear erythema)

Linear gingival erythema associated with HIV infection.

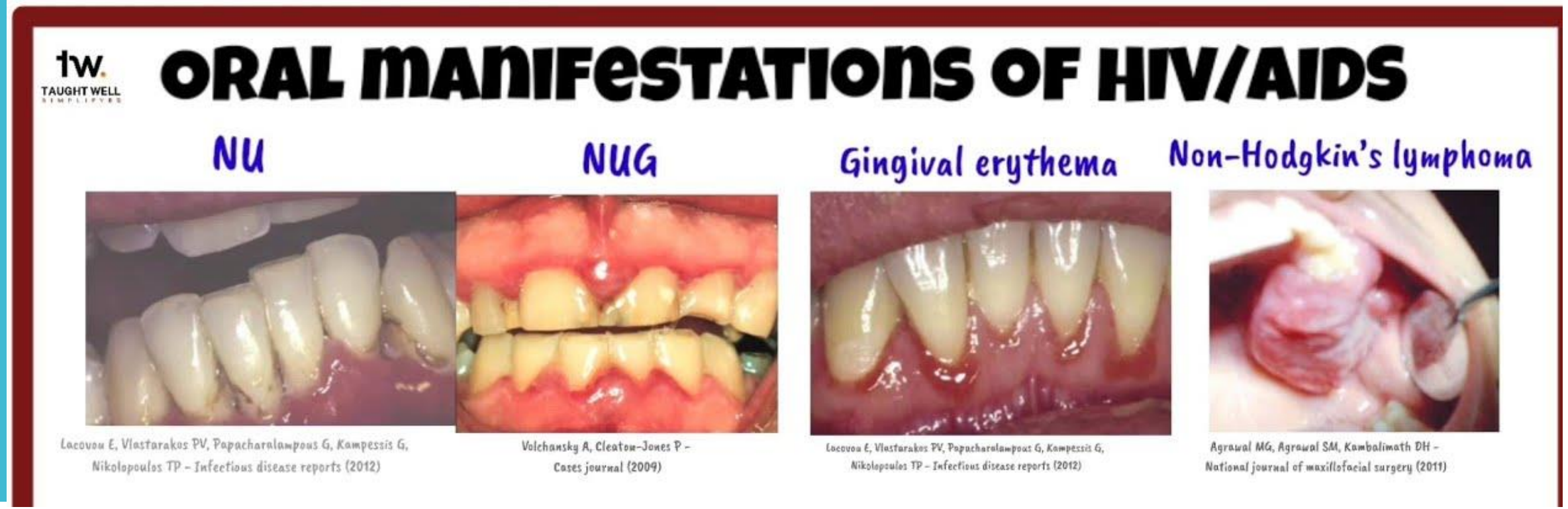
https://www.hiv.uw.edu/doc/874-3/linear-gingival-erythema.jpg

Oral Viral Lesions – Human Papilloma Virus (HPV)

Oral Viral Lesions
Human Papilloma Virus (HPV)

Papilloma on palatal gingiva of central incisor is cauliflower-like in appearance

HIV Manifestations In the Oral Cavity the Gingiva



III-Gingival Disease: Non-Dental-biofilm induced

- Fungal origin:
  - Candidiasis – Candida Albicans infection. (Most Common)
- Causes: topical steroids, decrease salivary flow (xerostomia)
  - Histoplasmosis (fungus histoplasma)

Fig. 16-5 Recurrent intraoral herpes infection. Ruptured vesicles of right palatal gingiva and mucosa.

Figure 4-9 A and B, Twenty-nine year-old male with primary herpetic infection and severe gingival inflammation. C and D, Six weeks postsystemic acyclovir.

Fig. 16-6 Herpes zoster of left palatal gingiva and mucosa. Irregular fibrin coated ulcerations with severe pain.

Gingival Diseases Herpes Zoster of left palatal gingival and mucosa

Chronic erythematous candidiasis

Pseudomembranous Candidiasis - HIV Patient

Streptococcal Gingivostomatitis

Hereditary Gingival Fibromatosis

- **2- <u>Genetic/developmental disorder:</u>**
- Hereditary gingival fibromatosis- Positive family history
- *<u>Gingival enlargement covers the teeth and delays eruption</u>*

3 – Inflammatory – Immune Conditions

A) Mucocutaneous lesions (LP- Lichen Planus, EM- Erythema Multiforme, PV-Pemphigus Vulgaris) have desquamative gingiva.

B) Allergic reactions (restorations – impression – toothpaste – mouth rinses, chewing gums)

Figure 4-11 Generalized severe allergic response of gingiva as a result of additive in chewing gum.

Traumatic lesions

Habits, chemical (tobacco, aspirin, bleaching agent), mechanical (toothbrush – orthodontic bands – clamps – crown), or thermal injury.

Figure 4-12 Self-inflicted gingival dehiscence induced via patient's fingernail.

6. Gingival pigmentation

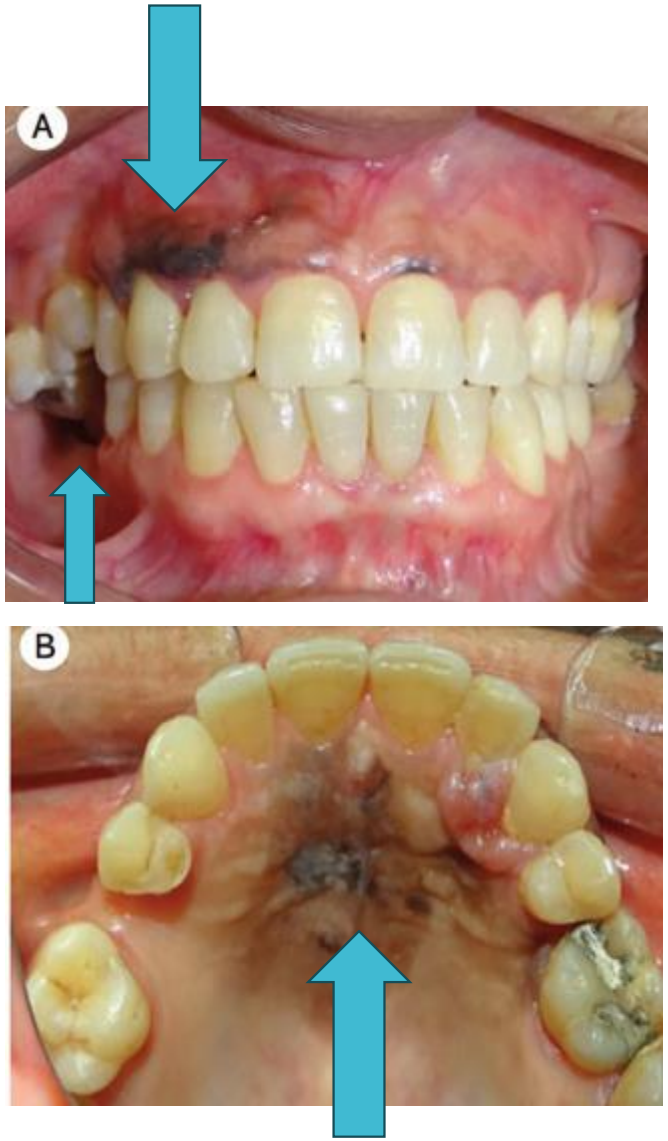
Physiologic gingival melanin Pigmentation

Amalgam Tattoo

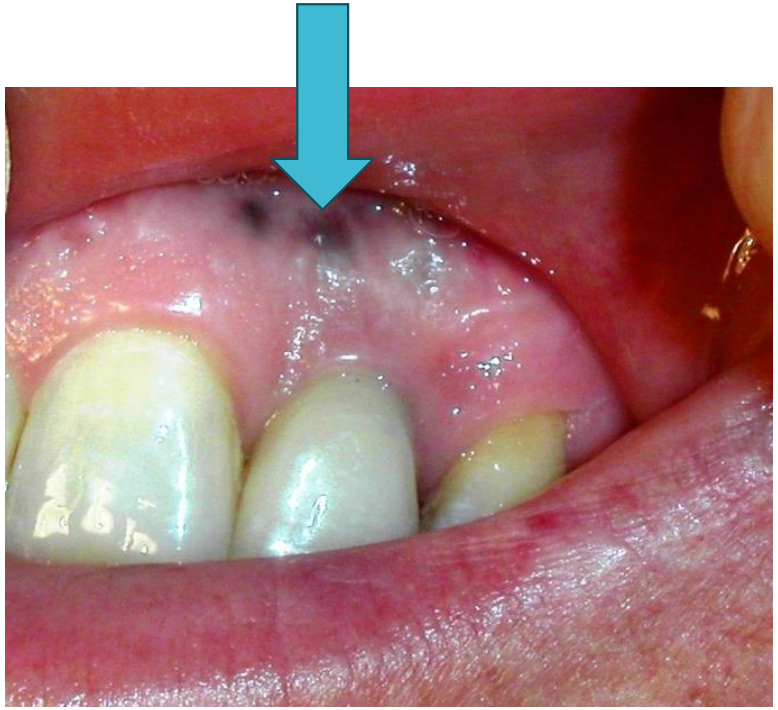
Sasaki, R., Taneda, S., & Okamoto, T. (2023). Gingival amalgam tattoo. *CMAJ : Canadian Medical Association journal = journal de l'Association medicale canadienne*, *195*(32), E1083–E1084. https://doi.org/10.1503/cmaj.230584

Melanoma of the oral cavity

(A-B) Multiple brownish macules and dark-nodular lesions with an irregular shape, located on upper gingiva, between the right maxillary first premolar (#14) and the right canine tooth (#13) and in the hard palate

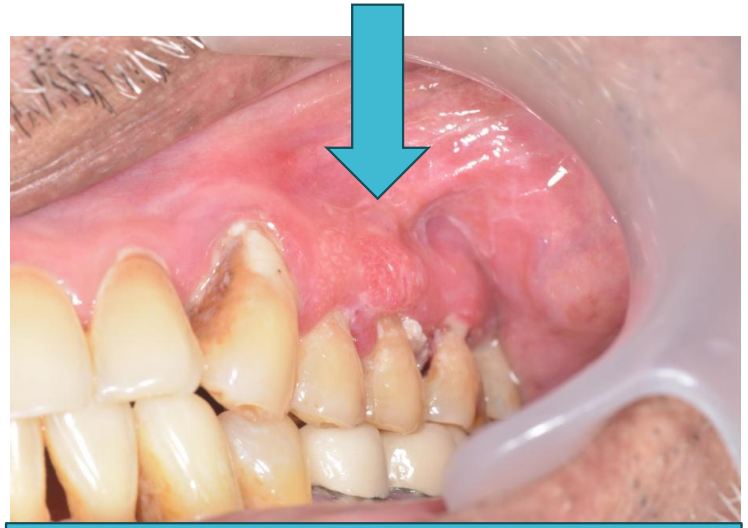


Rodrigues, B. T., Cunha, J. L., Albuquerque, D. M., Chagas, W. P., Freire, N. D., Agostini, M., Canedo, N. H., Albuquerque Júnior, R. L., de Sousa, S. F., Abrahão, A. C., Romañach, M. J., & Almeida, O. P. (2021). Primary melanoma of the oral cavity: A multi-institutional retrospective analysis in Brazil. *Medicina oral, patología oral y cirugía bucal*, *26*(3), e379–e386. https://doi.org/10.4317/medoral.24240



- Warszawik-Hendzel, O., Słowińska, M., Olszewska, M., & Rudnicka, L. (2014). Melanoma of the oral cavity: pathogenesis, dermoscopy, clinical features, staging and management. *Journal of dermatological case reports*, *8*(3), 60–66. https://doi.org/10.3315/jdcr.2014.1175

Gingival Squamous Cell Carcinoma

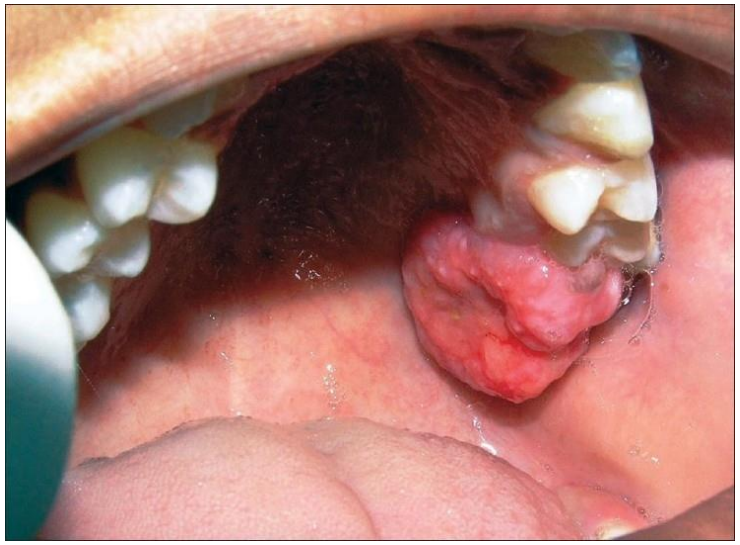


A 70-year-old male patient presented with a gingival lesion in maxilla region mimicking pyogenic granuloma without having a tobacco history. A thorough clinical, radiographical and histopathological examination was done and led to the diagnosis of GSCC and the treatment was initiated.

Kadhiresan, R., Reshmaa, R., Arunmozhi, U., Shanmugapriya, R. (2021). Rare presentation of gingival squamous cell carcinoma of maxillary posterior region mimicking pyogenic granuloma in a non-smoker – A case report. IP International Journal of Periodontology and Implantology, 6(3), 174-178. https://doi.org/10.18231/j.ijpi.2021.030

- Gingival Squamous cell carcinoma (GSCC) in maxilla is a rare malignant neoplasm especially when compared with mandible.
- The most common sites of oral carcinoma are being the lateral border of the tongue and the floor of the mouth which is followed by palate, buccal mucosa and rarely in gingiva.
- The clinical picture of oral carcinoma can be misdiagnosed for gingival overgrowth, desquamative lesions, traumatic ulcers or even pyogenic granuloma.
- Maxillary oral gingival carcinoma is a rare entity, especially in a non-smoker.

Reactive Process



- A 12-year-old girl reported with the chief complaint of soft tissue growth in the palate.
- Intraoral examination revealed a painless pedunculated, cauliflower-like rubbery mass on the palatal aspect of the maxillary left permanent molar extending towards the occlusal surface.

- The lesion was abnormally large about 2.5 cm mesiodistally and 1.5 cm buccopalatally and the side of the lesion facing the occlusal surface was focally ulcerated.
- The maxillary first permanent molar was buccally displaced.
- History revealed that the lesion started growing on its own since she first noticed it about a month back when it was a small nodule.
- The lesion was painless and occasionally bled on its own or when traumatized with toothbrush and in its present state was interfering with occlusion.

Poonacha, K. S., Shigli, A. L., & Shirol, D. (2010). Peripheral ossifying fibroma: A clinical report. *Contemporary clinical dentistry*, *1*(1), 54–56. https://doi.org/10.4103/0976-237X.62520

Severe Gingival Recession

Fig. 16-56 Severe gingival recession and wounding due to improper toothbrushing. Note the unaffected interdental papillae.

Thermal burn due to hot coffee (erosions + petechiae)

Gingival burn produced with 20% H_2O_2 (Hydrogen Peroxide)

Aspirin burn

Clinical appearance starts the examination

Describe before diagnosing

Color, contour, consistency, texture, plaque, recession, enlargement

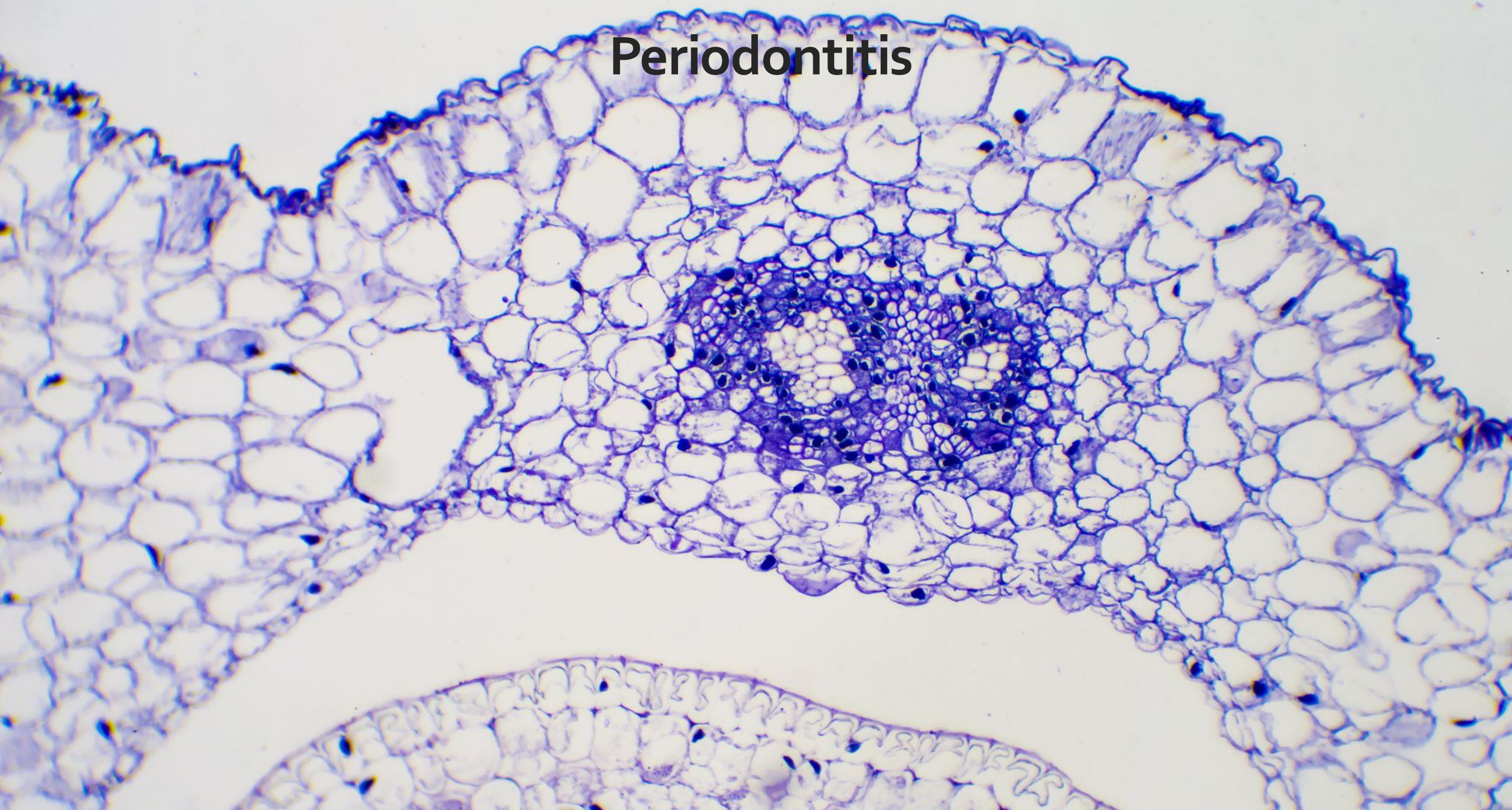
What measurements and history would you need before assigning a case definition?

Images: Mohamed Hamze, public domain; Shaimaa Abdellatif, CC BY-SA 4.0.

Erythema Multiforme (gingival erythema, ulcers, and crusted lips.)

Patient with Lichen Planus (LP) (reticular, desquamative gingivitis)

Periodontitis



Necrotizing Periodontal Diseases

1

A) Necrotizing gingivitis

2

B) Necrotizing periodontitis

3

C) Necrotizing stomatitis

Periodontitis

A) Based on severity and complexity of management

(Stage I, II, III, IV)

B) According to evidence and risk of rapid progression (Grade A, B, C)

C) According to extent and distribution (localized, generalized, molar incisor distribution)

Periodontitis as a manifestation of systemic diseases

- A) Hematologic disorders
  - Acquired neutropenia, Leukemias, others
- B) Genetic disorders
  - Papillon-Lefevre syndrome, Chediak-Higashi syndrome, Down syndrome, leukocyte adhesion deficiency syndrome, Familial & cyclic neutropenia, Histiocytosis syndrome, Glycogen storage diseases, infantile genetic agranulocytosis, Cohen syndrome, Ehlers-Danlos Syndrome, Hypophosphatasia, others.
- C)Not otherwise specified

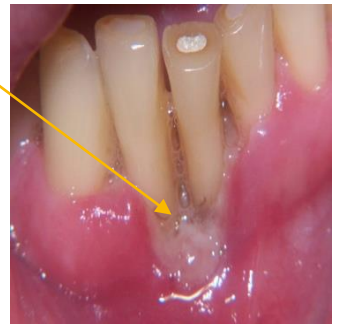
Necrotizing Periodontal Diseases:

Necrotizing gingivitis

- Sudden Onset
- The lesions appear **<u>punched out</u>** and **<u>crater-like</u>**, papilla necrosis; the necrotic tissue separates from the healthy gingiva, forming a **<u>gray pseudo membrane</u>**; and the gingiva shows linear erythema (redness)
- Causes: stress-smoking-malnutrition-immunosuppression and poor oral hygiene (OHI).
- Rx: antimicrobial therapy = plaque removal = OHI

Necrotizing Periodontitis

Necrotizing stomatitis



Category	Patients	Predisposing conditions	Clinical condition
Necrotizing periodontal diseases in chronically, severely compromised patients	In adults	HIV+/AIDS with CD4 counts < 200 and detectable viral load	NG, NP, NS, Noma. Possible progression
		Other severe systemic conditions (immunosuppression)	
	In children	Severe malnourishments[a]	
		Extreme living conditions[b]	
		Severe (viral) infections[c]	
Necrotizing periodontal diseases in temporarily and/or moderately compromised patients	In gingivitis patients	Uncontrolled factors: stress, nutrition, smoking, habits	Generalized NG. Possible progression to NP
		Previous NPD: residual craters	
		Local factors: root proximity, tooth malposition	Localized NG. Possible progression to NP
		Common predisposing factors for NPD (see paper)	NG. Infrequent progression
	In periodontitis patients		NP. Infrequent progression

Periodonti

How to diagnose and classify the periodontitis?

Identification of interdental CAL at two or more non-adjacent teeth

The attachment loss should be related ONLY to periodontitis while other etiologies such as recession, root fracture & defective restorations should be excluded.

Identification of the form of periodontitis

e.g., necrotizing, manifestation of systemic conditions, or periodontitis

Description of the presentation

Based on the 2017-18 AAP Staging and Grading System.

Steps for diagnosis & treatment

- Step 1 (initial overview of the case)
  - Screen full mouth radiographs, probing depths, recession, and missing teeth.
- Step 2 (Determine stage)
  - Determine maximum CAL or radiographic bone loss and bone loss pattern (horizontal/angular) → Stage I/II.
  - Check for tooth loss & determine case complexity (probing depths, furcation involvements, occlusion/function, need for extensive rehabilitation) → Stages III and IV.
- Step 3 (Determine grade)
  - History /risk of progression/age, risk factors and medical status and systemic inflammatory consideration.
  - Response to SRP and plaque control assessments.
- Step 4 (Treatment plan)
  - If stages I/II → standard periodontal treatment
  - If stages III/IV → complex or multidisciplinary treatment.

Periodontitis Staging: Severity and Complexity

- **Stage I**: **Initial periodontitis**
- Interdental CAL: 1–2 mm
- Radiographic bone loss: coronal third, <15%
- Tooth loss because of periodontitis: none
- Maximum probing depth: ≤4 mm
- Bone loss pattern: mostly horizontal

Periodontitis Staging: Severity and Complexity

- **Stage II: Moderate periodontitis**
- Interdental CAL: 3–4 mm
- Radiographic bone loss: coronal third, 15%–33%
- Tooth loss because of periodontitis: none
- Maximum probing depth: ≤5 mm
- Bone loss pattern: mostly horizontal

Periodontitis Staging: Severity and Complexity

- **Stage III**: **Severe periodontitis with potential for additional tooth loss**
- Interdental CAL: ≥5 mm
- Radiographic bone loss: middle third of the root and beyond
- Tooth loss because of periodontitis: ≤4 teeth
- Complexity may include PD ≥6 mm, vertical bone loss ≥3 mm, Class II or III furcation involvement, or moderate ridge defect

Periodontitis Staging: Severity and Complexity

- **Stage IV**: **Advanced periodontitis with extensive tooth loss or functional problems**
- Interdental CAL: ≥5 mm
- Radiographic bone loss: middle third of the root and beyond
- Tooth loss because of periodontitis: ≥5 teeth
- Stage III complexity plus rehabilitation or functional problems

Important: Mobility degree ≥2 may contribute to Stage IV complexity. The number of mobile teeth does not determine whether the case is Stage III or Stage IV.

	Periodontitis	Stage I	Stage II	Stage III	Stage IV
Severity	**Interdental CAL** *(at site of greatest loss)*	1 – 2 mm	3 – 4 mm	≥5 mm	≥5 mm
	RBL	Coronal third (<15%)	Coronal third (15% - 33%)	Extending to middle third of root and beyond	Extending to middle third of root and beyond
	Tooth loss *(due to periodontitis)*	No tooth loss		≤4 teeth	≥5 teeth
Complexity	**Local**	• Max. probing depth ≤4 mm • Mostly horizontal bone loss	• Max. probing depth ≤5 mm • Mostly horizontal bone loss	In addition to Stage II complexity: • Probing depths ≥6 mm • Vertical bone loss ≥3 mm • Furcation involvement Class II or III • Moderate ridge defects	In addition to Stage III complexity: • Need for complex rehabilitation due to: – Masticatory dysfunction – Secondary occlusal trauma (tooth mobility degree ≥2) – Severe ridge defects – Bite collapse, drifting, flaring – < 20 remaining teeth (10 opposing pairs)

Important: *Mobility degree ≥2 may contribute to Stage IV complexity. The number of mobile teeth does not determine whether the case is Stage III or Stage IV.*

According to evidence and risk of rapid progression

- Grade A: slow rate of progression
- Grade B: moderate rate of progression
- Grade C: rapid rate of progression
- Anticipated treatment response depend on case phenotype, smoking & hyperglycemia

	Progression		Grade A: Slow rate	Grade B: Moderate rate	Grade C: Rapid rate
Primary criteria *Whenever available, direct evidence should be used.*	Direct evidence of progression	Radiographic bone loss or CAL	No loss over 5 years	<2 mm over 5 years	≥2 mm over 5 years
	Indirect evidence of progression	% bone loss / age	<0.25	0.25 to 1.0	>1.0
		Case phenotype	Heavy biofilm deposits with low levels of destruction	Destruction commensurate with biofilm deposits	Destruction exceeds expectations given biofilm deposits; specific clinical patterns suggestive of periods of rapid progression and/or early onset disease
Grade modifiers	Risk factors	Smoking	Non-smoker	<10 cigarettes/day	≥10 cigarettes/day
		Diabetes	Normoglycemic/no diagnosis of diabetes	HbA1c <7.0% in patients with diabetes	HbA1c ≥7.0% in patients with diabetes

Atlas self-check

1. Gingival enlargement in a patient taking amlodipine: which 2017 category?
2. Gingivitis worsened by pregnancy: biofilm-induced or non-biofilm-induced?
3. Punched-out, necrotic papillae with pain and halitosis: which diagnostic family?
4. Desquamative gingiva with little plaque: what is your next step?

10 minutes

Atlas Self-Check Answers:

- (1) drug-influenced gingival enlargement;
- (2) biofilm-induced, mediated by a systemic risk factor (sex steroid hormones);
- (3) necrotizing periodontal diseases;
- (4) differential diagnosis and referral pathway.

References

- 1.Hiraki A, Nakamura S, Abe K, Takenoshita Y, Horinouchi Y, Shinohara M, et al. Numb chin syndrome as an initial symptom of acute lymphocytic leukemia: report of three cases. Oral Surg Oral Med Oral Pathol Oral Radiol Endod. 1997;83:555–61. doi: 10.1016/s1079-2104(97)90120-7. [DOI] [PubMed] [Google Scholar]
- 2.Dean AK, Ferguson JW, Marvan IS. Acute leukemia presenting as oral ulceration to a dental emergency service. Aust Dent J. 2003;48:195–7. doi: 10.1111/j.1834-7819.2003.tb00032.x. [DOI] [PubMed] [Google Scholar]
- 3.Bodey GP. Oral complications of myeloproliferative disease. J Postgrad Med. 1971;49:115–21. doi: 10.1080/00325481.1971.11696477. [DOI] [PubMed] [Google Scholar]
- 4.Katz J, Peretz B. Trismus in a 6 year old child: A manifestation of leukemia. J Clin Pediatr Dent. 2002;26:337–9. doi: 10.17796/jcpd.26.4.bk08734uj8q03245. [DOI] [PubMed] [Google Scholar]
- 5.Alirezaei S, Bakhshi M, Taheri J, Mafi A, Moghaddas O. Oral ulcerations as the first manifestations of acute leukemia. A case report. Open J of Stoma. 2013;3:507–9. [Google Scholar]

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Mirror and Finger Rests Maxillary and Mandibular

September 24, 2026 | 8:00–11:50 AM

DEN 624 • Periodontology I • Tanner College of Dental Medicine

Mirror And Finger Rests
In
Maxillary And Mandibular Sextants

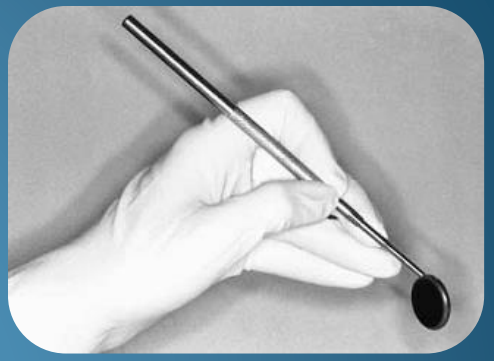
Overview

- Dental mirror: its types & surfaces
- Wrist position for instrumentation
- Fulcrum
- Mirror and finger rests in anterior sextants
- Mirror and finger rests in mandibular posterior sextants
- Mirror & finger rests in maxillary posterior sextants
- Conclusion
- References

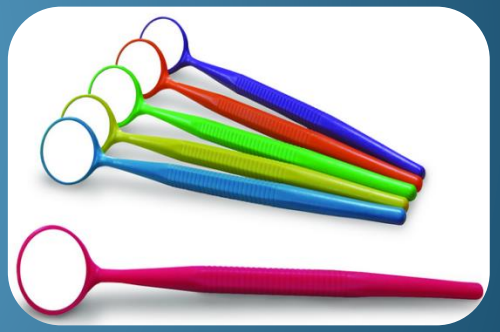
MIRROR & FINGER RESTS

Anterior Sextants

The Dental Mirror



- Types
  - Front surface mirror
  - Concave mirror
  - Plane mirror



- Dental/Mouth mirror

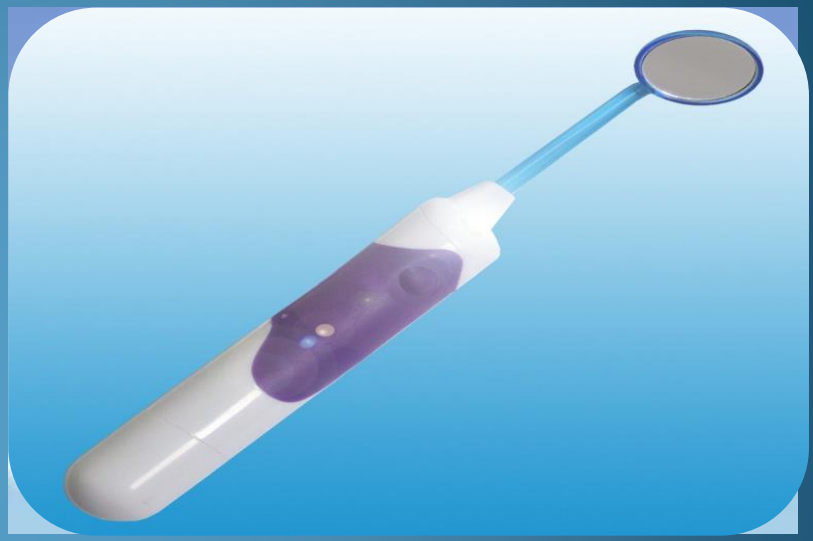
  working-end of a mirror has a reflecting mirrored surface used to view tooth surfaces that cannot be seen directly

Types of Mirror Surfaces

Type	Characteristics
Front surface	Reflecting surface is on front surface of glass Produces clear image & no distortion Good image quality Easily scratched
Concave	Reflecting surface on front surface of mirror lens Magnified image Not recommended as magnification distorts image
Plane (flat surface)	Reflecting surface is on back surface of mirror lens Produces double/ ghost image Not recommended as double image is distracting

Uses of Dental Mirror

1. Indirect Vision
2. Retraction
3. Indirect Illumination
4. Transillumination



Indirect Vision

To view a tooth surface or intraoral structure that cannot be seen directly

Retraction

Use of mirror head to hold the patient's cheek, lip, or tongue so that the clinician can view tooth surface that are otherwise hidden

Clinician's index finger or thumb is also used for retraction, esp lips

- Retraction
- Indirect vision
- Indirect illumination

Indirect illumination

to reflect light
onto a tooth surface
in a dark area of the mouth

Transillumination

Technique of directing light off the mirror surface and through the anterior teeth

Trans=through; Illumination=lighting up

Trans-illumination

A carious lesion appears as shadow when an anterior tooth is illuminated

- Unit light is positioned in such way that light beam is directed perpendicular to facial surfaces of anterior teeth

Transillumination technique

- Mirror is held behind the central incisors so that reflecting surface is parallel to lingual surfaces

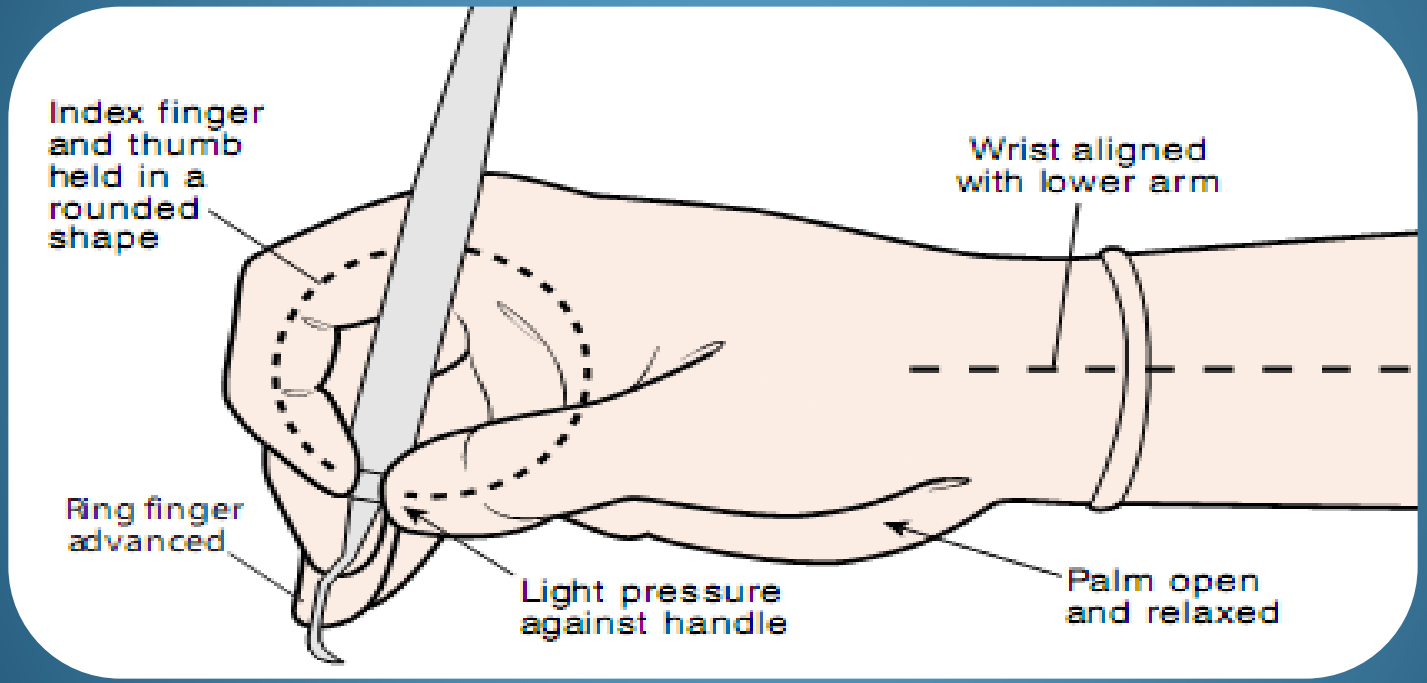
Mirror Use : Precautions

- Warm the reflecting surface against the patient's buccal mucosa
- Ask the patient to breathe through the nose
- Wipe the reflecting surface with a commercial defogging solution
- Wipe the reflecting surface with a gauze square moistened with a mouthwash

Avoid hitting the mirror head against the patient's teeth or resting the outer rim of the mirror head against the patient's gingival tissues

Wrist position for instrumentation

Neutral Wrist position

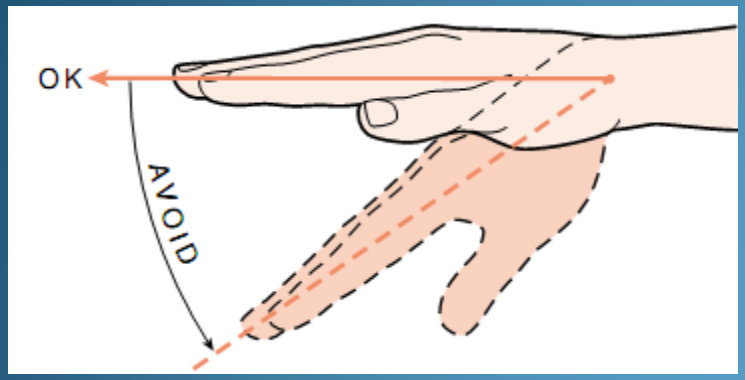


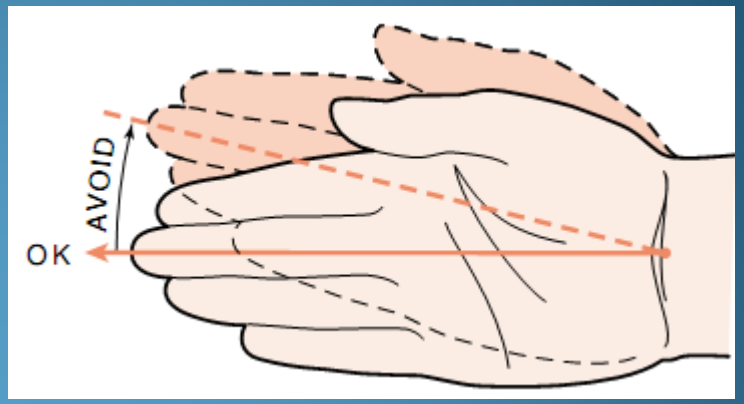
- ideal positioning of the wrist while performing work activities and is associated with decreased MS Injury

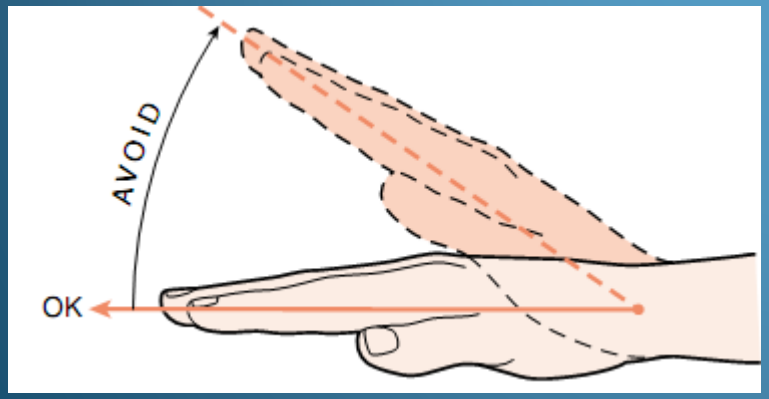
Neutral Hand Position

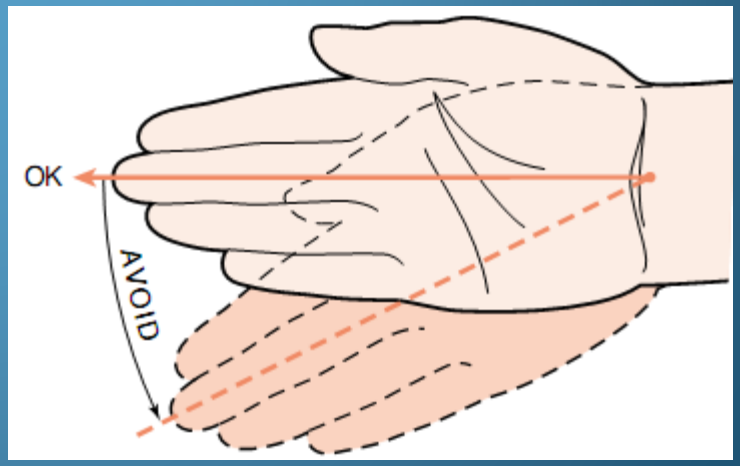
- Wrist aligned with the long axis of lower arm
- Little finger-side of palm rotated slightly downward
- Palm open & relaxed
- Thumb, middle, and index fingers held in rounded shape
- Light finger pressure against instrument handle
- Ring finger ahead of other fingers in grasp

Guideline of Neutral Wrist Position









The Fulcrum

- A finger rest used to stabilize the clinician's hand during periodontal instrumentation

- Functions:
  - "support beam" for hand during instrumentation
  - enables hand and instrument to move as a unit
  - allows precise control of stroke pressure and length during instrumentation

Types of Fulcrums

1. Intraoral
2. Extraoral
3. Advanced

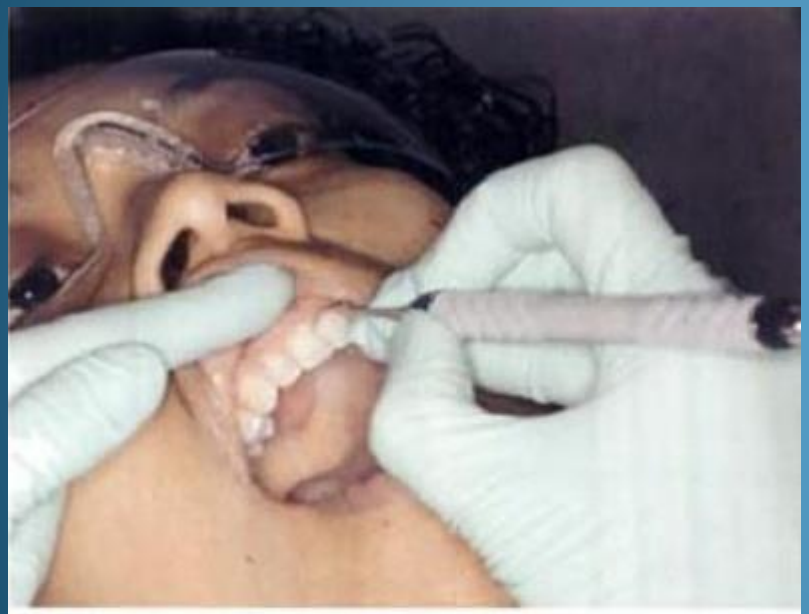
Intraoral

stabilization of clinician's hand by placing pad of ring finger on a tooth close to the working area

Extraoral

outside patient's mouth (chin/cheek)

Intraoral fulcrum



Intraoral Finger Rest. The pad of the ring finger is placed on a tooth near the tooth being instrumented.

Characteristics:

- Provides stable support for the hand
- Enables the hand and instrument to move as a unit
- Facilitates precise stroke pressure against the tooth surface
- Decreases injury due to unexpected patient movement

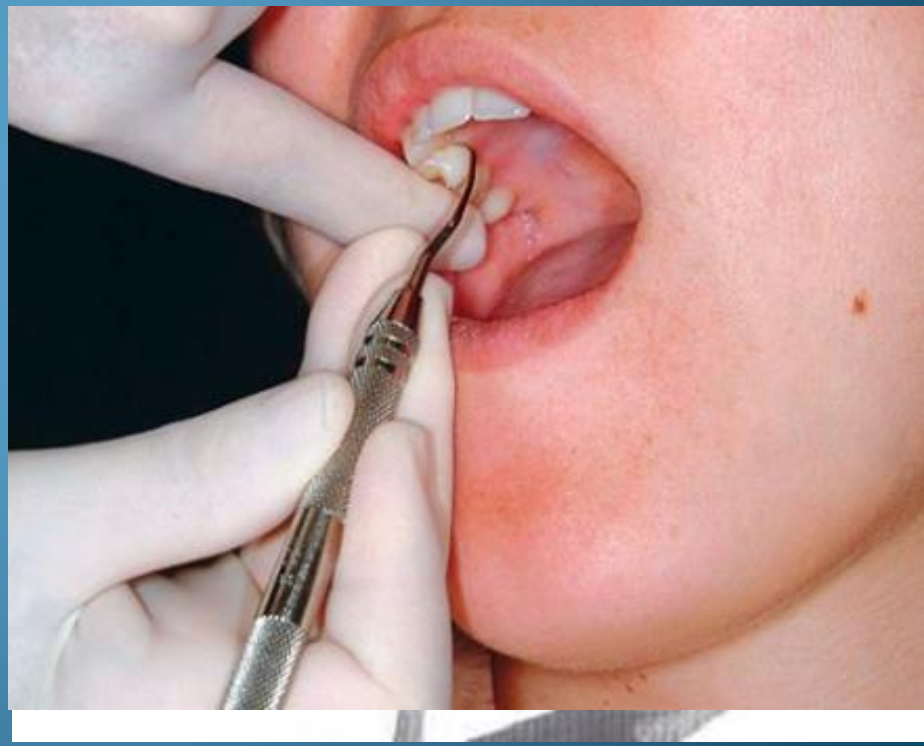
Types of intraoral fulcrums

1. Conventional

2. Opposite -arch

3. Cross - arch

4. Finger-on-finger



 Characteristics of intraoral fulcrum

1. Provides stable support for hand
2. Enables hand and instrument to move as a unit
3. Facilitates precise stroke pressure against tooth surface
4. Decrease likelihood of injury to patient & clinician if patient moves during instrumentation

Summary: Technique for Intraoral Fulcrum

Grasp	Hold the instrument in a modified pen grasp
Fulcrum	Keep ring finger straight, with the tip of the finger supporting the weight of the hand
Location	Finger rest near the tooth being instrumented Depending on the tooth instrumented and size of hand, the finger rest may be 1 to 4 teeth away from the tooth Should be out of “line of fire”
Rest	Finger rest on the same arch Fingertip of fulcrum finger rested on an incisal/occlusal surface or on occlusofacial/occlusolingual line angle of a tooth Teeth are saliva-covered, so more likely to slip if rest on facial/lingual surface Resting on mobile/large carious tooth avoided

Extraoral fulcrum

An extraoral fulcrum is a stabilizing point on patient's chin or cheek

- The clinician is using an extraoral fulcrum, i.e. ring finger rests on the patient's cheek to stabilize the dental mirror in the mouth

Extraoral Finger Rest. The ring finger rests on the patient's chin or cheek to stabilize the mirror.

Types of extraoral fulcrums

1. Palm- up

2. Palm- down

Extraoral fulcrums- not finger rests- instead of tip or pads of fingers, front or back surface of fingers on face

Reinforced type rests

1. Index finger reinforced rest 2. thumb reinforced rest

when precise control and pressure compromised by long distance between fulcrum & working end

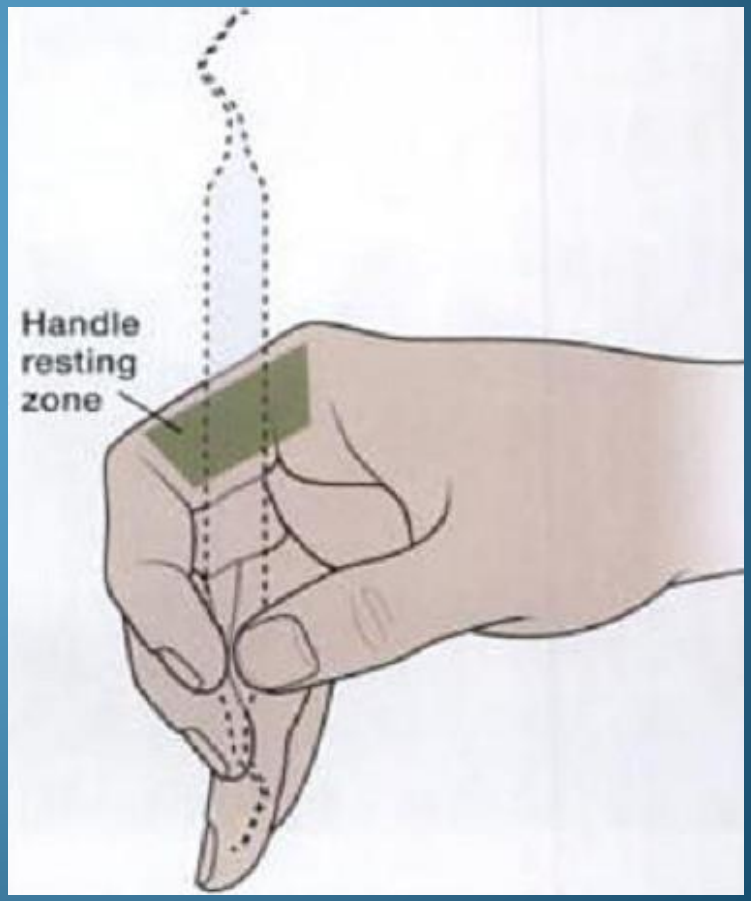
Benefits of reinforced instrumentation

- Allows both hands to work as a unit
- Provides more strength and power
- Enhances lateral pressure of blade on tooth surface
- Provides more stability to working hand
- Helps prevent hand/arm fatigue and injury during scaling

Mandibular Anterior Teeth

Handle Position

- Hold the hand in a palm down position
- Rest handle against index finger in shaded area



Facial aspect - Surface toward

Position overview

- 8 – 9 o’ clock

Retraction

- Retract lip with index finger/ thumb of left hand

Left canine mesial surface

- finger rest on occluso-facial line angle

Right canine distal surface

- Finger rest on incisal edge

Lingual aspect - Surfaces toward

Position overview

- 8 – 9 o’ clock

Mirror

- Retract tongue
- Indirect vision

Left canine mesial surface

- Finger rest on occluso-facial line angle

Right canine distal surface

- Finger rest on incisal edge

Facial aspect – Surfaces away

Position overview

- 12 o’ clock

Retraction

- Lip retracted with index finger/ thumb

Right canine mesial surface

- finger rest on occluso-facial line angle

Left canine distal surface

- Finger rest on incisal edge

Lingual aspect – Surfaces away

Position overview

- 12 o’ clock

Mirror

- Retract tongue
- Indirect vision

Right canine mesial surface

- finger rest on occluso-facial line angle

Left canine distal surface

- Finger rest on incisal edge

Anterior treatment areas - Mandible

Treatment area	Clock position	Patient's head
Facial surfaces Toward	8 – 9:00	Slightly toward Chin DOWN
Lingual surfaces Toward		
Facial surfaces Away	12:00	
Lingual surfaces Away		

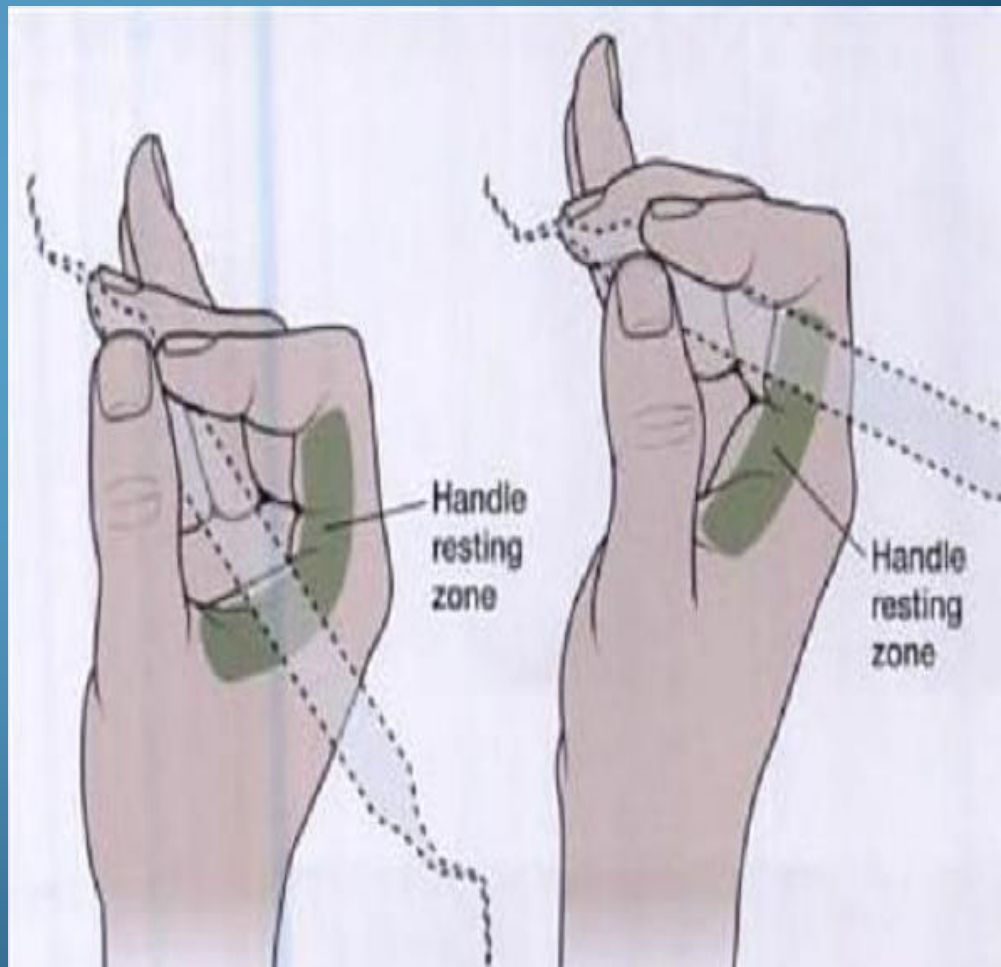
Anterior treatment areas – Mandible left handed

Treatment area	Clock position	Patient's head
Facial surfaces Toward NDH	4 – 3:00	Slightly toward Chin DOWN
Lingual surfaces Toward		
Facial surfaces Away	12:00	
Lingual surfaces Away		

Maxillary Anterior Teeth

Handle Position

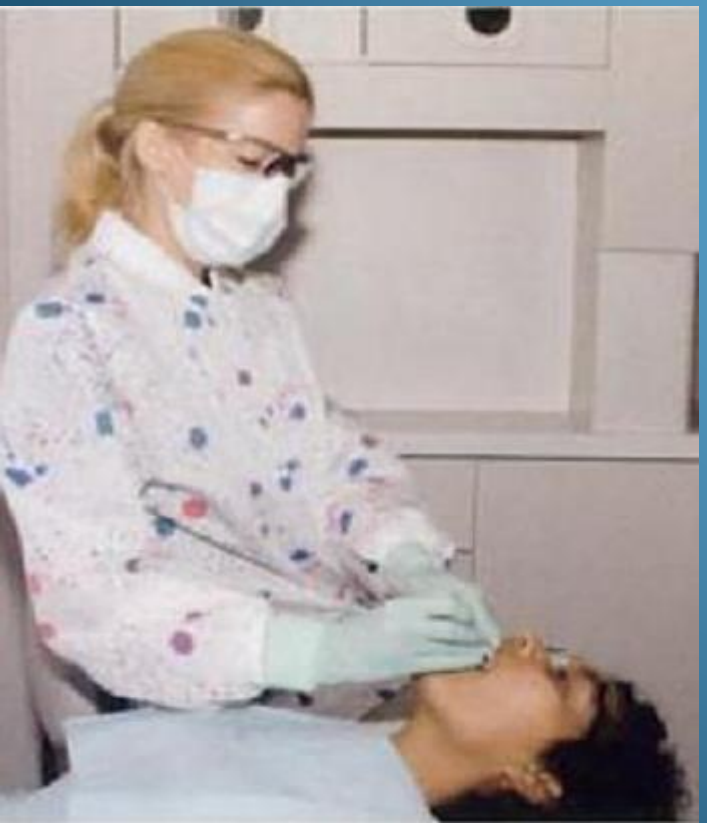
- Hold the hand in a palm up position
- Rest handle against index finger in shaded area



Facial aspect - Surfaces toward

Position overview

- 8 – 9 o’ clock



Retraction

- Retract lip with index finger/ thumb of left hand

Left canine mesial surface

- finger rest on occluso-facial line angle

Right canine distal surface

- Finger rest on incisal edge

Lingual aspect - Surfaces toward

Position overview

- 8 – 9 o’ clock

Mirror

- Retract tongue
- Indirect vision
- Indirect illumination

Left canine mesial surface

- finger rest on occluso-facial line angle

Right canine distal surface

- Finger rest on incisal edge

Facial aspect – Surfaces away

Position overview

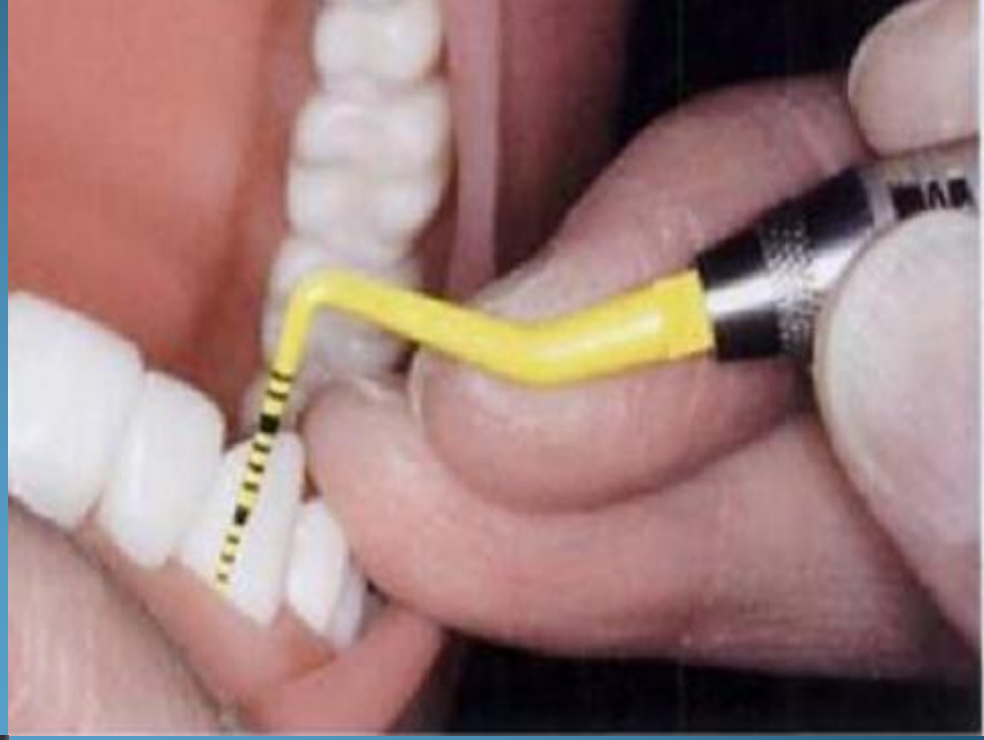
- 12 o’ clock

Retraction

- Lip retracted with index finger/ thumb

Right canine mesial surface

- finger rest on occlusal surface
- Should be able to see underside of middle & ring fingers



Left canine distal surface

- Finger rest on incisal edge

Lingual aspect – Surfaces away

Position overview

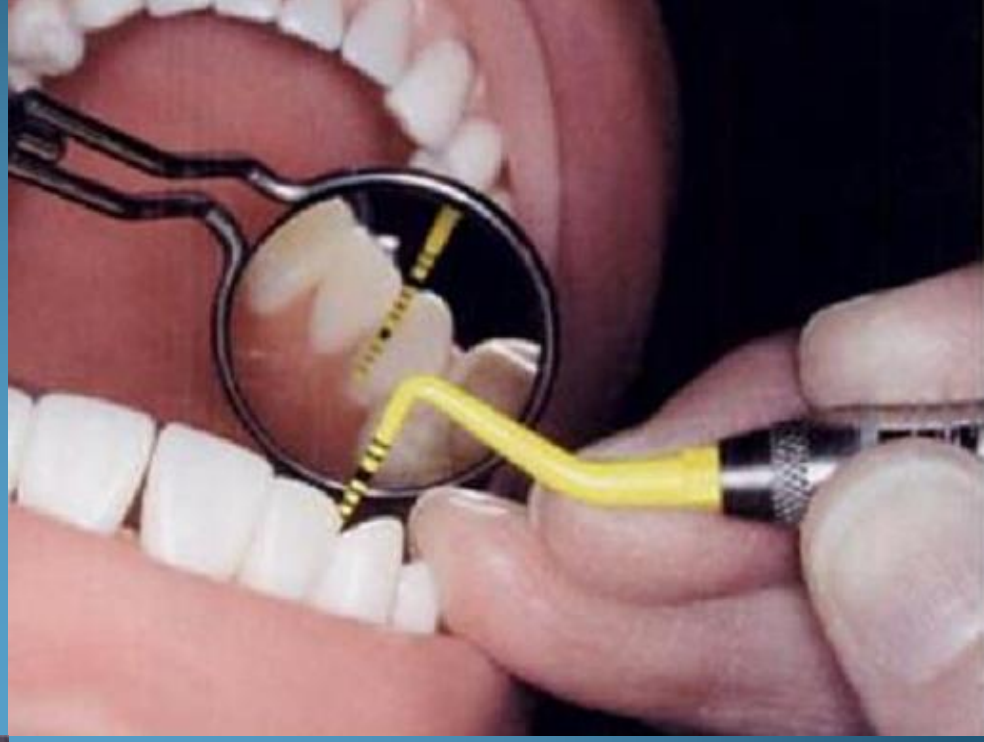
- 12 o’ clock

Mirror

- Mirror head positioned to visualize lingual surfaces

Right canine mesial surface

- finger rest on occlusal surface
- Should be able to see underside of middle & ring fingers



Left canine distal surface

- Finger rest on incisal edge

Anterior treatment areas - Maxilla

Treatment area	Clock position	Patient's head
Facial surfaces Toward	8 – 9:00	Slightly toward Chin UP
Lingual surfaces Toward		
Facial surfaces Away	12:00	
Lingual surfaces Away		

Anterior treatment areas – Maxilla left handed

Treatment area	Clock position	Patient's head
Facial surfaces Toward NDH	4 – 3:00	Slightly toward Chin UP
Lingual surfaces Toward		
Facial surfaces Away	12:00	
Lingual surfaces Away		

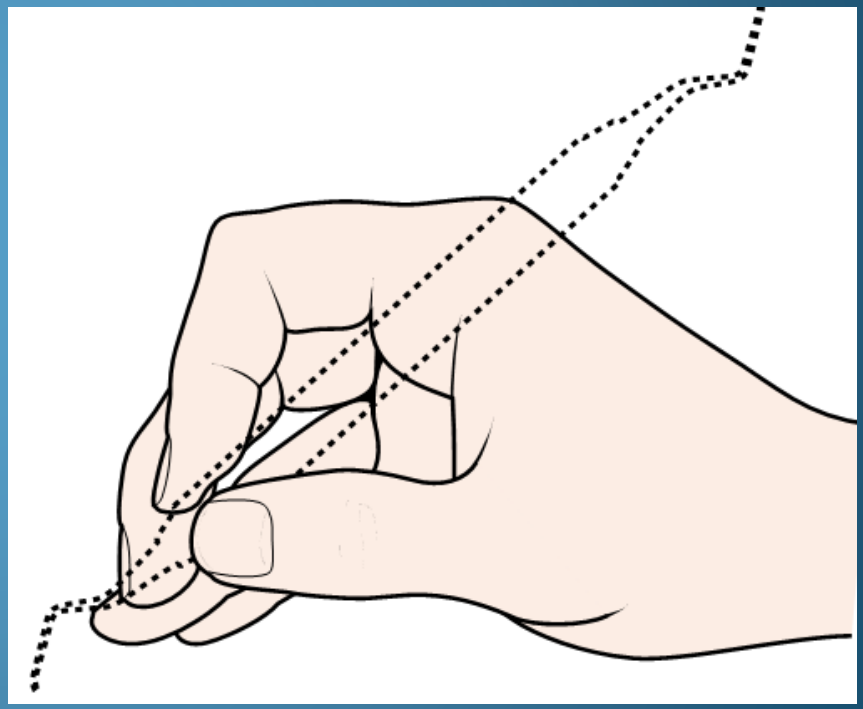
MIRROR & FINGER RESTS

Mandibular Posterior Sextants

Mandibular Posterior Teeth

Handle Position

- Hold the hand in a palm down position
- Rest handle against index finger somewhere betn 2nd & 3rd knuckles



Mandibular Right Posterior Sextant facial aspect

Position overview

- 9 o’ clock

Mirror

- Retract buccal mucosa
- Indirect vision esp distal surfaces

Second molar Facial aspect

- finger rest on occlusal surface

First premolar Facial aspect

- Finger rest on incisal surface of anteriors

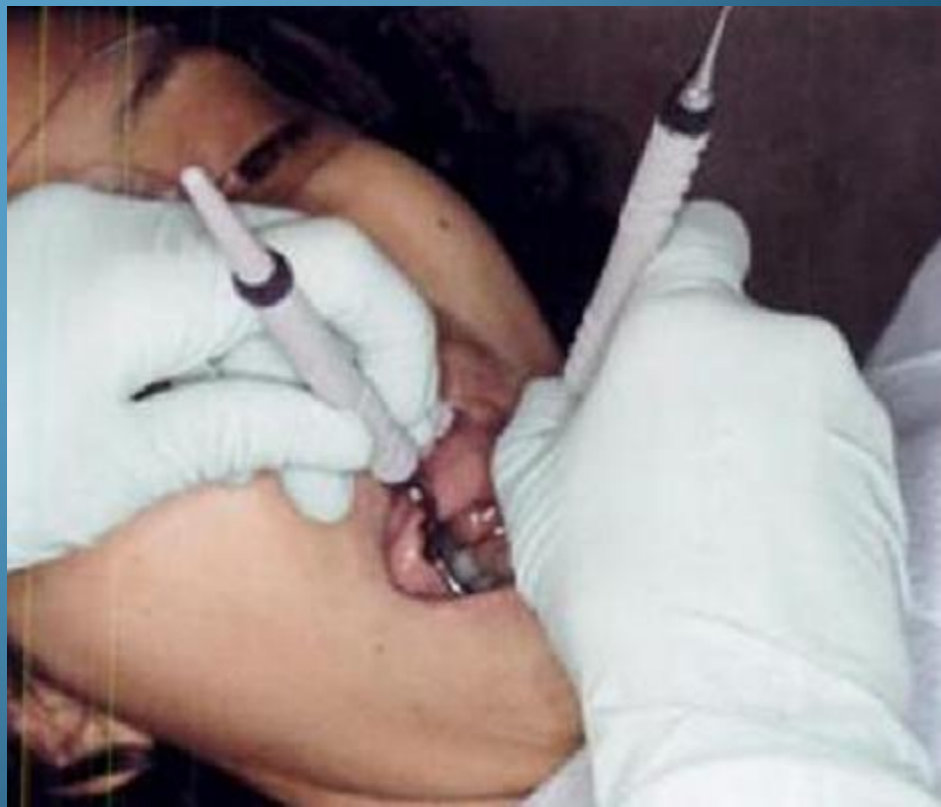
Mandibular Left Posterior Sextant lingual aspect

Position overview

- 9 o’ clock

Mirror

- Tongue retraction
- Indirect vision



Second molar Lingual aspe

- finger rest on occluso-fa line angle

First premolar Lingual aspect

- Finger rest on incisal edge of anteriors

Mandibular Left Posterior Sextant facial aspect

Position overview

- 10 – 11:00

Mirror

- Retract buccal mucosa

Second molar Facial aspect

- finger rest on occluso-facial line angle

First premolar Facial aspect

- Finger rest on incisal edge of anteriors

Mandibular Right Posterior Sextant lingual aspect

Position overview

- 10 – 11:00

Mirror

- Tongue retraction
- Indirect vision

Second molar Lingual aspect

- finger rest on occlusal surface

First premolar Lingual aspect

- Finger rest on incisal edge of anteriors

Posterior treatment areas - Mandible

Treatment area	Clock position	Patient's head
Right Facial Towards	9:00	Straight or slightly away Chin DOWN
Left Lingual Towards		
Right Lingual Away	10 - 11:00	Toward CHIN DOWN
Left Facial Away		

Posterior treatment areas – Mandible left handed

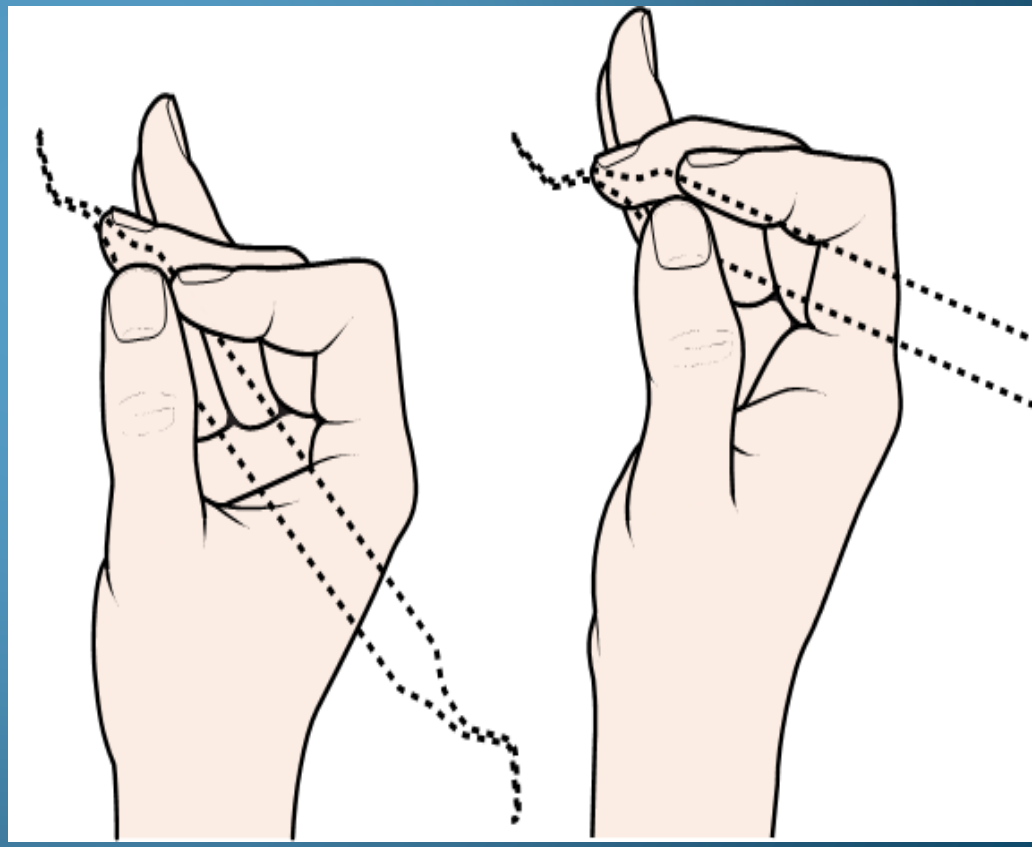
Treatment area	Clock position	Patient's head
Left Facial Towards	3:00	Straight or slightly away Chin DOWN
Right Lingual Towards		
Left Lingual Away	2 - 1:00	Toward CHIN DOWN
Right Facial Away		

MIRROR & FINGER RESTS

Maxillary Posterior Sextants

Maxillary Posterior Teeth
Handle Position

- Hold the hand in a palm-up position
- Rest handle against index finger somewhere behind 2nd knuckle to the “V” of hand



Maxillary Right Posterior Sextant facial aspect

Position overview

- 9 o’ clock

Mirror

- Retract buccal mucosa
- Indirect vision esp distal surfaces

Second molar Facial aspect

- finger rest on occlusal surface

First premolar Facial aspect

- Finger rest on incisal edge

Maxillary Left Posterior Sextant lingual aspect

Position overview

- 9 o’ clock

Mirror

- View distal surfaces

Second molar Lingual aspect

- finger rest on occluso-facial line angle

First premolar Lingual aspect

- Finger rest on occlusofacial line angle or incisal edge

Maxillary Left Posterior Sextant
Facial aspect

Position overview

- 10 – 11:00

Mirror

- Retract buccal mucosa down & away from teeth

Second molar Facial aspect

- finger rest on occlusal surface
- Can see underside of middle & ring fingers

First premolar Facial aspect

- Finger rest on incisal edge

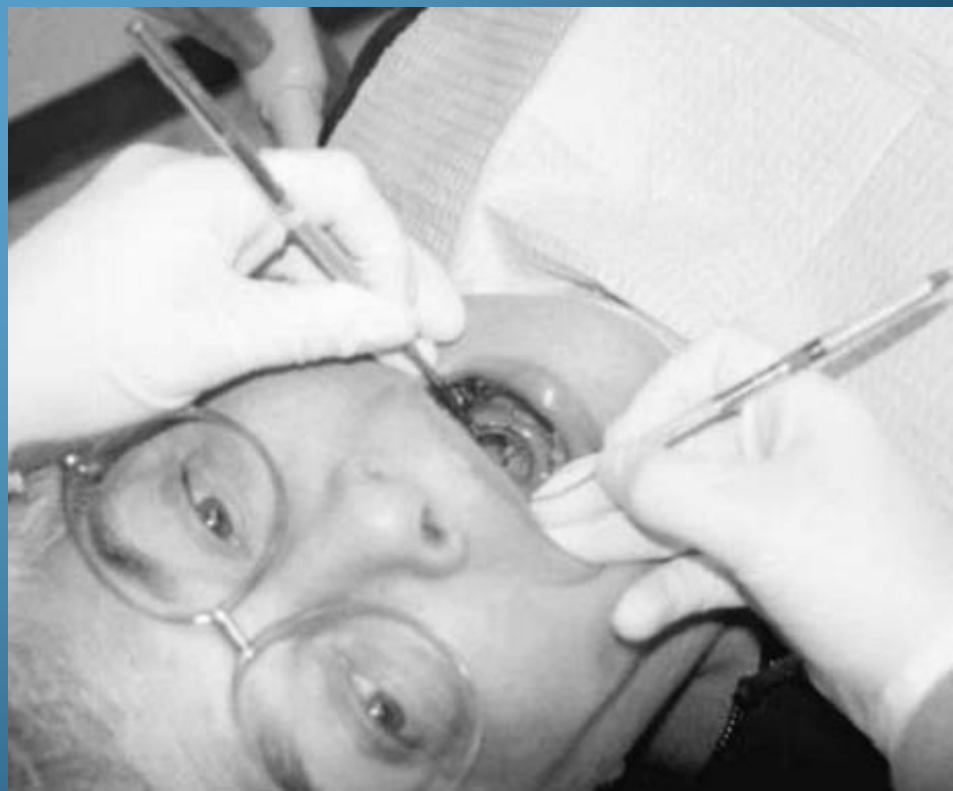
Maxillary Right Posterior Sextant
Lingual aspect

Position overview

- 10 – 11:00

Mirror

- Indirect Vision



Second molar Lingual aspect

- Finger rest on occlusal surface

First premolar Lingual aspect

- Finger rest on incisal edge

Posterior treatment areas - Maxilla

Treatment area	Clock position	Patient's head
Right Facial Towards	9:00	Straight or slightly away Chin UP
Left Lingual Towards		
Right Lingual Away	10 - 11:00	Toward CHIN UP
Left Facial Away		

Posterior treatment areas – Maxilla left handed

Treatment area	Clock position	Patient's head
Left Facial Towards	3:00	Straight or slightly away Chin DOWN
Right Lingual Towards		
Left Lingual Away	2 - 1:00	Toward CHIN DOWN
Right Facial Away		

Sequence for Establishing a Finger Rest

1.
   - ME
   - Assume a clock position for treatment area
2.
   - MY PATIENT
   - Establish patient chair & head position
3.
   - MY EQUIPMENT
   - Adjust unit light; pause & check clinician, patient & equipment position

4

- MY NONDOMINANT HAND
- Grasp mirror & establish proper position

5

- MY DOMINANT HAND
- Grasp the instrument & pause to evaluate finger placement

6

- MY FINGER REST
- Establish suitable finger rest for the first tooth to be treated

At last : Pause to evaluate finger rest

- Is the tip of ring finger on a secure tooth surface?
- Is ring finger straight, acting as support beam?
- Is my finger placement in grasp still correct?

Identified disorders, occupational risk factors and symptoms		
Disorders	**Occupational risk factors**	**Symptoms**
Tendonitis/tenosynovitis	Repetitive wrist motions Repetitive shoulder motions Sustained hyper extension of arms Prolonged load on shoulders	Pain, weakness, swelling, burning sensation or dull ache over affected area
Epicondylitis (elbow tendonitis)	Repeated or forceful rotation of the forearm and bending of the wrist at the same time	Same symptoms as tendonitis
Carpal tunnel syndrome	Repetitive wrist motions	Pain, numbness, tingling, burning sensations, wasting of muscles at base of thumb, dry palm
DeQuervain's disease	Repetitive hand twisting and forceful gripping	Pain at the base of thumb
Thoracic outlet syndrome	Prolonged shoulder flexion Extending arms above shoulder height Carrying loads on the shoulder	Pain, numbness, swelling of the hands
Tension neck syndrome	Prolonged restricted posture	Pain

Conclusion

- Periodontal therapy is directly affected by clinician's knowledge of suitable mirror and finger rests in various anterior and posterior sextants of both maxillary and mandibular arches

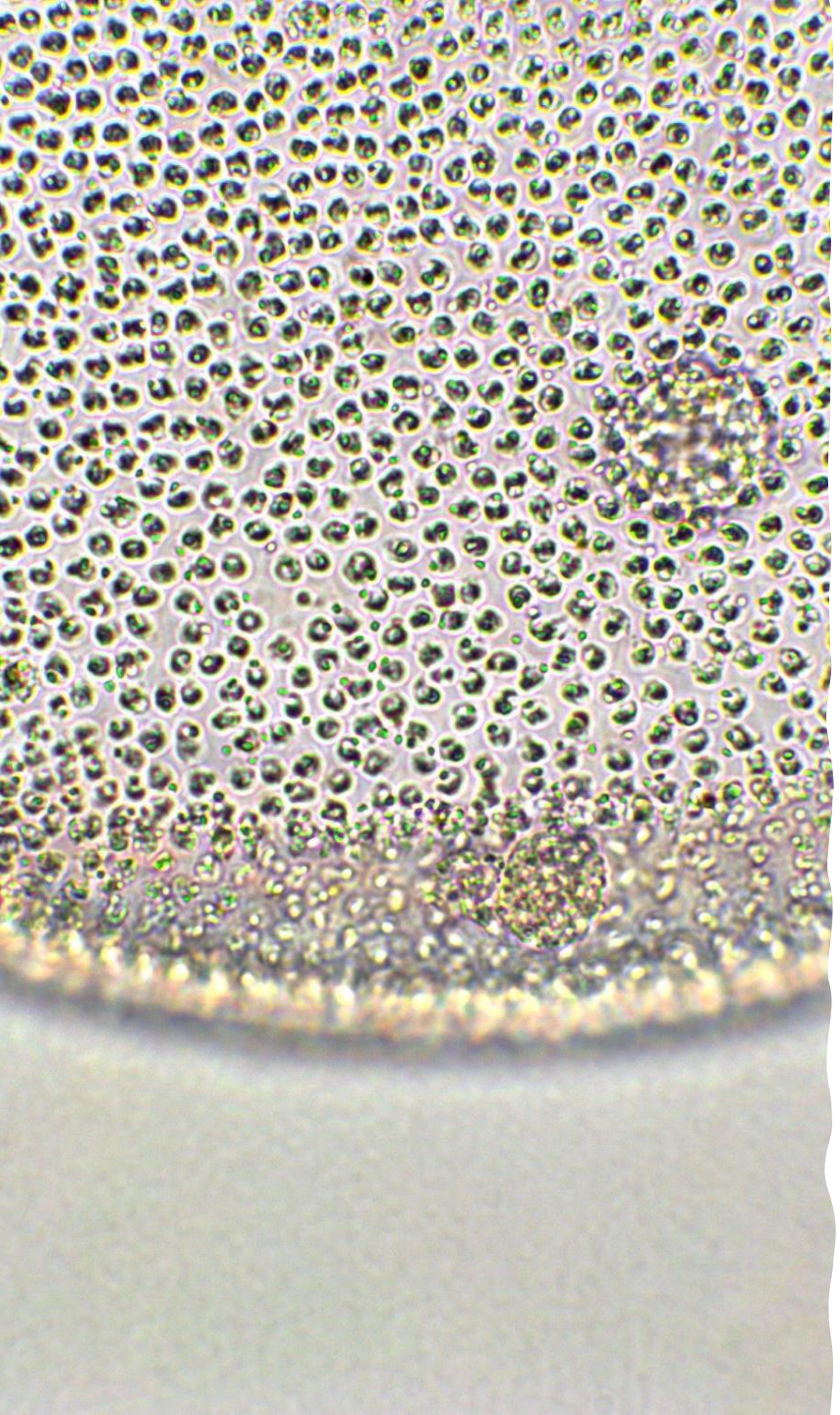
- A good finger rest prevents injury and laceration of the gingiva and surrounding tissues by poorly controlled instruments

- Not only appropriate mirror and finger rests but also proper clinician, patient, assistant and equipment positions are vital to effective instrumentation for undue fatigue and development of MSD to the operator

References

1. Jill Shiffer Nield-Gehrig, Fundamentals of periodontal instrumentation & advanced root instrumentation, 6th edition
2. Carranza's Clinical Periodontology, 9th edition
3. H. Dong, et al, The effects of finger rest positions on hand muscle load and pinch force in simulated dental hygiene work; Journal of Dental Education, Apr 2005
4. Gordon L. Pattison et al, Periodontal Instrumentation
5. Anna M. Pattison, Dimensions of Dental Hygiene, Oct 2004
6. Diane Millar, Reinforced Periodontal Instrumentation and Ergonomics for Dental Care Provider, Apr 2007

Thank You



ROLE OF DENTAL BIOFILM IN PATHOGENESIS OF PERIODONTAL DISEASE

Compiled and Presented by

Vickie E. Jones, DHSc, MS, RDH, FNAP

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  - Biofilm
- Dental plaque
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  - Dental Plaque formation
- Biofilm formation
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- Communication Between Biofilm Bacteria
- Transition From Health to Disease
- Microbial specificity of periodontal diseases
  - Gingivitis
  - Periodontitis
  - Localised Aggressive periodontitis
  - Necrotizing periodontal diseases
  - Abscesses of the periodontium
  - Periimplantitis
- Conclusion
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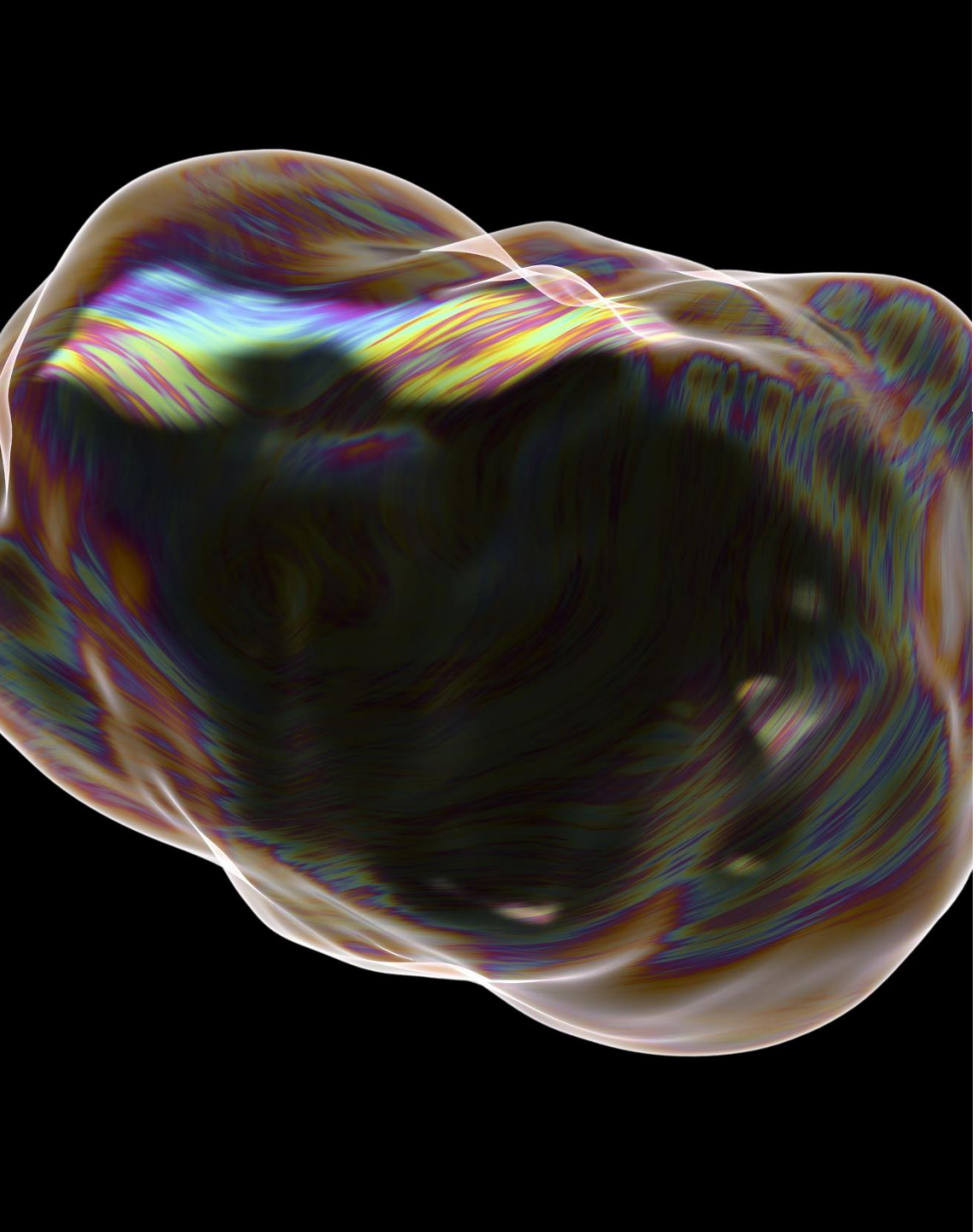
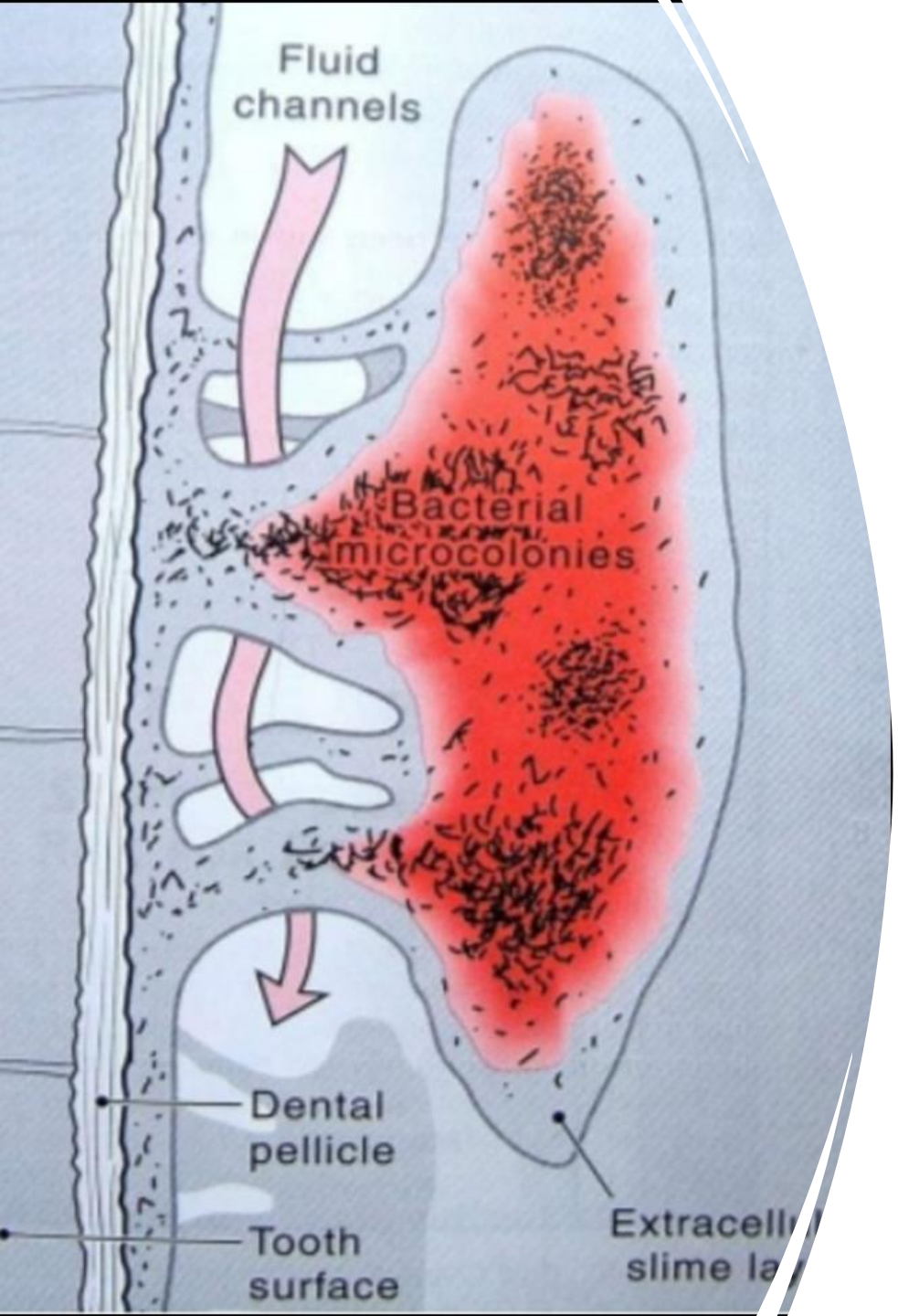


Introduction

- Most of periodontal disease occur due to increased bacterial mass or pathogenicity.
- Accumulation and metabolism of bacteria on oral surfaces is considered the primary cause of dental caries, gingivitis, periodontitis, peri-implant infections.
- The Dental biofilm has an undefinable microbial community associated with a tooth surface or any other hard, non-shedding surface. (Wilderer & Charaklis 1989)
- “Biofilms defined as matrix enclosed bacterial populations adherent to each other and/or to surfaces.” (Costerton 1994)



Biofilm structure

- Most biofilms are a dense layer of microbes which are bound together in a polysaccharide matrix with other organic and inorganic materials.
- A series of fluid channels penetrate the slime layer & facilitate the movement of nutrients & bacterial products throughout the biofilm.
- Fluid channels provide nutrients & oxygen for the bacterial microcolonies, waste products & enzymes within the biofilm structure.

Dental plaque

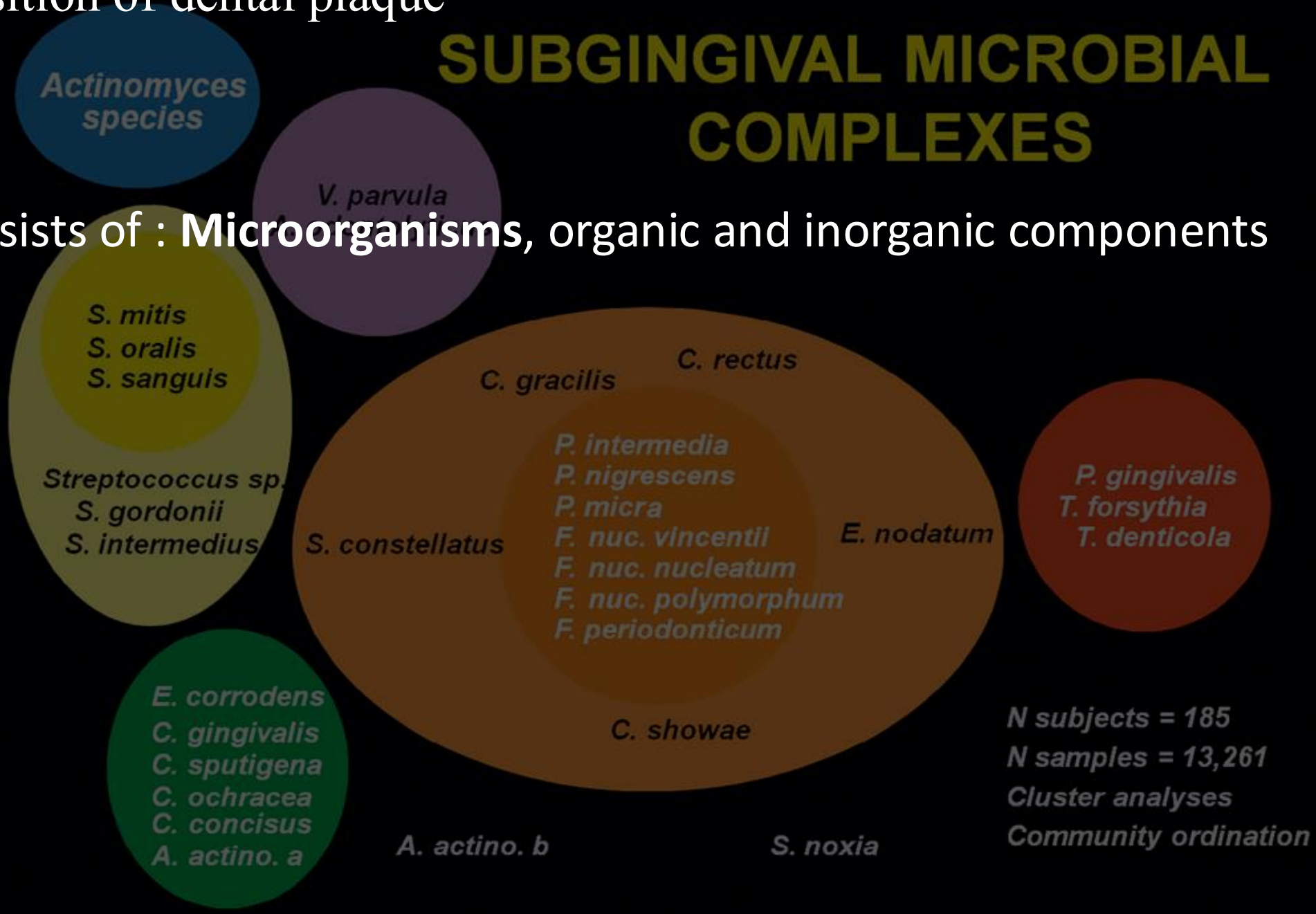
Structured, resilient, yellow-grayish substance adhered tenaciously to the intraoral hard surface, including removable and fixed restorations

G.V.Black (1836-1950) Coined the term *"Gelatinous Microbial Plaque"*

J.Leons Williams (1852-1932) First described *"Bacterial plaque"*

Composition of dental plaque

Consists of : **Microorganisms**, organic and inorganic components



- **Organic components :**
  - **Polysaccharides** produced by bacteria. They maintaining the integrity of the biofilm.
  - **Lipid material** from the membranes of disrupted bacterial and host cells, and possibly food debris.
  - **Glycoproteins** from saliva are an important component of the pellicle that initially coats a clean tooth surface.

- **Inorganic components:**

-Calcium and phosphorus (predominantly)

-Other minerals such as sodium, potassium, and fluoride.

Dental Plaque formation

The formation of the pellicle on the tooth surface

Initial adhesion/

attachment of bacteria

Colonization/

plaque maturation

1. Formation of the Pellicle

Surfaces in the oral cavity, including the hard and soft tissues, are coated with a layer of organic material known as the acquired pellicle.

Bacteria that adhere to tooth surfaces do not contact the enamel directly but interact with the acquired enamel pellicle.

Pellicle to be composed of two layers : a thin basal layer, difficult to remove even with harsh chemical and mechanical treatments, and a thicker globular layer, up to 1 µm or more, that is easier to detach.

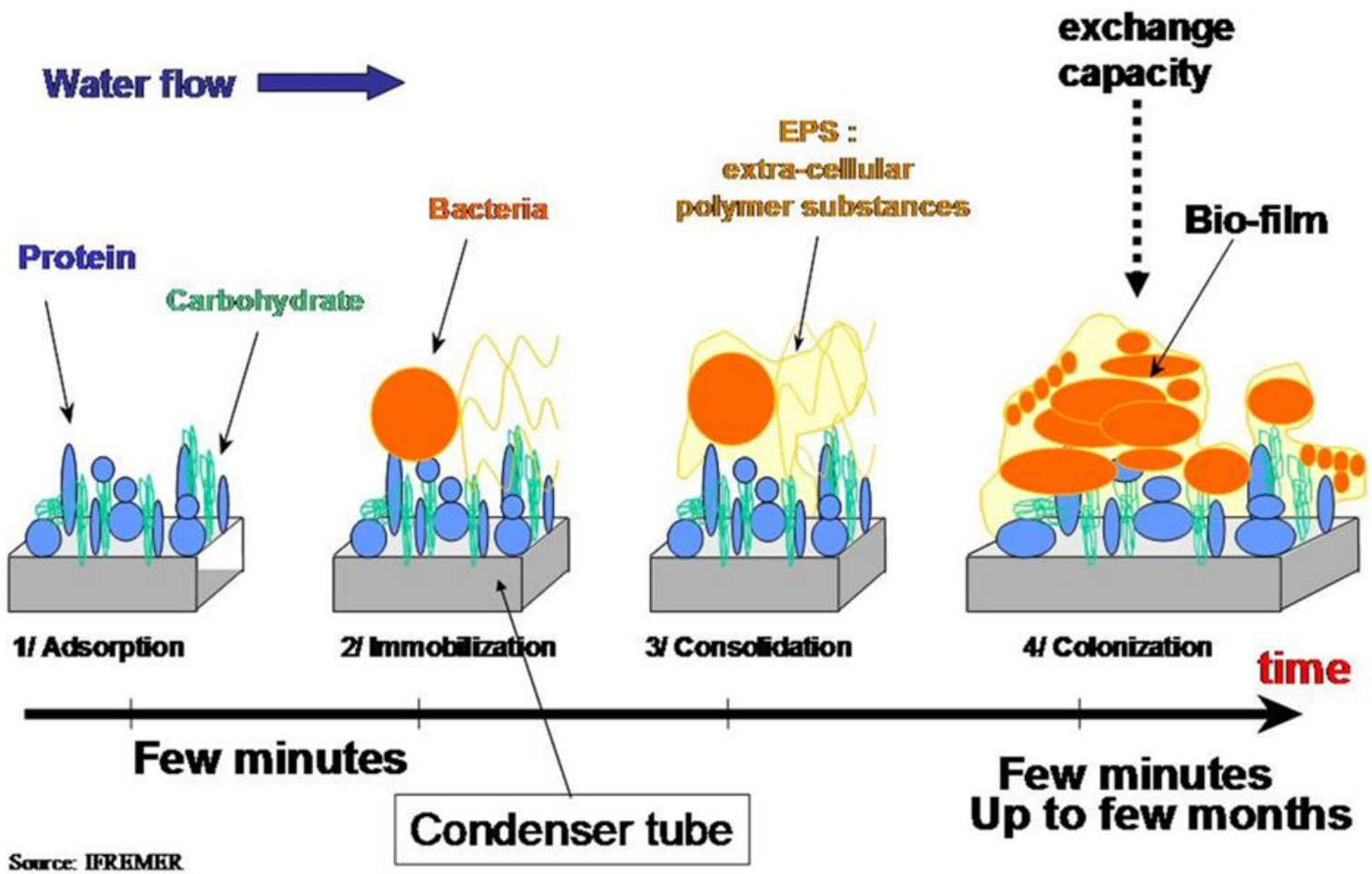
2. Initial Adhesion & Attachment of Bacteria

Phase 1	Transport to the Surface
Phase 2	Initial Adhesion
Phase 3	Strong Attachment
Phase 4	Colonization of the surface & biofilm formation

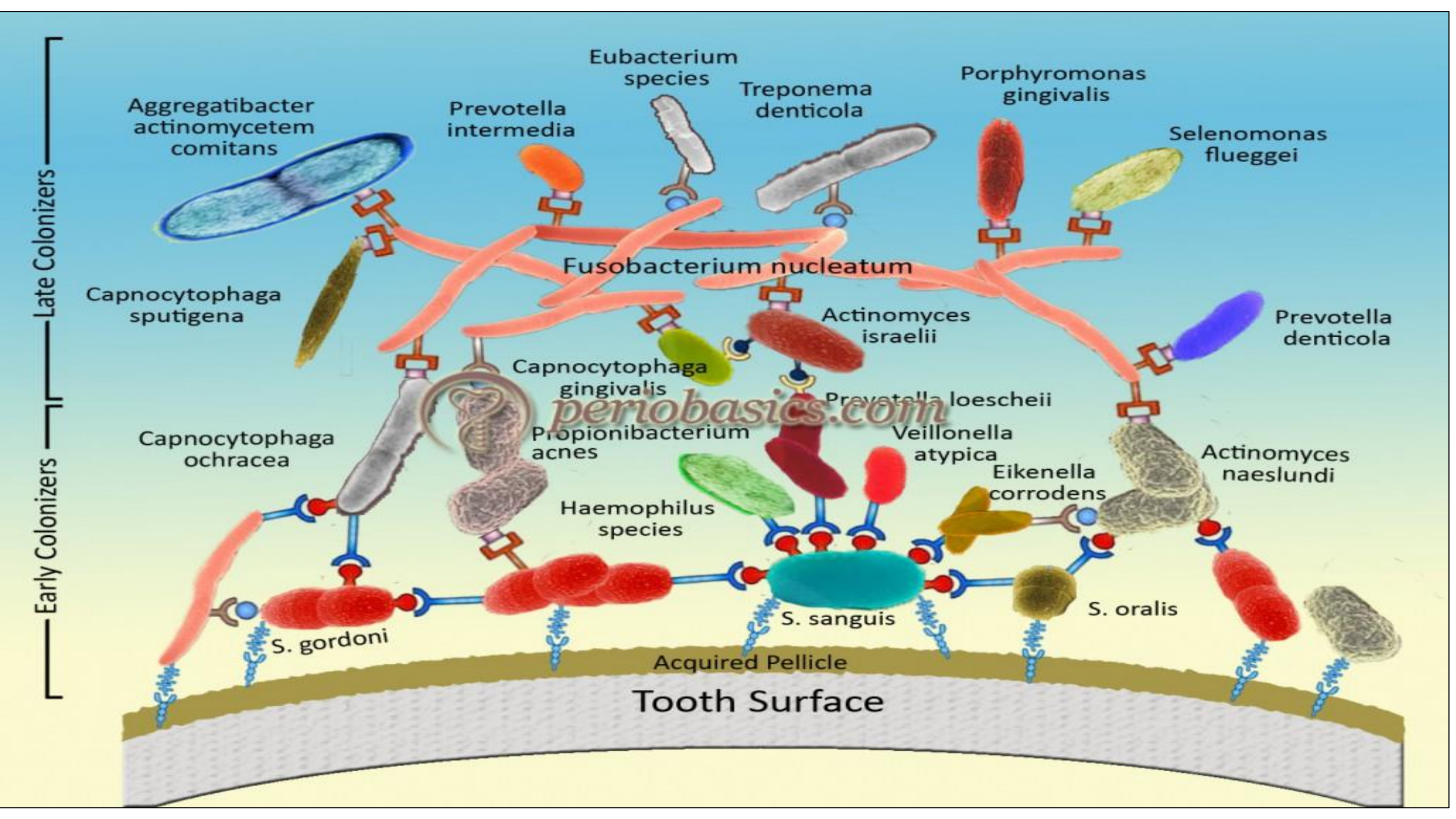
3. Colonization and Plaque Maturation

The primary colonizing bacteria adhered to the tooth surface provide new receptors for attachment by other bacteria as part of a process known as “coadhesion.”

The growth of adherent microorganisms, coadhesion leads to the development of microcolonies and eventually to a mature biofilm.



Biofilm formation


Late Colonizers
Early Colonizers
Aggregatibacter actinomycetem comitans
Prevotella intermedia
Eubacterium species
Treponema denticola
Porphyromonas gingivalis
Selenomonas flueggei
Fusobacterium nucleatum
Capnocytophaga sputigena
Actinomyces israelii
Prevotella denticola
Capnocytophaga gingivalis
Prevotella loescheii
perobasics.com
Capnocytophaga ochracea
Propionibacterium acnes
Veillonella atypica
Actinomyces naeslundi
Eikenella corrodens
Haemophilus species
S. gordoni
S. sanguis
S. oralis
Acquired Pellicle
Tooth Surface

Plaque hypothesis

- **<u>Nonspecific plaque hypothesis (Walter loesche, 1976) :</u>**

- Periodontal disease results from the “elaboration of noxious products by the entire plaque flora.”

- When small amounts of plaque are present, the noxious products are neutralized by the host. Similarly, large amounts of plaque would produce large amounts of noxious products, which would essentially overwhelm the host's defences.

- The current standard treatment of periodontitis by debridement (nonsurgical or surgical) and oral hygiene measures still focuses the nonspecific plaque hypothesis.

Specific plaque hypothesis

- Only certain plaque is pathogenic, its pathogenicity depends on the presence of specific microorganisms.
- Specific bacterial pathogens results in a periodontal disease because these organisms produce substances that mediate the destruction of host tissues.
- The association of Socranksy's “red complex” bacteria, *P. gingivalis*, *T. forsythia,* and *T. denticola* with periodontal disease was based on the analysis of 40 different bacteria in >13,000 plaque samples.

Ecologic Plaque Hypothesis

Both the total amount of dental plaque and the specific microbial composition of plaque may contribute to the transition from health to disease.

The host response may be brought about by the excessive accumulation of (nonspecific) dental plaque, or by plaque-independent host factors (e.g., the onset of an immune disorder, changes in hormonal balance such as in pregnancy), or environmental factors (e.g., smoking, diet).

(Marsh and co-workers ,1990s)

Biofilm Characteristics

Heterogeneous with open fluid channels.

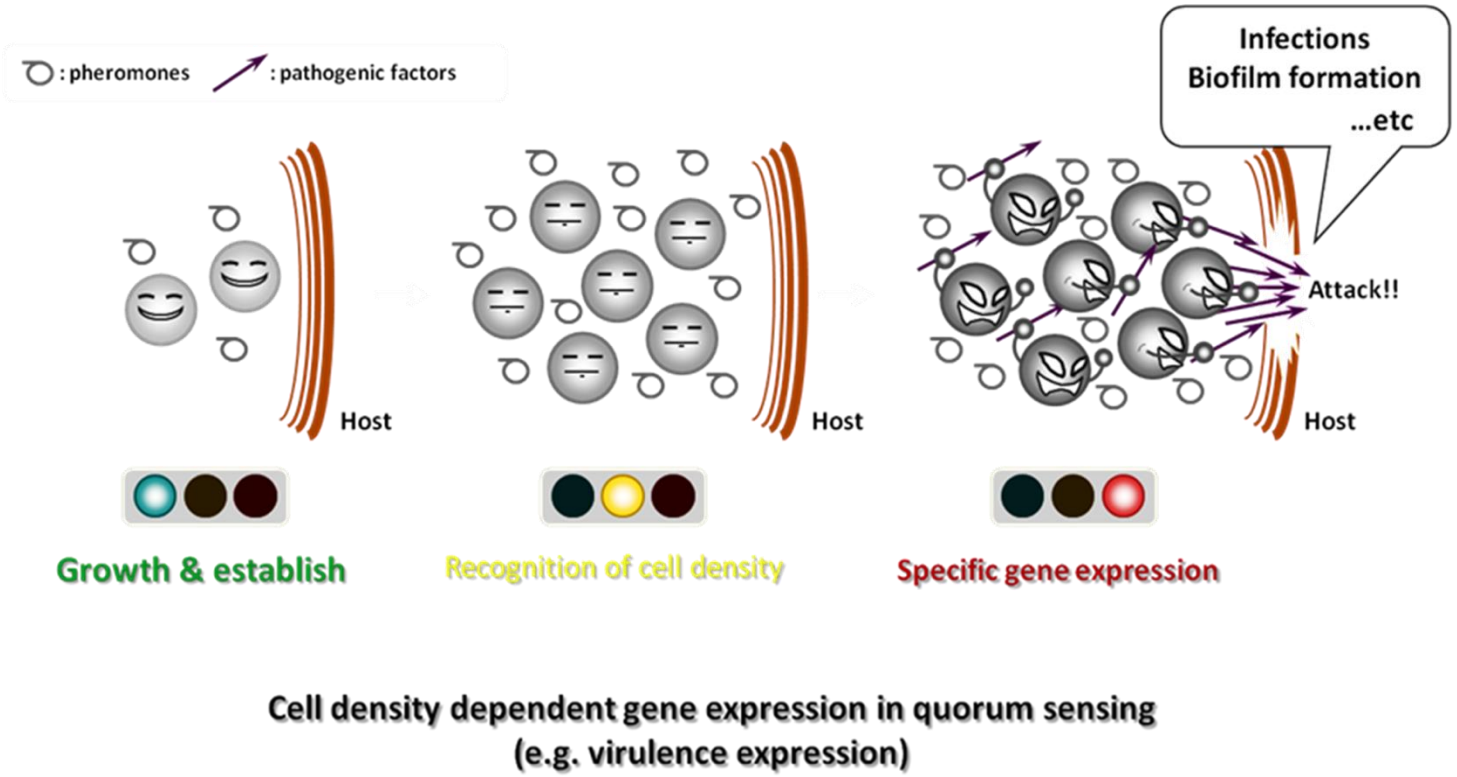
Bacteria are up to 1000 times more resistant to antimicrobial agents.

Oxygen and nutrient gradient.

Facilitates cell-cell signalling and deoxyribonucleic acid (DNA) exchange.

Communication Between Biofilm Bacteria

- **Quorum sensing**: signalling molecule that accumulates in the local environment and triggers a response such as a change in the expression of specific genes once they reach a critical threshold concentration.



(Fuqua *et al.*,1994)

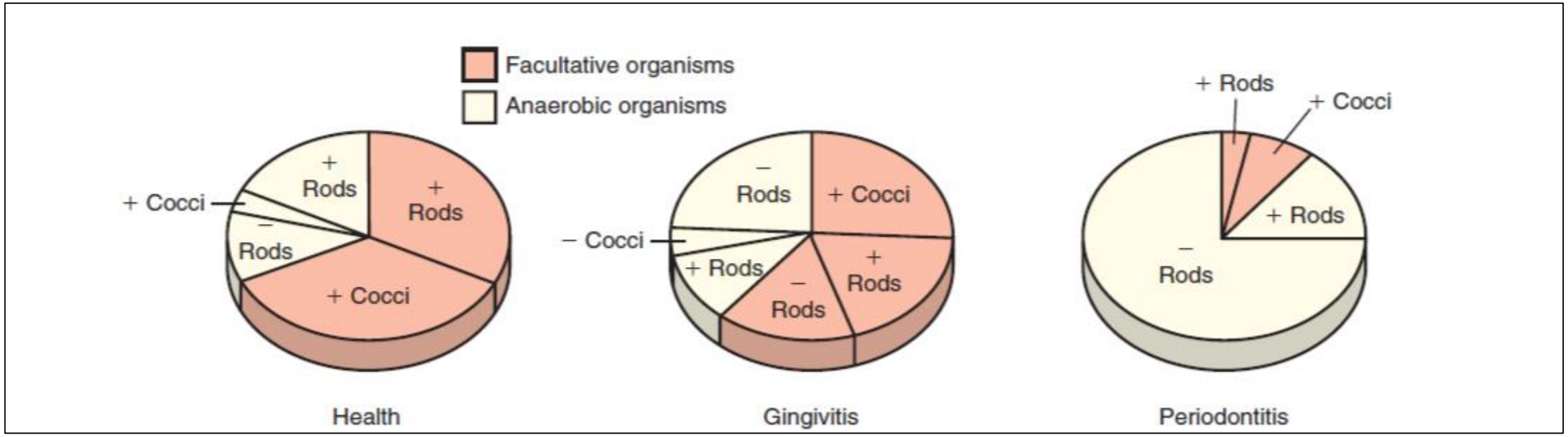
- There is some evidence that intercellular communication can occur after cell-cell contact and may not involve secreted signaling molecules.
- Two types of signaling molecules:

  a) Peptides released by gram-positive organisms during growth

  b) Universal signal molecule autoinducer 2 (AI-2)
- Play diverse roles in modulating the expression of genes for antibiotic resistance, encouraging the growth of beneficial species to the biofilm, and discouraging the growth of competitors.

Transition From Health to Disease

- Allogenic succession - change in the composition of a bacterial community as the result of external, nonmicrobial factors . eg. Smoking.
- Autogenic succession - change in the composition of a microbial community that arises from microbial activities or interbacterial interaction.



Culturing studies that represent the relative proportion of different morphotypes in subgingival samples in cases of periodontal health, gingivitis, and periodontitis

- Small amounts of bacterial plaque can be controlled by the body’s defense mechanisms without destruction.

- When dysbiosis happens (e.g.as a result of increased susceptibility, high bacterial load, or pathogenic infections), periodontal destruction might occur.

MICROBIAL SPECIFICITY OF PERIODONTAL DISEASES

- Gingivitis
- Periodontitis
- Localized Aggressive periodontitis
- Necrotizing periodontal diseases
- Abscesses of the periodontium
- Periimplantitis

	Bacterial spp.
Periodontal health	Streptococcus oralis, Streptococcus sanguis, Streptococcus mitis, Actinomyces naeslundii, Fusobacterium species, Prevotella nigrescens ,Veillonella species.
Gingivitis	Lactobacillus species, Actinomyces naeslundi, Streptococcus anginosus, Fusobacterium nucleatum, P. intermedia, Campylobacter species, Haemophilus species, Selenomonas Species, Treponema species.
Periodontitis	Eubacterium brachy, Eubacterium nodatum, Parvimonas micra, Peptostreptococcus stomatis, Tannerella forsythia, Fusobacterium nucleatun, Porphyromonas gingivalis, Prevotella intermedia, Campylobacter rectus, Treponema species.
Localised Aggressive periodontitis	Aggregatibacter actinomycetemcomitans, Porphyromonas gingivalis, Tannerella forsythia, Prevotella intermedia, Prevotella nigrescens, Eikenella corrodens, Selenomonas sputigena, F. nucleatum, Campylobacter rectus, Peptostreptococcus micros.
Necrotizing periodontal diseases	Fusobacterium nucleatum, P. intermedia, Spirochetes Treponema species
Abscesses of the periodontium	Fusobacterium nucleatum, Parvimonas micra, P. intermedia, Spirochetes Porphyromonas gingivalis
Periimplantitis	Capnocytophaga spp, Actinomycetemcomitans, Campyiobacter rectus, Fusobacterium nucleatum, Parvimonas micra, Prevotella intermedia, Enterobacteriaceae spp, P. Gingivalis, T. forsythia, Treponema denticola, Pseudomonas aeruginosa,Staphylococcus aureus, Staphylococcus epidermidis

Gingivitis

- Development of gingivitis has been experimental study by Harald Löe *and co-workers* :

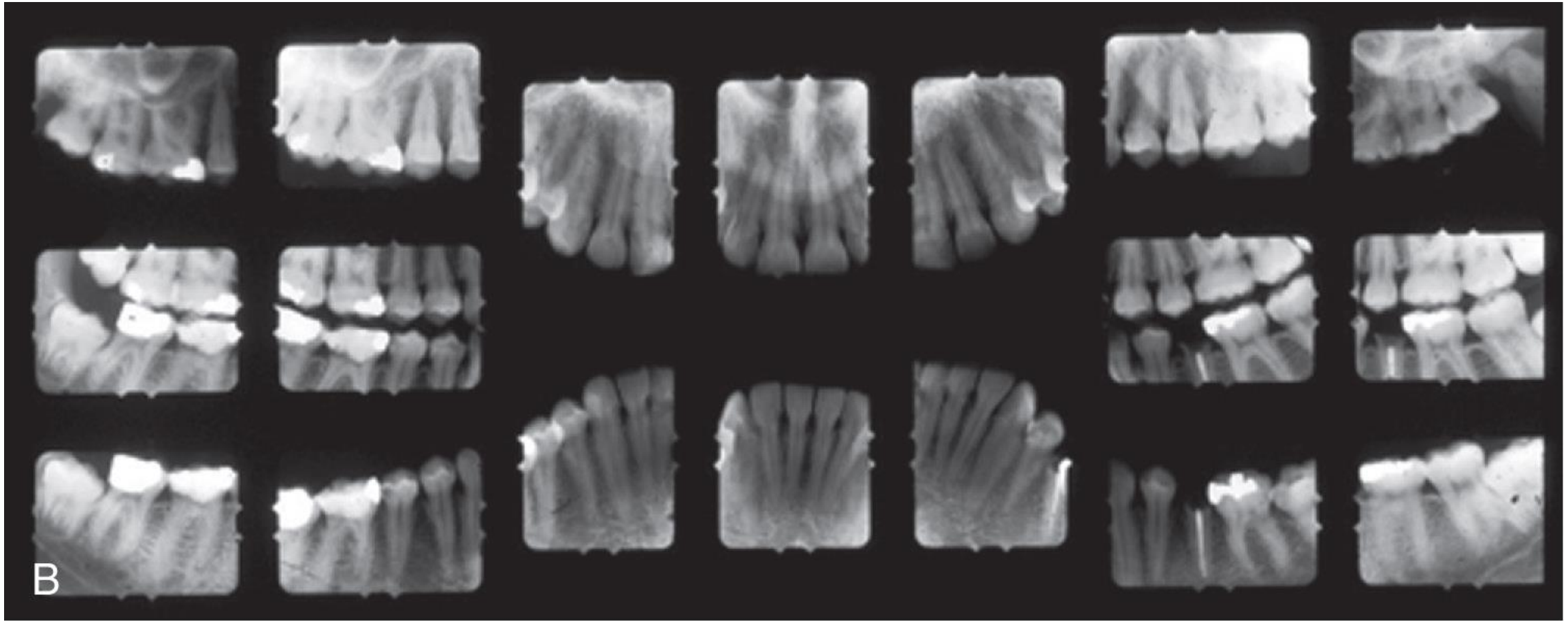
  - Periodontal health is first established in human subjects via cleaning and rigorous oral hygiene measures that are followed by abstinence from oral hygiene for 21 days.
  - After 8 hours without oral hygiene, bacteria found at concentrations of 10^3 to 10^4 per mm^2 of tooth surface, and they will increase in number by a factor of 100 to 1000 during next 24-hour period.
  - After 36 hours, plaque becomes clinically visible .

- The initial microbiota of experimental gingivitis consists of gram-positive rods, gram-positive cocci, and gram-negative cocci.

- The subgingival microbiota of dental-plaque–induced gingivitis (chronic gingivitis) .

Plaque-related gingivitis and marginal and papillary inflammation

1- to 4-mm probing depths and generalized zero clinical attachment loss


B

Gram-positive species –

-Streptococcus spp. (S. sanguinis, S. mitis, S. intermedius, S. oralis)
-Actinomyces spp. (A. oris, A. naeslundii)
-Eubacterium nodatum

Gram-negative species –

-Capnocytophaga spp. , Fusobacterium spp. , Prevotella spp.

Associated with periodontitis but in fewer numbers in gingivitis:

-P. gingivalis, P. intermedia, T. forsythia, Campylobacter rectus, Treponema spp., A. actinomycetemcomitans.

Gingivitis support the conclusion that disease development is associated with selected alterations in the microbial composition of dental plaque and not just the result of an accumulation of plaque.

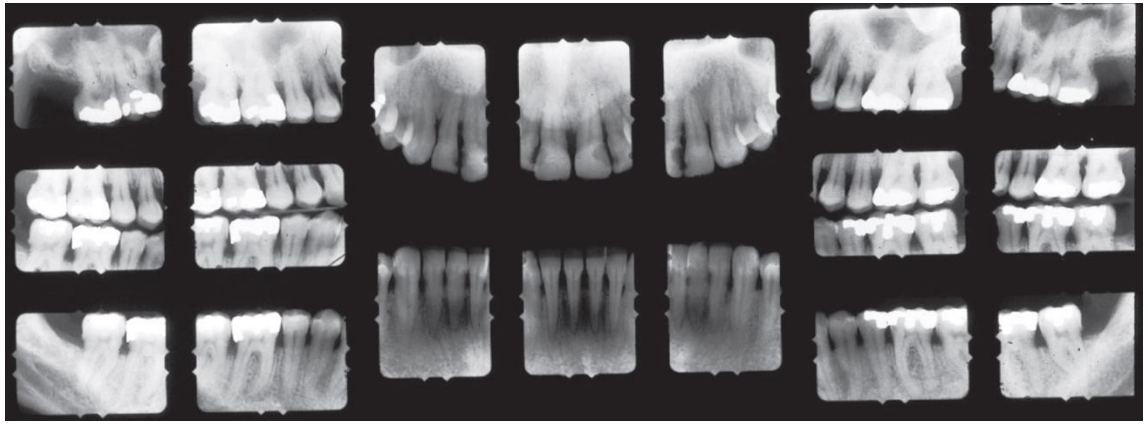
Chronic Periodontitis

Disease progression at mean rates 0.05 mm to 0.3 mm of attachment loss per year.

Microscopic examination of plaque from sites with chronic periodontitis has revealed elevated proportions of spirochetes.

The cultivation of plaque microorganisms from sites of chronic periodontitis reveals high percentages of anaerobic (90%) gram-negative (75%) bacteria.

Clinical image of plaque-related moderate chronic periodontitis



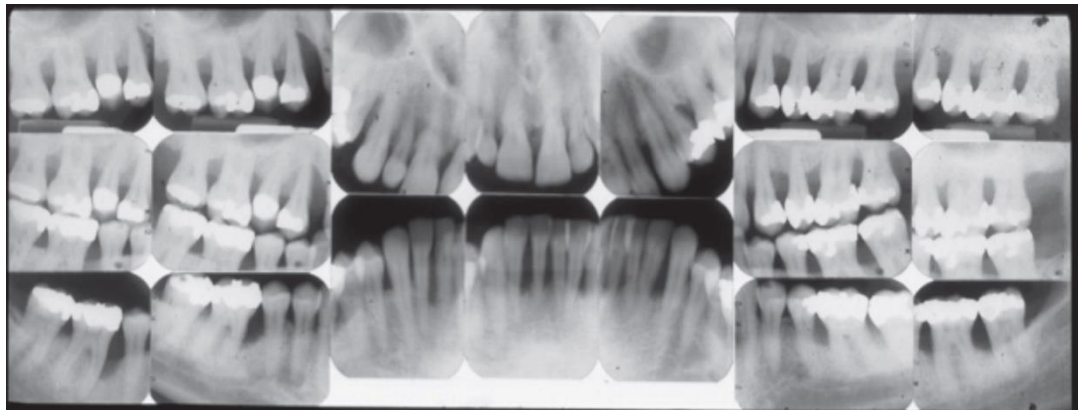
3 to 4 mm of clinical attachment loss

- High levels of -

P. gingivalis, T. forsythia,P. intermedia, P. nigrescens, F. nucleatum, A. actinomycetemcomitans , Treponema denticola, Selenomonas spp. and Enterobacter spp.

Clinical image of plaque-related severe advanced chronic periodontitis

- P. gingivalis and A. actinomycetemcomitans are known to invade host tissue cells, which may be significant in aggressive forms of periodontitis.



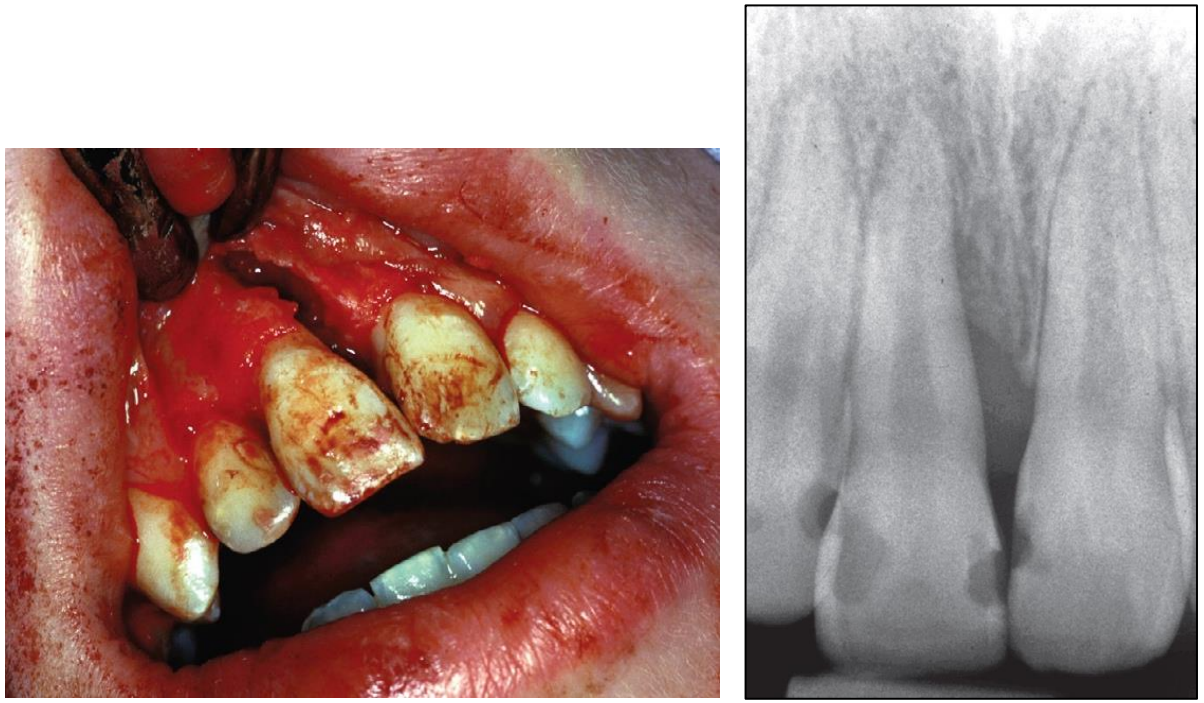
>5 mm of clinical attachment loss

Localized Aggressive Periodontitis

Characterized by rapid and severe attachment loss that occurs in individuals during or before puberty.

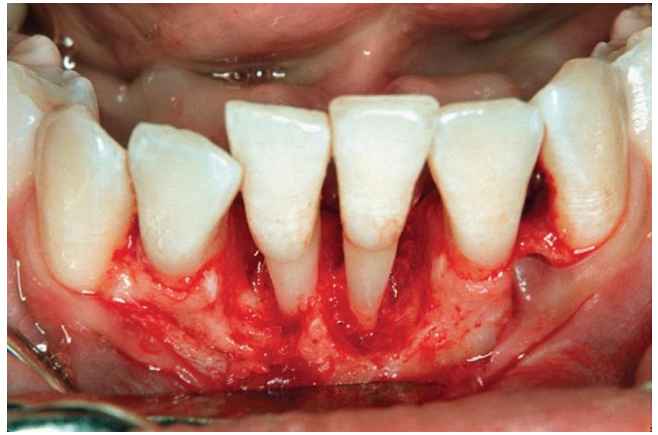
Previously known as localized juvenile periodontitis (LJP).

It is observed in females and it typically affects the permanent first molars and incisors



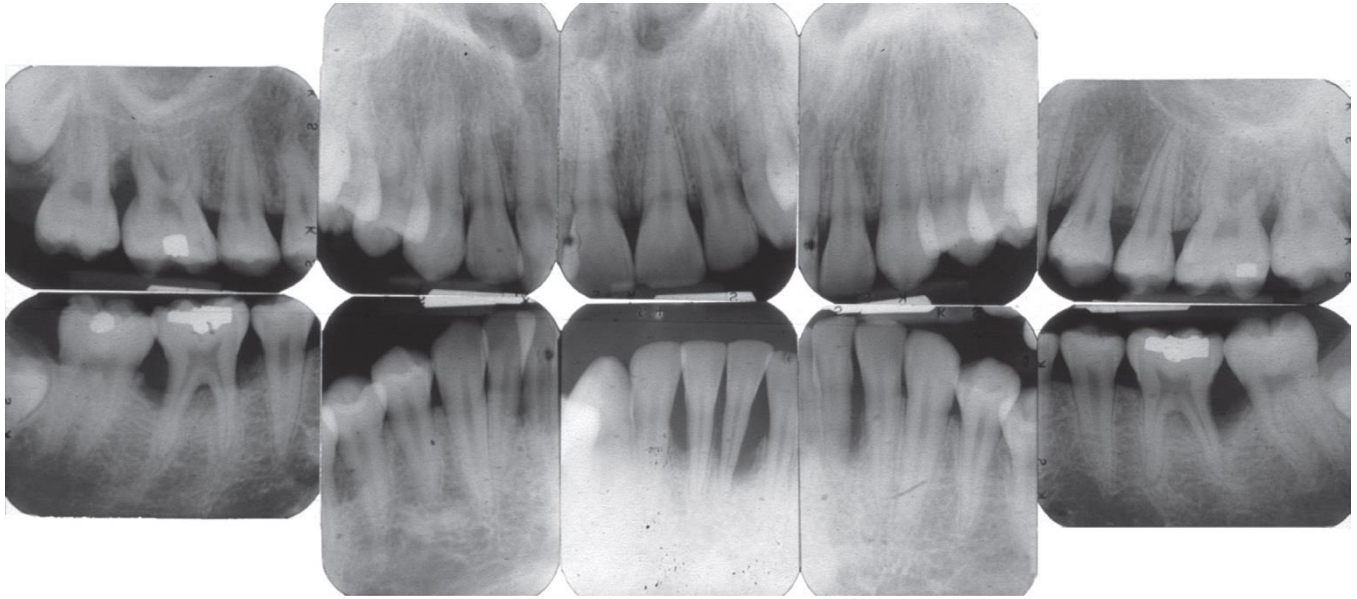
Clinical picture and intraoral radiograph of bone destruction in an adolescent patient with localized aggressive periodontitis.

Associated microbiota is predominantly composed of gram-negative, capnophilic (i.e., CO2-requiring), anaerobic rods.



Localized aggressive periodontitis in 15-year-old female patient

Microbiological studies indicate that
A.
actinomycetemcomitans, which is 90% of total cultivable microbiota.



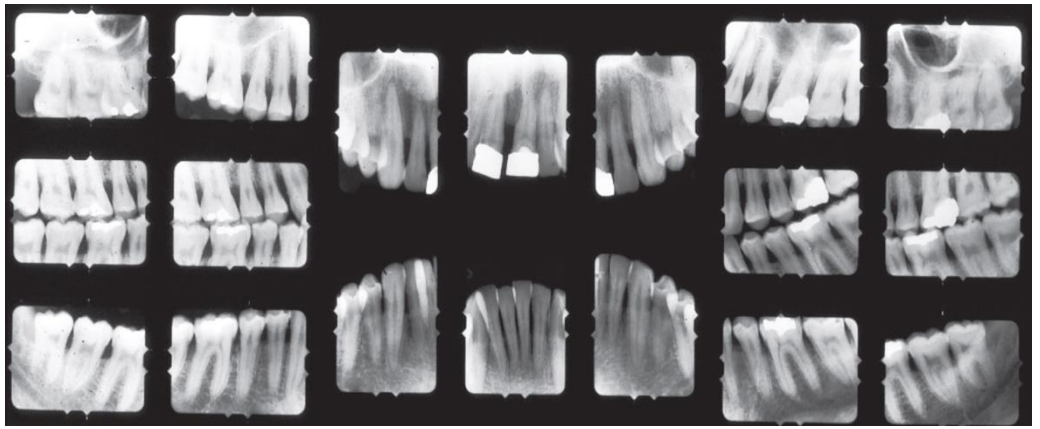
localized, vertical, angular bony defects affecting the mandibular incisors

Aggressive Periodontitis

(Generalized aggressive periodontitis, can have an early-onset periodontitis, or later in life and be rapidly progressive periodontitis)

- Can occurs at young age (20 to 40 yrs).
- Characterized by a severe gingivitis, a large number of deepened pockets, and a high tendency for bleeding on probing.
- Dominated by P. gingivalis, P. intermedia, T. forsythia, and Treponema spp. (T. denticola) and less by A. actinomycetemcomitans.

Clinical image of plaque-related aggressive periodontitis



1 to 7 mm of PD and 3 to 4 mm of clinical attachment loss .

Necrotizing Periodontal Diseases

Characterized by the necrosis of the marginal gingival tissue and the interdental papillae.

Presence of malodor and pain.

High levels of P. intermedia and especially of spirochetes and F. nucleatum.

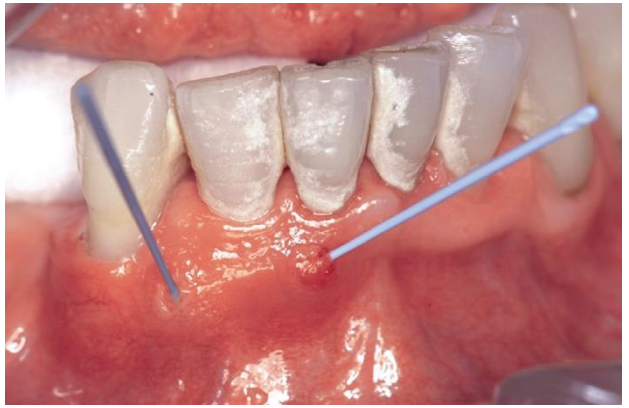
Clinical picture of lower front teeth with necrotizing gingivitis

Abscesses of the Periodontium

May also occur in the absence of periodontitis; may be associated with the impaction of a foreign object (e.g., a popcorn kernel, dental floss) or with endodontic problems.

Clinical symptoms - pain, swelling, suppuration, bleeding on probing, and mobility of the involved tooth.

microorganisms - F. nucleatum, P. intermedia, P. gingivalis, P. micra, and T. forsythia.



Clinical pictures of a patient with several periodontal abscesses. An intraoral radiograph showing the severity of the periodontal destruction. Gutta-percha points are inserted in the fistulae to show their course.

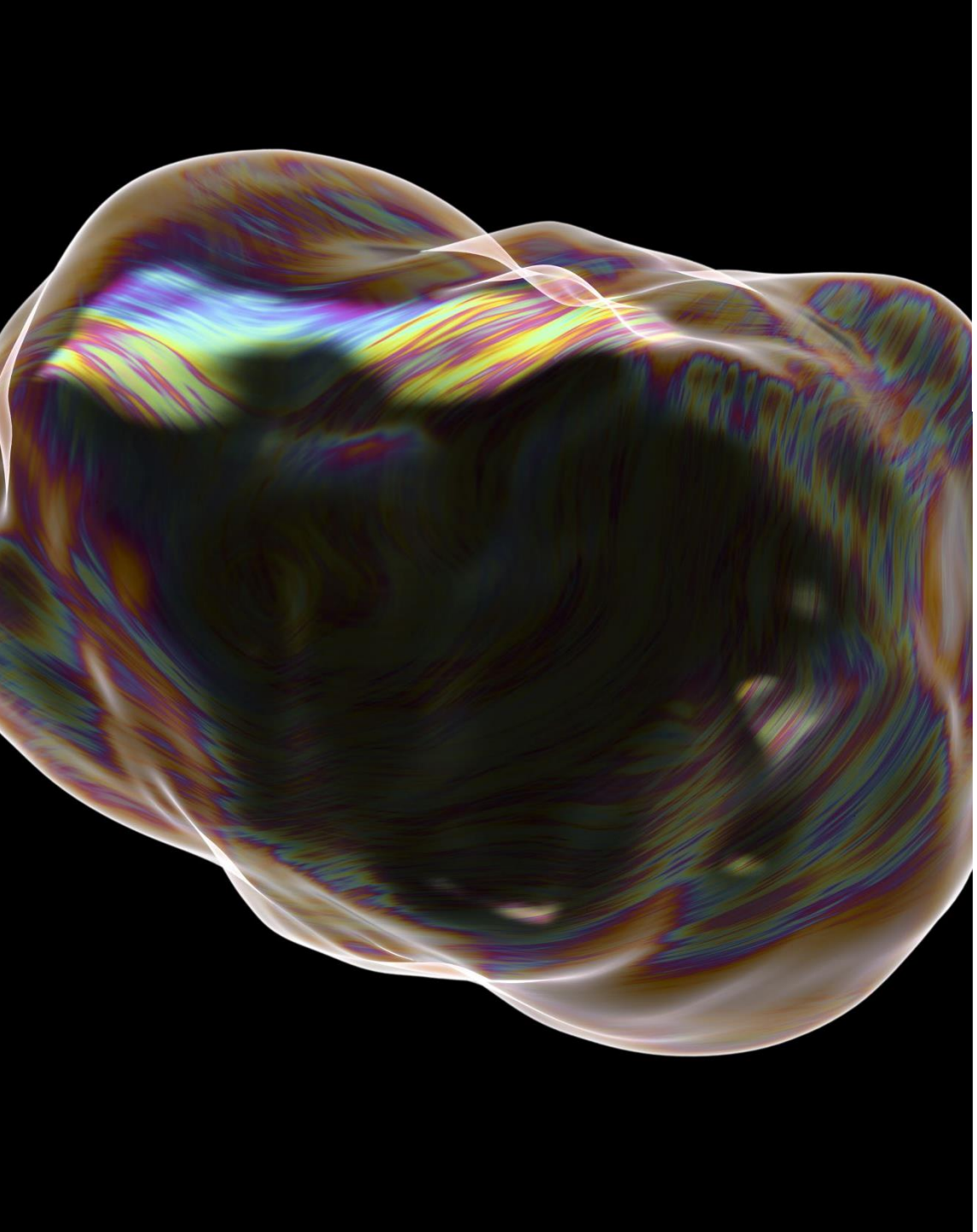
Periimplantitis

- Term refers to an “inflammatory process” - affects the tissues around an already osseointegrated implant and that results in the loss of supporting bone.

- Characterized by high proportions of A. actinomycetemcomitans,

P. gingivalis, T. forsythia, P. micra, C. rectus, F. nucleatum, P. intermedia, T. denticola, and Capnocytophaga.

Periimplantitis



Conclusion

- Gingivitis and periodontitis are caused by bacteria that colonize the gingival crevice and attach to the tooth surface.
- Plaque is highly organised but heterogeneous in structures
- Plaque is the main causative factor in causing periodontal diseases
- Removal of plaque and maintaining healthy oral environment is prerequisite for optimum health.

REFERENCES

- Clinical Periodontology- Newman, Takei, Klokkevold , Carranza 10th edition.
- Clinical periodontology and implant dentistry - Jan Lindhe, Niklaus P.Lang, Thorkild Karring 5th edition.
- Loe H, Theilade E, Jensen SB: Experimental gingivitis in man. J Periodontol 36:177, 1965.
- Theilade E, Wright WH, Jensen SB, et al: Experimental gingivitis in man. II. A longitudinal clinical and bacteriological investigation. J Periodontal Res 1:1, 1966.
- Brown LJ, Loe H: Prevalence, extent, severity and progression of periodontal disease. Periodontol 2000 2:57, 1993.
- Listgarten MA, Hellden L: Relative distribution of bacteria at clinically healthy and periodontally diseased sites in humans. J Clin Periodontol 5:115, 1978.

friends
OF HU-FRIEDY

Curettes

Clinical Application Guide

Hu-Friedy

Curettes

Hu-Friedy

Types of Curettes

A curette is a type of scaler that is designed for moderate calculus removal on supragingival and subgingival surfaces.

- Types of Curettes:
  - Universal
  - Area Specific (Gracey)
- All curettes share the same common elements:
  - Rounded back
  - Rounded toe
  - Semi-circular cross section

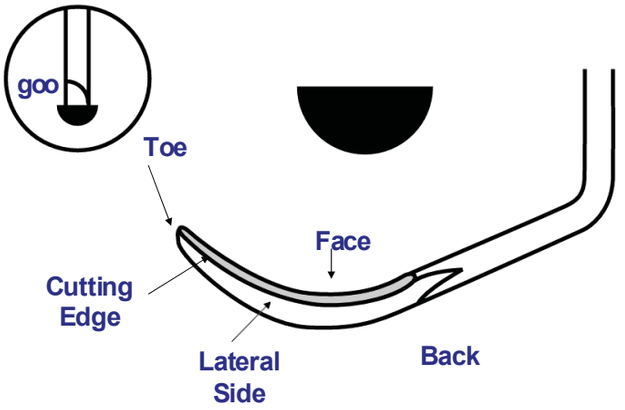
Hu-Friedy

Universal Curettes

- Universal curettes are designed for easy adaptation on all tooth surfaces (thus the name "universal").

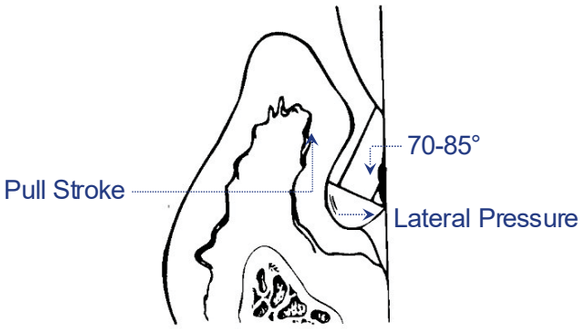
Universal Curettes

The blade of a universal curette has a round toe and back, and two cutting edges for scaling, making it an efficient design for scaling the entire mouth.



Universal Instrumentation

- The blade is correctly adapted when the toe is directed interproximally and the terminal shank is parallel to the tooth.
- To remove deposits, the cutting edge is applied to the tooth surface and the facial surface of the blade is tilted toward the tooth to achieve a 70-85° angle between the tooth and the blade.
- Apply lateral pressure against the tooth and pull upward while maintaining contact with the tooth.



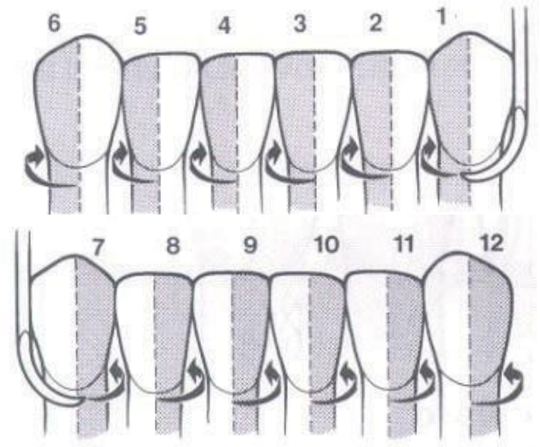
Anterior Universal

- To scale the facial surfaces, place the toe of the blade toward the proximal surface with the handle parallel to the tooth.
- Apply strokes to remove deposits from the midline of the tooth to the proximal surface.
- Work from canine to canine.
- Switch working ends and repeat for surfaces away from you.
- Repeat all of the above for the lingual surfaces.

SBH5/67

Universal Curettes

Anterior Universal Instrumentation Sequence

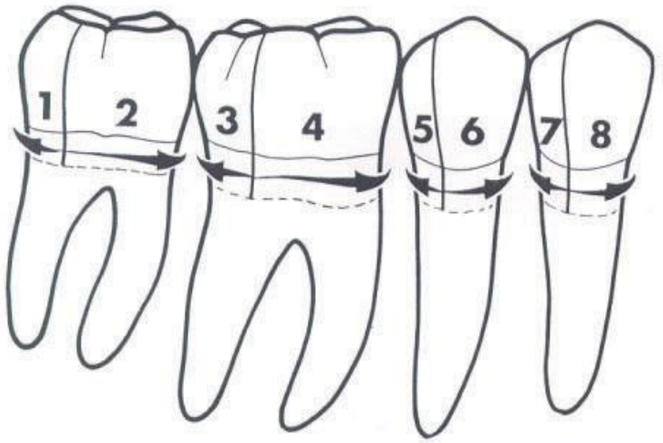


Posterior Universal

- Begin at the distal line angle of the most posterior tooth.
- Direct the toe of the blade toward the distal with the terminal shank angled slightly toward the tooth.
- Apply strokes from the line angle to the contact area.
- Next, turn the toe toward the mesial to scale the buccal and mesial surfaces.
- Continue this sequence to complete the posterior region.
- Switch ends and repeat from the lingual aspect.

Posterior Universal

Posterior Universal Instrumentation Sequence



SC4R/4L6

Area Specific (Gracey) Curettes

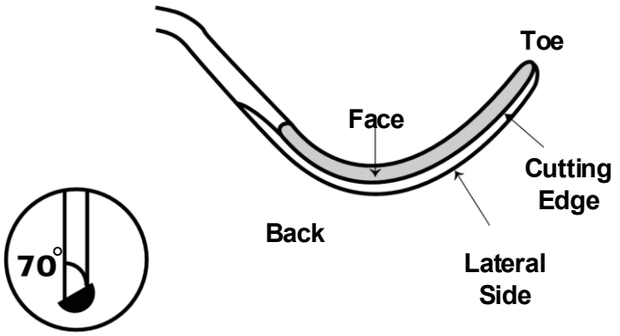
Gracey Curettes

- In the early 1940's, Dr. Clayton Gracey and Hu-Friedy introduced a set of instruments designed to be used on specific tooth surfaces that improve adaptation and deposit removal.
- The Gracey "area specific" designs have laid the groundwork for the new instruments of today.
- Today, the Gracey Curette family of instruments give clinicians many options for their treatment needs.

Hu-Friedy

Gracey Curettes

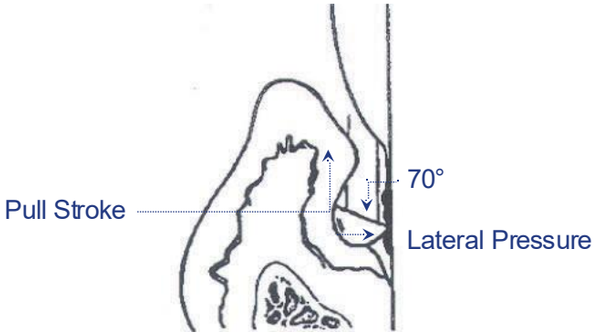
- The Gracey blade design is offset from the terminal shank at 70°.
- This creates one cutting edge which is referred to as the lower edge.



Hu-Friedy

Gracey Instrumentation

- The blade of a Gracey Curette is correctly adapted when the lower cutting edge is against the tooth and the terminal shank is parallel to the tooth surface being scaled.
- Apply lateral pressure against the tooth (root) and pull upward, maintaining the parallel shank.



Hu-Friedy

Gracey Curettes

The shank design is different for each Gracey pattern.

Hu-Friedy

Gracey Curettes

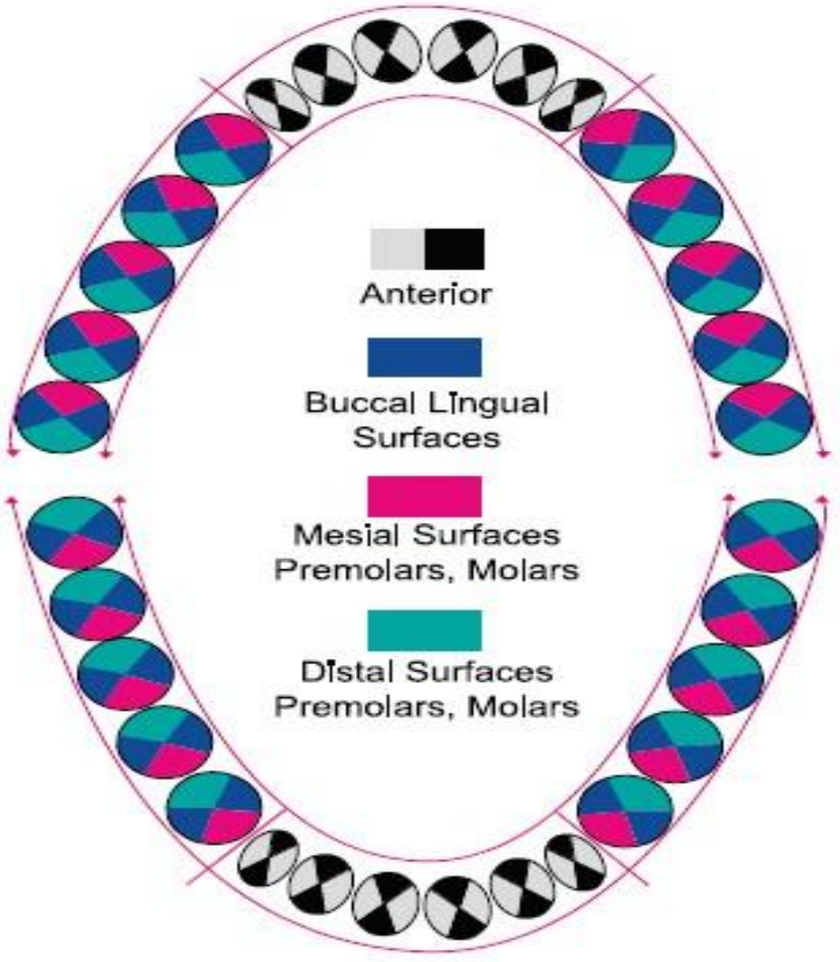
Gracey Curettes are available in:

- Standard
- Rigid
- After Five
- Mini Five

The following slides will explain each of these in more detail.

Standard Gracey Curettes

- Gracey Curettes are area specific to allow for deep scaling, root planing and periodontal debridement.
- The offset blade provides a perfect working angulation for the tooth surface.
- Gracey Curettes are used in a set to completely scale the dentition.



Gracey Chart

Gracey Design Comparisons

	Shank Design and Diameter	Blade Length	Blade Width	Available Patterns & Areas of Use (Chart on next slide)
Standard (Finishing) SG#/#	Standard	Standard	Standard	1/2, 3/4, 5/6, 7/8, 9/10, 11/12, 15/16, 13/14, 17/18*
Rigid SGR#/#R Rigid	Standard design, increased shank diameter	Standard	Standard	1/2, 3/4, 5/6, 7/8, 9/10, 11/12, 15/16, 13/14, 17/18*
After Five SRPG#/#	Longer terminal shank, standard diameter	Standard	Decreased by 10%	1/2, 3/4, 5/6, 7/8, 11/12, 15/16, 13/14
Rigid After Five SRPG#/#R	Longer terminal shank, increased diameter	Standard	Decreased by 10%	1/2, 3/4, 5/6, 7/8, 11/12, 15/16, 13/14
Mini Five SAS#/#	Longer terminal shank, standard diameter	Decreased by 50%	Decreased by 10%	1/2, 3/4, 5/6, 7/8, 11/12, 15/16, 13/14
Rigid Mini Five SAS#/#R	Longer terminal shank, increased diameter	Decreased by 50%	Decreased by 10%	1/2, 3/4, 5/6, 7/8, 11/12, 15/16, 13/14

* The 17/18 is a unique pattern, having a longer terminal shank and slightly shorter blade, so no AF or MF are available.

Rigid Gracey Curettes

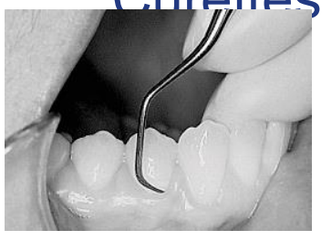
Standard Rigid

All Gracey designs are available with a wider taper - rigid shank.

The rigid shank may be preferred for heavier calculus removal.

Although the shank is wider, the blade width is the same as the standard Gracey.

Standard Gracey Curettes



7/8

11/12

15/16

17/18

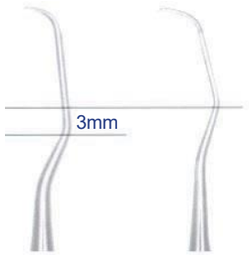
Rigid Gracey Curettes

- All Gracey designs are available with a wider taper - rigid shank.
- The rigid shank may be preferred for heavier calculus removal.
- Although the shank is wider, the blade width is the same as the standard Gracey.

Standard Rigid

After Five® Gracey Curettes

- Designed for instrumentation in deeper periodontal pockets.
- The terminal shank is elongated 3mm to provide better clearance around crowns, and superior access to root contours and pockets 5mm or more in depth.
- Blade thinned by 10% to ease gingival insertion and reduce tissue distention.



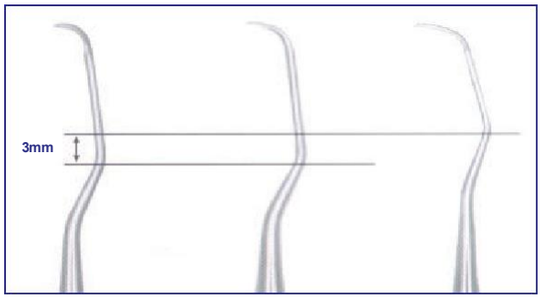
Gracey After Five Curettes

SRPG11/12

The change in the location of the shank bend permits deeper insertion into periodontal pockets. The thinned blade allows for easier insertion.

Mini Five® Curettes

- Designed with the same elongated terminal shank and thinned blades as the After Five Gracey Curettes.
- 50% shorter blade for access to smaller roots, narrow pockets, furcations, and developmental grooves.



Mini Five After Five Standard

Gracey Mini Five Curettes

SAS1/2

Mini Five reaching into a deep pocket on a narrow root.

Hu-Friedy Curettes

- Hu-Friedy offers an array of curettes.
- Many designs are available with different handle options and in Hu-Friedy's EverEdge Technology.
- Please refer to the Hu-Friedy catalog at www.hu-friedy.com for more information on all the curettes available.

Qualifying Question

- Thank you for taking the time to learn more about the clinical applications of curettes.
- To earn your Hu-Points towards a product evaluation, please exit and answer the following question on the Member Resources Page:

The After Five Gracey Curette has a terminal shank ____ longer and a blade thinned_____ than a Standard Gracey Curette.

c) 3mm, 10%
d) 3mm, 15%
e) 4mm, 10%
f) 4mm, 15%
g) 5mm, 15%

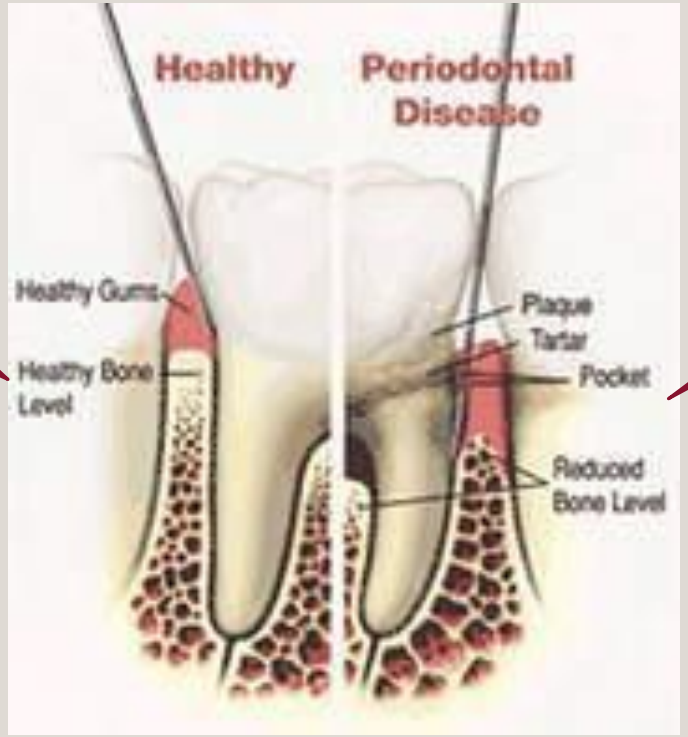
INSTRUMENTS UTILIZED IN PERIODONTAL THERAPY

COMPILED BY: VICKIE E. JONES, MDH, BSDH, RDH

TAKE A LOOK CLINICALLY-
NO CAVITIES THAT'S A GOOD THING, BUT HEALTHY TEETH NEED ???

A HEALTHY PERIODONTIUM

The periodontium is in a constant dynamic process...



....between health and disease

A BEAUTIFUL SMILE BEGINS WITH HEALTH ....***GUMS INCLUDED***

Periodontium is designed to support teeth.

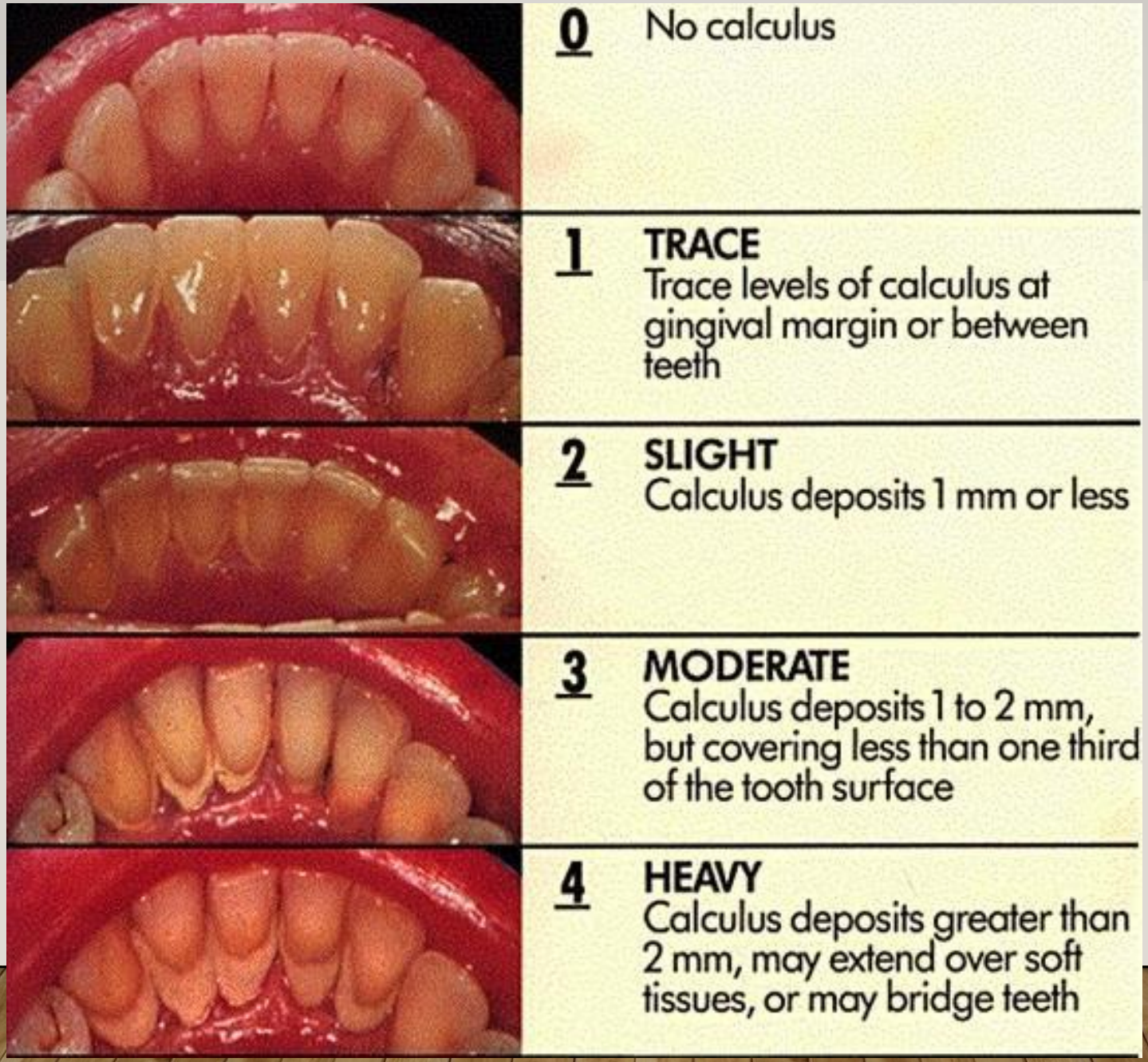
- If bacteria invade the periodontium.....
- There is a subclinical phase that we can't see...
- And if left undiagnosed and untreated...

INFLAMMATION AND BLEEDING GUMS OCCUR ---- KNOWN AS -***GINGIVITIS*** –

EARLY PERIODONTITIS
NOTICEABLY RECEDING GUMS AND POCKETS BETWEEN GUMS AND TEETH.

ADVANCED PERIODONTITIS DISEASE

TISSUE AND BONE THAT SUPPORT TEETH ARE DAMAGED CAUSING LOOSE TEETH. ULTIMATELY TOOTH LOSS

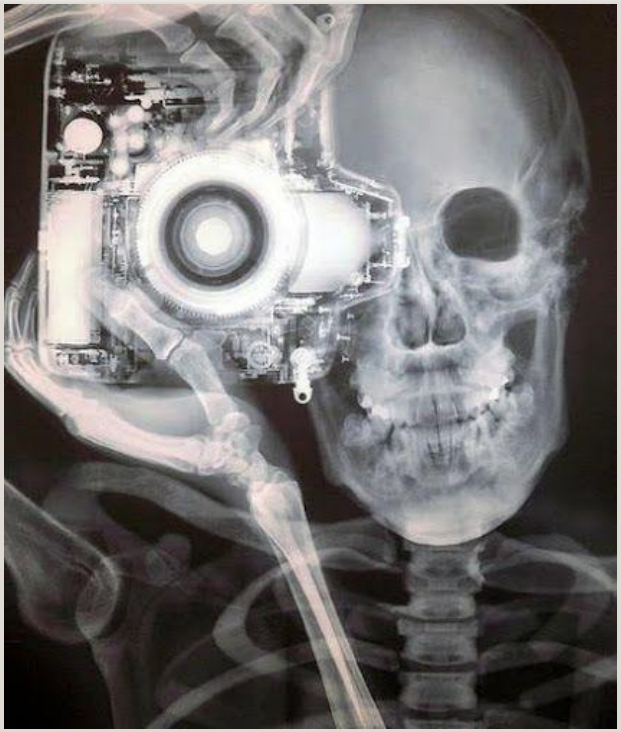

0 No calculus
1 TRACE
Trace levels of calculus at gingival margin or between teeth
2 SLIGHT
Calculus deposits 1 mm or less
3 MODERATE
Calculus deposits 1 to 2 mm, but covering less than one third of the tooth surface
4 HEAVY
Calculus deposits greater than 2 mm, may extend over soft tissues, or may bridge teeth

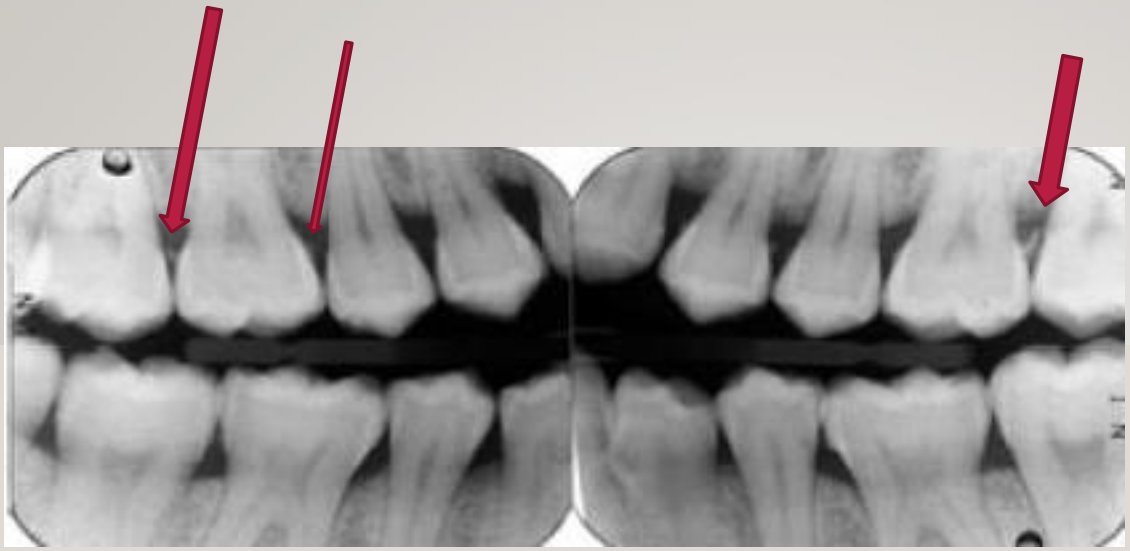
WHAT IS OUR PATIENTS' CURRENT PERIODONTAL STATUS…

- How do we know if our patients are healthy or what level of disease they have???
- Is calculus there???
  - If so, how much and where is it??
- How much attachment loss??

ROOT MORPHOLOGY IS COMPLICATED. ALL AREAS NEED ASSESSING.....HOW IS THIS POSSIBLE?

RADIOGRAPHS….ARE ONE PIECE OF THE PUZZLE …



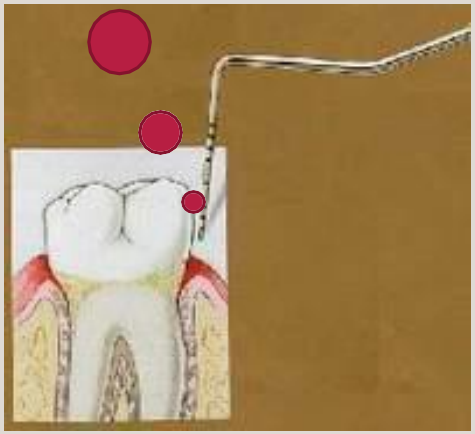


However, Radiographs are two-dimensional images of three-dimensional objects.

We need them but they only tell one part of the story…

INSTRUMENTS USED IN THE EXAMINATION OF THE PERIODONTIUM

Measures sulcus depths & Clinical Attachment Loss (CAL)



Williams Probe – Markings 1,2,3,5,7,8,9,& 10

Calculus detection

ODU -11/12 OLD DOMINION UNIVERSITY EXPLORER, (EXD - 11/12 EXPLORER) #6 SATIN STEEL® HANDLE EXD11/126

Furcation involvement

N COLOR-CODED NABERS PROBE PQ2N6

THERE ARE VARIOUS TYPES OF PROBES....

#5 HD CONE SOCKET MIRROR DOUBLE-SIDED

MIR5DSHD

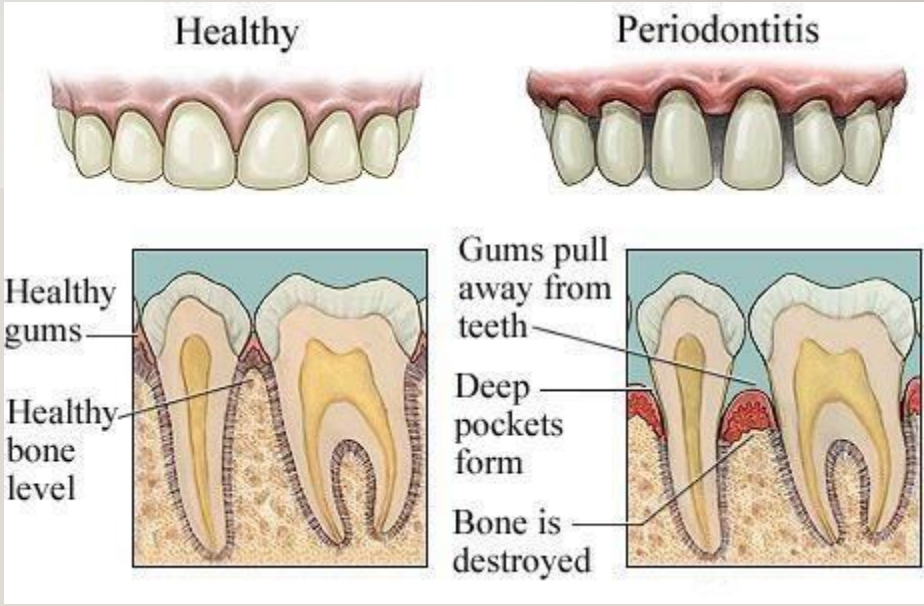
4 FRONT SURFACE MOUTH MIRROR

MIR4/3

EXD 11/12 EXPLORER EXAM INSTRUMENT

- Calculus detection
- Tip bent at 90 degree angle to lower shank
- Long, complex shank design

- http://cdcaexams.org/documents/media/Posterior_Explorer.mp4



What is the current phase of health or disease in your patient's mouth?

Bleeding upon probing is a sign of infection

https://www.cdcaexams.org/documents/media/Probe.mp4

THE EFFECT OF SMOKING ON GINGIVITIS

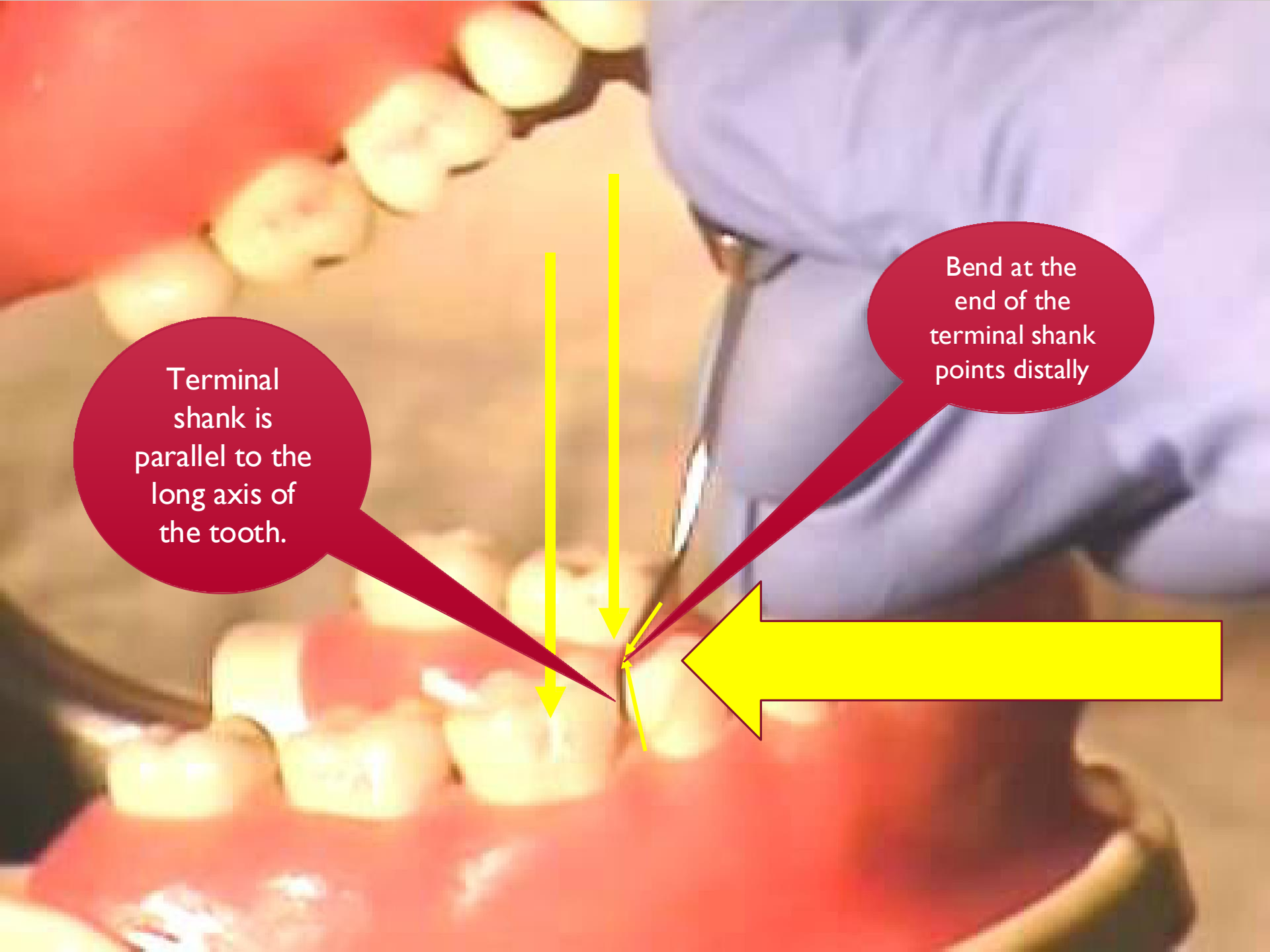
- Reduction of the gingival inflammation
- Reduction of bleeding on probing due to vasoconstrictive (effect of nicotine)

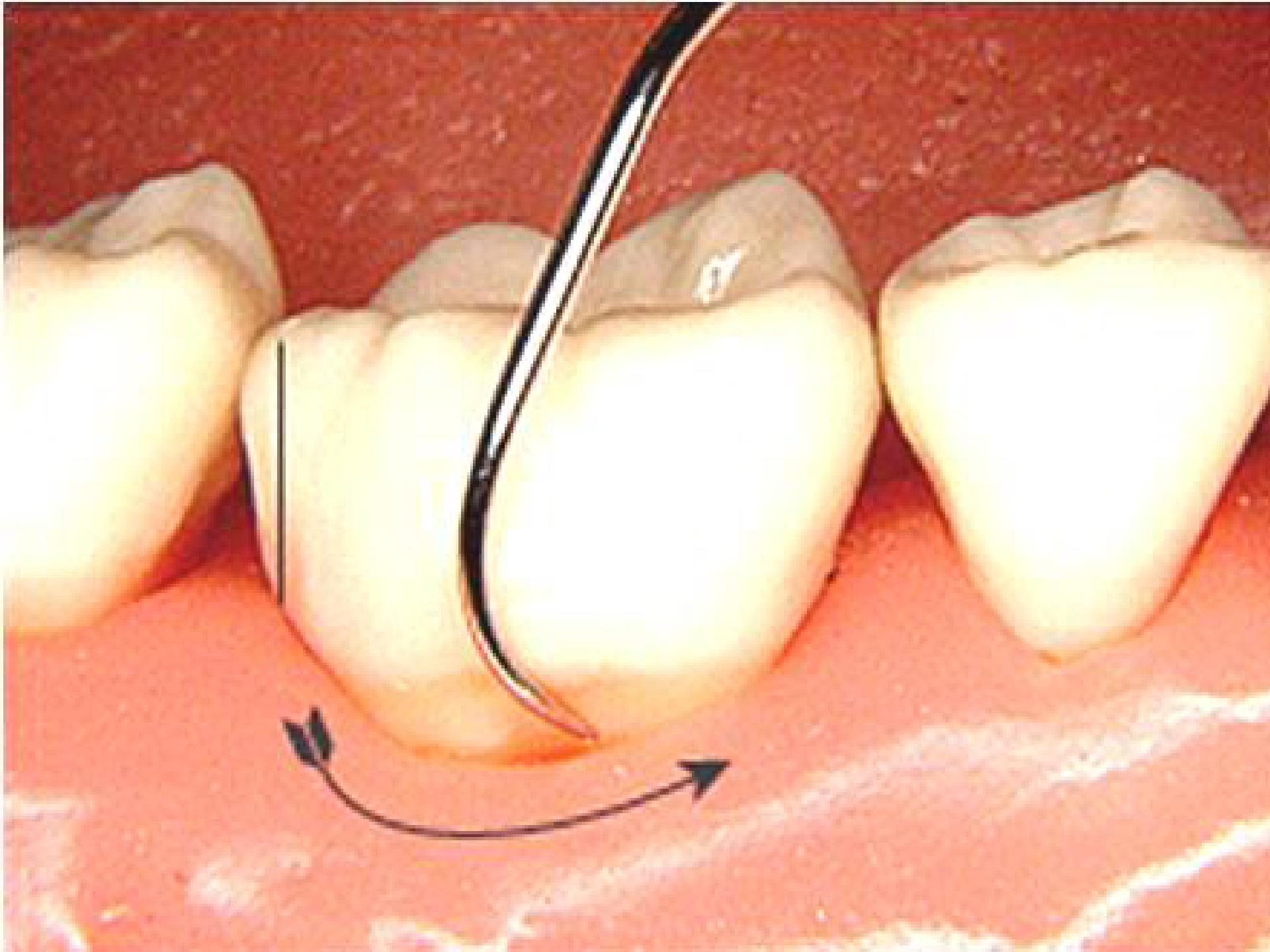
- **demonstration of**
  - **Periodontal Probing**
- FIND LINK FOR USING PROBE MAYBE SRTA OR CDCA

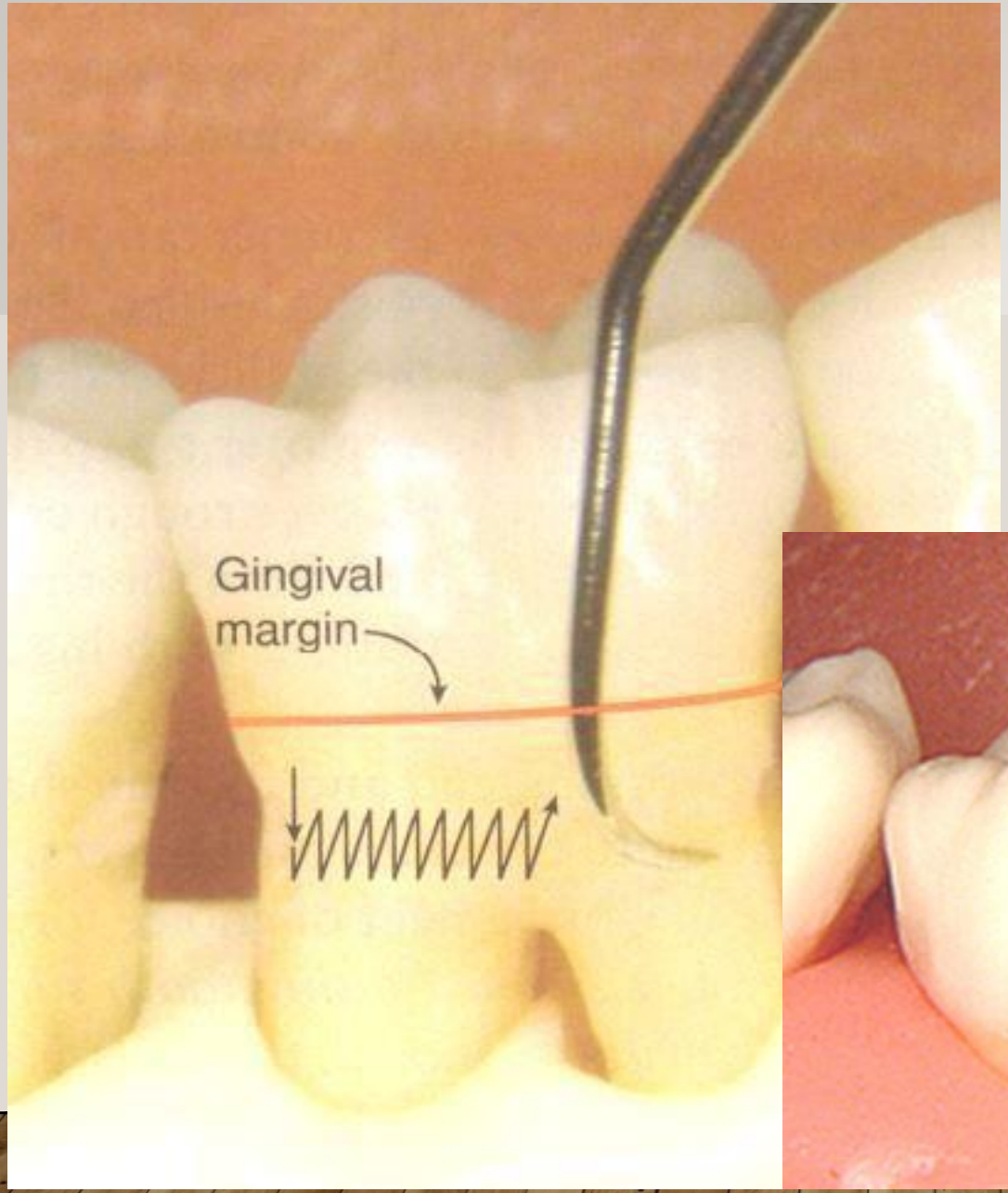
THE "BIG ~~BANG~~ THEORY"

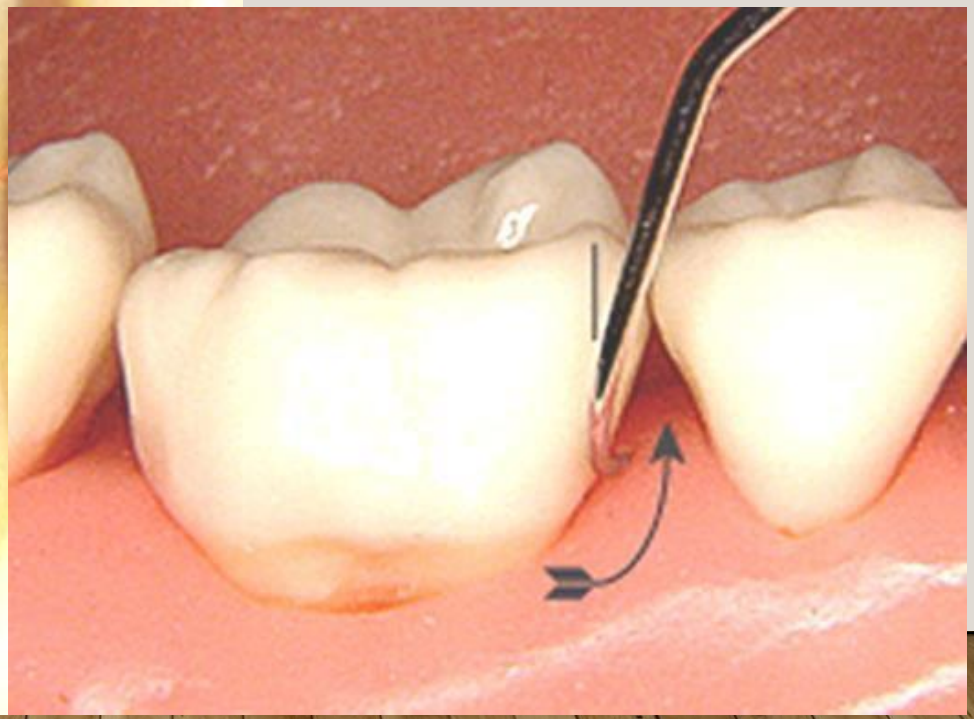
Note the direction of the last bend of the terminal shank
Note the direction of the handle as it exits the mouth

Terminal Shank


Terminal shank is parallel to the long axis of the tooth.
Bend at the end of the terminal shank points distally




Gingival
margin



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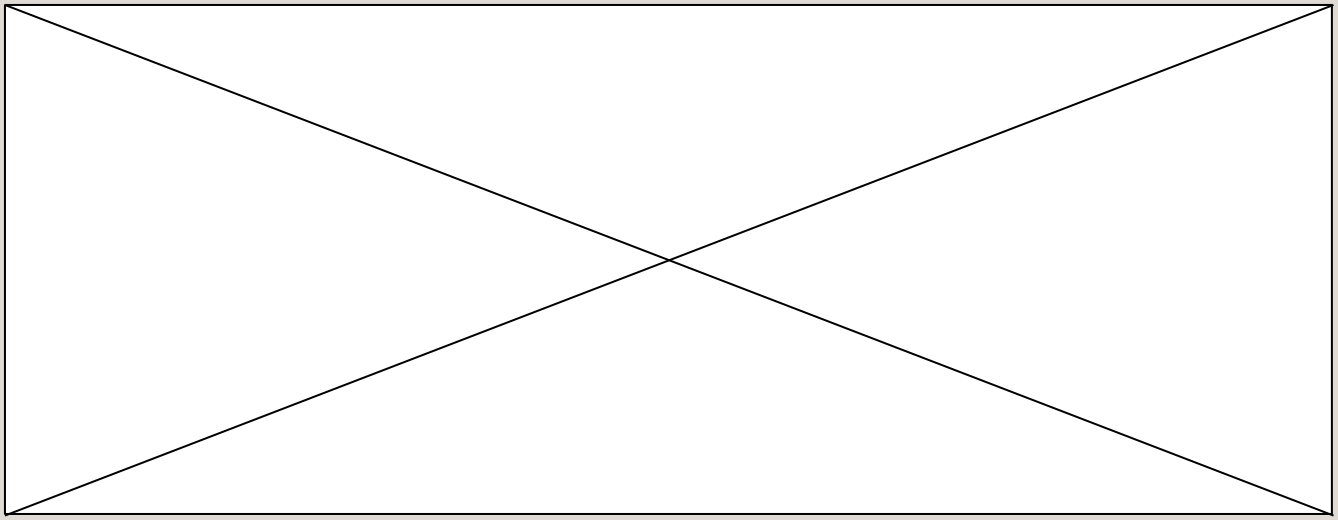

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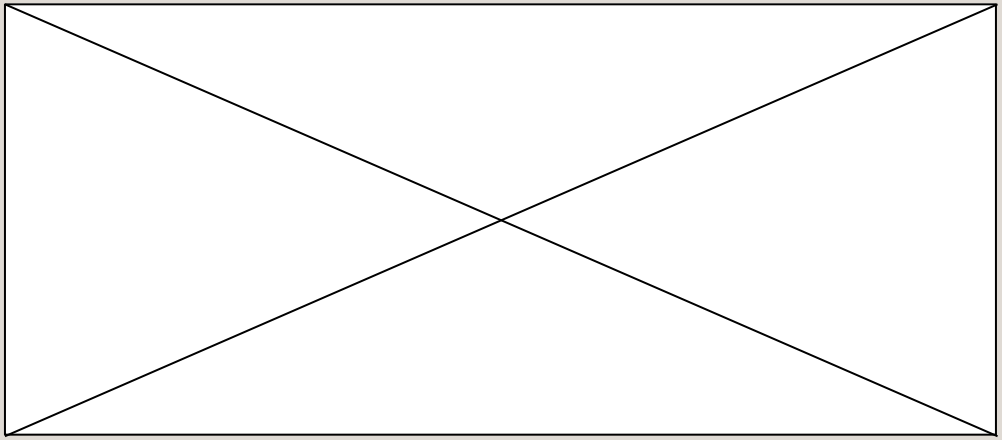


Use the 11/12 explorer and compressed air to determine if any type of calculus is present on the M, D, F, and L surfaces or if the surface is free of calculus. Floss may also help you to detect ledges or spicules of calculus

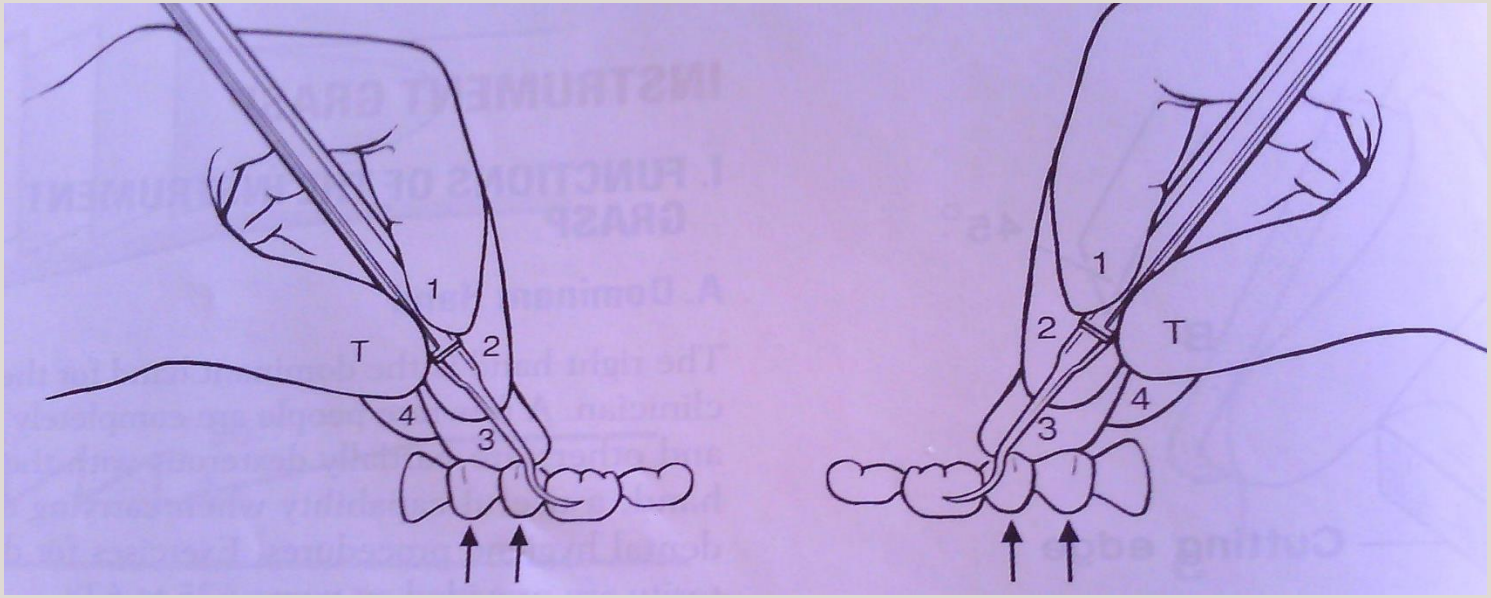
GRASP

- **The firm but light hold a dental clinician has on their instruments is referred to as "grasp".**
- **A proper grasp enables the clinician to maneuver the instrument around the tooth**
  - **For calculus detection**
  - **Correctly direct pressure when needed for calculus removal without damaging the periodontal tissues.**

THREE GRASPS USED

- **Three specific grasps are used. They are:**
  - **pen grasp**
  - **modified pen grasp**
  - **palm-thumb grasp.**
- **The *modified pen grasp* is the most useful.**

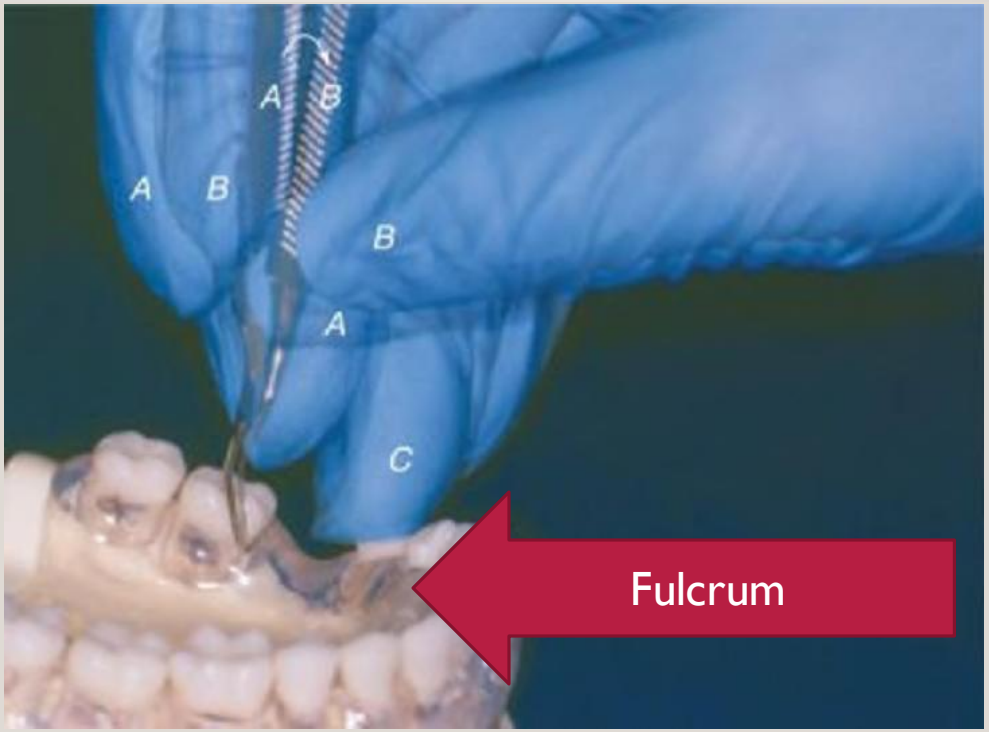
MODIFIED PEN GRASP



FULCRUM

- **The resting of the third finger on a firm intra- or extraoral site acts as a fulcrum.**
- **This improves control of the instrument and application of forces by the working end against the tooth.**

FULCRUM IS VERY IMPORTANT



EXTRAORAL FULCRUM

EXTENDED GRASP

INSTRUMENTS WITH CURVED SHANK

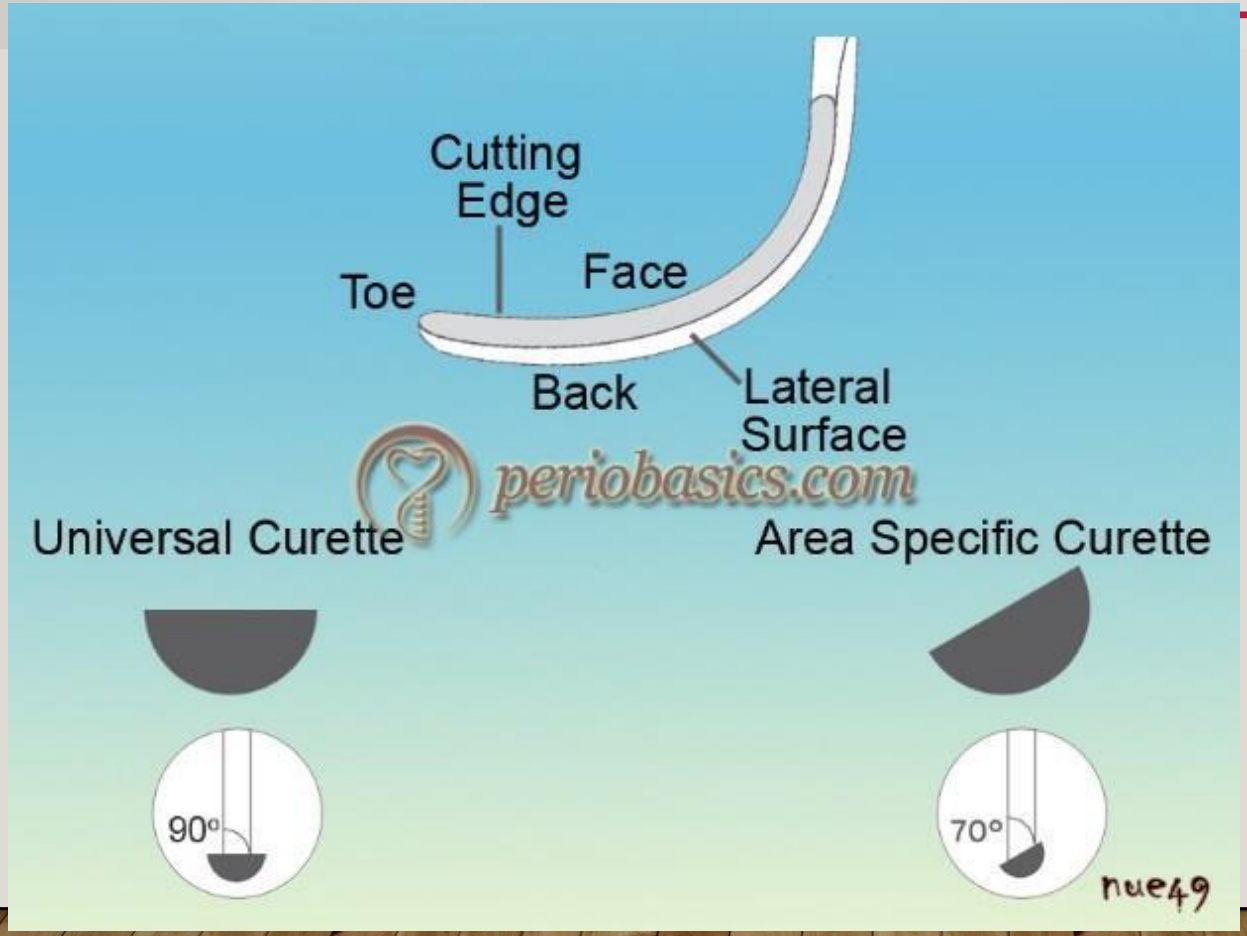
Periodontal scaling is:

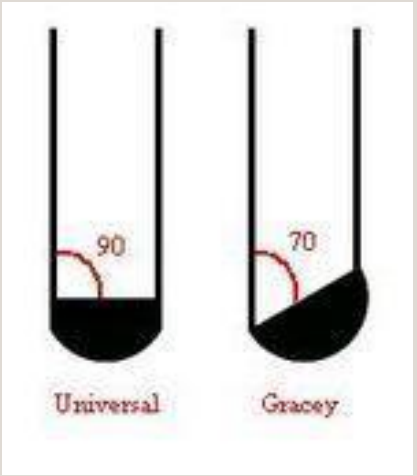
- Scaling and root planing, otherwise known as conventional periodontal therapy, non-surgical periodontal therapy, or deep-cleaning
- The process of removing or eliminating the etiologic agents –[dental plaque, its products(endotoxins) and calculus] – which cause inflammation, thus helping to establish a periodontium that is free of disease.

PERIODONTAL SCALING INSTRUMENTS

- Gracey and Universal Curettes and Sickle Scalers
- Based on instrument design, how they are used, where they are used and how they work, the following classification of periodontal scaling instruments is used.

CURETTE DESIGN





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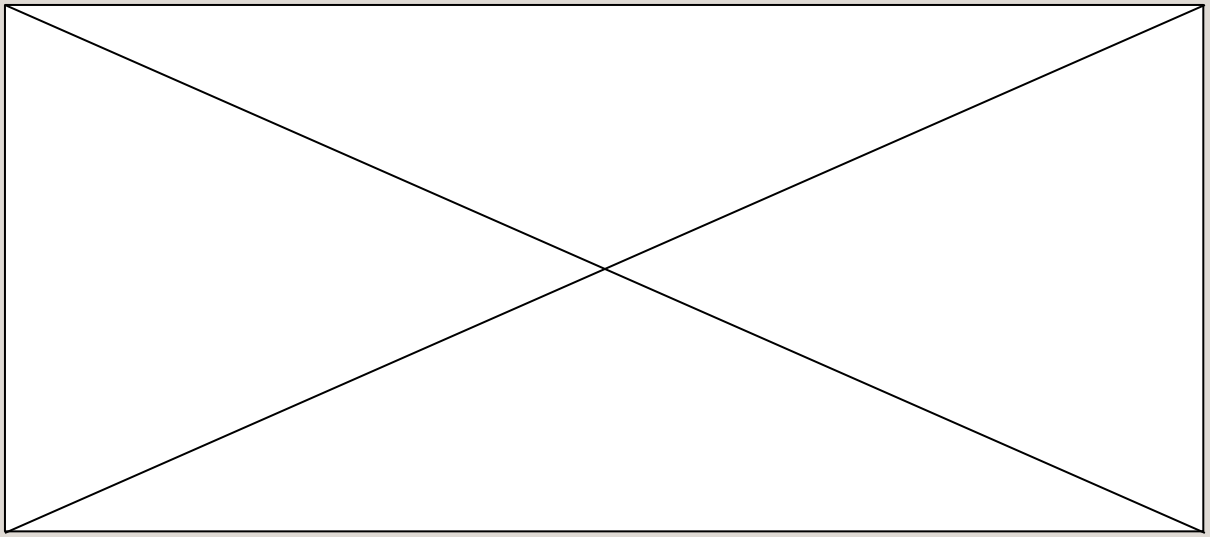
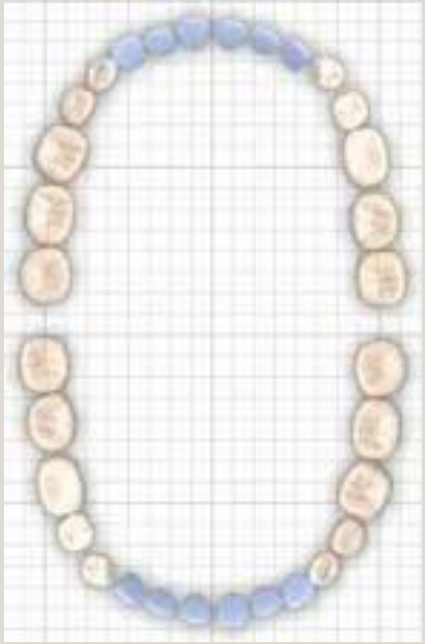


INSTRUMENT DESIGN OF UTHSC D-3 & D-4

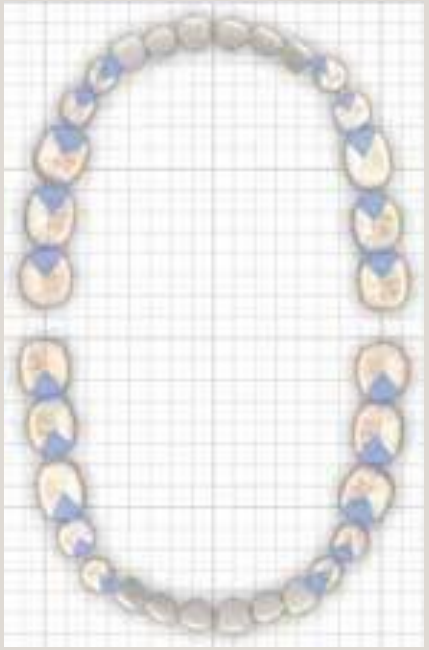
Maxillary and Mandibular Anterior



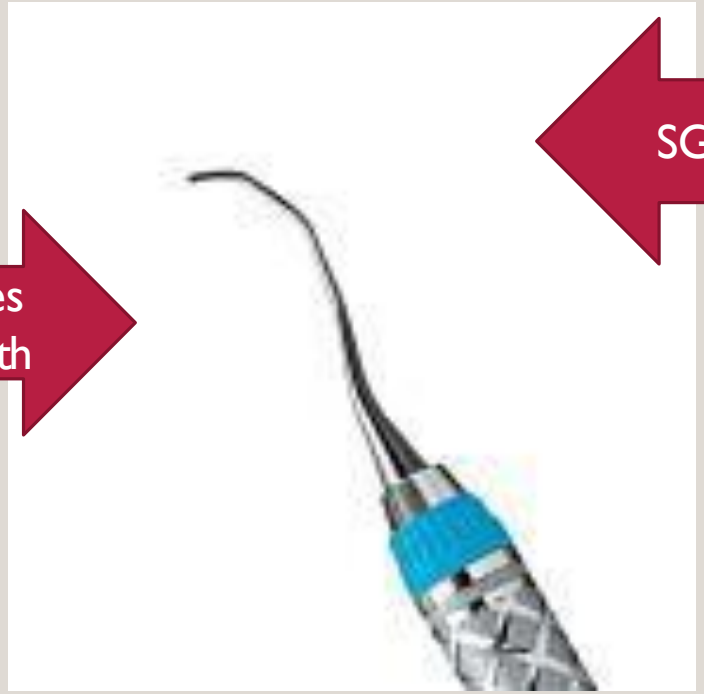
POSTERIOR GRACEY

Mesial surfaces of Posterior Teeth

SG11/129



POSTERIOR GRACEY



SG13/149

Distal Surfaces
Posterior Teeth

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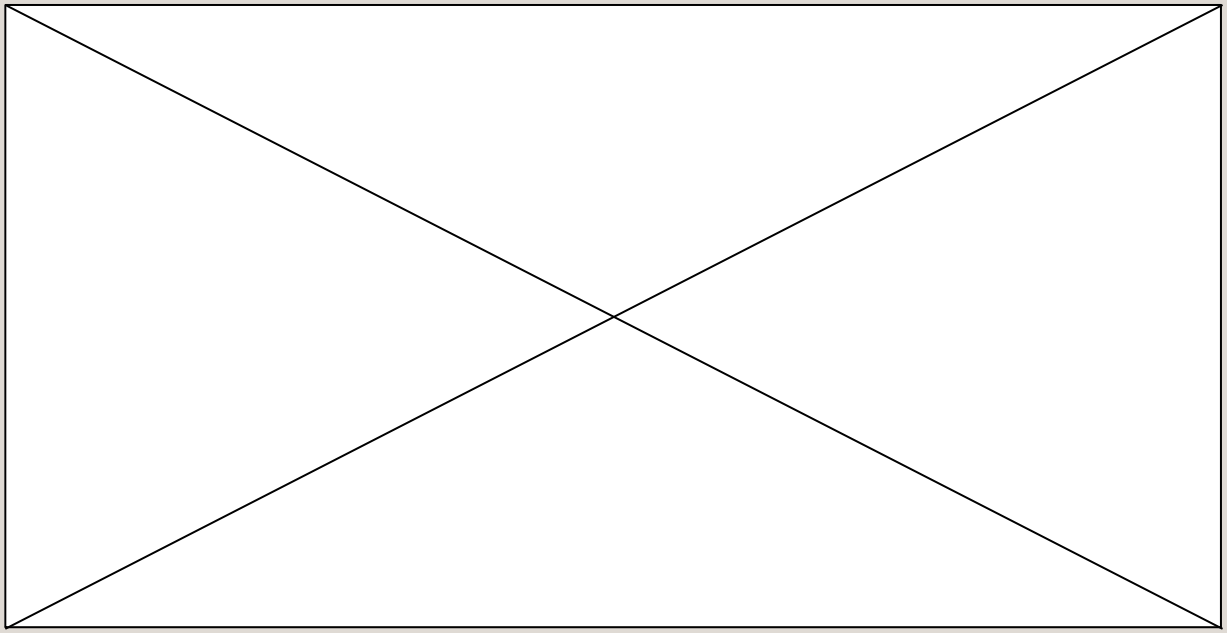

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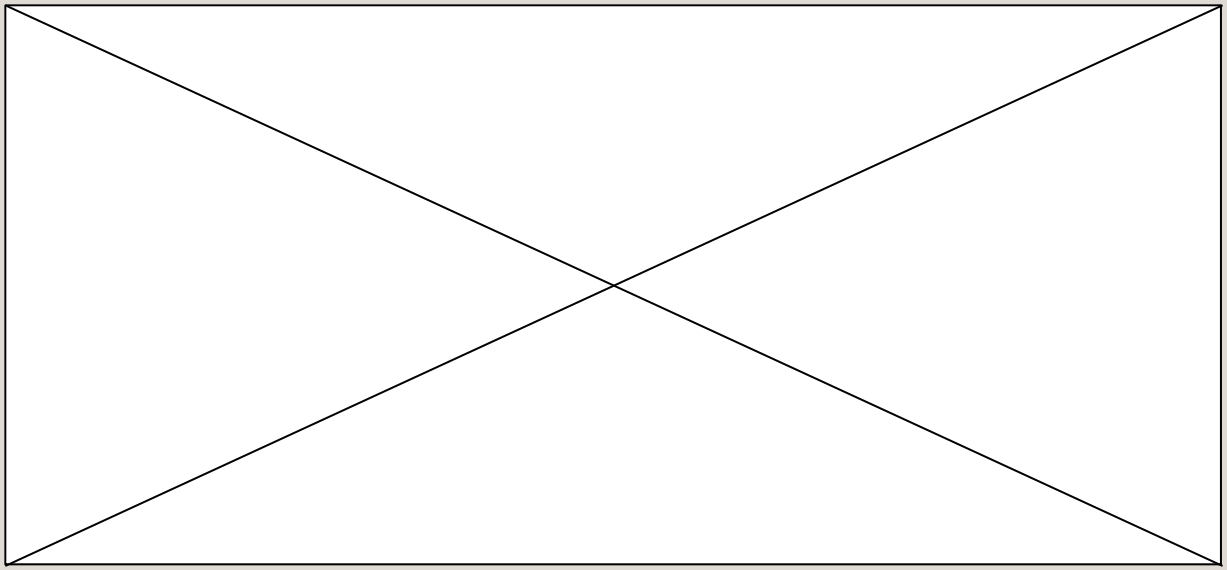
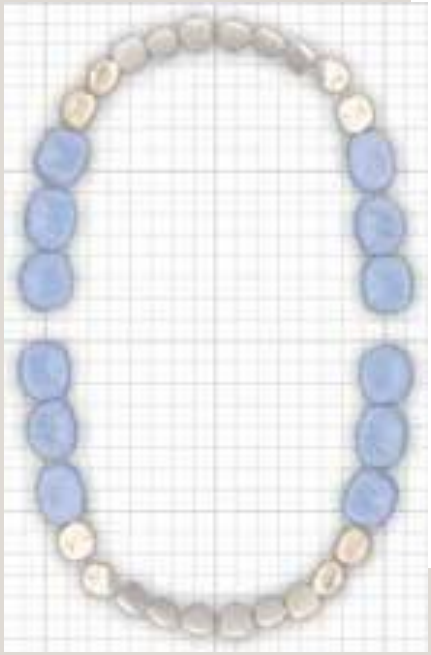
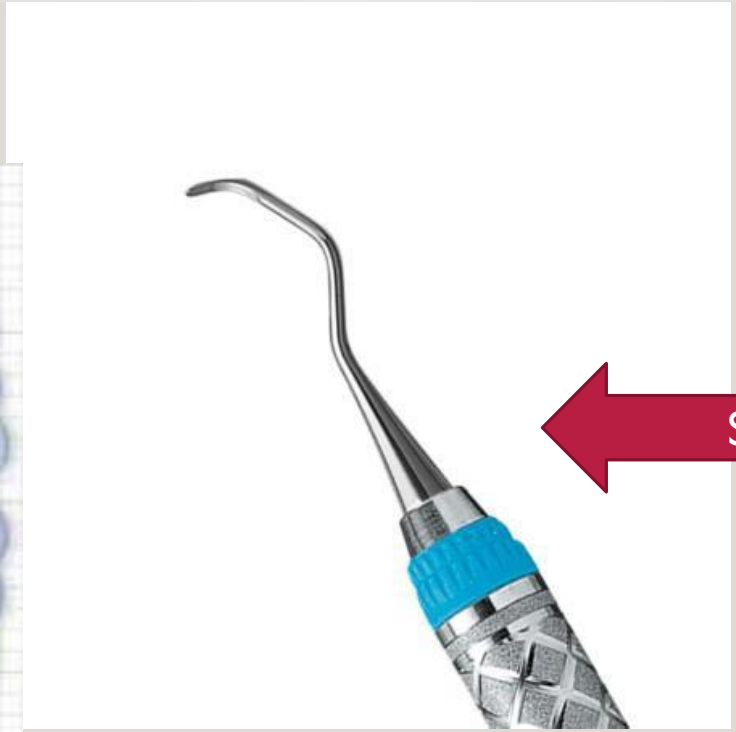


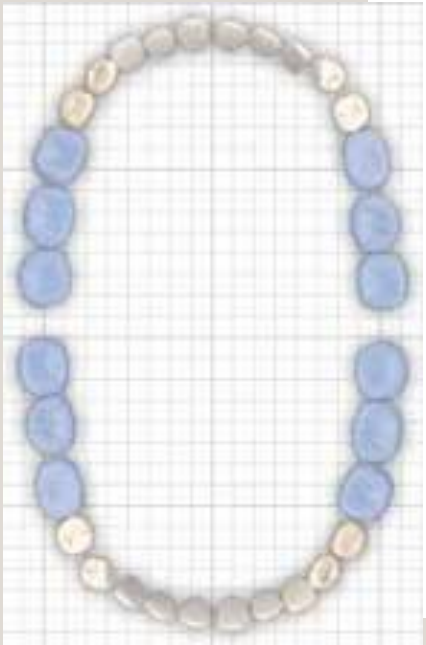
UNIVERSAL POSTERIOR CURETTE



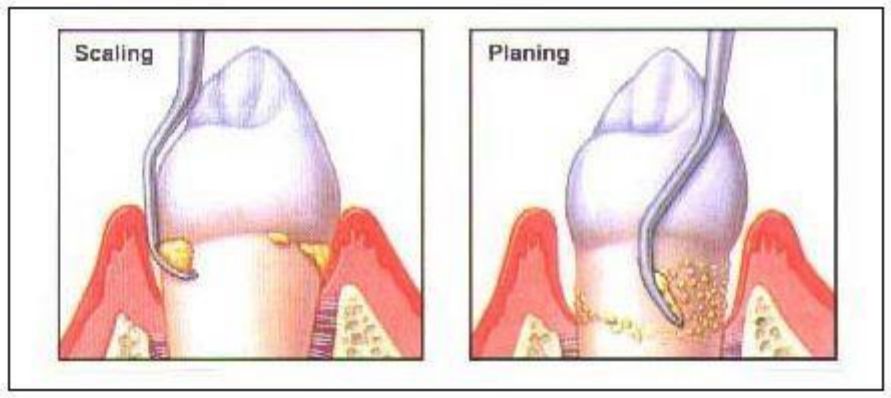


SC4R/4L9

UNIVERSAL CURETTE



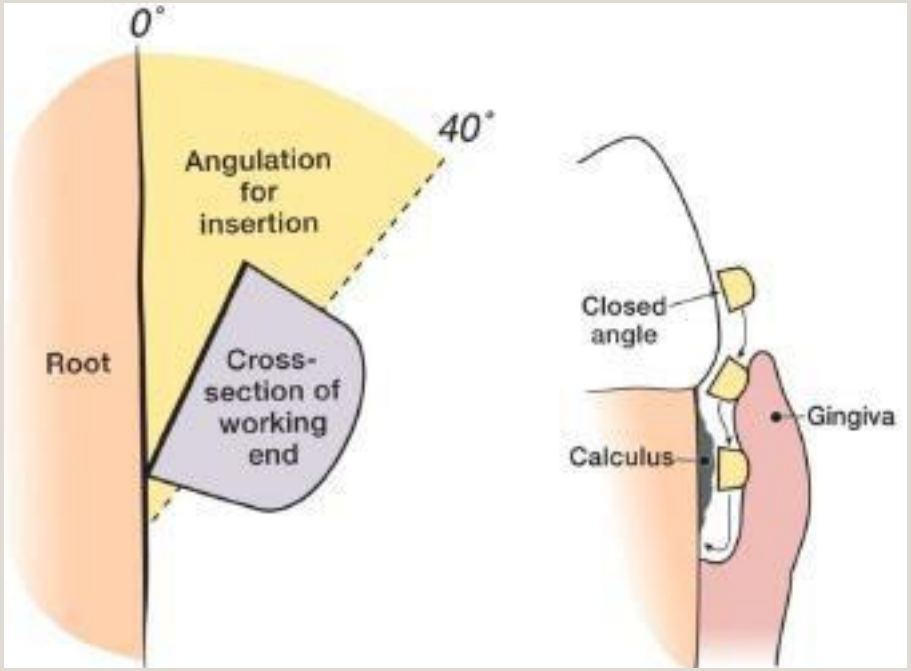
PERIODONTAL SCALING AND ROOT PLANING

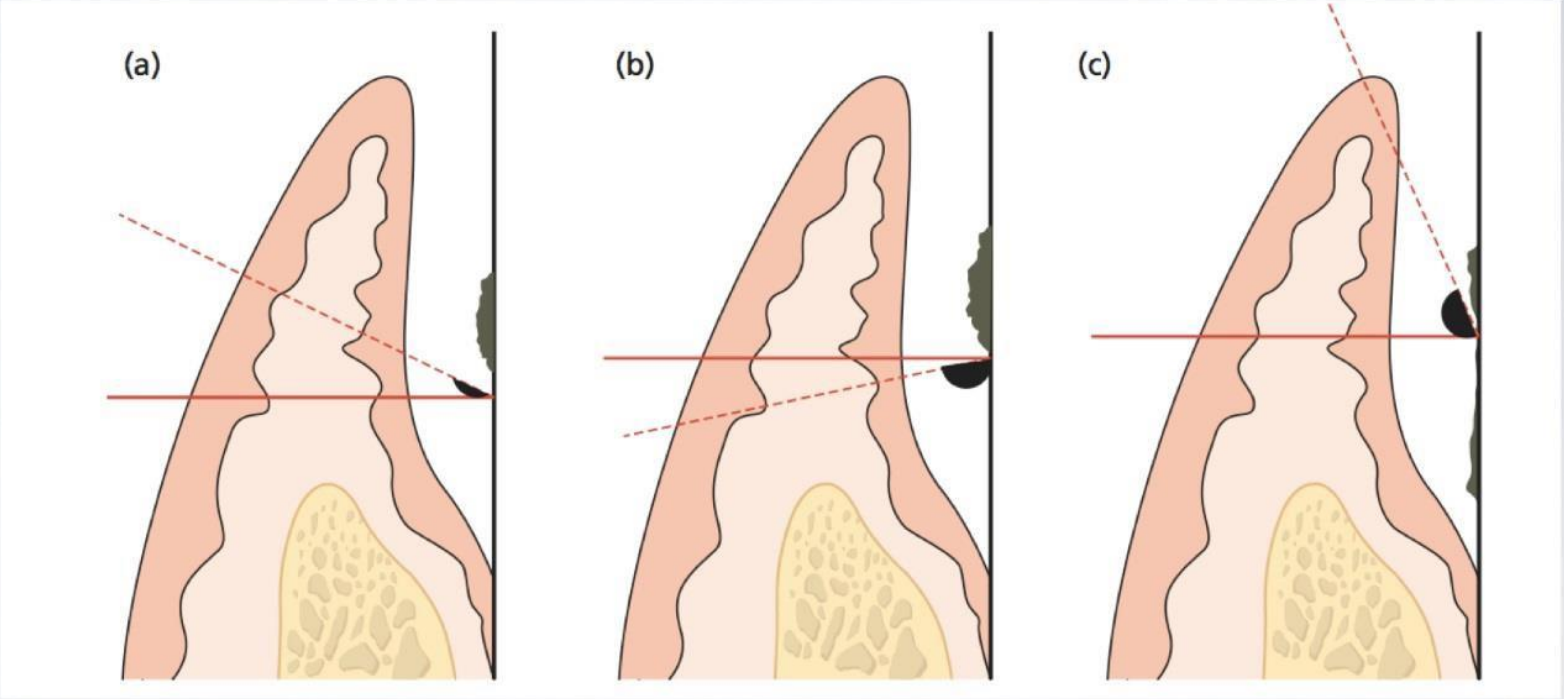


INSERTION

- **To avoid damaging the soft tissues when entering the sulcus with a large, sharp instrument, the working end is inserted into the sulcus with the face of the blade "closed" or flattened to the tooth surface.**
- **Once in the depth of the pocket or sulcus, the blade is opened 45 to 90 degrees for working strokes.**

ANGULATION


0°
40°
Angulation for insertion
Root
Cross-section of working end
Closed angle
Gingiva
Calculus



The effect of different angulations of the cutting edge of the curette to the tooth surface.

(a) Correct angle of application.

(b) Too obtuse angulation resulting in ineffective calculus removal and the possibility of cratering the surface.

(c) Too acute angulation resulting in ineffective calculus removal and burnishing of the calculus deposits.

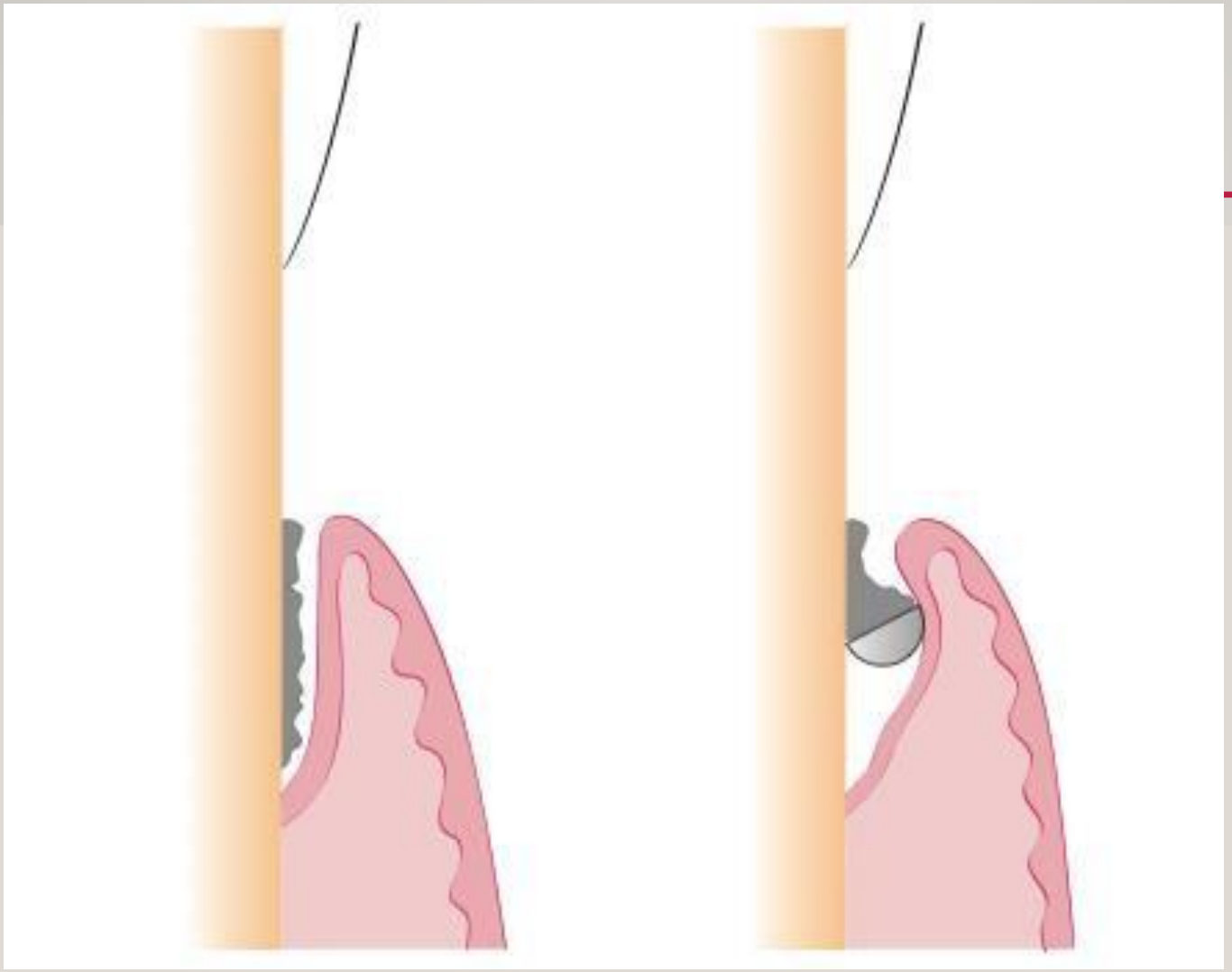
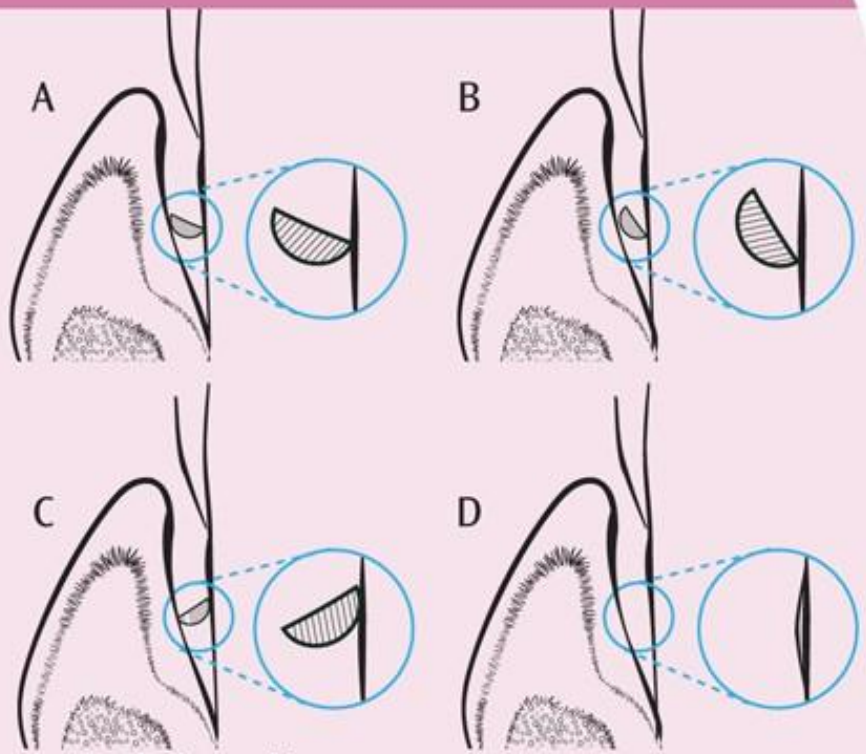


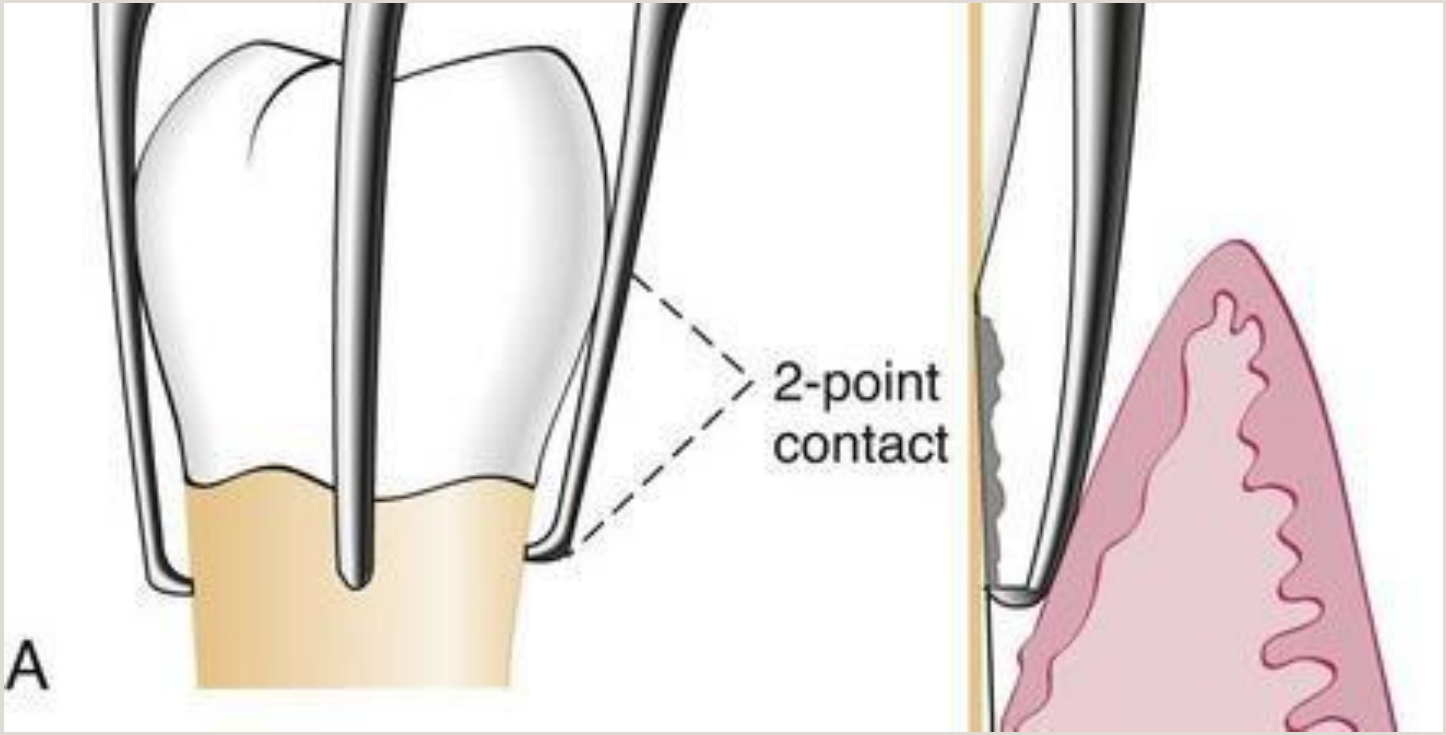
Figure 3.



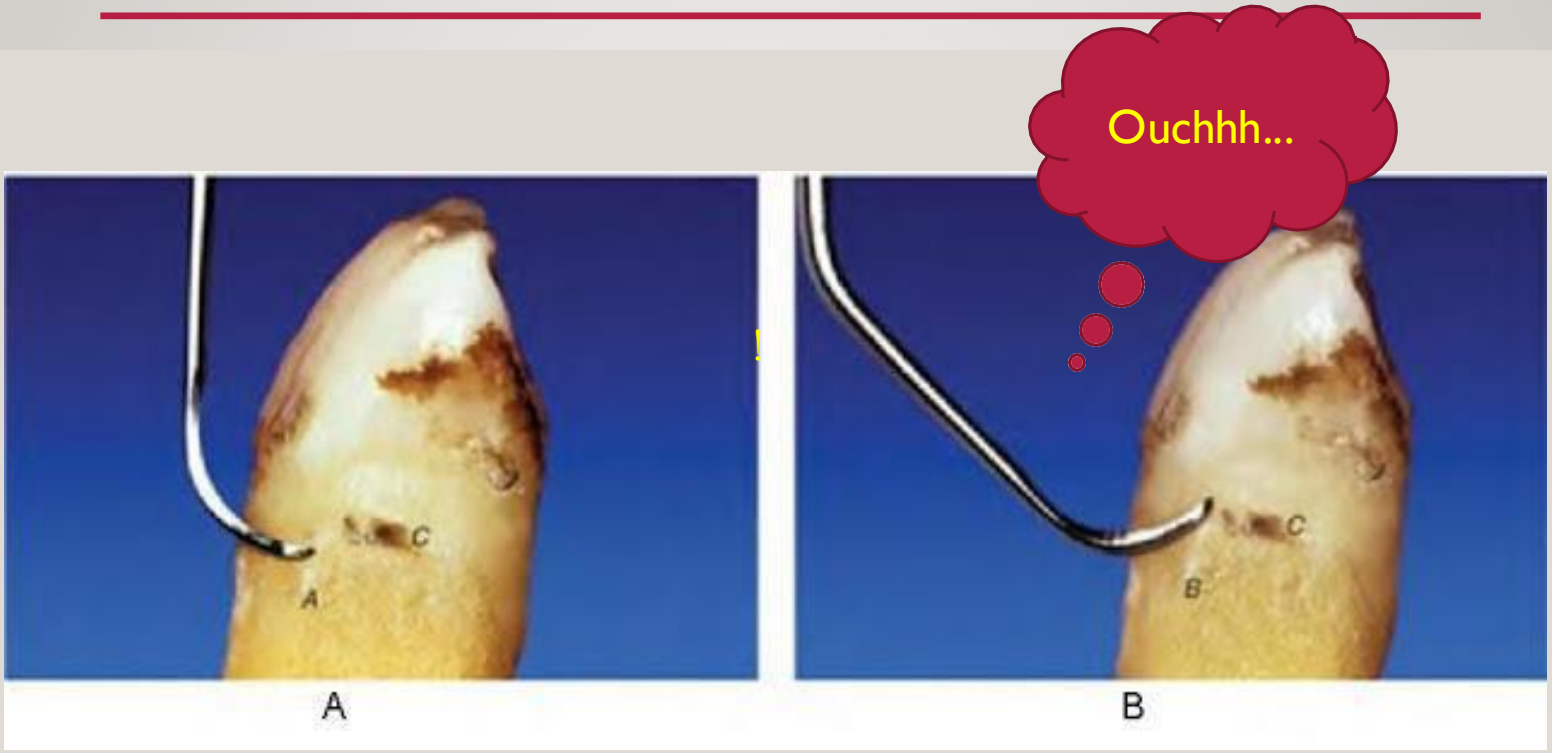
A. Blade working at angulation.
B. Error: blade closing on activation <45°.
C. Error: blade opening on activation >90°.
D. Burnished calculus now undetectable.

Dimensions OF DENTAL HYGIENE NOVEMBER 2008

ADAPTATION

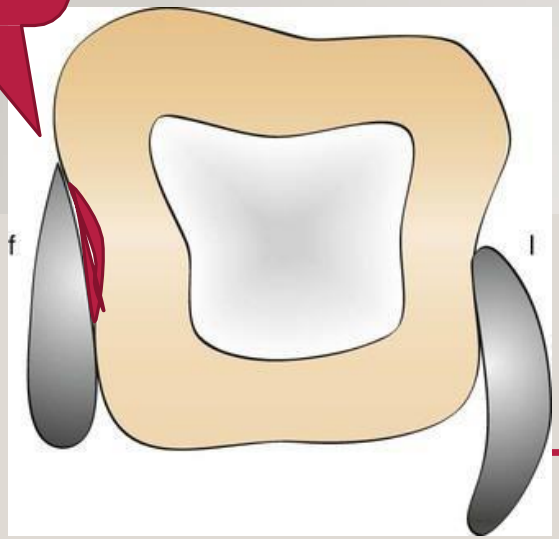


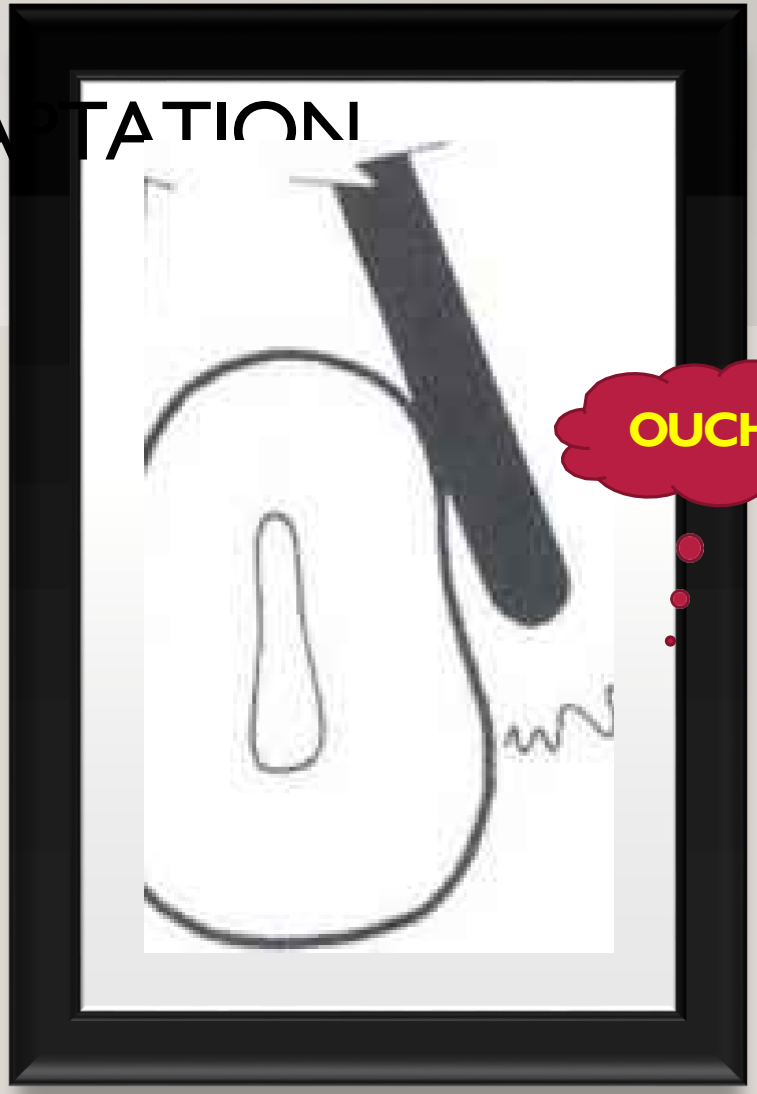
SUBGINGIVAL CALCULUS REMOVAL



ADAPTATION

We must chose the right instrument





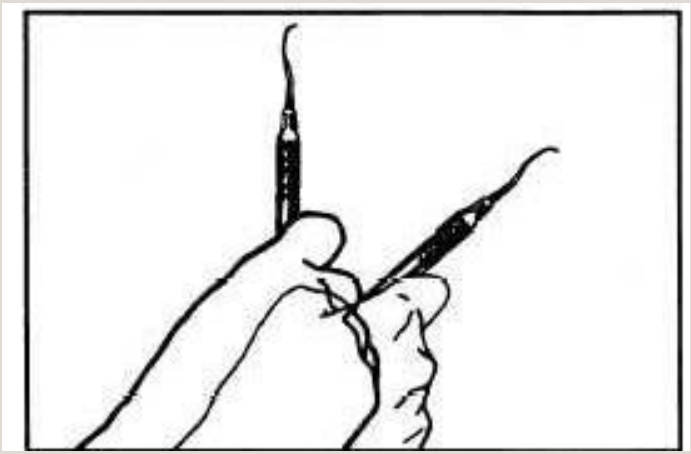
This refers to the placement of the working end of the instrument in the correct relationship to the tooth.

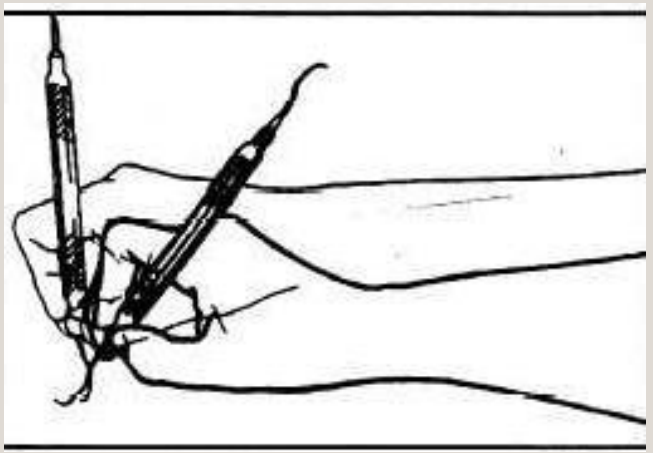
CARPAL TUNNEL SYNDROME MAY OCCUR IF

WRIST AND ARM MOTION

- Side to side
- Up and down
- Activated by pivoting fulcrum finger
- Wrist must be straight to activate stroke movement of instrument

ACTIVATION





STROKES

- **Exploratory strokes provide light tactile feedback from the instrument tip.**

WORKING STROKE

- **Working strokes provide controlled pressure against a tooth surface using an appropriate cutting angulation.**

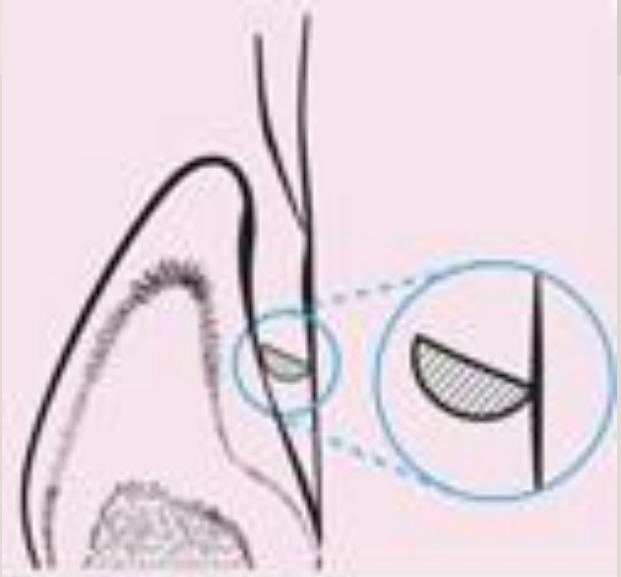
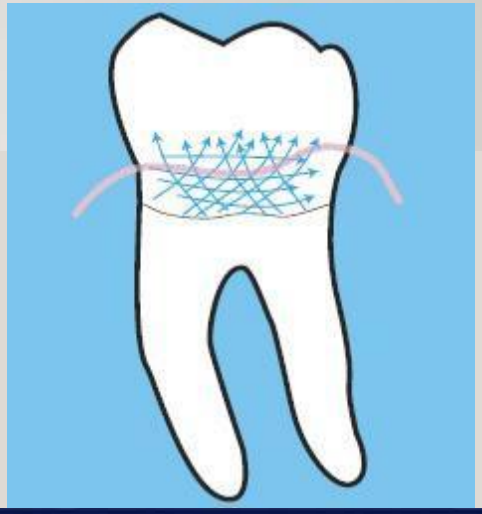


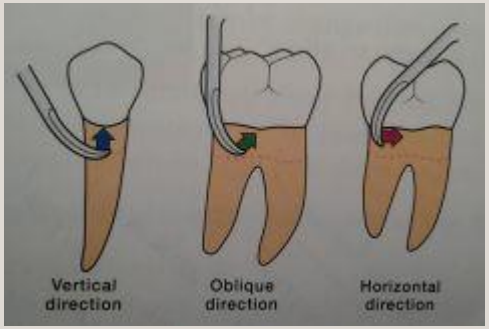
Figure 1: Surface-to-blade angulation for correct working stroke: 70°.

WORKING STROKES

- **Working strokes are ideally a series of overlapping strokes covering the entire tooth surface.**



WORKING STROKES



STROKES

- **Scaling strokes are short pulling strokes made with firm pressure to remove increment from root and enamel surfaces.**
- **Root planning strokes are made to remove deposits and smooth the root surface.**

ROOT PLANING STROKE

- Used for final smoothing & planing of the root surfaces

- **Check your work by ending with an exploratory strokes provide light tactile feedback from the instrument tip.**

GRACEY CURETTES

4. Area-specific shank parallel to surfa
angulation.[6]

- When adapting the Gracey instruments to the teeth to perform scaling, the cutting edge must first be identified and placed against the surface to be scaled and the terminal shank should be parallel to that surface.
- In this position, the ideal working angulation for calculus removal is achieved.

GRACEY CURETTES

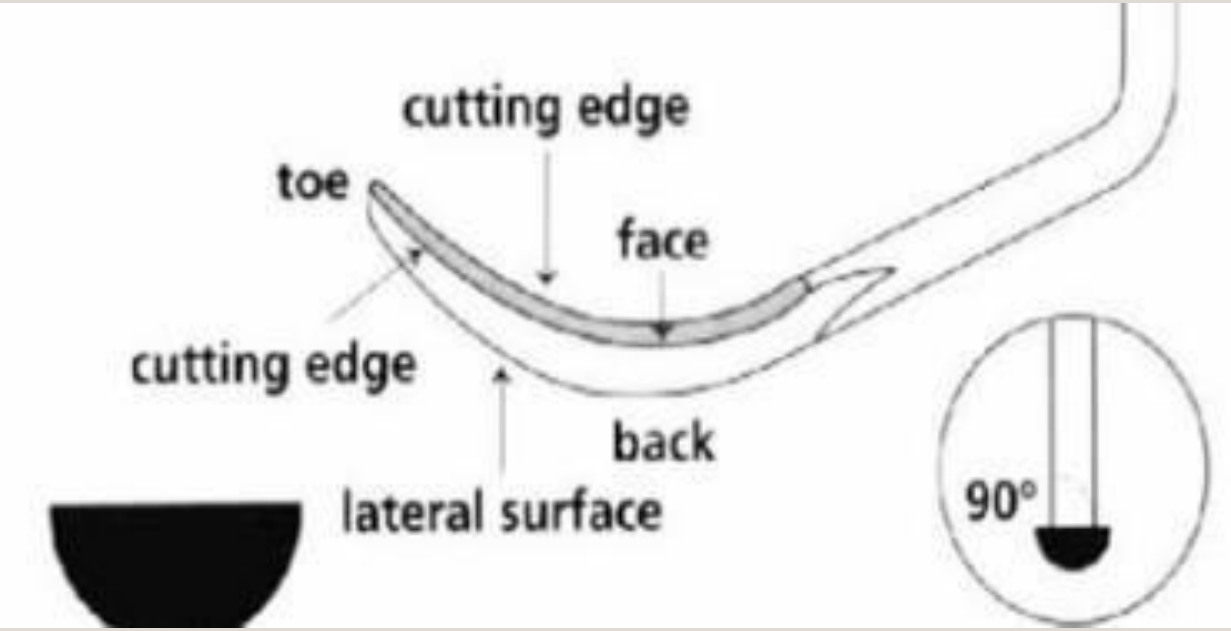
Figure 7a: Area specific curet in place at stroke onset, terminal shank parallel.

Figure 7b: Terminal shank still parallel to surface as stroke ends.

Figure 7c: Error: Note ending terminal shank position. Lateral rotation on fulcrum closes blade angulation during stroke.

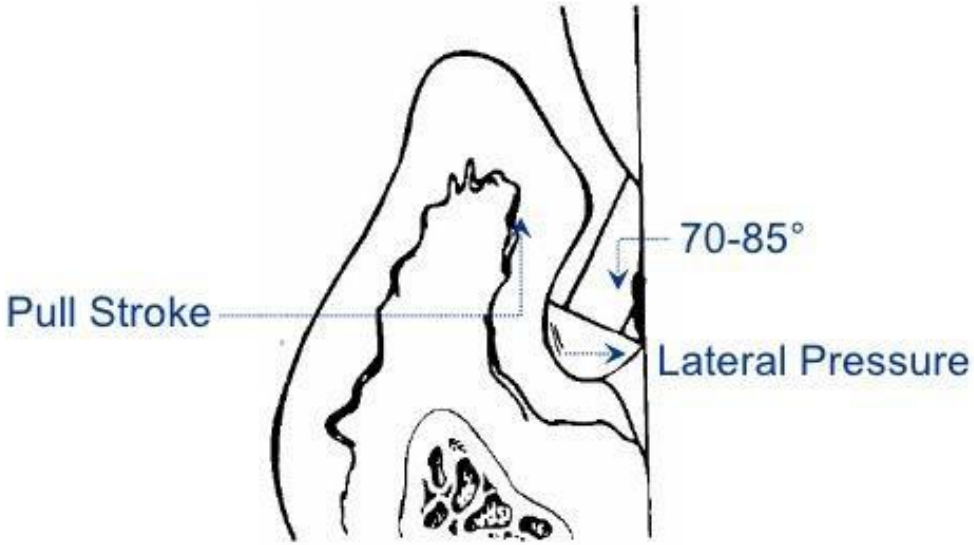
COLUMBIA CURETTES (UNIVERSAL)

UNIVERSAL CURETTE DESIGN



Universal Instrumentation

- The blade is correctly adapted when the toe is directed interproximally and the terminal shank is parallel to the tooth.
- To remove deposits, the cutting edge is applied to the tooth surface and the facial surface of the blade is tilted toward the tooth to achieve a 70-85° angle between the tooth and the blade.
- Apply lateral pressure against the tooth and pull upward while maintaining contact with the tooth.



Hu-Friedy

PERIODONTAL SCALING INSTRUMENTS

IN THE CATEGORY OF AREA SPECIFIC CURETTES A SUB CLASSIFICATION EXISTS.

- Modifications to Gracey Design
  - After five or extended series
  - Mini five series
  - Langer Rigid series
- Furcation series

SICKLE SCALERS

- **The sickle scaler, primarily used for supragingival calculus removal, is a very useful instrument.**
- **It is often the first instrument used to remove large, heavy deposits thus improving access to subgingival area for other instruments.**

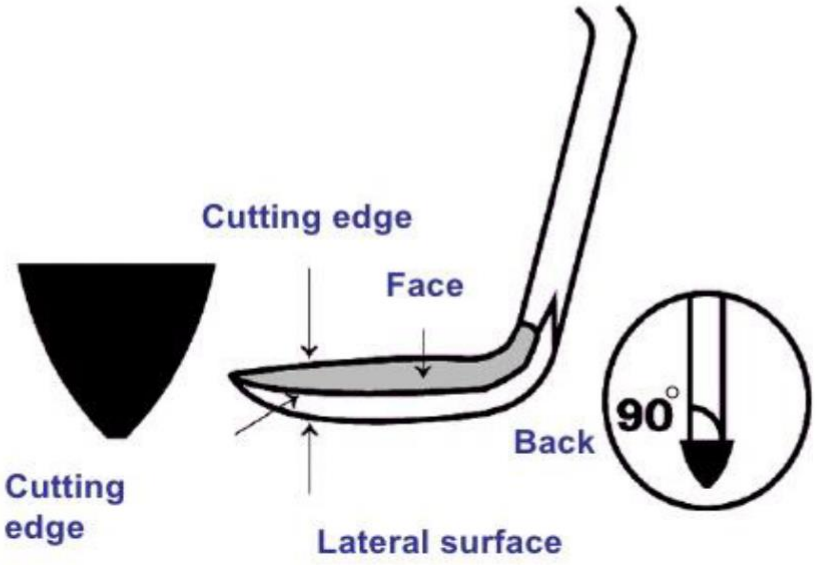
SICKLE SCALERS

- **Sickles have common features making them suited to the removal of heavy calculus deposits and working interproximally, around the contact areas of teeth.**
  - **A flat blade cut at a 90 degree angle to the shank.**
  - **Cutting edges on both sides of the face of the blade.**
  - **The face of the blade is triangular tapering to a strong, sharp point at the toe.**
  - **The face of the blade may be straight or curved.**

SICKLE DESIGN

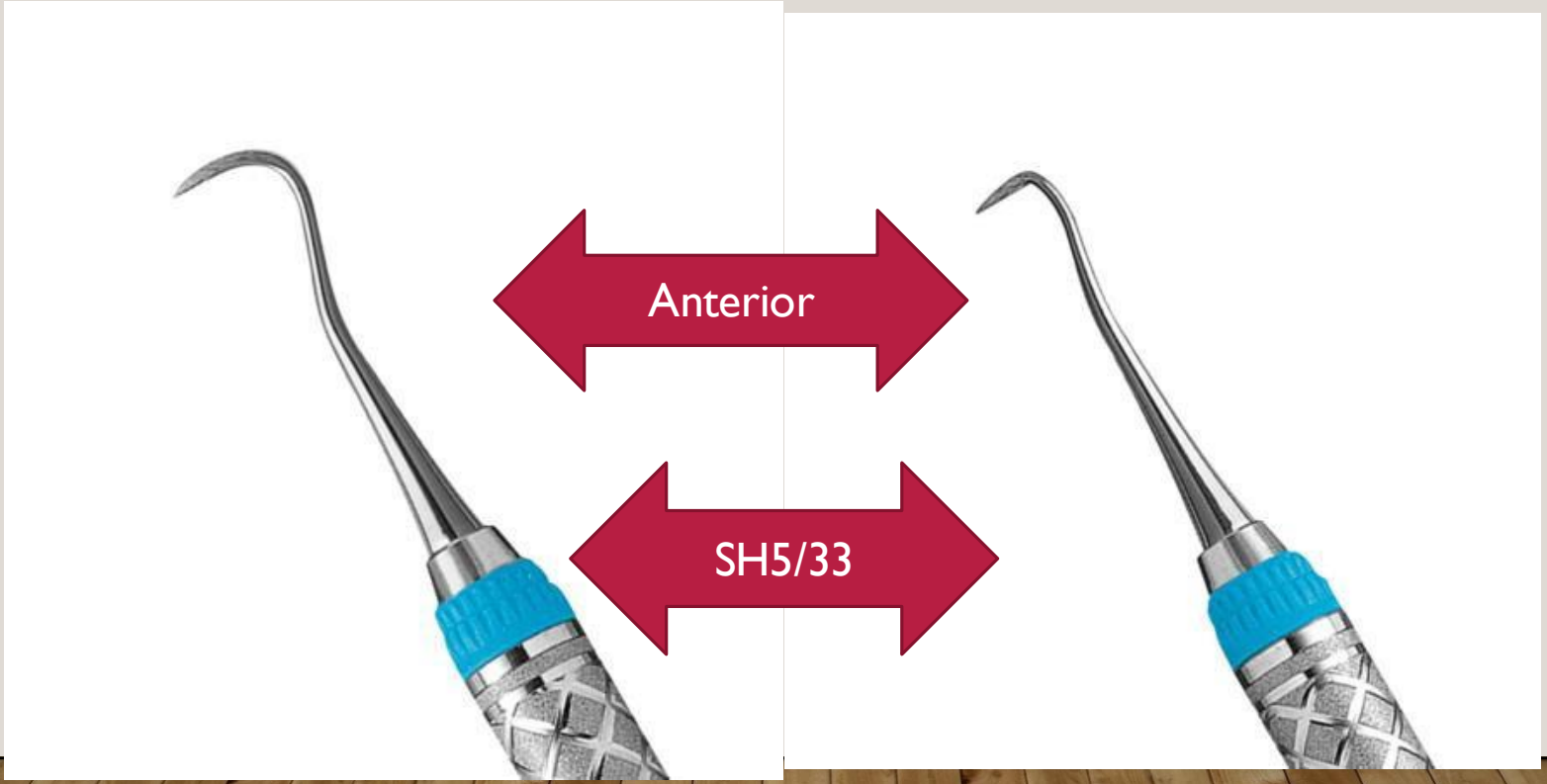
Sickle Scalers – Straight Blade

- Two cutting edges on a straight blade that end in a sharp point.
- Also known as a Jacquette Scaler.

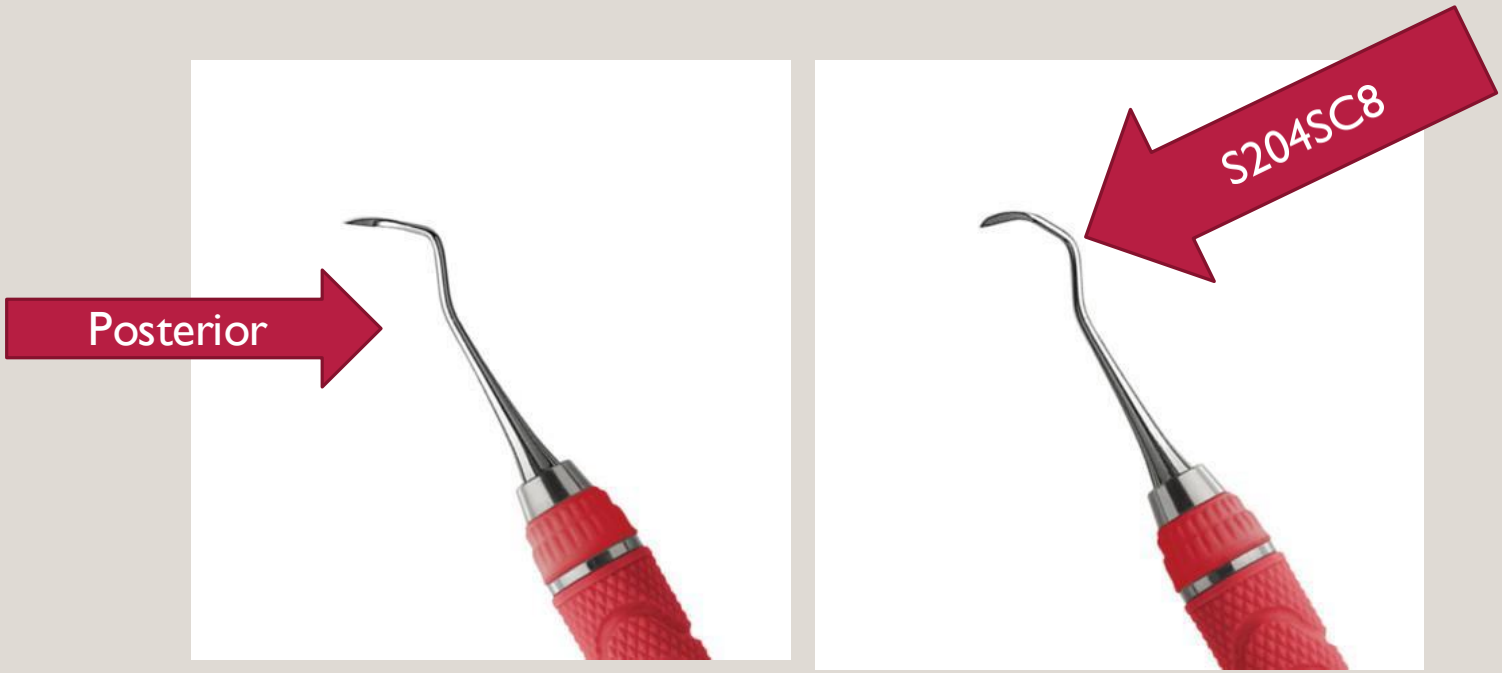


Hu-Friedy

H3 JACQUETTE SCALER

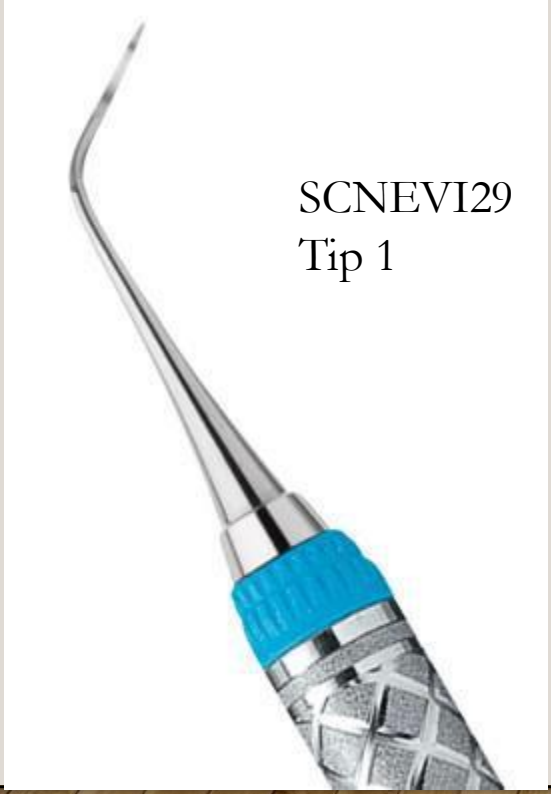


SICKLE SCALER



I NEVI® ANTERIOR SCALER

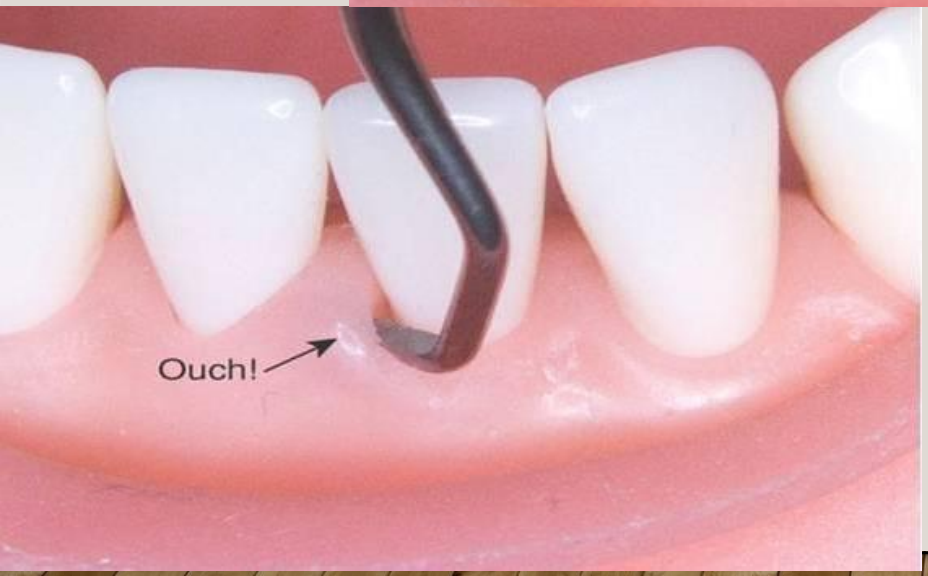
2 NEVI® POSTERIOR SCALER



ADAPTATION OF SICKLE

It is the sharp pointed toe and the first 1-2 mm of the lateral cutting edges that perform the work.

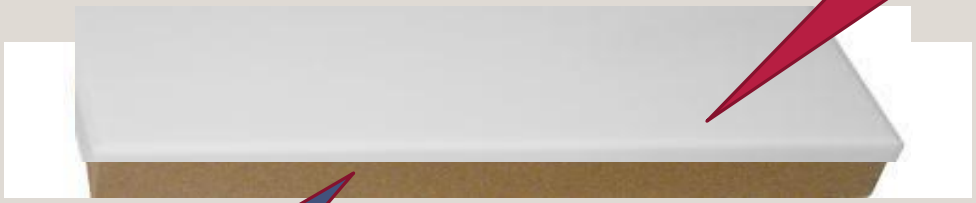
Line
angle


Ouch!

SICKLE SCALERS

Test Stick

Ceramic Stone



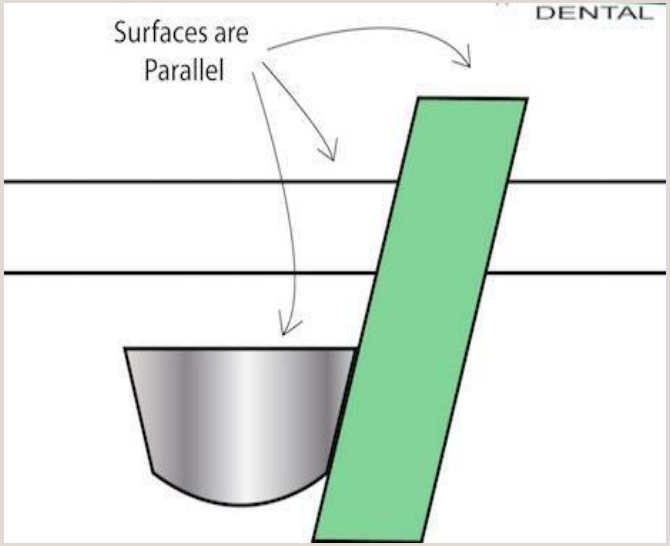
Fine side of Stone for slightly dull instruments

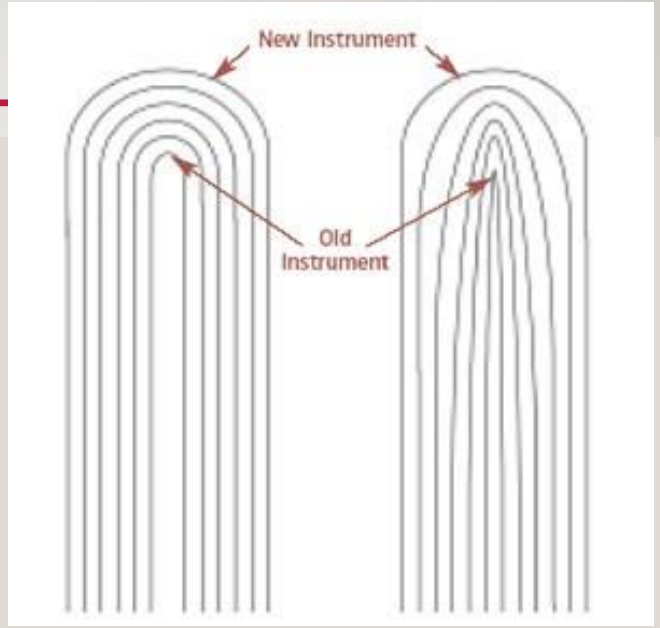
The **coarse** side of stone for really dull instruments

HTTPS://WWW.YOUTUBE.COM/WATCH?V=Y4S_Q_ADK38

- **Hu-Friedy's It's About Time Sharpening [FULL]**
- **You will only be as effective as your instruments are. Sharp instruments are much more effective than dull ones.**
- **WORK SMARTER NOT HARDER….**

SHARPENING


Surfaces are
Parallel
DENTAL

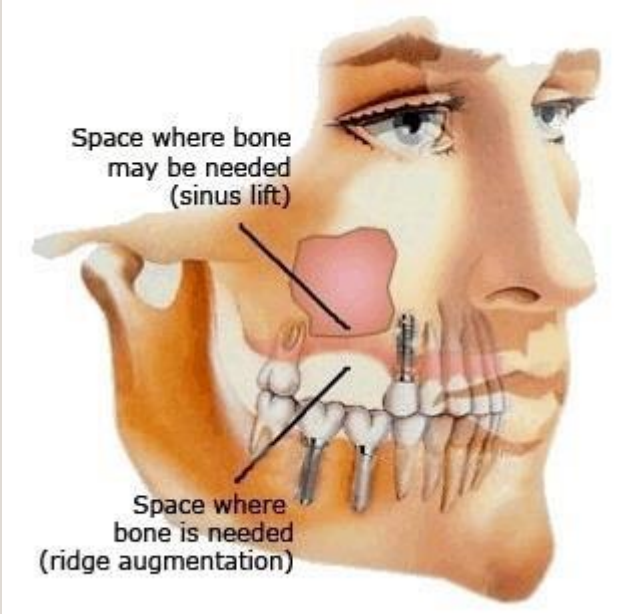

New Instrument
Old
Instrument

GRACEY CURETTE EVEREDGE BLADE/HANDLE

Gracey Curette
Double ended
#1/2 Mini Five

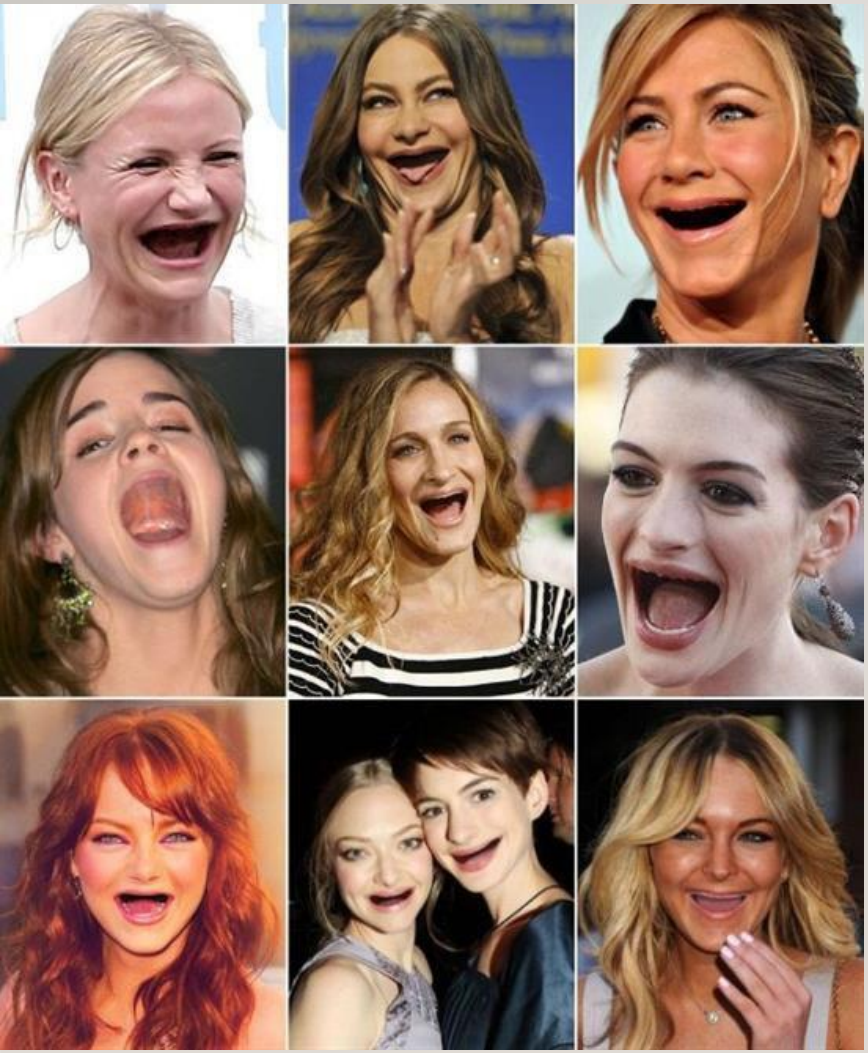
Gracey Curette Double End #1/2
W# 9 Everedge Handle

EVEREDGE® TECHNOLOGY SCALERS AND CURETTES


Space where bone
may be needed
(sinus lift)
Space where
bone is needed
(ridge augmentation)

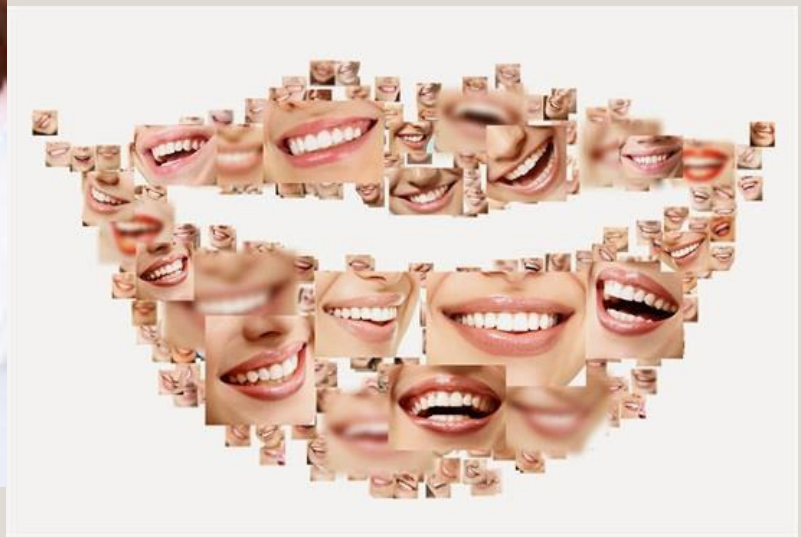
Even the most beautiful

The most handsome....

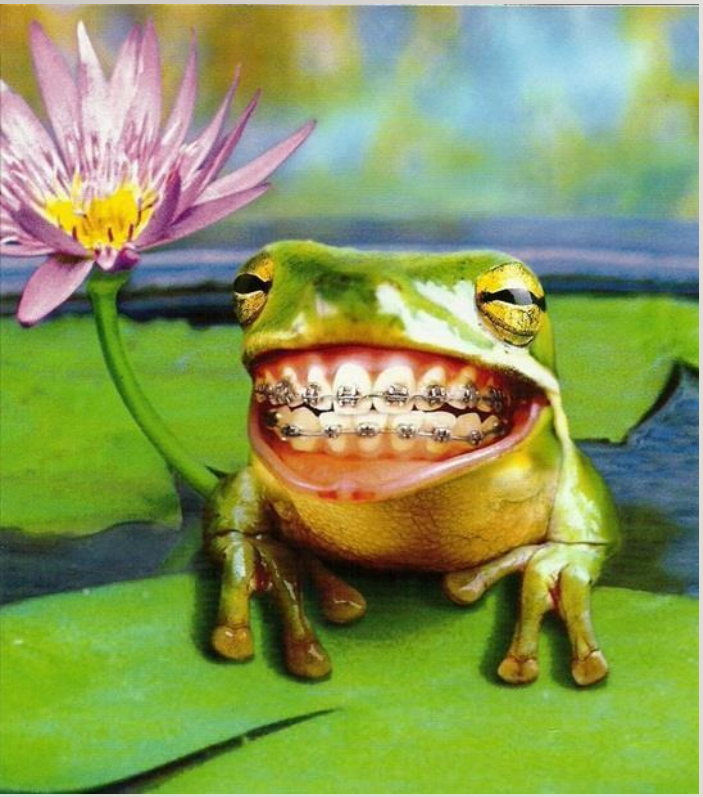


"A"
Listers...

Need their teeth
VIA 9GAG.COM
9GAG is your best source of fun.



SO WHAT DO WE DO TO PREVENT TOOTH LOSS DUE TO PERIODONTAL DISEASE



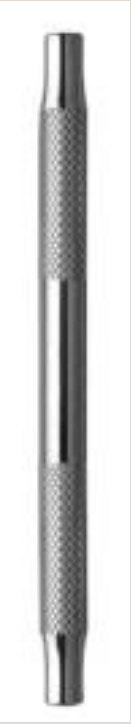
- The following slides depict more specific instruments commonly used in nonsurgical periodontal instrumentation.

½ VISION CURVETTE®

For anterior and premolar surfaces and interproximal areas

#6 Satin Steel Handle

#4 Handle



13/14 VISION CURVETTE®

- 50% shorter, curved blades.
- 5 mm and 10 mm shank markings.
- Blade indentification mark (+) on handle.
- Identification bands on handle. One band on 11/12, two bands on 13/14 (on #4 handle only).

SUB-0 VISION CURVETTE®

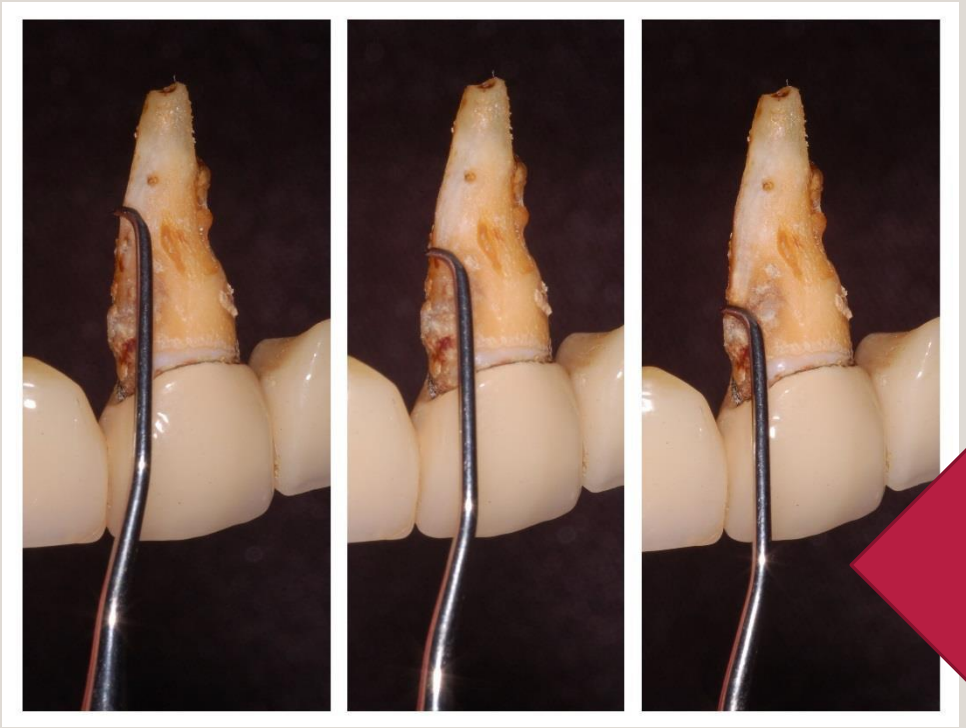
- Long shank reaches deep into the sulcus, adapting to the facial and lingual root surfaces
- 50% shorter, curved blades.
- 5 mm and 10 mm shank markings.
- Blade indentification mark (+) on handle.
- Identification bands on handle. One band on 11/12, two bands on 13/14 (on #4 handle only).

FURCATION SERIES

- Mesial-Distal Quetin Furcation Curette

EXTENDED 3MM

NON-SURGICAL PERIODONTAL TREATMENT. MAKING IT WORK



Mini Five Series Note the longer terminal shank (3mm longer)

University of Pikeville

UPIKE

Mirror and Finger Rests Maxillary and Mandibular

September 24, 2026 | 8:00–11:50 AM

DEN 624 • Periodontology I • Tanner College of Dental Medicine

Mirror And Finger Rests
In
Maxillary And Mandibular Sextants

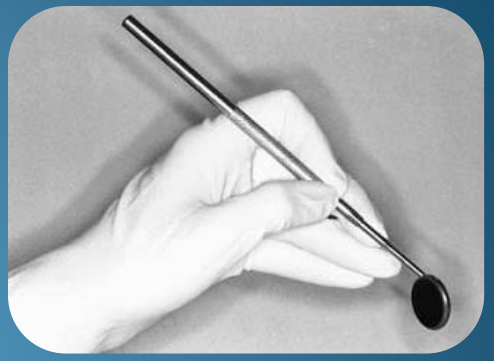
Overview

- Dental mirror: its types & surfaces
- Wrist position for instrumentation
- Fulcrum
- Mirror and finger rests in anterior sextants
- Mirror and finger rests in mandibular posterior sextants
- Mirror & finger rests in maxillary posterior sextants
- Conclusion
- References

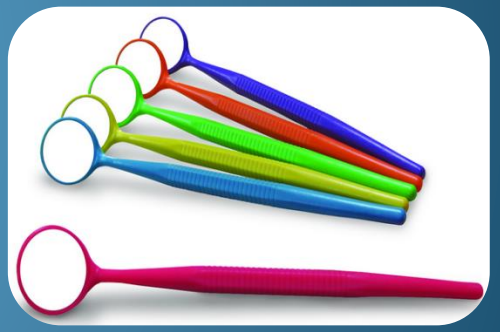
MIRROR & FINGER RESTS

Anterior Sextants

The Dental Mirror



- Types
  - Front surface mirror
  - Concave mirror
  - Plane mirror



- Dental/Mouth mirror

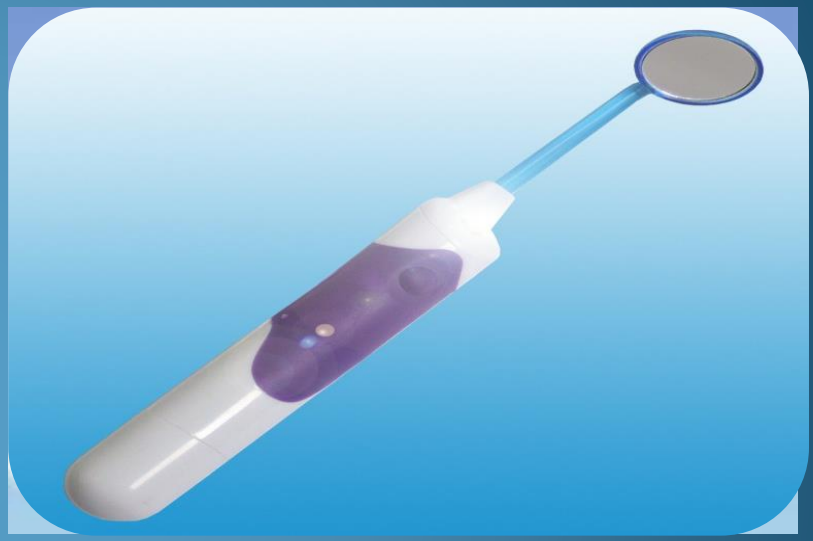
  working-end of a mirror has a reflecting mirrored surface used to view tooth surfaces that cannot be seen directly

Types of Mirror Surfaces

Type	Characteristics
Front surface	Reflecting surface is on front surface of glass Produces clear image & no distortion Good image quality Easily scratched
Concave	Reflecting surface on front surface of mirror lens Magnified image Not recommended as magnification distorts image
Plane (flat surface)	Reflecting surface is on back surface of mirror lens Produces double/ ghost image Not recommended as double image is distracting

Uses of Dental Mirror

1. Indirect Vision
2. Retraction
3. Indirect Illumination
4. Transillumination



Indirect Vision

To view a tooth surface or intraoral structure that cannot be seen directly

Retraction

Use of mirror head to hold the patient's cheek, lip, or tongue so that the clinician can view tooth surface that are otherwise hidden

Clinician's index finger or thumb is also used for retraction, esp lips

- Retraction
- Indirect vision
- Indirect illumination

Indirect illumination

to reflect light
onto a tooth surface
in a dark area of the mouth

Transillumination

Technique of directing light off the mirror surface and through the anterior teeth

Trans=through; Illumination=lighting up

Trans-illumination

A carious lesion appears as shadow when an anterior tooth is illuminated

- Unit light is positioned in such way that light beam is directed perpendicular to facial surfaces of anterior teeth

Transillumination technique

- Mirror is held behind the central incisors so that reflecting surface is parallel to lingual surfaces

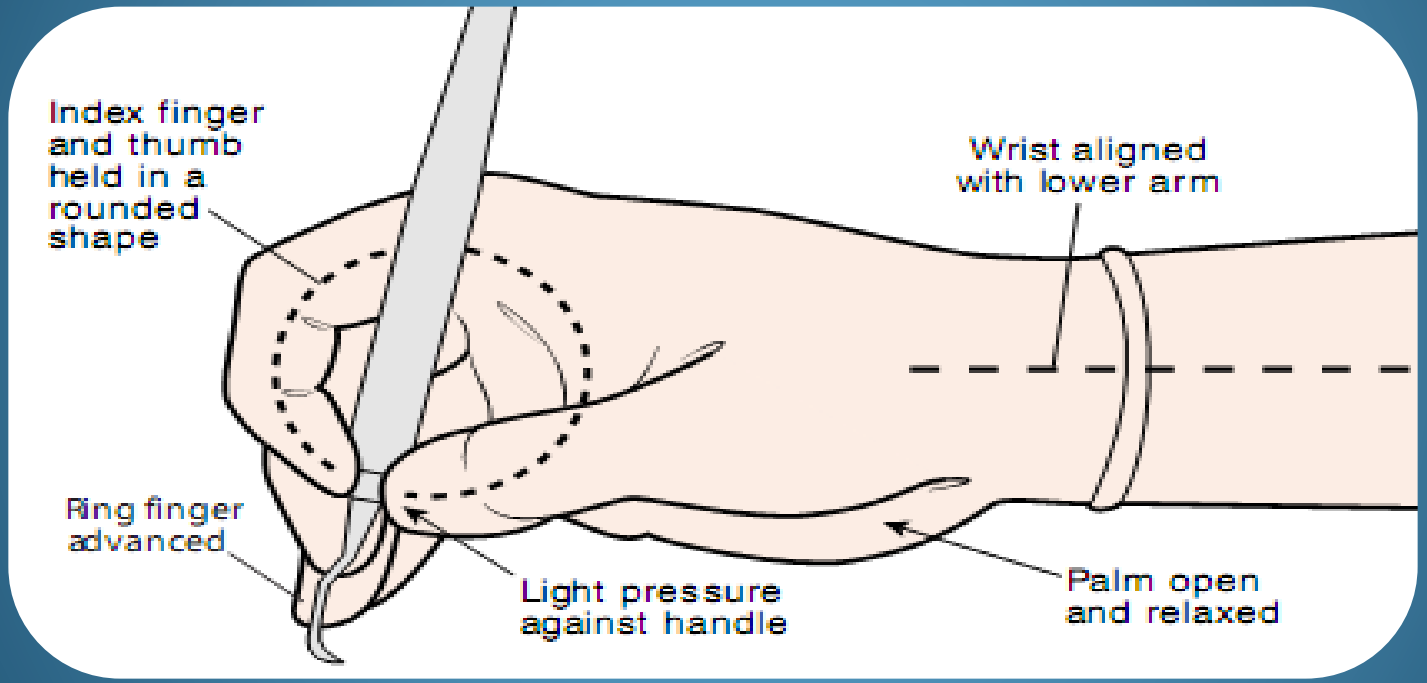
Mirror Use : Precautions

- Warm the reflecting surface against the patient's buccal mucosa
- Ask the patient to breathe through the nose
- Wipe the reflecting surface with a commercial defogging solution
- Wipe the reflecting surface with a gauze square moistened with a mouthwash

Avoid hitting the mirror head against the patient's teeth or resting the outer rim of the mirror head against the patient's gingival tissues

Wrist position for instrumentation

Neutral Wrist position

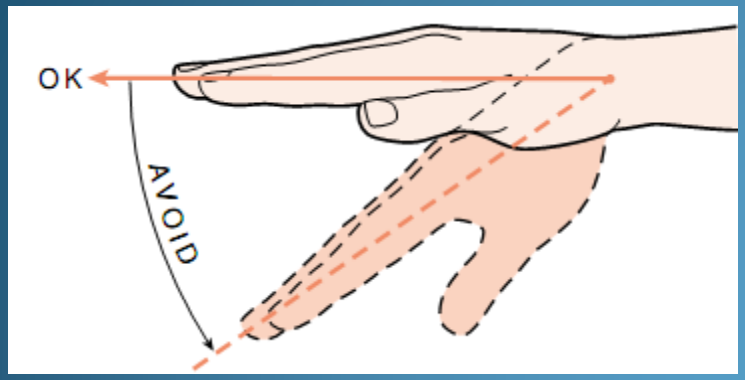


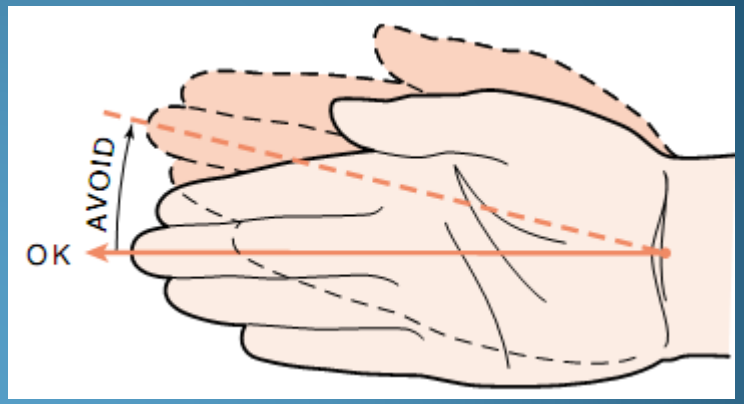
- ideal positioning of the wrist while performing work activities and is associated with decreased MS Injury

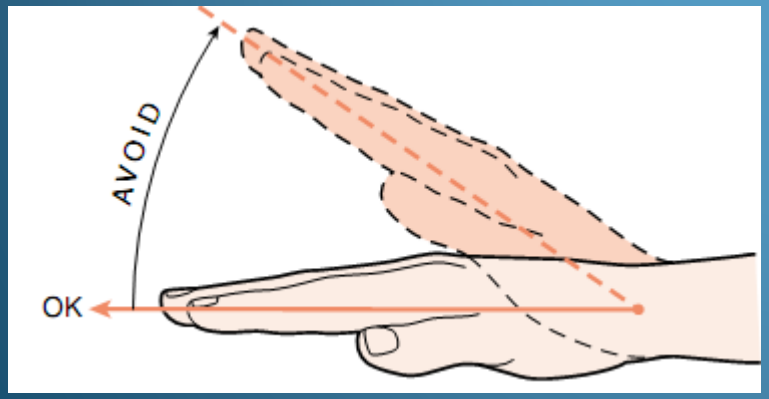
Neutral Hand Position

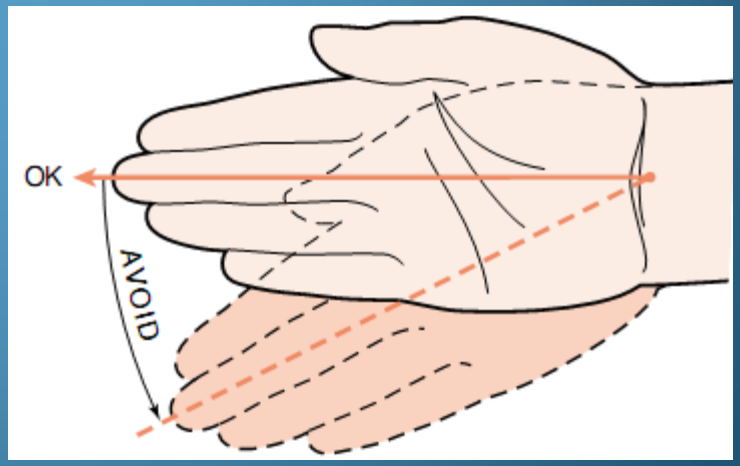
- Wrist aligned with the long axis of lower arm
- Little finger-side of palm rotated slightly downward
- Palm open & relaxed
- Thumb, middle, and index fingers held in rounded shape
- Light finger pressure against instrument handle
- Ring finger ahead of other fingers in grasp

Guideline of Neutral Wrist Position









The Fulcrum

- A finger rest used to stabilize the clinician's hand during periodontal instrumentation

- Functions:
  - "support beam" for hand during instrumentation
  - enables hand and instrument to move as a unit
  - allows precise control of stroke pressure and length during instrumentation

Types of Fulcrums

1. Intraoral
2. Extraoral
3. Advanced

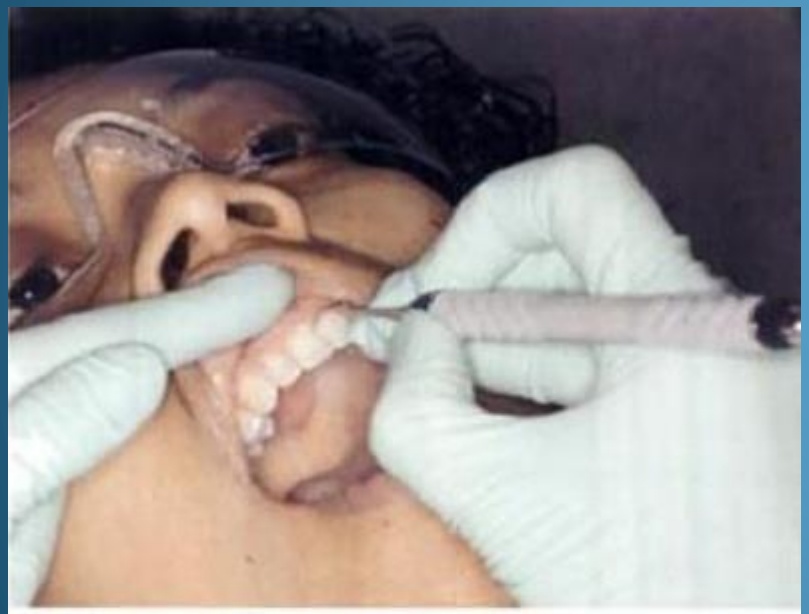
Intraoral

stabilization of clinician's hand by placing pad of ring finger on a tooth close to the working area

Extraoral

outside patient's mouth (chin/cheek)

Intraoral fulcrum



Intraoral Finger Rest. The pad of the ring finger is placed on a tooth near the tooth being instrumented.

Characteristics:

- Provides stable support for the hand
- Enables the hand and instrument to move as a unit
- Facilitates precise stroke pressure against the tooth surface
- Decreases injury due to unexpected patient movement

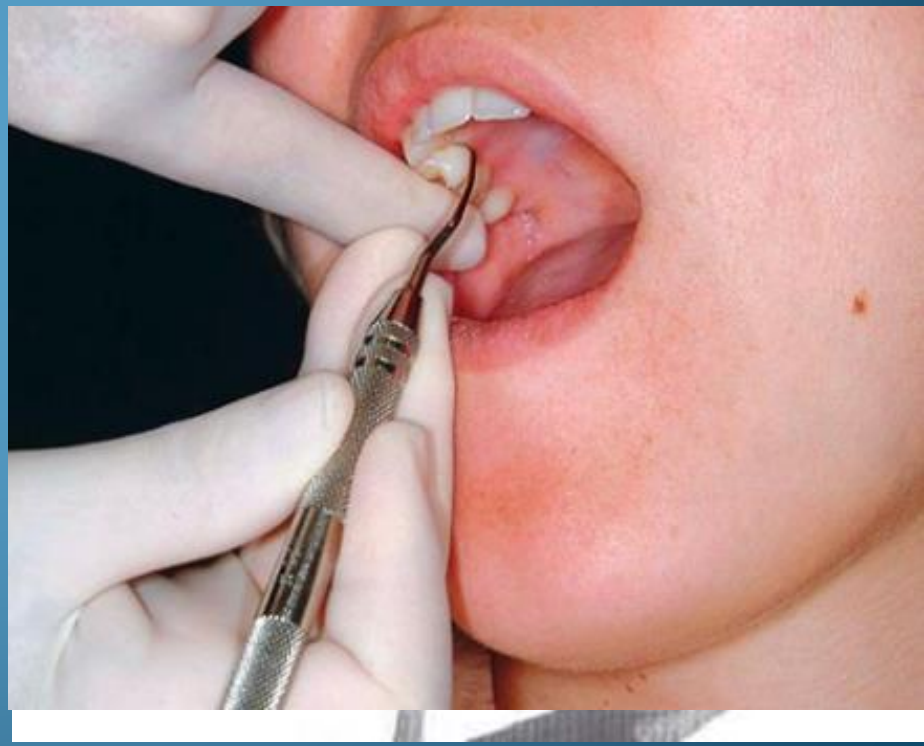
Types of intraoral fulcrums

1. Conventional

2. Opposite -arch

3. Cross - arch

4. Finger-on-finger



 Characteristics of intraoral fulcrum

1. Provides stable support for hand
2. Enables hand and instrument to move as a unit
3. Facilitates precise stroke pressure against tooth surface
4. Decrease likelihood of injury to patient & clinician if patient moves during instrumentation

Summary: Technique for Intraoral Fulcrum

Grasp	Hold the instrument in a modified pen grasp
Fulcrum	Keep ring finger straight, with the tip of the finger supporting the weight of the hand
Location	Finger rest near the tooth being instrumented Depending on the tooth instrumented and size of hand, the finger rest may be 1 to 4 teeth away from the tooth Should be out of “line of fire”
Rest	Finger rest on the same arch Fingertip of fulcrum finger rested on an incisal/occlusal surface or on occlusofacial/occlusolingual line angle of a tooth Teeth are saliva-covered, so more likely to slip if rest on facial/lingual surface Resting on mobile/large carious tooth avoided

Extraoral fulcrum

An extraoral fulcrum is a stabilizing point on patient's chin or cheek

- The clinician is using an extraoral fulcrum, i.e. ring finger rests on the patient's cheek to stabilize the dental mirror in the mouth

Extraoral Finger Rest. The ring finger rests on the patient's chin or cheek to stabilize the mirror.

Types of extraoral fulcrums

1. Palm- up

2. Palm- down

Extraoral fulcrums- not finger rests- instead of tip or pads of fingers, front or back surface of fingers on face

Reinforced type rests

1. Index finger reinforced rest 2. thumb reinforced rest

when precise control and pressure compromised by long distance between fulcrum & working end

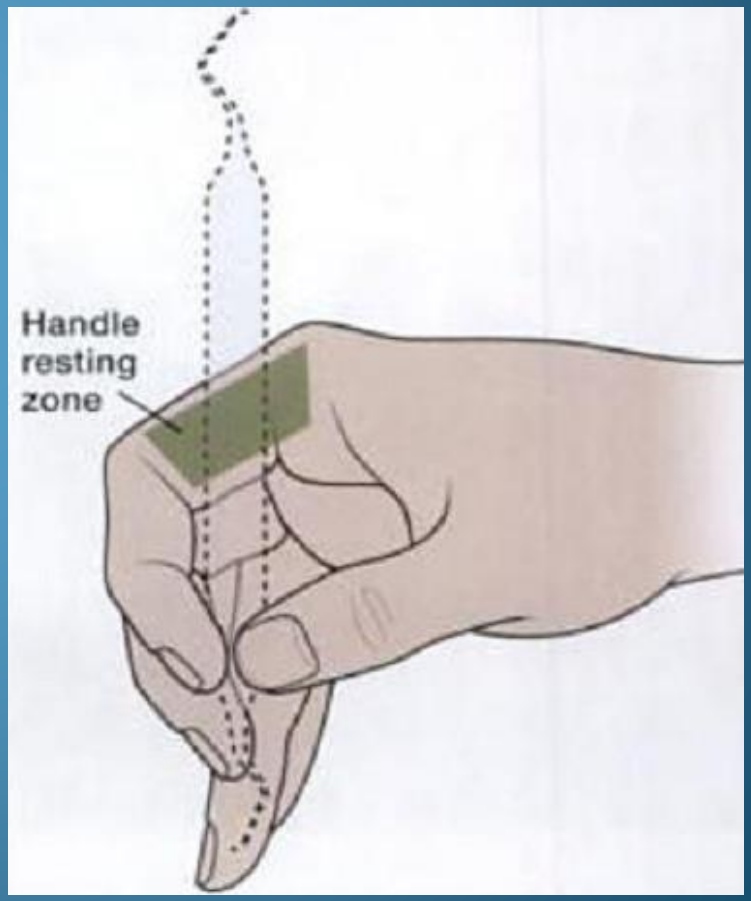
Benefits of reinforced instrumentation

- Allows both hands to work as a unit
- Provides more strength and power
- Enhances lateral pressure of blade on tooth surface
- Provides more stability to working hand
- Helps prevent hand/arm fatigue and injury during scaling

Mandibular Anterior Teeth

Handle Position

- Hold the hand in a palm down position
- Rest handle against index finger in shaded area



Facial aspect - Surface toward

Position overview

- 8 – 9 o’ clock

Retraction

- Retract lip with index finger/ thumb of left hand

Left canine mesial surface

- finger rest on occluso-facial line angle

Right canine distal surface

- Finger rest on incisal edge

Lingual aspect - Surfaces toward

Position overview

- 8 – 9 o’ clock

Mirror

- Retract tongue
- Indirect vision

Left canine mesial surface

- Finger rest on occluso-facial line angle

Right canine distal surface

- Finger rest on incisal edge

Facial aspect – Surfaces away

Position overview

- 12 o’ clock

Retraction

- Lip retracted with index finger/ thumb

Right canine mesial surface

- finger rest on occluso-facial line angle

Left canine distal surface

- Finger rest on incisal edge

Lingual aspect – Surfaces away

Position overview

- 12 o’ clock

Mirror

- Retract tongue
- Indirect vision

Right canine mesial surface

- finger rest on occluso-facial line angle

Left canine distal surface

- Finger rest on incisal edge

Anterior treatment areas - Mandible

Treatment area	Clock position	Patient's head
Facial surfaces Toward	8 – 9:00	Slightly toward Chin DOWN
Lingual surfaces Toward		
Facial surfaces Away	12:00	
Lingual surfaces Away		

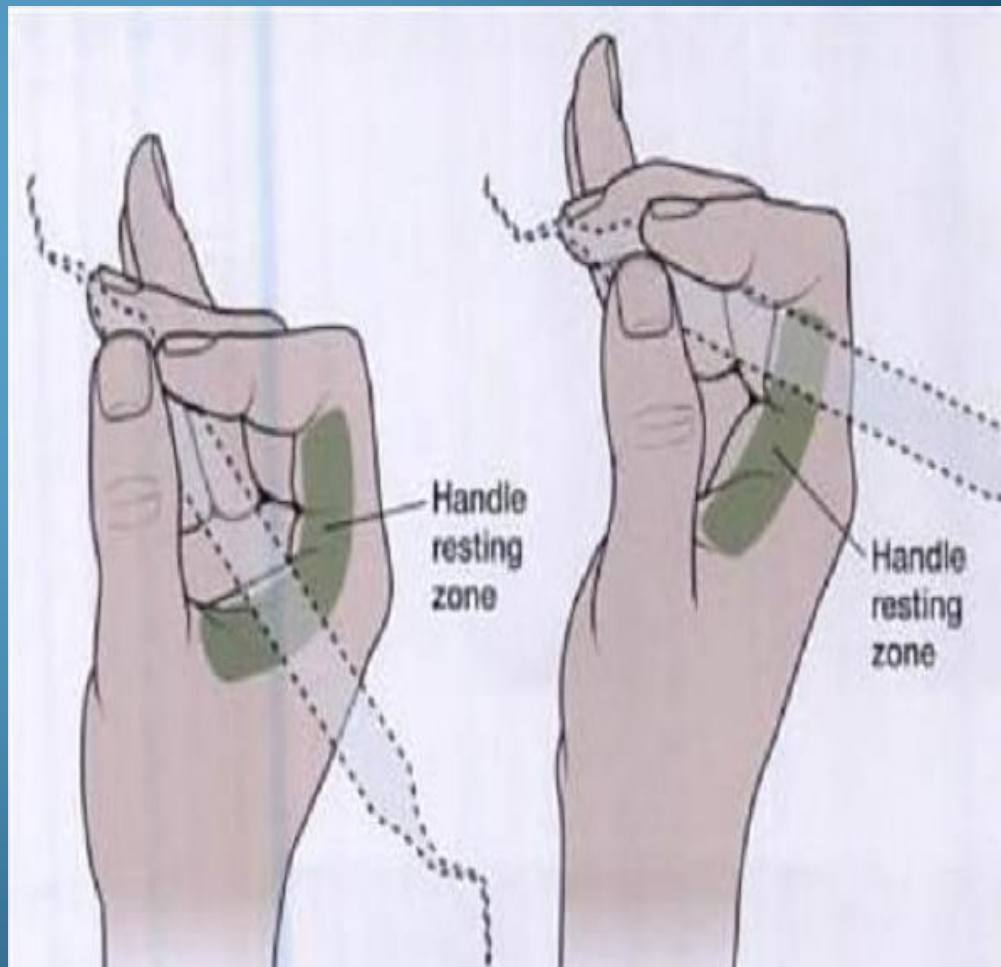
Anterior treatment areas – Mandible left handed

Treatment area	Clock position	Patient's head
Facial surfaces Toward NDH	4 – 3:00	Slightly toward Chin DOWN
Lingual surfaces Toward		
Facial surfaces Away	12:00	
Lingual surfaces Away		

Maxillary Anterior Teeth

Handle Position

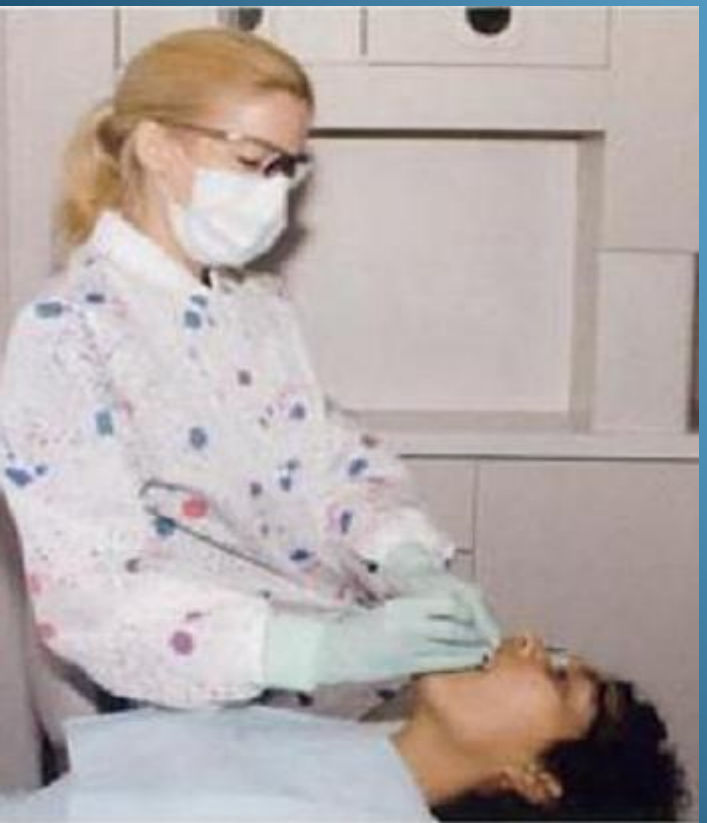
- Hold the hand in a palm up position
- Rest handle against index finger in shaded area



Facial aspect - Surfaces toward

Position overview

- 8 – 9 o’ clock



Retraction

- Retract lip with index finger/ thumb of left hand

Left canine mesial surface

- finger rest on occluso-facial line angle

Right canine distal surface

- Finger rest on incisal edge

Lingual aspect - Surfaces toward

Position overview

- 8 – 9 o’ clock

Mirror

- Retract tongue
- Indirect vision
- Indirect illumination

Left canine mesial surface

- finger rest on occluso-facial line angle

Right canine distal surface

- Finger rest on incisal edge

Facial aspect – Surfaces away

Position overview

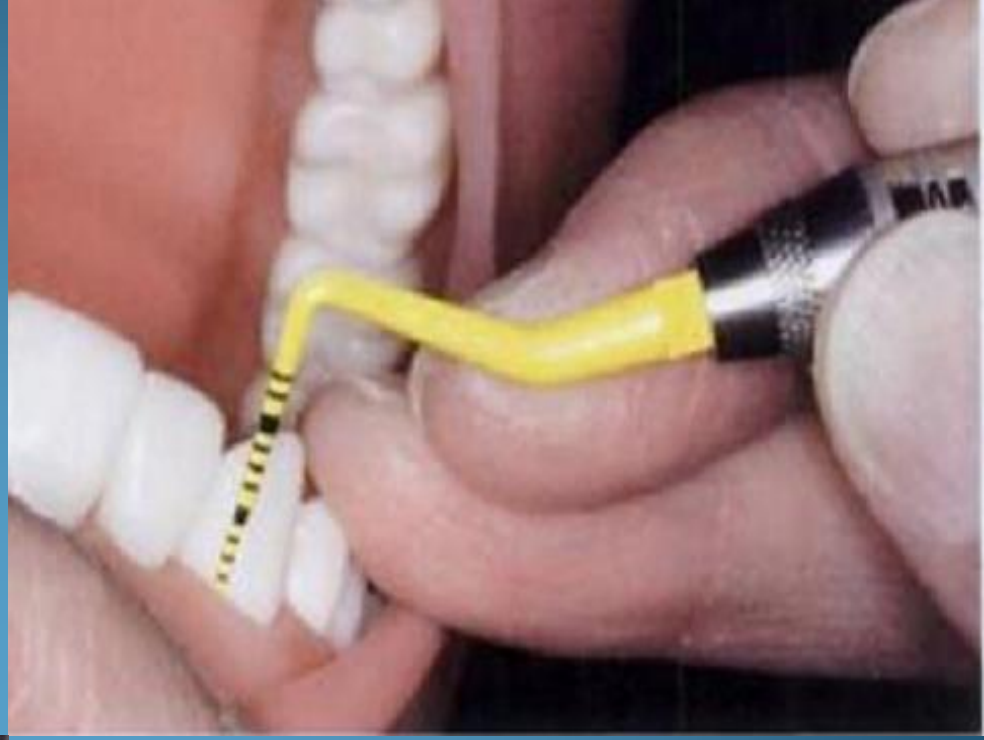
- 12 o’ clock

Retraction

- Lip retracted with index finger/ thumb

Right canine mesial surface

- finger rest on occlusal surface
- Should be able to see underside of middle & ring fingers



Left canine distal surface

- Finger rest on incisal edge

Lingual aspect – Surfaces away

Position overview

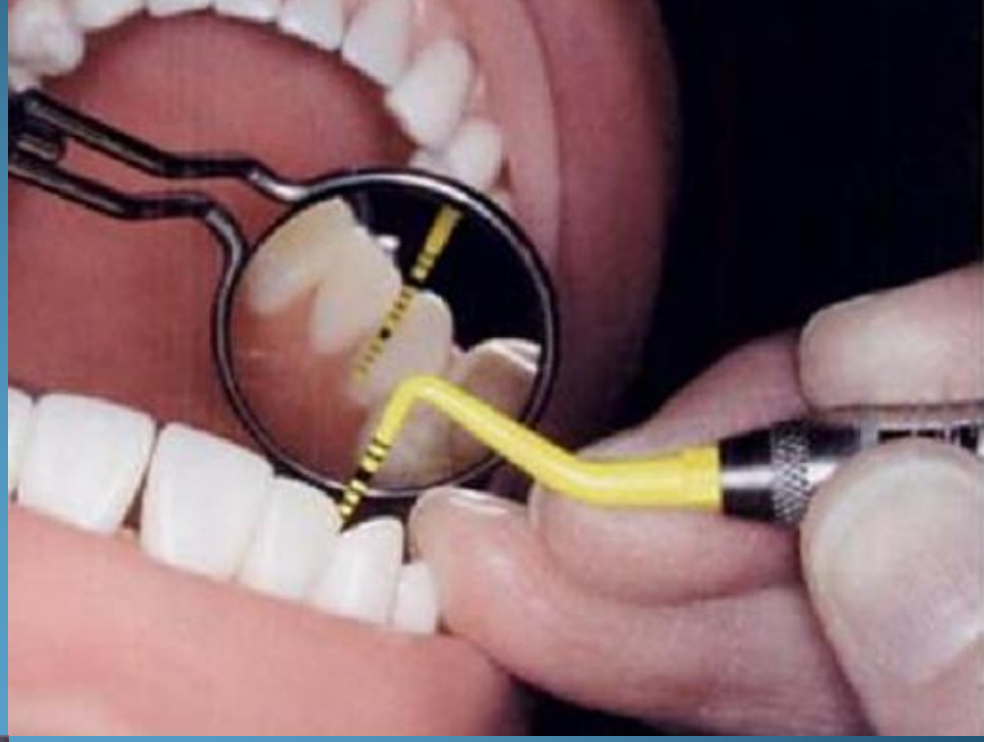
- 12 o’ clock

Mirror

- Mirror head positioned to visualize lingual surfaces

Right canine mesial surface

- finger rest on occlusal surface
- Should be able to see underside of middle & ring fingers



Left canine distal surface

- Finger rest on incisal edge

Anterior treatment areas - Maxilla

Treatment area	Clock position	Patient's head
Facial surfaces Toward	8 – 9:00	Slightly toward Chin UP
Lingual surfaces Toward		
Facial surfaces Away	12:00	
Lingual surfaces Away		

Anterior treatment areas – Maxilla left handed

Treatment area	Clock position	Patient's head
Facial surfaces Toward NDH	4 – 3:00	Slightly toward Chin UP
Lingual surfaces Toward		
Facial surfaces Away	12:00	
Lingual surfaces Away		

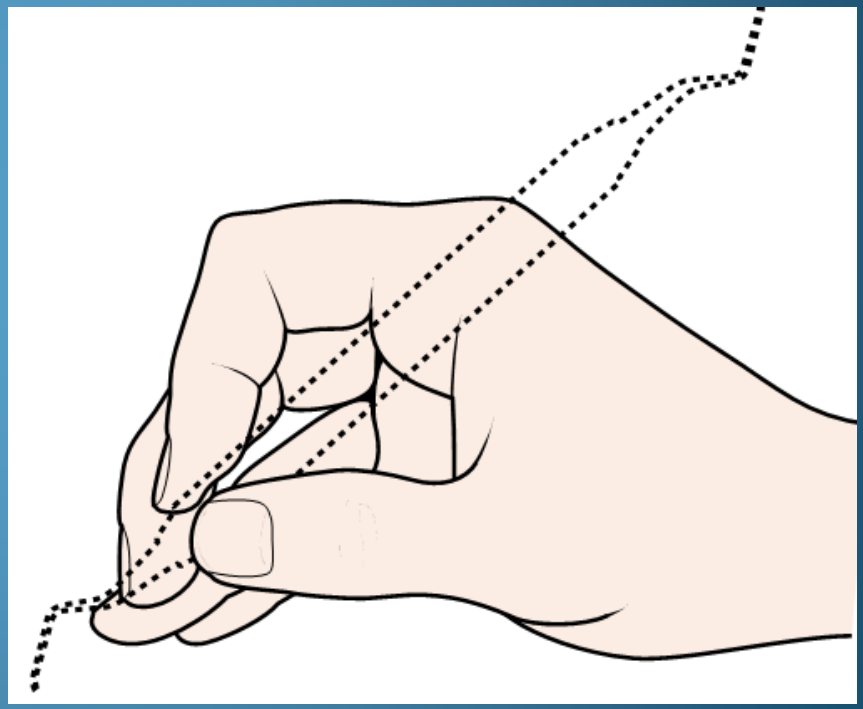
MIRROR & FINGER RESTS

Mandibular Posterior Sextants

Mandibular Posterior Teeth

Handle Position

- Hold the hand in a palm down position
- Rest handle against index finger somewhere betn 2nd & 3rd knuckles



Mandibular Right Posterior Sextant facial aspect

Position overview

- 9 o’ clock

Mirror

- Retract buccal mucosa
- Indirect vision esp distal surfaces

Second molar Facial aspect

- finger rest on occlusal surface

First premolar Facial aspect

- Finger rest on incisal surface of anteriors

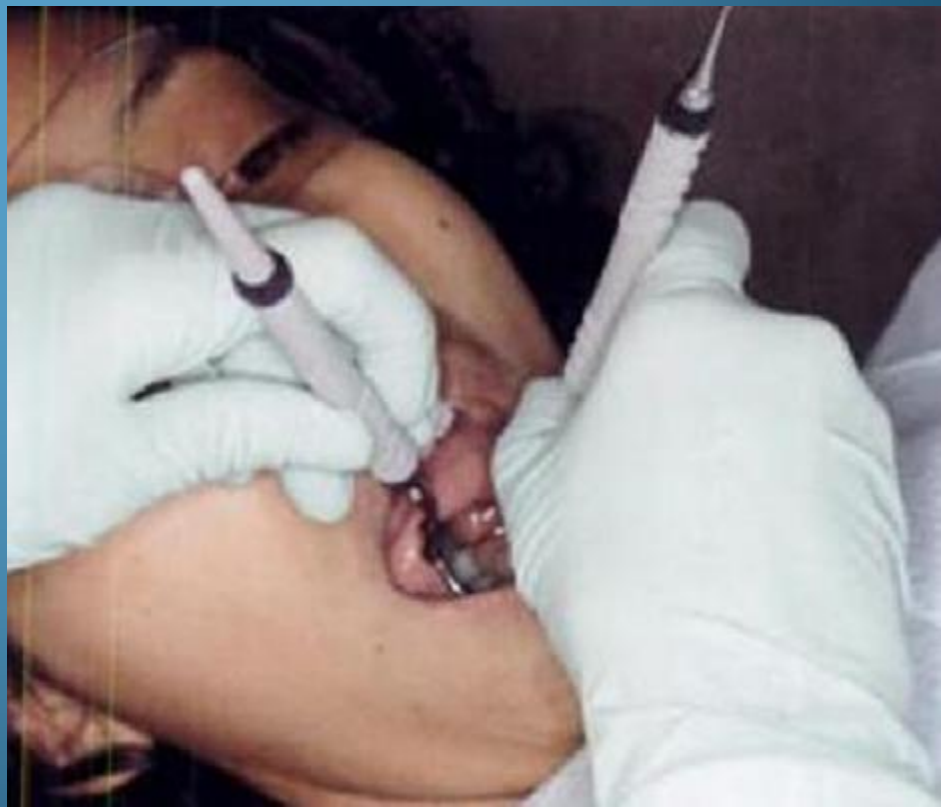
Mandibular Left Posterior Sextant lingual aspect

Position overview

- 9 o’ clock

Mirror

- Tongue retraction
- Indirect vision



Second molar Lingual aspe

- finger rest on occluso-fa line angle

First premolar Lingual aspect

- Finger rest on incisal edge of anteriors

Mandibular Left Posterior Sextant facial aspect

Position overview

- 10 – 11:00

Mirror

- Retract buccal mucosa

Second molar Facial aspect

- finger rest on occluso-facial line angle

First premolar Facial aspect

- Finger rest on incisal edge of anteriors

Mandibular Right Posterior Sextant lingual aspect

Position overview

- 10 – 11:00

Mirror

- Tongue retraction
- Indirect vision

Second molar Lingual aspect

- finger rest on occlusal surface

First premolar Lingual aspect

- Finger rest on incisal edge of anteriors

Posterior treatment areas - Mandible

Treatment area	Clock position	Patient's head
Right Facial Towards	9:00	Straight or slightly away Chin DOWN
Left Lingual Towards		
Right Lingual Away	10 - 11:00	Toward CHIN DOWN
Left Facial Away		

Posterior treatment areas – Mandible left handed

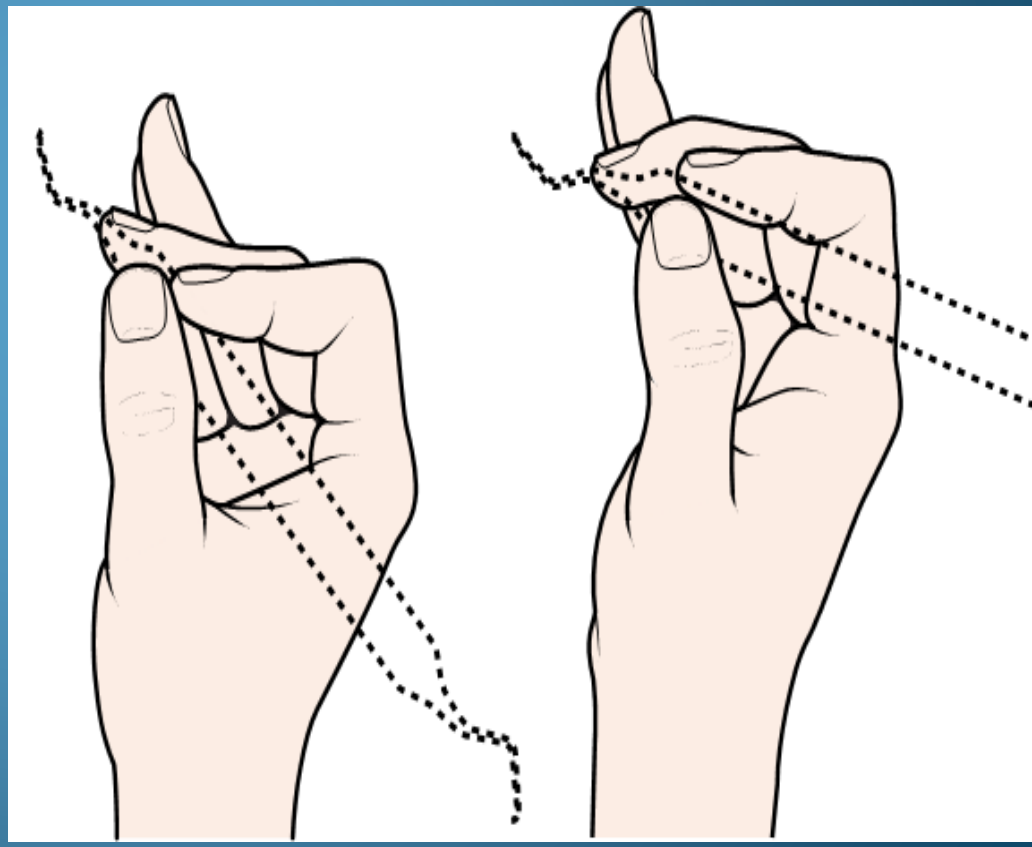
Treatment area	Clock position	Patient's head
Left Facial Towards	3:00	Straight or slightly away Chin DOWN
Right Lingual Towards		
Left Lingual Away	2 - 1:00	Toward CHIN DOWN
Right Facial Away		

MIRROR & FINGER RESTS

Maxillary Posterior Sextants

Maxillary Posterior Teeth
Handle Position

- Hold the hand in a palm-up position
- Rest handle against index finger somewhere behind 2nd knuckle to the “V” of hand



Maxillary Right Posterior Sextant facial aspect

Position overview

- 9 o’ clock

Mirror

- Retract buccal mucosa
- Indirect vision esp distal surfaces

Second molar Facial aspect

- finger rest on occlusal surface

First premolar Facial aspect

- Finger rest on incisal edge

Maxillary Left Posterior Sextant lingual aspect

Position overview

- 9 o’ clock

Mirror

- View distal surfaces

Second molar Lingual aspect

- finger rest on occluso-facial line angle

First premolar Lingual aspect

- Finger rest on occlusofacial line angle or incisal edge

Maxillary Left Posterior Sextant
Facial aspect

Position overview

- 10 – 11:00

Mirror

- Retract buccal mucosa down & away from teeth

Second molar Facial aspect

- finger rest on occlusal surface
- Can see underside of middle & ring fingers

First premolar Facial aspect

- Finger rest on incisal edge

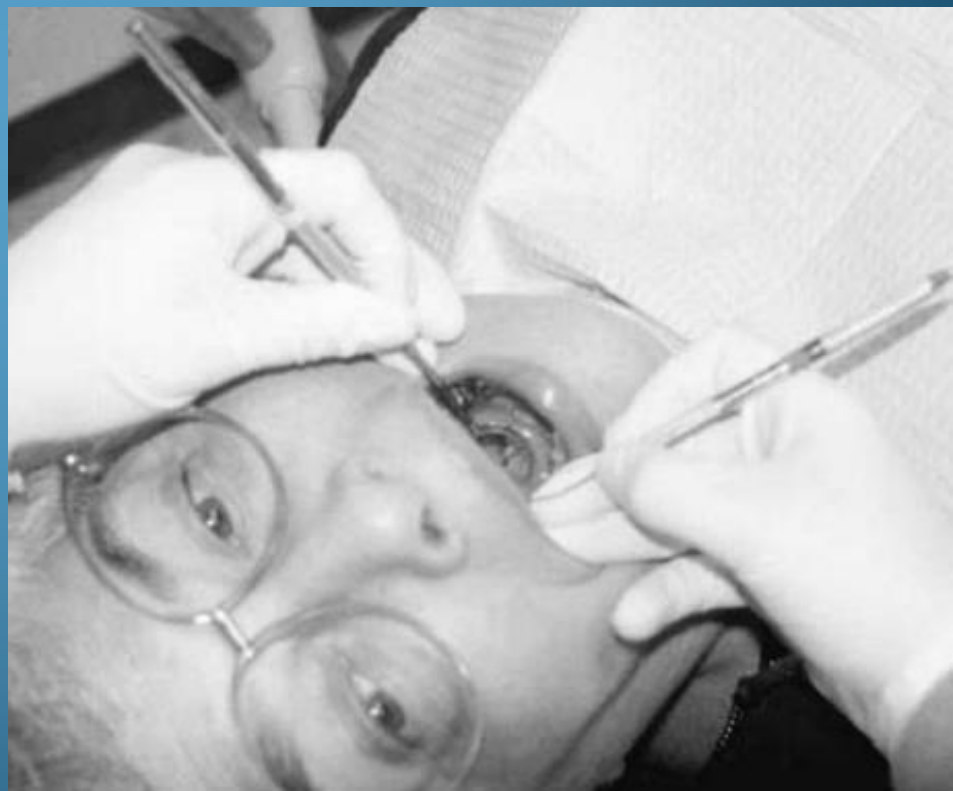
Maxillary Right Posterior Sextant
Lingual aspect

Position overview

- 10 – 11:00

Mirror

- Indirect Vision



Second molar Lingual aspect

- Finger rest on occlusal surface

First premolar Lingual aspect

- Finger rest on incisal edge

Posterior treatment areas - Maxilla

Treatment area	Clock position	Patient's head
Right Facial Towards	9:00	Straight or slightly away Chin UP
Left Lingual Towards		
Right Lingual Away	10 - 11:00	Toward CHIN UP
Left Facial Away		

Posterior treatment areas – Maxilla left handed

Treatment area	Clock position	Patient's head
Left Facial Towards	3:00	Straight or slightly away Chin DOWN
Right Lingual Towards		
Left Lingual Away	2 - 1:00	Toward CHIN DOWN
Right Facial Away		

Sequence for Establishing a Finger Rest

1.
   - ME
   - Assume a clock position for treatment area
2.
   - MY PATIENT
   - Establish patient chair & head position
3.
   - MY EQUIPMENT
   - Adjust unit light; pause & check clinician, patient & equipment position

4

- MY NONDOMINANT HAND
- Grasp mirror & establish proper position

5

- MY DOMINANT HAND
- Grasp the instrument & pause to evaluate finger placement

6

- MY FINGER REST
- Establish suitable finger rest for the first tooth to be treated

At last : Pause to evaluate finger rest

- Is the tip of ring finger on a secure tooth surface?
- Is ring finger straight, acting as support beam?
- Is my finger placement in grasp still correct?

Identified disorders, occupational risk factors and symptoms		
Disorders	**Occupational risk factors**	**Symptoms**
Tendonitis/tenosynovitis	Repetitive wrist motions Repetitive shoulder motions Sustained hyper extension of arms Prolonged load on shoulders	Pain, weakness, swelling, burning sensation or dull ache over affected area
Epicondylitis (elbow tendonitis)	Repeated or forceful rotation of the forearm and bending of the wrist at the same time	Same symptoms as tendonitis
Carpal tunnel syndrome	Repetitive wrist motions	Pain, numbness, tingling, burning sensations, wasting of muscles at base of thumb, dry palm
DeQuervain's disease	Repetitive hand twisting and forceful gripping	Pain at the base of thumb
Thoracic outlet syndrome	Prolonged shoulder flexion Extending arms above shoulder height Carrying loads on the shoulder	Pain, numbness, swelling of the hands
Tension neck syndrome	Prolonged restricted posture	Pain

Conclusion

- Periodontal therapy is directly affected by clinician's knowledge of suitable mirror and finger rests in various anterior and posterior sextants of both maxillary and mandibular arches

- A good finger rest prevents injury and laceration of the gingiva and surrounding tissues by poorly controlled instruments

- Not only appropriate mirror and finger rests but also proper clinician, patient, assistant and equipment positions are vital to effective instrumentation for undue fatigue and development of MSD to the operator

References

1. Jill Shiffer Nield-Gehrig, Fundamentals of periodontal instrumentation & advanced root instrumentation, 6th edition
2. Carranza's Clinical Periodontology, 9th edition
3. H. Dong, et al, The effects of finger rest positions on hand muscle load and pinch force in simulated dental hygiene work; Journal of Dental Education, Apr 2005
4. Gordon L. Pattison et al, Periodontal Instrumentation
5. Anna M. Pattison, Dimensions of Dental Hygiene, Oct 2004
6. Diane Millar, Reinforced Periodontal Instrumentation and Ergonomics for Dental Care Provider, Apr 2007

Thank You